

Principal contribution terms of the PolyBench kernels

The full reuse-distance term list per kernel: distances, populations, orders, and what each term contributes to misses and data movement

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How to read this document

The analyzer bins every warm access (every reuse) of a kernel by its **reuse distance** (RD): the number of distinct cache lines touched in the reuse window, at block granularity 8 doubles = one 64-byte line. Arrays follow the padded layout of the paper’s evaluation: every row starts on a fresh line (only the innermost subscript is blocked). Each bin is a **principal contribution term** — a pair

$$(V(n), M(n))$$

of a **distance** $V(n)$ (cache lines) and a **population** $M(n)$ (number of accesses), both exact functions of the problem size n (all program parameters bound to n ; convolution fixes its filter extent at 9). A term states, completely and exactly:

- **Miss contribution.** The term’s $M(n)$ accesses hit if the cache holds at least $V(n)$ lines and miss otherwise: the term contributes $M(n)$ misses on the cache range $C < V(n)$ and none above. Its *boundary* is $C^* = V(n)$ lines (64 $V(n)$ bytes); its *portion* is $M(n)/A(n)$ of all accesses. The kernel’s entire miss-ratio curve is the sum of these step contributions — nothing else.
- **Data-movement (DMD) contribution.** Under the square-root cost model the term contributes $M(n)\sqrt{V(n)}$, of order $n^{\deg M + \deg V/2}$; the kernel’s DMD spectrum is the ordered list of these orders with coefficients.

Two kinds appear. A **level** has one distance for its whole population (split per residue class of an iterator mod 8 where the line boundary makes classes differ — classes with identical polynomials are merged). A **ramp** (family) spans a distance *range* $[V_{\min}(n), V_{\max}(n)]$ that grows with a loop iterator (triangular kernels): it contributes misses that taper off as the cache grows through the range; its population and range bounds are exact polynomials, and its order is $\deg M + \deg V_{\max}/2$.

Why a term with vanishing portion still matters. A population $\Theta(n^2)$ term in a $\Theta(n^3)$ -access kernel has portion $\Theta(1/n)$, so the *total* miss ratio above the lower boundaries thins as n grows. The term itself does not: on its active cache range it is the entire miss traffic, its absolute misses grow as n^2 , and its boundary is exactly the cliff a cache-size sweep measures. The terms — not any single aggregate — are the objects to reason with.

Trust gate. Every kernel below ran with **exact Barvinok counting** (method = exact): bin populations conserve to the integer ($\$ \Sigma M = \$$ warm accesses, verified at two sizes per kernel and shown in each header), and the reconstructed distance histograms match a brute-force trace interpreter bin-for-bin at aligned and unaligned sizes (30/30 spot checks). Polynomials are exact on the anchor residue class $n \equiv 0 \pmod{h}$ ($h = 8$ or 16 , per kernel); other classes differ only

in lower-order boundary constants. Ramp coefficients are evaluated at the anchor sizes and quoted to three digits; ramp *orders* come from the exact degrees. Populations below 10^{-6} of the accesses are aggregated into their order row but omitted from the tables.

Suite overview

Access order a , DMD order d , headroom $d - a$; top spectrum entries. Single-shot / infinite-repeat per kernel.

kernel	model	a	d	headroom	leading terms
heat-3d	single	4	5.5	+1.5	$0.75 \cdot n^{5.5}, 1.06 \cdot n^5$
2mm	inf	3	4	+1	$0.0884 \cdot n^4, 2.12 \cdot n^{3.5}$
2mm	single	3	4	+1	$0.0884 \cdot n^4, 2.12 \cdot n^{3.5}$
3mm	inf	3	4	+1	$0.133 \cdot n^4, 3.18 \cdot n^{3.5}$
3mm	single	3	4	+1	$0.133 \cdot n^4, 3.18 \cdot n^{3.5}$
atax	inf	2	3	+1	$0.0442 \cdot n^3, 0.22 \cdot n^{2.5}$
bicg	inf	2	3	+1	$0.0442 \cdot n^3, 0.153 \cdot n^{2.5}$
cholesky	inf	3	4	+1	$0.00347 \cdot n^4, 0.00607 \cdot n^{3.5}$
cholesky	single	3	4	+1	$0.00347 \cdot n^4, 0.00607 \cdot n^{3.5}$
convolution	inf	2	3	+1	$0.125 \cdot n^3, 1.12 \cdot n^{2.5}$
correlation	inf	3	4	+1	$0.0153 \cdot n^4, 1.33 \cdot n^{3.5}$
correlation	single	3	4	+1	$0.0161 \cdot n^4, 1.33 \cdot n^{3.5}$
covariance	inf	3	4	+1	$0.0161 \cdot n^4, 1.33 \cdot n^{3.5}$
covariance	single	3	4	+1	$0.0161 \cdot n^4, 1.33 \cdot n^{3.5}$
doitgen	inf	4	5	+1	$0.0442 \cdot n^5, 1.1 \cdot n^{4.5}$
doitgen	single	4	5	+1	$0.0442 \cdot n^5, 1.06 \cdot n^{4.5}$
fdtd	inf	3	4	+1	$0.617 \cdot n^4, 0.253 \cdot n^{3.5}$
fdtd	single	3	4	+1	$0.616 \cdot n^4, 0.253 \cdot n^{3.5}$
floyd_warshall	inf	3	4	+1	$0.0442 \cdot n^4, 0.0625 \cdot n^{3.5}$
floyd_warshall	single	3	4	+1	$0.0442 \cdot n^4, 0.0625 \cdot n^{3.5}$
gemm	inf	3	4	+1	$0.0442 \cdot n^4, 0.0625 \cdot n^{3.5}$
gemm	single	3	4	+1	$0.0442 \cdot n^4, 0.0625 \cdot n^{3.5}$
gemver	inf	2	3	+1	$0.114 \cdot n^3, 1.28 \cdot n^{2.5}$
gemver	single	2	3	+1	$0.0722 \cdot n^3, 1.28 \cdot n^{2.5}$
gesummv	inf	2	3	+1	$0.125 \cdot n^3, 0.0765 \cdot n^{2.5}$
gramschmidt	inf	3	4	+1	$0.016 \cdot n^4, 2.74 \cdot n^{3.5}$
gramschmidt	single	3	4	+1	$0.0161 \cdot n^4, 2.74 \cdot n^{3.5}$
jacobi-2d	inf	3	4	+1	$0.5 \cdot n^4, 0.707 \cdot n^{3.5}$
jacobi-2d	single	3	4	+1	$0.5 \cdot n^4, 0.707 \cdot n^{3.5}$
lu	inf	3	4	+1	$0.00643 \cdot n^4, 0.22 \cdot n^{3.5}$
lu	single	3	4	+1	$0.00643 \cdot n^4, 0.221 \cdot n^{3.5}$
lu_decomp	inf	3	4	+1	$0.0101 \cdot n^4, 0.0166 \cdot n^{3.5}$
lu_decomp	single	3	4	+1	$0.01 \cdot n^4, 0.0166 \cdot n^{3.5}$
mvt	inf	2	3	+1	$0.0721 \cdot n^3, 1.12 \cdot n^{2.5}$
mvt	single	2	3	+1	$0.036 \cdot n^3, 1.12 \cdot n^{2.5}$
seidel-2d	inf	3	4	+1	$0.0884 \cdot n^4, 0.306 \cdot n^{3.5}$
seidel-2d	single	3	4	+1	$0.0884 \cdot n^4, 0.306 \cdot n^{3.5}$
symm	inf	3	4	+1	$0.0466 \cdot n^4, 1.08 \cdot n^{3.5}$
symm	single	3	4	+1	$0.0466 \cdot n^4, 1.08 \cdot n^{3.5}$
syr2k	inf	3	4	+1	$0.0466 \cdot n^4, 1.08 \cdot n^{3.5}$
syr2k	single	3	4	+1	$0.0466 \cdot n^4, 1.08 \cdot n^{3.5}$
syrk	inf	3	4	+1	$0.0166 \cdot n^4, 0.416 \cdot n^{3.5}$
syrk	single	3	4	+1	$0.0166 \cdot n^4, 0.416 \cdot n^{3.5}$

kernel	model	a	d	headroom	leading terms
trisolve	inf	2	3	+1	$0.0156 \cdot n^3, 0.0226 \cdot n^{2.5}$
trmm	inf	3	4	+1	$0.0169 \cdot n^4, 1.05 \cdot n^{3.5}$
trmm	single	3	4	+1	$0.0169 \cdot n^4, 1.05 \cdot n^{3.5}$
atax	single	2	2.5	+0.5	$0.219 \cdot n^{2.5}, 11.7 \cdot n^2$
bicg	single	2	2.5	+0.5	$0.153 \cdot n^{2.5}, 12.8 \cdot n^2$
convolution	single	2	2.5	+0.5	$1.12 \cdot n^{2.5}, 394 \cdot n^2$
gesummv	single	2	2.5	+0.5	$0.0765 \cdot n^{2.5}, 15.1 \cdot n^2$
jacobi-1d	inf	2	2.5	+0.5	$0.25 \cdot n^{2.5}, 9.4 \cdot n^2$
jacobi-1d	single	2	2.5	+0.5	$0.25 \cdot n^{2.5}, 9.4 \cdot n^2$
trisolve	single	2	2.5	+0.5	$0.0226 \cdot n^{2.5}, 2.88 \cdot n^2$

Per-kernel principal terms

2mm — infinite-repeat [exact]

Accesses $A(n) = 8 \cdot n^3 + 3 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0884 \cdot n^4 + 2.12 \cdot n^{3.5} + 10.7 \cdot n^3 + 7.95 \cdot n^{2.5} + 13.2 \cdot n^2 + 9.33 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.0137	read E[i6, i5] (i0=0, i4=0); read E[i6, i5] (i0=0)
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.0137	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.00195	read E[i6, i5] (i0=0, i4=0); read E[i6, i5] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.00195	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0957	read E[i6, i5] (i0=0, i4=0); read E[i6, i5] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0957	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0137	read E[i6, i5] (i0=0, i4=0); read E[i6, i5] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0137	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.0137	read A[i4, i6] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.0137	read B[i1, i3] (i0=0, i1=0, i2=0); read B[i1, i3] (i0=0)
$n^{3.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.00195	read A[i4, i6] (i0=0)
$n^{3.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.00195	read B[i1, i3] (i0=0, i1=0, i2=0); read B[i1, i3] (i0=0)
n^3	3.03	level	3	$(7/4) \cdot n^3$	0.219	read B[i1, i3] (i0=0); read A[i4, i6] (i0=0)
n^3	3.03	level	3	$(7/4) \cdot n^3$	0.219	write A[i1, i2] (i0=0); write D[i4, i5] (i0=0)
n^3	1.75	level	1	$(7/4) \cdot n^3 + (21/8) \cdot n^2$	0.219	write A[i1, i2] (i0=0); read A[i1, i2] (i0=0) (+3)
n^3	0.433	level	3	$(1/4) \cdot n^3$	0.0312	write A[i1, i2] (i0=0); write D[i4, i5] (i0=0, i6=0) (+1)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.109/n	read E[i6, i5] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.109/n	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.109/n	read E[i6, i5] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.109/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^3	0.25	level	1	$(1/4) \cdot n^3 + (1/8) \cdot n^2$	0.0312	read A[i1, i2] (i0=0); write D[i4, i5] (i0=0) (+2)
n^3	0.0988	level	$(5/8) \cdot n^2$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0156/n	read B[i1, i3] (i0=0, i1=0, i2=0); read B[i1, i3] (i0=0, i2=0)
n^3	0.0988	level	$(5/8) \cdot n^2$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0156/n	read D[i4, i5] (i0=0, i4=0); read D[i4, i5] (i0=0)
n^3	0.0884	level	$(1/2) \cdot n^2 + 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0156/n	read A[i4, i6] (i0=0, i5=0)
n^3	0.0884	level	$(1/2) \cdot n^2 + 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0156/n	write A[i1, i2] (i0=0)
n^3	0.0773	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0137/n	read E[i6, i5] (i0=0, i4=0)
n^3	0.0773	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0137/n	read C[i3, i2] (i0=0, i1=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0156/n	read E[i6, i5] (i0=0, i4=0); read E[i6, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read E[i6, i5] (i0=0, i4=0); read E[i6, i5] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0156/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read E[i6, i5] (i0=0, i4=0, i6=0); read E[i6, i5] (i0=0, i6=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read C[i3, i2] (i0=0, i1=0, i3=0); read C[i3, i2] (i0=0, i3=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read E[i6, i5] (i0=0, i5=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read E[i6, i5] (i0=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.011	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00195/n$	read E[i6, i5] (i0=0, i4=0)
n^3	0.011	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00195/n$	read C[i3, i2] (i0=0, i1=0)
$n^{2.5}$	1.86	level	$(9/8) \cdot n + 1$	$(7/4) \cdot n^2$	$0.219/n$	read C[i3, i2] (i0=0); read E[i6, i5] (i0=0, i4=0) (+1)
$n^{2.5}$	1.86	level	$(9/8) \cdot n - 6$	$(7/4) \cdot n^2$	$0.219/n$	read B[i1, i3] (i0=0); read A[i4, i6] (i0=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	$0.109/n$	read E[i6, i5] (i0=0, i4=0, i6=0); read E[i6, i5] (i0=0, i6=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	$0.109/n$	read C[i3, i2] (i0=0, i1=0, i3=0); read C[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.109/n$	write A[i1, i2] (i0=0); read A[i4, i6] (i0=0, i6=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.109/n$	read B[i1, i3] (i0=0, i1=0, i2=0, i3=0); read B[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.265	level	$(9/8) \cdot n - 5$	$(1/4) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read B[i1, i3] (i0=0); read A[i4, i6] (i0=0)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0156/n$	read A[i4, i6] (i0=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0156/n$	read B[i1, i3] (i0=0, i1=0, i2=0, i3=0); read B[i1, i3] (i0=0, i3=0)
n^2	1.58	level	$(5/8) \cdot n^2$	$2 \cdot n$	$0.25 \cdot n^{-2}$	read B[i1, i3] (i0=0); read D[i4, i5] (i0=0, i4=0, i5=0) (+1)
n^2	0.791	level	$(5/8) \cdot n^2$	n	$0.125 \cdot n^{-2}$	read B[i1, i3] (i0=0, i1=0, i2=0); read B[i1, i3] (i0=0, i2=0)
n^2	0.791	level	$(5/8) \cdot n^2$	n	$0.125 \cdot n^{-2}$	read D[i4, i5] (i0=0, i4=0); read D[i4, i5] (i0=0)
n^2	0.707	level	$(1/2) \cdot n^2 + (-1/8) \cdot n + 1$	$n - 2$	$0.125 \cdot n^{-2}$	read A[i4, i6] (i0=0, i5=0)
n^2	0.707	level	$(1/2) \cdot n^2 + (-1/8) \cdot n + 1$	$n - 2$	$0.125 \cdot n^{-2}$	write A[i1, i2] (i0=0, i2=0)
n^2	0.707	level	$(1/2) \cdot n^2 + 1$	$n - 2$	$0.125 \cdot n^{-2}$	write A[i1, i2] (i0=0)
n^2	0.707	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	n	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0, i5=0, i6=0); read E[i6, i5] (i0=0, i4=0, i5=0)
n^2	0.707	level	$(1/2) \cdot n^2 + 1$	$n - 2$	$0.125 \cdot n^{-2}$	read A[i4, i6] (i0=0, i5=0, i6=0)
n^2	0.619	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	$(7/8) \cdot n - 1$	$0.109 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0)
n^2	0.617	ramp	$(1/2) \cdot n^2 + (1/8) \cdot n + 1 \rightarrow (1/2) \cdot n^2 + (1/4) \cdot n$	$(7/8) \cdot n - 1$	$0.109 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.617	ramp	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$ → $(1/2) \cdot n^2 + (1/4) \cdot n$	$(7/8) \cdot n - 1$	$0.109 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0, i3=0); read C[i3, i2] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0, i5=0); read E[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0, i5=0); read E[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.125 \cdot n^{-2}$	read B[i1, i3] (i0=0); read E[i6, i5] (i0=0, i5=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0, i2=0, i3=0); read C[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (29/8) \cdot n + (15/8)$	$(1/8) \cdot n + (-9/8)$	$0.0156 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0156 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (15/4) \cdot n + (7/4)$	$(1/8) \cdot n + (-9/8)$	$0.0156 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read E[i6, i5] (i0=0, i4=0, i6=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0, i3=0)
n^2	0.0862	ramp	$(1/2) \cdot n^2 + (1/8) \cdot n + 2 \rightarrow (1/2) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 1$	$0.0156 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0, i2=0)
n^2	0.0862	ramp	$(1/2) \cdot n^2 + (1/8) \cdot n + 1 \rightarrow (1/2) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n - 1$	$0.0156 \cdot n^{-2}$	read C[i3, i2] (i0=0, i1=0)
n^2	0.0782	ramp	$(3/8) \cdot n^2 + n + 1 \rightarrow (1/2) \cdot n^2 - 2 \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read A[i4, i6] (i0=0, i5=0)
n^2	0.0782	ramp	$(3/8) \cdot n^2 + n + 1 \rightarrow (1/2) \cdot n^2 - 2 \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	write A[i1, i2] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0728	ramp	$(3/8) \cdot n^2 + 10 \rightarrow$ $(3/8) \cdot n^2 + n - 14$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	write A[i1, i2] (i0=0)
n^2	0.0728	ramp	$(3/8) \cdot n^2 + 9 \rightarrow$ $(3/8) \cdot n^2 + n - 15$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read A[i4, i6] (i0=0, i4=0, i5=0)
n^1	0.707	level	$(1/2) \cdot n^2 - n + 1$	1	$0.125 \cdot n^{-3}$	write A[i1, i2] (i0=0, i1=0)
n^1	0.707	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.125 \cdot n^{-3}$	read E[i6, i5] (i0=0, i4=0)
n^1	0.707	level	$(1/2) \cdot n^2 + (1/8) \cdot n$	1	$0.125 \cdot n^{-3}$	read C[i3, i2] (i0=0, i1=0)
n^1	0.707	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.125 \cdot n^{-3}$	read E[i6, i5] (i0=0, i4=0, i6=0)
n^1	0.707	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	1	$0.125 \cdot n^{-3}$	read C[i3, i2] (i0=0, i1=0, i3=0)
n^1	0.707	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	1	$0.125 \cdot n^{-3}$	read C[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.707	level	$(1/2) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.125 \cdot n^{-3}$	read C[i3, i2] (i0=0, i1=0, i2=0, i3=0)
n^1	0.707	level	$(1/2) \cdot n^2 - n + 1$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i5=0, i6=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (7/8) \cdot n - 7$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i4=0, i5=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i5=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.125 \cdot n^{-3}$	write A[i1, i2] (i0=0, i1=0, i2=0); read A[i4, i6] (i0=0)
n^1	0.612	level	$(3/8) \cdot n^2 + 1$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i4=0, i5=0, i6=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (7/8) \cdot n - 6$	1	$0.125 \cdot n^{-3}$	write A[i1, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.612	level	$(3/8) \cdot n^2 + 2$	1	$0.125 \cdot n^{-3}$	write A[i1, i2] (i0=0)

Two chained matmuls: the intermediate C and the result E each contribute a verbatim copy of gemm's n^4 term ($0.0387 + 0.0055$ each, i.e. 0.0442 per matmul), so the total leading coefficient 0.0884 is exactly twice gemm's at the same boundaries. Everything else mirrors gemm per stage.

2mm — single-shot [exact]

Accesses $A(n) = 8 \cdot n^3 + 3 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0884 \cdot n^4 + 2.12 \cdot n^{3.5} + 10.3 \cdot n^3 + 7.95 \cdot n^{2.5} + 6.85 \cdot n^2 + 2.54 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.0137	read E[i6, i5] (i0=0)
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.0137	read C[i3, i2] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.00195	read E[i6, i5] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.00195	read C[i3, i2] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0957	read E[i6, i5] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0957	read C[i3, i2] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0137	read E[i6, i5] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0137	read C[i3, i2] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.0137	read A[i4, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 +$ $(-7/4) \cdot n^2$	0.0137	read B[i1, i3] (i0=0)
$n^{3.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 +$ $(-3/8) \cdot n^2$	0.00195	read A[i4, i6] (i0=0)
$n^{3.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 +$ $(-3/8) \cdot n^2$ $+ 2 \cdot n$	0.00195	read B[i1, i3] (i0=0)
n^3	3.46	level	3	$2 \cdot n^3$	0.25	read B[i1, i3] (i0=0); write A[i1, i2] (i0=0) (+2)
n^3	2	level	1	$2 \cdot n^3 +$ n^2	0.25	read A[i1, i2] (i0=0); write D[i4, i5] (i0=0) (+2)
n^3	1.52	level	3	$(7/8) \cdot n^3$	0.109	read A[i4, i6] (i0=0)
n^3	1.52	level	3	$(7/8) \cdot n^3$	0.109	read B[i1, i3] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 +$ $(1/4) \cdot n + 2$ $\rightarrow$	$(7/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 1	0.109/n	read E[i6, i5] (i0=0, i5=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 +$ $(3/8) \cdot n + 1$ $(1/8) \cdot n^2 +$ $(1/4) \cdot n + 2$ $\rightarrow$	$(7/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 1	0.109/n	read C[i3, i2] (i0=0, i2=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 +$ $(3/8) \cdot n + 1$ $(1/8) \cdot n^2 +$ $(1/4) \cdot n + 2$ $\rightarrow$	$(7/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 1	0.109/n	read E[i6, i5] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 +$ $(3/8) \cdot n + 1$ $(1/8) \cdot n^2 +$ $(1/4) \cdot n + 2$ $\rightarrow$	$(7/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 1	0.109/n	read C[i3, i2] (i0=0)
n^3	0.0884	level	$(1/2) \cdot n^2 +$ 1	$(1/8) \cdot n^2 +$ $(-9/4) \cdot n + 4$	0.0156/n	read A[i4, i6] (i0=0, i5=0)
n^3	0.0442	level	$(1/8) \cdot n^2 +$ $(5/4) \cdot n +$ $(29/8)$	$(1/8) \cdot n^2 +$ $(-5/4) \cdot n +$ $(9/8)$	0.0156/n	read E[i6, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read E[i6, i5] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0156/n	read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read E[i6, i5] (i0=0, i6=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0156/n	read C[i3, i2] (i0=0, i3=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read E[i6, i5] (i0=0, i5=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read C[i3, i2] (i0=0, i2=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read E[i6, i5] (i0=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0156/n	read C[i3, i2] (i0=0)
$n^{2.5}$	1.86	level	$(9/8) \cdot n + 1$	$(7/4) \cdot n^2$	0.219/n	read C[i3, i2] (i0=0); read E[i6, i5] (i0=0)
$n^{2.5}$	1.86	level	$(9/8) \cdot n - 6$	$(7/4) \cdot n^2$	0.219/n	read B[i1, i3] (i0=0); read A[i4, i6] (i0=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	0.109/n	read E[i6, i5] (i0=0, i6=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	0.109/n	read C[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	0.109/n	read A[i4, i6] (i0=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.109/n$	read B[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.265	level	$(9/8) \cdot n - 5$	$(1/4) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read B[i1, i3] (i0=0); read A[i4, i6] (i0=0)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0156/n$	read A[i4, i6] (i0=0, i6=0)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0156/n$	read B[i1, i3] (i0=0, i3=0)
n^2	1.75	level	1	$(7/4) \cdot n^2$	$0.219/n$	write A[i1, i2] (i0=0); read D[i4, i5] (i0=0)
n^2	0.707	level	$(1/2) \cdot n^2 + (-1/8) \cdot n + 1$	$n - 2$	$0.125 \cdot n^{-2}$	read A[i4, i6] (i0=0, i5=0)
n^2	0.707	level	$(1/2) \cdot n^2 + 1$	$n - 2$	$0.125 \cdot n^{-2}$	read A[i4, i6] (i0=0, i5=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.125 \cdot n^{-2}$	read E[i6, i5] (i0=0, i5=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.125 \cdot n^{-2}$	read C[i3, i2] (i0=0, i2=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0782	ramp	$(3/8) \cdot n^2 + n + 1 \rightarrow (1/2) \cdot n^2 - 2 \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read A[i4, i6] (i0=0, i5=0)
n^2	0.0728	ramp	$(3/8) \cdot n^2 + 9 \rightarrow (3/8) \cdot n^2 + n - 15$	$(1/8) \cdot n - 2$	$0.0156 \cdot n^{-2}$	read A[i4, i6] (i0=0, i4=0, i5=0)
n^1	0.707	level	$(1/2) \cdot n^2 - n + 1$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i5=0, i6=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (7/8) \cdot n - 7$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i4=0, i5=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i5=0)
n^1	0.612	level	$(3/8) \cdot n^2 + 1$	1	$0.125 \cdot n^{-3}$	read A[i4, i6] (i0=0, i4=0, i5=0, i6=0)

Two chained matmuls: the intermediate C and the result E each contribute a verbatim copy of gemm's n^4 term (0.0387 + 0.0055 each, i.e. 0.0442 per matmul), so the total leading coefficient 0.0884 is exactly twice gemm's at the same boundaries. Everything else mirrors gemm per stage.

3mm — infinite-repeat [exact]

Accesses $A(n) = 12 \cdot n^3 + 3 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.133 \cdot n^4 + 3.18 \cdot n^{3.5} + 16.2 \cdot n^3 + 11.9 \cdot n^{2.5} + 21.5 \cdot n^2 + 18.3 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.00911	read D[i9, i8] (i0=0)
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.00911	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.00911	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.0013	read D[i9, i8] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.0013	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.0013	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0638	read D[i9, i8] (i0=0, i7=0); read D[i9, i8] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0638	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0638	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.00911	read D[i9, i8] (i0=0, i7=0); read D[i9, i8] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.00911	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.00911	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.00911	read A[i7, i9] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.00911	read E[i4, i6] (i0=0, i4=0, i5=0); read E[i4, i6] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.00911	read B[i1, i3] (i0=0, i1=0, i2=0); read B[i1, i3] (i0=0)
$n^{3.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.0013	read A[i7, i9] (i0=0)
$n^{3.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.0013	read E[i4, i6] (i0=0, i4=0, i5=0); read E[i4, i6] (i0=0)
$n^{3.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.0013	read B[i1, i3] (i0=0, i1=0, i2=0); read B[i1, i3] (i0=0)
n^3	4.55	level	3	$(21/8) \cdot n^3$	0.219	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0) (+1)
n^3	4.55	level	3	$(21/8) \cdot n^3$	0.219	write A[i1, i2] (i0=0); write D[i4, i5] (i0=0) (+1)
n^3	2.62	level	1	$(21/8) \cdot n^3 + (21/8) \cdot n^2$	0.219	write A[i1, i2] (i0=0); read A[i1, i2] (i0=0) (+5)
n^3	0.65	level	3	$(3/8) \cdot n^3$	0.0312	write A[i1, i2] (i0=0); write D[i4, i5] (i0=0) (+2)
n^3	0.375	level	1	$(3/8) \cdot n^3$	0.0312	read A[i1, i2] (i0=0); read D[i4, i5] (i0=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read D[i9, i8] (i0=0, i8=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read F[i6, i5] (i0=0, i4=0, i5=0); read F[i6, i5] (i0=0, i5=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read D[i9, i8] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^3	0.234	level	$(7/8) \cdot n^2$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0, i4=0, i5=0) (+1)
n^3	0.117	level	$(7/8) \cdot n^2$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0104/n	write G[i7, i8] (i0=0, i7=0); write G[i7, i8] (i0=0)
n^3	0.108	level	$(3/4) \cdot n^2 + 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0104/n	read A[i7, i9] (i0=0, i8=0)
n^3	0.0955	ramp	$(1/2) \cdot n^2 + (5/2) \cdot n - 2$ → $(3/4) \cdot n^2 - 2$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0104/n	write D[i4, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0947	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.00911/n	read F[i6, i5] (i0=0, i4=0)
n^3	0.0947	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.00911/n	read C[i3, i2] (i0=0, i1=0)
n^3	0.0884	level	$(1/2) \cdot n^2 + 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0104/n	write A[i1, i2] (i0=0)
n^3	0.0547	ramp	$(1/8) \cdot n^2 + (21/8) \cdot n - 2 \rightarrow (3/8) \cdot n^2 + (1/8) \cdot n - 2$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.00911/n	read D[i9, i8] (i0=0, i7=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0104/n	read D[i9, i8] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read D[i9, i8] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0104/n	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0104/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read D[i9, i8] (i0=0, i9=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read F[i6, i5] (i0=0, i4=0, i6=0); read F[i6, i5] (i0=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read C[i3, i2] (i0=0, i1=0, i3=0); read C[i3, i2] (i0=0, i3=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read D[i9, i8] (i0=0, i8=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read F[i6, i5] (i0=0, i4=0, i5=0); read F[i6, i5] (i0=0, i5=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read D[i9, i8] (i0=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^3	0.0135	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.0013/n	read F[i6, i5] (i0=0, i4=0)
n^3	0.0135	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.0013/n	read C[i3, i2] (i0=0, i1=0)
n^3	0.00764	ramp	$(1/8) \cdot n^2 + (33/8) \cdot n - 14 \rightarrow (3/8) \cdot n^2 + (-3/4) \cdot n - 9$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.0013/n	read D[i9, i8] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	2.78	level	$(9/8) \cdot n + 1$	$(21/8) \cdot n^2$	$0.219/n$	read C[i3, i2] (i0=0); read F[i6, i5] (i0=0) (+2)
$n^{2.5}$	2.78	level	$(9/8) \cdot n - 6$	$(21/8) \cdot n^2$	$0.219/n$	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0) (+1)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	$0.0729/n$	read D[i9, i8] (i0=0, i7=0, i9=0); read D[i9, i8] (i0=0, i9=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	$0.0729/n$	read F[i6, i5] (i0=0, i4=0, i6=0); read F[i6, i5] (i0=0, i6=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	$0.0729/n$	read C[i3, i2] (i0=0, i1=0, i3=0); read C[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.0729/n$	write A[i1, i2] (i0=0); read A[i7, i9] (i0=0, i9=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.0729/n$	read E[i4, i6] (i0=0, i4=0, i5=0, i6=0); read E[i4, i6] (i0=0, i6=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.0729/n$	read B[i1, i3] (i0=0, i1=0, i2=0, i3=0); read B[i1, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.398	level	$(9/8) \cdot n - 5$	$(3/8) \cdot n^2 - 3 \cdot n$	$0.0312/n$	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0) (+1)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0104/n$	read A[i7, i9] (i0=0, i9=0)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0104/n$	read E[i4, i6] (i0=0, i4=0, i5=0, i6=0); read E[i4, i6] (i0=0, i6=0)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0104/n$	read B[i1, i3] (i0=0, i1=0, i2=0, i3=0); read B[i1, i3] (i0=0, i3=0)
n^2	2.81	level	$(7/8) \cdot n^2$	$3 \cdot n$	$0.25 \cdot n^{-2}$	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0) (+2)
n^2	0.935	level	$(7/8) \cdot n^2$	n	$0.0833 \cdot n^{-2}$	read E[i4, i6] (i0=0, i4=0, i5=0); read E[i4, i6] (i0=0, i5=0)
n^2	0.935	level	$(7/8) \cdot n^2$	n	$0.0833 \cdot n^{-2}$	read B[i1, i3] (i0=0, i1=0, i2=0); read B[i1, i3] (i0=0, i2=0)
n^2	0.935	level	$(7/8) \cdot n^2$	n	$0.0833 \cdot n^{-2}$	write G[i7, i8] (i0=0, i7=0); write G[i7, i8] (i0=0)
n^2	0.866	level	$(3/4) \cdot n^2 + (-1/8) \cdot n + 1$	$n - 2$	$0.0833 \cdot n^{-2}$	read A[i7, i9] (i0=0, i8=0)
n^2	0.866	level	$(3/4) \cdot n^2 + 1$	$n - 2$	$0.0833 \cdot n^{-2}$	read A[i7, i9] (i0=0, i8=0, i9=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.824	ramp	$(5/8) \cdot n^2 + (1/4) \cdot n \rightarrow (3/4) \cdot n^2 + (-1/8) \cdot n$	$n - 2$	$0.0833 \cdot n^2$	write D[i4, i5] (i0=0, i5=0)
n^2	0.786	ramp	$(1/2) \cdot n^2 + (3/2) \cdot n - 1 \rightarrow (3/4) \cdot n^2 + (-1/4) \cdot n + 2$	$n - 2$	$0.0833 \cdot n^2$	write D[i4, i5] (i0=0)
n^2	0.755	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 1 \rightarrow (3/4) \cdot n^2 + (1/4) \cdot n$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^2$	read F[i6, i5] (i0=0, i4=0, i5=0)
n^2	0.755	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 1 \rightarrow (3/4) \cdot n^2 + (1/4) \cdot n$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^2$	read C[i3, i2] (i0=0, i1=0, i2=0)
n^2	0.755	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 1 \rightarrow (3/4) \cdot n^2 + (1/4) \cdot n$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^2$	read F[i6, i5] (i0=0, i4=0)
n^2	0.755	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 1 \rightarrow (3/4) \cdot n^2 + (1/4) \cdot n$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^2$	read C[i3, i2] (i0=0, i1=0)
n^2	0.707	level	$(1/2) \cdot n^2 + (-1/8) \cdot n + 1$	$n - 2$	$0.0833 \cdot n^2$	write A[i1, i2] (i0=0, i2=0)
n^2	0.707	level	$(1/2) \cdot n^2 + 1$	$n - 2$	$0.0833 \cdot n^2$	write A[i1, i2] (i0=0)
n^2	0.487	ramp	$(1/4) \cdot n^2 + (3/8) \cdot n \rightarrow (3/8) \cdot n^2$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^2$	read D[i9, i8] (i0=0, i7=0)
n^2	0.433	ramp	$(1/8) \cdot n^2 + (13/8) \cdot n - 1 \rightarrow (3/8) \cdot n^2 + (-1/4) \cdot n + 3$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^2$	read D[i9, i8] (i0=0, i7=0, i8=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i4=0); read F[i6, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i1=0); read C[i3, i2] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0, i9=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i4=0, i6=0); read F[i6, i5] (i0=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i1=0, i3=0); read C[i3, i2] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0, i8=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0, i8=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i4=0, i5=0); read F[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i4=0, i5=0); read F[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i1=0, i2=0); read C[i3, i2] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.0833 \cdot n^2$	read B[i1, i3] (i0=0); read D[i9, i8] (i0=0, i8=0, i9=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i4=0, i5=0, i6=0); read F[i6, i5] (i0=0, i5=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i1=0, i2=0, i3=0); read C[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.108	level	$(3/4) \cdot n^2 + 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read A[i7, i9] (i0=0, i7=0, i8=0)
n^2	0.108	level	$(3/4) \cdot n^2 + (11/2) \cdot n + (7/4)$	$(1/8) \cdot n + (-9/8)$	$0.0104 \cdot n^2$	read F[i6, i5] (i0=0, i4=0)
n^2	0.108	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read F[i6, i5] (i0=0, i4=0)
n^2	0.108	level	$(3/4) \cdot n^2 + (11/2) \cdot n + (7/4)$	$(1/8) \cdot n + (-9/8)$	$0.0104 \cdot n^2$	read C[i3, i2] (i0=0, i1=0)
n^2	0.108	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read C[i3, i2] (i0=0, i1=0)
n^2	0.108	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read F[i6, i5] (i0=0, i4=0, i6=0)
n^2	0.108	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read C[i3, i2] (i0=0, i1=0, i3=0)
n^2	0.106	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 2$ → $(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^2$	read F[i6, i5] (i0=0, i4=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.106	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 2$ → $(3/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^2$	read C[i3, i2] (i0=0, i1=0, i2=0)
n^2	0.106	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 1$ → $(3/4) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^2$	read F[i6, i5] (i0=0, i4=0)
n^2	0.106	ramp	$(3/4) \cdot n^2 + (1/8) \cdot n + 1$ → $(3/4) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^2$	read C[i3, i2] (i0=0, i1=0)
n^2	0.103	ramp	$(3/4) \cdot n^2 + 2$ → $(3/4) \cdot n^2 + (1/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	write D[i4, i5] (i0=0)
n^2	0.0982	ramp	$(5/8) \cdot n^2 + n + 1$ → $(3/4) \cdot n^2 - 2 \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read A[i7, i9] (i0=0, i8=0)
n^2	0.0884	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	write D[i4, i5] (i0=0, i4=0)
n^2	0.0782	ramp	$(3/8) \cdot n^2 + n + 1$ → $(1/2) \cdot n^2 - 2 \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	write A[i1, i2] (i0=0, i1=0)
n^2	0.0728	ramp	$(3/8) \cdot n^2 + 10$ → $(3/8) \cdot n^2 + n - 14$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	write A[i1, i2] (i0=0)
n^2	0.0727	ramp	$(3/8) \cdot n^2 + (1/8) \cdot n + 2$ → $(3/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read D[i9, i8] (i0=0, i7=0, i9=0)
n^2	0.068	ramp	$(1/4) \cdot n^2 + (9/8) \cdot n$ → $(3/8) \cdot n^2 + (-7/8) \cdot n$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^2$	read D[i9, i8] (i0=0, i7=0)
n^2	0.0605	ramp	$(1/8) \cdot n^2 + (25/8) \cdot n - 7$ → $(3/8) \cdot n^2 - 2 \cdot n + 11$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^2$	read D[i9, i8] (i0=0, i7=0, i8=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read D[i9, i8] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.866	level	$(3/4) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.0833 \cdot n^3$	read A[i7, i9] (i0=0, i7=0, i8=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/8) \cdot n$	1	$0.0833 \cdot n^3$	read F[i6, i5] (i0=0, i4=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/8) \cdot n$	1	$0.0833 \cdot n^3$	read C[i3, i2] (i0=0, i1=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	1	$0.0833 \cdot n^3$	read F[i6, i5] (i0=0, i4=0, i6=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	1	$0.0833 \cdot n^3$	read C[i3, i2] (i0=0, i1=0, i3=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	1	$0.0833 \cdot n^3$	read F[i6, i5] (i0=0, i4=0, i5=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.0833 \cdot n^3$	read F[i6, i5] (i0=0, i4=0, i5=0, i6=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/4) \cdot n$	1	$0.0833 \cdot n^3$	read C[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.0833 \cdot n^3$	read C[i3, i2] (i0=0, i1=0, i2=0, i3=0)
n^1	0.866	level	$(3/4) \cdot n^2$	1	$0.0833 \cdot n^3$	write D[i4, i5] (i0=0, i5=0)
n^1	0.866	level	$(3/4) \cdot n^2 + 1$	1	$0.0833 \cdot n^3$	read A[i7, i9] (i0=0, i7=0, i8=0, i9=0)
n^1	0.866	level	$(3/4) \cdot n^2 + 1$	1	$0.0833 \cdot n^3$	write D[i4, i5] (i0=0)
n^1	0.866	level	$(3/4) \cdot n^2 - n + 1$	1	$0.0833 \cdot n^3$	read A[i7, i9] (i0=0, i8=0, i9=0)
n^1	0.791	level	$(5/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.0833 \cdot n^3$	read A[i7, i9] (i0=0, i8=0)
n^1	0.707	level	$(1/2) \cdot n^2 - n + 1$	1	$0.0833 \cdot n^3$	write A[i1, i2] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.707	level	$(1/2) \cdot n^2 + (1/8) \cdot n$	1	$0.0833 \cdot n^3$	write D[i4, i5] (i0=0, i4=0, i5=0); write D[i4, i5] (i0=0, i5=0)
n^1	0.707	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	1	$0.0833 \cdot n^3$	write D[i4, i5] (i0=0, i4=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.0833 \cdot n^3$	write A[i1, i2] (i0=0, i1=0, i2=0); read A[i7, i9] (i0=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n$	1	$0.0833 \cdot n^3$	read D[i9, i8] (i0=0, i7=0, i9=0)
n^1	0.612	level	$(3/8) \cdot n^2 + 2$	1	$0.0833 \cdot n^3$	read A[i7, i9] (i0=0, i7=0, i8=0, i9=0); read D[i9, i8] (i0=0, i7=0, i8=0, i9=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (7/8) \cdot n - 6$	1	$0.0833 \cdot n^3$	write A[i1, i2] (i0=0, i2=0)
n^1	0.612	level	$(3/8) \cdot n^2 + 2$	1	$0.0833 \cdot n^3$	write A[i1, i2] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.0833 \cdot n^3$	read D[i9, i8] (i0=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (21/8)$	1	$0.0833 \cdot n^3$	read D[i9, i8] (i0=0, i7=0, i8=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	1	$0.0833 \cdot n^3$	read D[i9, i8] (i0=0, i7=0, i8=0)

Three chained matmuls: three copies of gemm's n^4 term, total coefficient $3 \times 0.0442 = 0.133$, at the same $(1/8)n^2$ -line boundary. The 1:2:3 gemm:2mm:3mm ratio is exact in the term table.

3mm — single-shot [exact]

Accesses $A(n) = 12 \cdot n^3 + 3 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.133 \cdot n^4 + 3.18 \cdot n^{3.5} + 15.4 \cdot n^3 + 11.9 \cdot n^{2.5} + 11 \cdot n^2 + 5.67 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.00911	read D[i9, i8] (i0=0)
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.00911	read F[i6, i5] (i0=0)
n^4	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-127/64) \cdot n^2 + (31/8) \cdot n - 2$	0.00911	read C[i3, i2] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.0013	read D[i9, i8] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.0013	read F[i6, i5] (i0=0)
n^4	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/64) \cdot n^3 + (-25/64) \cdot n^2 + (19/8) \cdot n - 2$	0.0013	read C[i3, i2] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0638	read D[i9, i8] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0638	read F[i6, i5] (i0=0)
$n^{3.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^3 + (-7/8) \cdot n^2$	0.0638	read C[i3, i2] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.00911	read D[i9, i8] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.00911	read F[i6, i5] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^3 + (-7/8) \cdot n^2$	0.00911	read C[i3, i2] (i0=0)
$n^{3.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.00911	read A[i7, i9] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^3.5$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.00911	read E[i4, i6] (i0=0)
$n^3.5$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.00911	read B[i1, i3] (i0=0)
$n^3.5$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.0013	read A[i7, i9] (i0=0)
$n^3.5$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.0013	read E[i4, i6] (i0=0)
$n^3.5$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.0013	read B[i1, i3] (i0=0)
n^3	5.2	level	3	$3 \cdot n^3$	0.25	read B[i1, i3] (i0=0); write A[i1, i2] (i0=0) (+5)
n^3	3.03	level	3	$(7/4) \cdot n^3$	0.146	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0)
n^3	3	level	1	$3 \cdot n^3$	0.25	read A[i1, i2] (i0=0); read D[i4, i5] (i0=0) (+2)
n^3	1.52	level	3	$(7/8) \cdot n^3$	0.0729	read A[i7, i9] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read D[i9, i8] (i0=0, i8=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read F[i6, i5] (i0=0, i5=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read C[i3, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read D[i9, i8] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read F[i6, i5] (i0=0)
n^3	0.308	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.0729/n	read C[i3, i2] (i0=0)
n^3	0.108	level	$(3/4) \cdot n^2 + 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0104/n	read A[i7, i9] (i0=0, i8=0)
n^3	0.0547	ramp	$(1/8) \cdot n^2 + (21/8) \cdot n - 2$ → $(3/8) \cdot n^2 + (1/8) \cdot n - 2$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.00911/n	read D[i9, i8] (i0=0, i7=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0104/n	read D[i9, i8] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read D[i9, i8] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0104/n	read F[i6, i5] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read F[i6, i5] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	0.0104/n	read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read C[i3, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read D[i9, i8] (i0=0, i9=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read F[i6, i5] (i0=0, i6=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0104/n	read C[i3, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read D[i9, i8] (i0=0, i8=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read F[i6, i5] (i0=0, i5=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read C[i3, i2] (i0=0, i2=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read D[i9, i8] (i0=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read F[i6, i5] (i0=0)
n^3	0.0431	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0104/n	read C[i3, i2] (i0=0)
n^3	0.00764	ramp	$(1/8) \cdot n^2 + (33/8) \cdot n - 14$ → $(3/8) \cdot n^2 + (-3/4) \cdot n - 9$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.0013/n	read D[i9, i8] (i0=0, i7=0)
$n^{2.5}$	2.78	level	$(9/8) \cdot n + 1$	$(21/8) \cdot n^2$	0.219/n	read C[i3, i2] (i0=0); read F[i6, i5] (i0=0) (+1)
$n^{2.5}$	2.78	level	$(9/8) \cdot n - 6$	$(21/8) \cdot n^2$	0.219/n	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0) (+1)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	0.0729/n	read D[i9, i8] (i0=0, i9=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	0.0729/n	read F[i6, i5] (i0=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2$	$0.0729/n$	read C[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.0729/n$	read A[i7, i9] (i0=0, i9=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.0729/n$	read E[i4, i6] (i0=0, i6=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^2$	$0.0729/n$	read B[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.398	level	$(9/8) \cdot n - 5$	$(3/8) \cdot n^2 - 3 \cdot n$	$0.0312/n$	read B[i1, i3] (i0=0); read E[i4, i6] (i0=0) (+1)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0104/n$	read A[i7, i9] (i0=0, i9=0)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0104/n$	read E[i4, i6] (i0=0, i6=0)
$n^{2.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^2 - n$	$0.0104/n$	read B[i1, i3] (i0=0, i3=0)
n^2	2.62	level	1	$(21/8) \cdot n^2$	$0.219/n$	write A[i1, i2] (i0=0); write D[i4, i5] (i0=0) (+1)
n^2	0.866	level	$(3/4) \cdot n^2 + (-1/8) \cdot n + 1$	$n - 2$	$0.0833 \cdot n^{-2}$	read A[i7, i9] (i0=0, i8=0)
n^2	0.866	level	$(3/4) \cdot n^2 + 1$	$n - 2$	$0.0833 \cdot n^{-2}$	read A[i7, i9] (i0=0, i8=0, i9=0)
n^2	0.487	ramp	$(1/4) \cdot n^2 + (3/8) \cdot n \rightarrow (3/8) \cdot n^2$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^{-2}$	read D[i9, i8] (i0=0, i7=0)
n^2	0.433	ramp	$(1/8) \cdot n^2 + (13/8) \cdot n - 1 \rightarrow (3/8) \cdot n^2 + (-1/4) \cdot n + 3$	$(7/8) \cdot n - 1$	$0.0729 \cdot n^{-2}$	read D[i9, i8] (i0=0, i7=0, i8=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^{-2}$	read D[i9, i8] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^{-2}$	read F[i6, i5] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^{-2}$	read C[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0, i9=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0, i8=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0, i8=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.0833 \cdot n^2$	read D[i9, i8] (i0=0, i8=0, i9=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.0833 \cdot n^2$	read F[i6, i5] (i0=0, i5=0, i6=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$n - 1$	$0.0833 \cdot n^2$	read C[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.108	level	$(3/4) \cdot n^2 + 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read A[i7, i9] (i0=0, i7=0, i8=0)
n^2	0.0982	ramp	$(5/8) \cdot n^2 + n + 1 \rightarrow (3/4) \cdot n^2 - 2 \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read A[i7, i9] (i0=0, i8=0)
n^2	0.0727	ramp	$(3/8) \cdot n^2 + (1/8) \cdot n + 2 \rightarrow (3/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^2$	read D[i9, i8] (i0=0, i7=0, i9=0)
n^2	0.068	ramp	$(1/4) \cdot n^2 + (9/8) \cdot n \rightarrow (3/8) \cdot n^2 + (-7/8) \cdot n$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^2$	read D[i9, i8] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0605	ramp	$(1/8) \cdot n^2 + (25/8) \cdot n - 7$ $\rightarrow$ $(3/8) \cdot n^2 - 2 \cdot n + 11$	$(1/8) \cdot n - 1$	$0.0104 \cdot n^{-2}$	read D[i9, i8] (i0=0, i7=0, i8=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0104 \cdot n^{-2}$	read D[i9, i8] (i0=0, i7=0)
n^1	0.866	level	$(3/4) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.0833 \cdot n^{-3}$	read A[i7, i9] (i0=0, i7=0, i8=0)
n^1	0.866	level	$(3/4) \cdot n^2 + 1$	1	$0.0833 \cdot n^{-3}$	read A[i7, i9] (i0=0, i7=0, i8=0, i9=0)
n^1	0.866	level	$(3/4) \cdot n^2 - n + 1$	1	$0.0833 \cdot n^{-3}$	read A[i7, i9] (i0=0, i8=0, i9=0)
n^1	0.791	level	$(5/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.0833 \cdot n^{-3}$	read A[i7, i9] (i0=0, i8=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n$	1	$0.0833 \cdot n^{-3}$	read D[i9, i8] (i0=0, i7=0, i9=0)
n^1	0.612	level	$(3/8) \cdot n^2 + 2$	1	$0.0833 \cdot n^{-3}$	read A[i7, i9] (i0=0, i7=0, i8=0, i9=0); read D[i9, i8] (i0=0, i7=0, i8=0, i9=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.0833 \cdot n^{-3}$	read D[i9, i8] (i0=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (21/8)$	1	$0.0833 \cdot n^{-3}$	read D[i9, i8] (i0=0, i7=0, i8=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	1	$0.0833 \cdot n^{-3}$	read D[i9, i8] (i0=0, i7=0, i8=0)

Three chained matmuls: three copies of gemm's n^4 term, total coefficient $3 \times 0.0442 = 0.133$, at the same $(1/8)n^2$ -line boundary. The 1:2:3 gemm:2mm:3mm ratio is exact in the term table.

atax — infinite-repeat [exact]

Accesses $A(n) = 8 \cdot n^2 + 2 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^3 + 0.22 \cdot n^{2.5} + 12.5 \cdot n^2 + 3.77 \cdot n^{1.5} + 3.71 \cdot n^1 + 1.47 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0156	read C[i2, i3] (i0=0)
$n^{2.5}$	0.069	ramp	$(3/8) \cdot n + 3 \rightarrow (1/2) \cdot n$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0137	read D[i3] (i0=0, i2=0); read D[i3] (i0=0)
$n^{2.5}$	0.0687	ramp	$(3/8) \cdot n + 2 \rightarrow (1/2) \cdot n - 1$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0137	read A[i4] (i0=0, i2=1); read A[i4] (i0=0)
$n^{2.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0156	read C[i2, i4] (i0=0)
$n^{2.5}$	0.00989	ramp	$(3/8) \cdot n + 4 \rightarrow (1/2) \cdot n + 1$	$(1/64) \cdot n^2 + (-1/4) \cdot n$	0.00195	read D[i3] (i0=0, i2=0); read D[i3] (i0=0)
$n^{2.5}$	0.0096	ramp	$(3/8) \cdot n + 3 \rightarrow (1/2) \cdot n$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00195	read A[i4] (i0=0)
n^2	3.25	level	3	$(15/8) \cdot n^2$	0.234	read D[i3] (i0=0, i2=0); write A[i4] (i0=0, i2=0) (+3)
n^2	3.03	level	3	$(7/4) \cdot n^2$	0.219	write A[i4] (i0=0, i2=0); read C[i2, i4] (i0=0) (+1)
n^2	2.84	level	3	$(105/64) \cdot n^2 + (7/8) \cdot n$	0.205	write B[i2] (i0=0); read B[i2] (i0=0, i4=0) (+1)
n^2	0.875	level	1	$(7/8) \cdot n^2$	0.109	read A[i4] (i0=0, i2=0); read A[i4] (i0=0)
n^2	0.875	level	1	$(7/8) \cdot n^2 + (7/8) \cdot n$	0.109	write A[i1] (i0=0); write B[i2] (i0=0) (+2)
n^2	0.406	level	3	$(15/64) \cdot n^2 + (1/8) \cdot n$	0.0293	write B[i2] (i0=0); read B[i2] (i0=0, i4=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$n - 2$	$0.125/n$	read C[i2, i3] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$n - 2$	$0.125/n$	read C[i2, i3] (i0=0, i3=0)
n^2	0.25	level	4	$(1/8) \cdot n^2 - n$	0.0156	read B[i2] (i0=0)
n^2	0.125	level	1	$(1/8) \cdot n^2$	0.0156	read B[i2] (i0=0, i3=0); read B[i2] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0156/n$	read C[i2, i3] (i0=0, i2=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (21/8)$	$(1/8) \cdot n + (-9/8)$	$0.0156/n$	read C[i2, i3] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0156/n$	read C[i2, i3] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (-5/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0156/n$	write B[i2] (i0=0)
$n^{1.5}$	0.619	level	$(1/2) \cdot n + 1$	$(7/8) \cdot n$	$0.109/n$	read D[i3] (i0=0, i2=0); read D[i3] (i0=0)
$n^{1.5}$	0.619	level	$(1/2) \cdot n$	$(7/8) \cdot n$	$0.109/n$	read A[i4] (i0=0, i2=1, i4=0); read A[i4] (i0=0, i4=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + 1$	n	$0.125/n$	read D[i3] (i0=0, i2=0); read A[i4] (i0=0, i2=0) (+2)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 2$	$(7/8) \cdot n$	$0.109/n$	read D[i3] (i0=0, i2=0, i3=0); read D[i3] (i0=0, i3=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 2$	n	$0.125/n$	read C[i2, i4] (i0=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 2$	n	$0.125/n$	read C[i2, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.0884	level	$(1/2) \cdot n + 2$	$(1/8) \cdot n$	$0.0156/n$	read D[i3] (i0=0, i2=0); read D[i3] (i0=0)
$n^{1.5}$	0.0884	level	$(1/2) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0156/n$	read A[i4] (i0=0, i4=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n$	$0.0156/n$	read D[i3] (i0=0, i2=0, i3=0); read D[i3] (i0=0, i3=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0156/n$	read D[i3] (i0=0, i2=0); read A[i4] (i0=0, i2=0)
$n^{1.5}$	0.0514	ramp	$(1/8) \cdot n + 2$ $\rightarrow (1/4) \cdot n -$ 1	$(1/8) \cdot n - 2$	$0.0156/n$	write A[i1] (i0=0)
n^1	1.24	level	2	$(7/8) \cdot n$	$0.109/n$	write B[i2] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read C[i2, i3] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read C[i2, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read C[i2, i3] (i0=0, i2=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(5/4) \cdot n +$ $(21/8)$	1	$0.125 \cdot n^{-2}$	read C[i2, i3] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read C[i2, i3] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(-5/8) \cdot n$	1	$0.125 \cdot n^{-2}$	write B[i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(-5/8) \cdot n$	1	$0.125 \cdot n^{-2}$	write B[i2] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 1$	1	$0.125 \cdot n^{-2}$	read D[i3] (i0=0, i2=0); read A[i4] (i0=0, i2=0, i4=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.125 \cdot n^{-2}$	write A[i1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.354	level	$(1/8) \cdot n$	1	$0.125 \cdot n^{-2}$	write A[i1] (i0=0); write B[i2] (i0=0)

Infinite repeat adds the single decisive term: the matrix wraparound at distance $(1/8)n^2 + (3/8)n$ with population $n^2/8$ and coefficient 0.0442 — the same constant as gemm’s matrix term. This is the closed form of the Krylov-solver argument: across iterations the matrix is genuine reuse, worth keeping resident; single-shot files it as unavoidable cold traffic.

atax — single-shot [exact]

Accesses $A(n) = 8 \cdot n^2 + 2 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order $n^{2.5}$, headroom **+0.5**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.219 \cdot n^{2.5} + 11.7 \cdot n^2 + 3.72 \cdot n^{1.5} + 2.11 \cdot n^1 + 0.612 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.069	ramp	$(3/8) \cdot n + 3$ $\rightarrow (1/2) \cdot n$	$(7/64) \cdot n^2$ $+ (-7/4) \cdot n$	0.0137	read D[i3] (i0=0)
$n^{2.5}$	0.0687	ramp	$(3/8) \cdot n + 2$ $\rightarrow (1/2) \cdot n - 1$	$(7/64) \cdot n^2$ $+ (-7/4) \cdot n$	0.0137	read A[i4] (i0=0)
$n^{2.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0156	read C[i2, i4] (i0=0)
$n^{2.5}$	0.00963	ramp	$(3/8) \cdot n + 4$ $\rightarrow (1/2) \cdot n + 1$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n + 2$	0.00195	read D[i3] (i0=0)
$n^{2.5}$	0.0096	ramp	$(3/8) \cdot n + 3$ $\rightarrow (1/2) \cdot n$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n + 2$	0.00195	read A[i4] (i0=0)
n^2	4.98	level	3	$(23/8) \cdot n^2$ $+ n$	0.359	write B[i2] (i0=0); read B[i2] (i0=0, i4=0) (+2)
n^2	4.55	level	3	$(21/8) \cdot n^2$	0.328	read C[i2, i3] (i0=0); read D[i3] (i0=0) (+1)
n^2	1	level	1	n^2	0.125	read B[i2] (i0=0, i3=0); read B[i2] (i0=0)
n^2	0.875	level	1	$(7/8) \cdot n^2$	0.109	read A[i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.25	level	4	$(1/8) \cdot n^2 - n$	0.0156	read C[i2, i4] (i0=0, i4=0); read B[i2] (i0=0)
$n^{1.5}$	0.619	level	$(1/2) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read D[i3] (i0=0)
$n^{1.5}$	0.619	level	$(1/2) \cdot n$	$(7/8) \cdot n$	0.109/n	read A[i4] (i0=0, i4=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + 1$	n	0.125/n	read A[i4] (i0=0, i2=0); read A[i4] (i0=0)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 2$	$(7/8) \cdot n$	0.109/n	read D[i3] (i0=0, i3=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 2$	n	0.125/n	read C[i2, i4] (i0=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 2$	n	0.125/n	read C[i2, i4] (i0=0, i4=0)
$n^{1.5}$	0.0884	level	$(1/2) \cdot n + 2$	$(1/8) \cdot n - 1$	0.0156/n	read D[i3] (i0=0)
$n^{1.5}$	0.0884	level	$(1/2) \cdot n + 1$	$(1/8) \cdot n - 1$	0.0156/n	read A[i4] (i0=0, i4=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n - 1$	0.0156/n	read D[i3] (i0=0, i3=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 1$	$(1/8) \cdot n - 2$	0.0156/n	read A[i4] (i0=0, i2=0)
n^1	1.24	level	2	$(7/8) \cdot n$	0.109/n	write B[i2] (i0=0)
n^1	0.875	level	1	$(7/8) \cdot n$	0.109/n	write A[i1] (i0=0); write B[i2] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 1$	1	$0.125 \cdot n^{-2}$	read A[i4] (i0=0, i2=0, i4=0)

Single-shot, the matrix C is streamed once and never re-touched: no n^2 -distance term exists. The top terms are vector reuses — read D[i3]/read A[i4] ramps between $(3/8)n$ and $(1/2)n$ lines and the row-window level at $(1/4)n + 2$ — giving $d = 2.5$, headroom +0.5. The right transformation at this scope is fusion/interchange of the two nests.

bicg — infinite-repeat [exact]

Accesses $A(n) = 7 \cdot n^2 + n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^3 + 0.153 \cdot n^{2.5} + 13.8 \cdot n^2 + 2.45 \cdot n^{1.5} + 6.2 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0442	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0179	read D[i1, i2] (i0=0)
$n^{2.5}$	0.067	level	$(3/8) \cdot n + 2$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0156	read D[i1, i2] (i0=0, i1=0); read E[i2] (i0=0, i1=0) (+1)
$n^{2.5}$	0.067	level	$(3/8) \cdot n + 1$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0156	read B[i2] (i0=0, i1=0); read B[i2] (i0=0, i1=1) (+1)
$n^{2.5}$	0.00957	level	$(3/8) \cdot n + 4$	$(1/64) \cdot n^2 + (-1/4) \cdot n$	0.00223	read E[i2] (i0=0, i1=0); read E[i2] (i0=0)
$n^{2.5}$	0.00957	level	$(3/8) \cdot n + 3$	$(1/64) \cdot n^2 + (-1/4) \cdot n$	0.00223	read B[i2] (i0=0, i1=0); read B[i2] (i0=0)
n^2	3.03	level	3	$(7/4) \cdot n^2 + (-7/8) \cdot n$	0.25	write B[i2] (i0=0, i1=0); read B[i2] (i0=0) (+1)
n^2	1.96	level	5	$(7/8) \cdot n^2$	0.125	read E[i2] (i0=0, i1=0); read E[i2] (i0=0)
n^2	1.96	level	5	$(7/8) \cdot n^2$	0.125	read D[i1, i2] (i0=0)
n^2	1.96	level	5	$(7/8) \cdot n^2$	0.125	read C[i1] (i0=0, i2=0); read C[i1] (i0=0)
n^2	1.75	level	4	$(7/8) \cdot n^2$	0.125	read A[i1] (i0=0, i2=0); read A[i1] (i0=0)
n^2	1.24	level	2	$(7/8) \cdot n^2$	0.125	write A[i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$n - 2$	$0.143/n$	read D[i1, i2] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$n - 2$	$0.143/n$	read D[i1, i2] (i0=0, i2=0)
n^2	0.306	level	6	$(1/8) \cdot n^2 - n$	0.0179	read C[i1] (i0=0)
n^2	0.25	level	4	$(1/8) \cdot n^2$	0.0179	read A[i1] (i0=0, i2=0); read A[i1] (i0=0)
n^2	0.217	level	3	$(1/8) \cdot n^2 + (7/8) \cdot n$	0.0179	read B[i2] (i0=0, i1=0); write B[i2] (i0=0, i1=0) (+1)
n^2	0.177	level	2	$(1/8) \cdot n^2$	0.0179	write A[i1] (i0=0, i2=0); write A[i1] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.0179/n$	read D[i1, i2] (i0=0, i1=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (11/8) \cdot n + (7/2)$	$(1/8) \cdot n + (-9/8)$	$0.0179/n$	read D[i1, i2] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.0179/n$	read D[i1, i2] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0179/n$	read C[i1] (i0=0, i2=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (-1/2) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0179/n$	write A[i1] (i0=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + 2$	n	$0.143/n$	read B[i2] (i0=0, i1=0); read D[i1, i2] (i0=0, i1=0) (+3)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + 2$	n	$0.143/n$	read B[i2] (i0=0, i1=0, i2=0); read D[i1, i2] (i0=0, i1=0, i2=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	$0.125/n$	read B[i2] (i0=0, i1=0); read B[i2] (i0=0, i1=1) (+1)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	$0.125/n$	read B[i2] (i0=0, i1=0, i2=0); read B[i2] (i0=0, i1=1, i2=0) (+1)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n$	$0.0179/n$	read E[i2] (i0=0, i1=0); read E[i2] (i0=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 4$	$(1/8) \cdot n$	$0.0179/n$	read E[i2] (i0=0, i1=0, i2=0); read E[i2] (i0=0, i2=0)
n^1	2.14	level	6	$(7/8) \cdot n$	$0.125/n$	read C[i1] (i0=0, i2=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.125/n$	write A[i1] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.143 \cdot n^{-2}$	read D[i1, i2] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.143 \cdot n^{-2}$	read D[i1, i2] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 1$	1	$0.143 \cdot n^{-2}$	read C[i1] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.143 \cdot n^{-2}$	read D[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (11/8) \cdot n + (7/2)$	1	$0.143 \cdot n^{-2}$	read D[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.143 \cdot n^{-2}$	read D[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 1$	1	$0.143 \cdot n^{-2}$	read C[i1] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n - 1$	1	$0.143 \cdot n^{-2}$	write A[i1] (i0=0, i1=0)

n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(-1/2) \cdot n - 1$	$0.143 \cdot n^{-2}$	write A[i1] (i0=0)
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The matrix wraparound term appears ($0.0442 \cdot n^3$ at the $(1/8)n^2$ -line boundary), lifting d to 3.0, headroom +1.0.

bicg — single-shot [exact]

Accesses $A(n) = 7 \cdot n^2 + n$ (exact on $n \equiv 0 \pmod{8}$); DMD order $n^{2.5}$, headroom **+0.5**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.153 \cdot n^{2.5} + 12.8 \cdot n^2 + 2.45 \cdot n^{1.5} + 3.02 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.067	level	$(3/8) \cdot n + 2$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0156	read E[i2] (i0=0)
$n^{2.5}$	0.067	level	$(3/8) \cdot n + 1$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0156	read B[i2] (i0=0)
$n^{2.5}$	0.00957	level	$(3/8) \cdot n + 4$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00223	read E[i2] (i0=0)
$n^{2.5}$	0.00957	level	$(3/8) \cdot n + 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00223	read B[i2] (i0=0)
n^2	3.91	level	5	$(7/4) \cdot n^2$	0.25	read D[i1, i2] (i0=0); read E[i2] (i0=0)
n^2	2	level	4	n^2	0.143	read B[i2] (i0=0, i2=0); read A[i1] (i0=0, i2=0) (+1)
n^2	1.96	level	5	$(7/8) \cdot n^2$	0.125	read E[i2] (i0=0, i2=0); read C[i1] (i0=0)
n^2	1.73	level	3	n^2	0.143	read B[i2] (i0=0, i2=0); write B[i2] (i0=0)
n^2	1.52	level	3	$(7/8) \cdot n^2$	0.125	read B[i2] (i0=0)
n^2	1.41	level	2	n^2	0.143	write A[i1] (i0=0)
n^2	0.306	level	6	$(1/8) \cdot n^2 - n$	0.0179	read E[i2] (i0=0, i2=0); read C[i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.612	level	$(3/8) \cdot n + 2$	$n - 1$	$0.143/n$	read B[i2] (i0=0); read E[i2] (i0=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + 2$	$n - 1$	$0.143/n$	read B[i2] (i0=0, i2=0); read E[i2] (i0=0, i2=0)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	$0.125/n$	read B[i2] (i0=0)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	$0.125/n$	read B[i2] (i0=0, i2=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n - 1$	$0.0179/n$	read E[i2] (i0=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 4$	$(1/8) \cdot n - 1$	$0.0179/n$	read E[i2] (i0=0, i2=0)
n^1	2.14	level	6	$(7/8) \cdot n$	$0.125/n$	read C[i1] (i0=0, i2=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.125/n$	write A[i1] (i0=0)

Same structure as atax: matrix streamed once, vector ramps only (d = 2.5, +0.5).

cholesky — infinite-repeat [exact]

Accesses $A(n) = (2/3) \cdot n^3 + n^2 + (1/3) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.00347 \cdot n^4 + 0.00607 \cdot n^{3.5} + 1.17 \cdot n^3 + 0.298 \cdot n^{2.5} + 26.5 \cdot n^2 + 0.611 \cdot n^{1.5} + 77.1 \cdot n^1 + 172 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00268	ramp	$39 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(49/3072) \cdot n^3$ $+ (-$ $45/64) \cdot n^2$ $+ (479/48) \cdot n -$ 45	0.0239	read A[i2, i3] (i0=0)
n^4	0.00037	ramp	$64 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(7/3072) \cdot n^3$ $+ (-1/8) \cdot n^2$ $+ (107/48) \cdot n -$ 13	0.00342	read A[i2, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00037	ramp	55 → (1/16)· n^2 + (-1/4)· n - 1	(7/3072)· n^3 + (- 15/128)· n^2 + (23/12)· n - 10	0.00342	read A[i2, i3] (i0=0)
n^4	4.95e-05	ramp	89 → (1/16)· n^2 + (-1/4)· n - 1	(1/3072)· n^3 + (- 3/128)· n^2 + (13/24)· n - 4	0.000488	read A[i2, i3] (i0=0)
$n^{3.5}$	0.00455	ramp	8 → (1/4)· n	(1/64)· n^3 + (- 129/128)· n^2 + (341/16)· n - 147	0.0234	read A[i1, i3] (i0=0)
$n^{3.5}$	0.00079	ramp	8 → (1/4)· n	(1/384)· n^3 + (-9/64)· n^2 + (59/24)· n - 14	0.00391	read A[i1, i3] (i0=0)
$n^{3.5}$	0.000736	ramp	9 → (1/4)· n - 1	(1/384)· n^3 + (- 25/128)· n^2 + (229/48)· n - 38	0.00391	read A[i1, i3] (i0=0)
n^3	0.253	level	3	(7/48)· n^3 + (- 133/32)· n^2 + (931/24)· n - 119	0.219	read A[i2, i3] (i0=0)
n^3	0.221	level	3	(49/384)· n^3 + (- 469/128)· n^2 + (1687/48)· n - 112	0.191	read A[i1, i3] (i0=0)
n^3	0.217	level	3	(1/8)· n^3 + (-27/8)· n^2 + (125/4)· n - 98	0.188	read A[i1, i2] (i0=0, i3=0); read A[i1, i2] (i0=0)
n^3	0.109	level	1	(7/64)· n^3 + (- 351/128)· n^2 + (369/16)· n - 65	0.164	write A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0631	level	3	$(7/192) \cdot n^3 + (-133/128) \cdot n^2 + (413/48) \cdot n - 21$	0.0547	read A[i1, i2] (i0=0)
n^3	0.0618	ramp	$18 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(49/128) \cdot n^2 + (-131/16) \cdot n + 43$	0.574/n	read A[i2, i3] (i0=0, i3=0)
n^3	0.0618	ramp	$18 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(49/128) \cdot n^2 + (-131/16) \cdot n + 43$	0.574/n	read A[i2, i3] (i0=0)
n^3	0.0316	level	3	$(7/384) \cdot n^3 + (-15/128) \cdot n^2 + (-5/48) \cdot n - 1$	0.0273	read A[i1, i3] (i0=0)
n^3	0.0182	level	1	$(7/384) \cdot n^3 + (9/64) \cdot n^2 + (-8/3) \cdot n + 3$	0.0273	write A[i1, i2] (i0=0)
n^3	0.0182	level	1	$(7/384) \cdot n^3 + (-63/128) \cdot n^2 + (175/48) \cdot n - 7$	0.0273	write A[i1, i2] (i0=0)
n^3	0.0156	level	1	$(1/64) \cdot n^3 + (-33/128) \cdot n^2 + (17/16) \cdot n$	0.0234	write A[i1, i2] (i0=0)
n^3	0.0137	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(7/128) \cdot n^2 + (-39/16) \cdot n + 27$	0.082/n	read A[i1, i2] (i0=0, i3=0)
n^3	0.00902	level	3	$(1/192) \cdot n^3 + (-11/128) \cdot n^2 + (77/48) \cdot n - 11$	0.00781	read A[i1, i2] (i0=0)
n^3	0.00871	ramp	$32 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n - 1$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	0.082/n	read A[i2, i2] (i0=0)
n^3	0.00866	ramp	$35 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n - 1$	$(7/128) \cdot n^2 + (-25/16) \cdot n + 11$	0.082/n	read A[i2, i3] (i0=0, i3=0)
n^3	0.0086	ramp	$29 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	0.082/n	read A[i2, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0086	ramp	$29 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	$0.082/n$	read A[i2, i3] (i0=0)
n^3	0.00837	ramp	$60 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(7/128) \cdot n^2 + (-37/16) \cdot n + 24$	$0.082/n$	read A[i2, i3] (i0=0)
n^3	0.00836	ramp	$59 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n - 1$	$(7/128) \cdot n^2 + (-37/16) \cdot n + 24$	$0.082/n$	read A[i2, i3] (i0=0)
n^3	0.00718	ramp	$64 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(3/64) \cdot n^2 - 2 \cdot n + 21$	$0.0703/n$	read A[i2, i3] (i0=0)
n^3	0.00521	level	1	$(1/192) \cdot n^3 + (-3/128) \cdot n^2 + (563/48) \cdot n - 37$	0.00781	read A[i1, i1] (i0=0, i1=0); read A[i2, i2] (i0=0, i1=1, i2=0) (+9)
n^3	0.00195	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	$(1/128) \cdot n^2 + (-29/64) \cdot n + (825/128)$	$0.0117/n$	read A[i1, i2] (i0=0, i3=0)
n^3	0.00195	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.0117/n$	read A[i1, i2] (i0=0, i3=0)
n^3	0.00119	ramp	$60 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.0117/n$	read A[i2, i3] (i0=0)
n^3	0.00119	ramp	$56 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.0117/n$	read A[i2, i3] (i0=0)
n^3	0.00118	ramp	$54 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.0117/n$	read A[i2, i3] (i0=0, i3=0)
n^3	0.00118	ramp	$54 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.0117/n$	read A[i2, i2] (i0=0)
n^3	0.00114	ramp	$89 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.0117/n$	read A[i2, i3] (i0=0)
n^3	0.00113	ramp	$88 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.0117/n$	read A[i2, i3] (i0=0)
$n^{2.5}$	0.114	ramp	$6 \rightarrow (1/4) \cdot n$	$(7/16) \cdot n^2 + (-63/4) \cdot n + 141$	$0.656/n$	read A[i1, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.0981	ramp	$6 \rightarrow (1/4) \cdot n$	$(3/8) \cdot n^2 + (-105/8) \cdot n + 114$	$0.562/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0168	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	$0.0938/n$	read A[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.0168	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	$0.0938/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0162	ramp	$4 \rightarrow (1/4) \cdot n - 2$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0938/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0162	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(1/16) \cdot n^2 + (-21/8) \cdot n + 27$	$0.0938/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0148	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(3/64) \cdot n^2 + (-15/8) \cdot n + 18$	$0.0703/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.00247	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.0117/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.00234	ramp	$8 \rightarrow (1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.0117/n$	read A[i1, i4] (i0=0)
n^2	3.79	level	3	$(35/16) \cdot n^2 + (-147/4) \cdot n + 155$	$3.28/n$	read A[i2, i3] (i0=0)
n^2	2.27	level	3	$(21/16) \cdot n^2 + (-161/8) \cdot n + 77$	$1.97/n$	read A[i1, i3] (i0=0)
n^2	1.86	level	2	$(21/16) \cdot n^2 + (-35/4) \cdot n + 6$	$1.97/n$	read A[i2, i3] (i0=0)
n^2	1.62	level	3	$(15/16) \cdot n^2 + (-125/8) \cdot n + 65$	$1.41/n$	read A[i1, i2] (i0=0)
n^2	1.56	level	1	$(25/16) \cdot n^2 - 15 \cdot n + 40$	$2.34/n$	write A[i1, i2] (i0=0)
n^2	1.33	level	2	$(15/16) \cdot n^2 - 10 \cdot n + 30$	$1.41/n$	read A[i1, i2] (i0=0)
n^2	1.33	level	2	$(15/16) \cdot n^2 + (-13/4) \cdot n + 2$	$1.41/n$	read A[i2, i2] (i0=0); read A[i1, i2] (i0=0)
n^2	1.31	level	1	$(21/16) \cdot n^2 + (-97/8) \cdot n + 33$	$1.97/n$	read A[i1, i3] (i0=0)
n^2	1.08	level	3	$(5/8) \cdot n^2 + (-55/8) \cdot n + 15$	$0.938/n$	read A[i1, i2] (i0=0)
n^2	1.06	level	2	$(3/4) \cdot n^2 + (-9/4) \cdot n$	$1.12/n$	read A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.875	level	1	$(7/8) \cdot n^2 + (-63/8) \cdot n + 28$	$1.31/n$	write A[i1, i2] (i0=0); write A[i1, i1] (i0=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-63/8) \cdot n + 35$	$0.656/n$	read A[i1, i3] (i0=0)
n^2	0.707	level	2	$(1/2) \cdot n^2 + (-19/2) \cdot n + 46$	$0.75/n$	read A[i1, i1] (i0=0, i1=0); read A[i1, i2] (i0=0, i1=1, i2=0) (+2)
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-57/8) \cdot n + 33$	$0.562/n$	read A[i1, i2] (i0=0)
n^2	0.56	ramp	$6 \rightarrow (1/16) \cdot n^2 + (3/8) \cdot n + 1$	$(35/8) \cdot n - 30$	$6.56 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.541	level	2	$(49/128) \cdot n^2 + (-105/16) \cdot n + 28$	$0.574/n$	read A[i1, i4] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2 + (-15/8) \cdot n + 2$	$0.656/n$	read A[i1, i3] (i0=0)
n^2	0.375	level	1	$(3/8) \cdot n^2 + (-9/8) \cdot n$	$0.562/n$	write A[i1, i2] (i0=0)
n^2	0.336	ramp	$5 \rightarrow (1/16) \cdot n^2 + (3/8) \cdot n + 1$	$(21/8) \cdot n - 26$	$3.94 \cdot n^{-2}$	read A[i2, i3] (i0=0, i2=1, i3=0); read A[i2, i3] (i0=0, i2=7, i3=0) (+1)
n^2	0.312	level	1	$(5/16) \cdot n^2 + (-5/8) \cdot n$	$0.469/n$	write A[i1, i2] (i0=0)
n^2	0.312	level	1	$(5/16) \cdot n^2 + (-45/8) \cdot n + 25$	$0.469/n$	write A[i1, i2] (i0=0)
n^2	0.219	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 9$	$1.31 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.219	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 16$	$1.31 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^2	0.219	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 16$	$1.31 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.188	level	1	$(3/16) \cdot n^2 + (-15/8) \cdot n + 3$	$0.281/n$	write A[i1, i2] (i0=0)
n^2	0.128	ramp	$29 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$n - 17$	$1.5 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.125	level	1	$(1/8) \cdot n^2 + (-9/8) \cdot n + 8$	$0.188/n$	write A[i1, i2] (i0=0); write A[i1, i1] (i0=0)
n^2	0.113	ramp	$14 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 8$	$1.31 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.112	ramp	$16 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 9$	$1.31 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.112	ramp	$33 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 15$	$1.31 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.112	ramp	$12 \rightarrow (1/16) \cdot n^2 + (3/8) \cdot n$	$(7/8) \cdot n - 8$	$1.31 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	$0.0938/n$	read A[i1, i3] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0938/n$	read A[i1, i2] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0938/n$	read A[i1, i2] (i0=0, i3=0)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-13/8) \cdot n + 11$	$0.0938/n$	read A[i1, i2] (i0=0, i1=1, i2=0); read A[i1, i2] (i0=0, i1=2, i2=0) (+2)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0938/n$	read A[i1, i2] (i0=0)
n^2	0.0967	ramp	$16 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(3/4) \cdot n - 7$	$1.12 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0964	ramp	$36 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(3/4) \cdot n - 13$	$1.12 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0964	ramp	$36 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(3/4) \cdot n - 13$	$1.12 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0964	ramp	$36 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(3/4) \cdot n - 13$	$1.12 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (1/4) \cdot n$	$0.0938/n$	read A[i1, i2] (i0=0, i2=1, i3=0); read A[i1, i2] (i0=0)
n^2	0.0786	ramp	$11 \rightarrow$ $(1/16) \cdot n^2$ $+ (-3/8) \cdot n +$ 1	$(5/8) \cdot n - 5$	$0.938 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.0773	level	2	$(7/128) \cdot n^2$ $+ (-21/16) \cdot n$ $+ 7$	$0.082/n$	read A[i1, i4] (i0=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2$ $+ (-5/8) \cdot n +$ 1	$0.0938/n$	write A[i1, i2] (i0=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2$ $+ (-5/8) \cdot n +$ 1	$0.0938/n$	write A[i1, i2] (i0=0, i3=0)
n^2	0.047	ramp	$11 \rightarrow$ $(1/16) \cdot n^2$ $+ (-3/8) \cdot n +$ 1	$(3/8) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i2, i3] (i0=0, i2=1, i3=0); read A[i2, i3] (i0=0, i2=7, i3=0) (+1)
n^2	0.0312	level	$(1/16) \cdot n^2$ $+ (1/2) \cdot n +$ $(7/16)$	$(1/8) \cdot n +$ $(-17/8)$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.0312	level	$(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.0312	level	$(1/16) \cdot n^2$ $+ (1/2) \cdot n +$ $(7/16)$	$(1/8) \cdot n +$ $(-33/8)$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0312	level	$(1/16) \cdot n^2$ $+ (1/2) \cdot n +$ $(3/4)$	$(1/8) \cdot n +$ $(-13/4)$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0312	level	$(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(1/8) \cdot n - 4$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0312	level	$(1/16) \cdot n^2$ $+ (1/2) \cdot n +$ $(7/16)$	$(1/8) \cdot n +$ $(-25/8)$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^2	0.0312	level	$(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=8, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0312	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 4$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0312	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	$(1/8) \cdot n + (-33/8)$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0312	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 4$	$0.188 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0312	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	$(1/8) \cdot n + (-25/8)$	$0.188 \cdot n^{-2}$	read A[i1, i1] (i0=0, i4=0)
n^2	0.0312	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i1, i1] (i0=0, i4=0)
n^2	0.0159	ramp	$14 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n$	$(1/8) \cdot n - 1$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0159	ramp	$33 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0159	ramp	$32 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.0158	ramp	$12 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n$	$(1/8) \cdot n - 1$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0158	ramp	$30 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0158	ramp	$29 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0157	ramp	$55 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0157	ramp	$55 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.0157	ramp	$55 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0157	ramp	$27 \rightarrow (1/16) \cdot n^2 + (-3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0)
$n^{1.5}$	0.291	ramp	$6 \rightarrow (1/4) \cdot n$	$(7/8) \cdot n - 15$	$1.31 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.187	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(3/4) \cdot n - 12$	$1.12 \cdot n^{-2}$	read A[i1, i4] (i0=0)
$n^{1.5}$	0.0418	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=0)
$n^{1.5}$	0.0312	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0)
$n^{1.5}$	0.0308	ramp	$6 \rightarrow (1/8) \cdot n + 2$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0)
$n^{1.5}$	0.0296	ramp	$3 \rightarrow (1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	14	level	4	$7 \cdot n - 98$	$10.5 \cdot n^{-2}$	read A[i1, i3] (i0=0, i3=0)
n^1	7.58	level	3	$(35/8) \cdot n - 36$	$6.56 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^1	7.07	level	2	$5 \cdot n - 30$	$7.5 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^1	5.25	level	1	$(21/4) \cdot n - 21$	$7.88 \cdot n^{-2}$	write A[i1, i1] (i0=0)
n^1	3.71	level	2	$(21/8) \cdot n - 6$	$3.94 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^1	3.71	level	2	$(21/8) \cdot n - 21$	$3.94 \cdot n^{-2}$	read A[i1, i1] (i0=0)
n^1	2.83	level	2	$2 \cdot n - 16$	$3 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	2.62	level	1	$(21/8) \cdot n$	$3.94 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	2.62	level	1	$(21/8) \cdot n$	$3.94 \cdot n^{-2}$	read A[i1, i1] (i0=0)
n^1	2.12	level	2	$(3/2) \cdot n - 12$	$2.25 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	2	level	4	$n - 8$	$1.5 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=2, i2=0); read A[i2, i3] (i0=0, i1=3, i2=1, i3=0) (+2)
n^1	1.75	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	7	$10.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	1.5	level	1	$(3/2) \cdot n - 12$	$2.25 \cdot n^{-2}$	read A[i1, i1] (i0=0); write A[i1, i1] (i0=0)
n^1	1.41	level	2	$n - 9$	$1.5 \cdot n^{-2}$	read A[i1, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	5	$7.5 \cdot n^{-3}$	read A[i1, i1] (i0=0, i1=0); read A[i1, i2] (i0=0, i1=1, i2=0) (+3)
n^1	1.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	5	$7.5 \cdot n^{-3}$	read A[i1, i1] (i0=0, i1=0); read A[i1, i2] (i0=0, i1=1, i2=0) (+3)
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$1.31 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$1.31 \cdot n^{-2}$	read A[i1, i1] (i0=0)
n^1	1.24	level	2	$(7/8) \cdot n + (-63/8)$	$1.31 \cdot n^{-2}$	read A[i1, i1] (i0=0, i4=0)
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$1.31 \cdot n^{-2}$	read A[i1, i1] (i0=0, i4=0)
n^1	1.24	level	2	$(7/8) \cdot n - 1$	$1.31 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^1	1	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	4	$6 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	1	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	4	$6 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.875	level	1	$(7/8) \cdot n - 7$	$1.31 \cdot n^{-2}$	write A[i1, i1] (i0=0)
n^1	0.75	level	1	$(3/4) \cdot n - 6$	$1.12 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	0.5	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	2	$3 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=16, i3=0); read A[i1, i1] (i0=0, i1=8, i4=0)
n^1	0.5	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	2	$3 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=16, i3=0); read A[i1, i1] (i0=0, i1=8, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (3/4)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (3/4)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (3/4)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	1	$1.5 \cdot n^{-3}$	read A[i1, i1] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i1] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (3/4)$	1	$1.5 \cdot n^{-3}$	read A[i1, i1] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n$	1	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	1	$(1/4) \cdot n - 2$	$0.375 \cdot n^{-2}$	read A[i1, i1] (i0=0); write A[i1, i1] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (7/16)$	1	$1.5 \cdot n^{-3}$	read A[i1, i1] (i0=0, i4=0)
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^0	14	level	4	7	$10.5 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0)
n^0	6.63	level	11	2	$3 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=9, i2=7, i3=0); read A[i2, i3] (i0=0, i1=9, i2=8, i3=0)
n^0	6	level	1	6	$9 \cdot n^{-3}$	read A[i1, i1] (i0=0, i4=0)
n^0	5.39	level	29	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=17, i2=15, i3=0)
n^0	5.29	level	28	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=17, i2=8, i3=0)
n^0	5.1	level	26	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	5.1	level	26	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	5	level	25	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.9	level	24	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.9	level	24	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	4.8	level	23	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.69	level	22	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.69	level	22	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.58	level	21	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.47	level	20	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.47	level	20	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.36	level	19	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.24	level	18	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.24	level	18	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.12	level	17	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4	level	16	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4	level	16	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	3.87	level	15	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	3.74	level	14	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	3.61	level	13	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=10, i2=8, i3=0)
n^0	3.16	level	10	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=9, i2=0)
n^0	3	level	9	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	3	level	9	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.83	level	8	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.83	level	8	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.83	level	8	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2.65	level	7	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.65	level	7	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.65	level	7	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2.45	level	6	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.45	level	6	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.45	level	6	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2.24	level	5	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.24	level	5	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.24	level	5	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2	level	4	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2	level	4	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	1.73	level	3	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)

The factorization re-reads the trailing triangle: read A[i2,i3] ramps from tens of lines up to $(1/16)n^2 + (1/2)n$ (the triangular footprint), population $\sim n^3/60$, coefficient $0.0027 +$ smaller siblings — headroom +1.0 with the *smallest* leading coefficients in the suite. The term list

quantifies exactly how latent cholesky's tiling payoff is: the n^4 term exists, but its boundary crossover against the n^3 bulk sits far beyond the sizes any flat-miss-curve sweep reaches.

cholesky — single-shot [exact]

Accesses $A(n) = (2/3) \cdot n^3 + n^2 + (1/3) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.00347 \cdot n^4 + 0.00607 \cdot n^{3.5} + 1.15 \cdot n^3 + 0.298 \cdot n^{2.5} + 27.6 \cdot n^2 + 0.611 \cdot n^{1.5} + 53.7 \cdot n^1 + 173 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00268	ramp	$39 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(49/3072) \cdot n^3 + (-45/64) \cdot n^2 + (479/48) \cdot n - 45$	0.0239	read A[i2, i3] (i0=0)
n^4	0.00037	ramp	$64 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(7/3072) \cdot n^3 + (-1/8) \cdot n^2 + (107/48) \cdot n - 13$	0.00342	read A[i2, i3] (i0=0)
n^4	0.00037	ramp	$55 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(7/3072) \cdot n^3 + (-15/128) \cdot n^2 + (23/12) \cdot n - 10$	0.00342	read A[i2, i3] (i0=0)
n^4	4.95e-05	ramp	$89 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/3072) \cdot n^3 + (-3/128) \cdot n^2 + (13/24) \cdot n - 4$	0.000488	read A[i2, i3] (i0=0)
$n^{3.5}$	0.00455	ramp	$8 \rightarrow (1/4) \cdot n$	$(1/64) \cdot n^3 + (-129/128) \cdot n^2 + (341/16) \cdot n - 147$	0.0234	read A[i1, i3] (i0=0)
$n^{3.5}$	0.00079	ramp	$8 \rightarrow (1/4) \cdot n$	$(1/384) \cdot n^3 + (-9/64) \cdot n^2 + (59/24) \cdot n - 14$	0.00391	read A[i1, i3] (i0=0)
$n^{3.5}$	0.000736	ramp	$9 \rightarrow (1/4) \cdot n - 1$	$(1/384) \cdot n^3 + (-25/128) \cdot n^2 + (229/48) \cdot n - 38$	0.00391	read A[i1, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.505	level	3	$(7/24) \cdot n^3 + (-133/16) \cdot n^2 + (931/12) \cdot n - 238$	0.438	read A[i1, i3] (i0=0); read A[i2, i3] (i0=0)
n^3	0.289	level	3	$(1/6) \cdot n^3 - 5 \cdot n^2 + (299/6) \cdot n - 164$	0.25	read A[i2, i3] (i0=0, i1=2, i2=1, i3=0); read A[i1, i2] (i0=0)
n^3	0.146	level	1	$(7/48) \cdot n^3 + (-119/32) \cdot n^2 + (371/12) \cdot n - 84$	0.219	write A[i1, i2] (i0=0)
n^3	0.0618	ramp	$18 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(49/128) \cdot n^2 + (-131/16) \cdot n + 43$	0.574/n	read A[i2, i3] (i0=0, i3=0)
n^3	0.0618	ramp	$18 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(49/128) \cdot n^2 + (-131/16) \cdot n + 43$	0.574/n	read A[i2, i3] (i0=0)
n^3	0.0208	level	1	$(1/48) \cdot n^3 + (-9/32) \cdot n^2 + (11/12) \cdot n$	0.0312	write A[i1, i2] (i0=0)
n^3	0.00871	ramp	$32 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n - 1$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	0.082/n	read A[i2, i2] (i0=0)
n^3	0.00866	ramp	$35 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n - 1$	$(7/128) \cdot n^2 + (-25/16) \cdot n + 11$	0.082/n	read A[i2, i3] (i0=0, i3=0)
n^3	0.0086	ramp	$29 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	0.082/n	read A[i2, i3] (i0=0, i3=0)
n^3	0.0086	ramp	$29 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	0.082/n	read A[i2, i3] (i0=0)
n^3	0.00837	ramp	$60 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(7/128) \cdot n^2 + (-37/16) \cdot n + 24$	0.082/n	read A[i2, i3] (i0=0)
n^3	0.00836	ramp	$59 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n - 1$	$(7/128) \cdot n^2 + (-37/16) \cdot n + 24$	0.082/n	read A[i2, i3] (i0=0)
n^3	0.00718	ramp	$64 \rightarrow (1/16) \cdot n^2 + (1/2) \cdot n$	$(3/64) \cdot n^2 - 2 \cdot n + 21$	0.0703/n	read A[i2, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00119	ramp	$60 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n + 3$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.0117/n	read A[i2, i3] (i0=0)
n^3	0.00119	ramp	$56 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n + 3$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.0117/n	read A[i2, i3] (i0=0)
n^3	0.00118	ramp	$54 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.0117/n	read A[i2, i3] (i0=0, i3=0)
n^3	0.00118	ramp	$54 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.0117/n	read A[i2, i2] (i0=0)
n^3	0.00114	ramp	$89 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.0117/n	read A[i2, i3] (i0=0)
n^3	0.00113	ramp	$88 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.0117/n	read A[i2, i3] (i0=0)
$n^{2.5}$	0.114	ramp	$6 \rightarrow (1/4) \cdot n$	$(7/16) \cdot n^2 + (-63/4) \cdot n + 141$	0.656/n	read A[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.0981	ramp	$6 \rightarrow (1/4) \cdot n$	$(3/8) \cdot n^2 + (-105/8) \cdot n + 114$	0.562/n	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0168	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0938/n	read A[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.0168	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0938/n	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0162	ramp	$4 \rightarrow (1/4) \cdot n - 2$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	0.0938/n	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0162	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(1/16) \cdot n^2 + (-21/8) \cdot n + 27$	0.0938/n	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0148	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(3/64) \cdot n^2 + (-15/8) \cdot n + 18$	0.0703/n	read A[i1, i4] (i0=0)
$n^{2.5}$	0.00247	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.0117/n	read A[i1, i4] (i0=0)
$n^{2.5}$	0.00234	ramp	$8 \rightarrow (1/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.0117/n	read A[i1, i4] (i0=0)
n^2	3.79	level	3	$(35/16) \cdot n^2 + (-259/8) \cdot n + 119$	3.28/n	read A[i2, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	3.03	level	3	$(7/4) \cdot n^2 + (-49/2) \cdot n + 84$	$2.62/n$	read A[i1, i3] (i0=0)
n^2	3.03	level	3	$(7/4) \cdot n^2 + (-49/2) \cdot n + 84$	$2.62/n$	read A[i1, i2] (i0=0)
n^2	2.62	level	1	$(21/8) \cdot n^2 + (-189/8) \cdot n + 56$	$3.94/n$	write A[i1, i2] (i0=0)
n^2	2.47	level	2	$(7/4) \cdot n^2 - 14 \cdot n + 35$	$2.62/n$	read A[i1, i2] (i0=0)
n^2	1.86	level	2	$(21/16) \cdot n^2 + (-49/8) \cdot n$	$1.97/n$	read A[i2, i3] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2 - 14 \cdot n + 35$	$2.62/n$	read A[i1, i3] (i0=0)
n^2	1.41	level	2	$n^2 - 11 \cdot n + 47$	$1.5/n$	read A[i2, i2] (i0=0, i1=2, i2=0); read A[i1, i2] (i0=0) (+1)
n^2	0.866	level	3	$(1/2) \cdot n^2 + (-19/2) \cdot n + 45$	$0.75/n$	read A[i1, i2] (i0=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-63/8) \cdot n + 35$	$0.656/n$	read A[i1, i2] (i0=0, i3=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-63/8) \cdot n + 35$	$0.656/n$	read A[i1, i3] (i0=0)
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-63/8) \cdot n + 35$	$0.656/n$	read A[i1, i4] (i0=0)
n^2	0.56	ramp	$6 \rightarrow (1/16) \cdot n^2 + (3/8) \cdot n + 1$	$(35/8) \cdot n - 30$	$6.56 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.541	level	2	$(49/128) \cdot n^2 + (-7/16) \cdot n$	$0.574/n$	read A[i2, i2] (i0=0)
n^2	0.5	level	1	$(1/2) \cdot n^2 + (19/2) \cdot n - 36$	$0.75/n$	read A[i2, i2] (i0=0, i1=1, i2=0); read A[i1, i3] (i0=0, i3=0) (+4)
n^2	0.438	level	1	$(7/16) \cdot n^2 - 7 \cdot n + 28$	$0.656/n$	write A[i1, i2] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2 - 7 \cdot n + 28$	$0.656/n$	write A[i1, i1] (i0=0)
n^2	0.375	level	1	$(3/8) \cdot n^2 + (-3/8) \cdot n$	$0.562/n$	write A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.336	ramp	$5 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/8) \cdot n +$ 1	$(21/8) \cdot n -$ 26	$3.94 \cdot n^{-2}$	read A[i2, i3] (i0=0, i2=1, i3=0); read A[i2, i3] (i0=0, i2=7, i3=0) (+1)
n^2	0.128	ramp	$29 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$n - 17$	$1.5 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.113	ramp	$14 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(7/8) \cdot n - 8$	$1.31 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.112	ramp	$16 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(7/8) \cdot n - 9$	$1.31 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.112	ramp	$33 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(7/8) \cdot n - 15$	$1.31 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.112	ramp	$12 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/8) \cdot n$	$(7/8) \cdot n - 8$	$1.31 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0)
n^2	0.0967	ramp	$16 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(3/4) \cdot n - 7$	$1.12 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0964	ramp	$36 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(3/4) \cdot n - 13$	$1.12 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0964	ramp	$36 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(3/4) \cdot n - 13$	$1.12 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.0964	ramp	$36 \rightarrow$ $(1/16) \cdot n^2$ $+ (1/2) \cdot n$	$(3/4) \cdot n - 13$	$1.12 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0786	ramp	$11 \rightarrow$ $(1/16) \cdot n^2$ $+ (-3/8) \cdot n +$ 1	$(5/8) \cdot n - 5$	$0.938 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.0773	level	2	$(7/128) \cdot n^2$ $+ (-7/16) \cdot n$	$0.082/n$	read A[i2, i2] (i0=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2$ $+ (-1/2) \cdot n$	$0.0938/n$	write A[i1, i2] (i0=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2$ $+ (-1/2) \cdot n$	$0.0938/n$	write A[i1, i1] (i0=0)
n^2	0.047	ramp	$11 \rightarrow$ $(1/16) \cdot n^2$ $+ (-3/8) \cdot n +$ 1	$(3/8) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i2, i3] (i0=0, i2=1, i3=0); read A[i2, i3] (i0=0, i2=7, i3=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0159	ramp	$14 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n$	$(1/8) \cdot n - 1$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0159	ramp	$33 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0159	ramp	$32 \rightarrow (1/16) \cdot n^2 + (-1/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.0158	ramp	$12 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n$	$(1/8) \cdot n - 1$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0158	ramp	$30 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0158	ramp	$29 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0157	ramp	$55 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0157	ramp	$55 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0, i3=0)
n^2	0.0157	ramp	$55 \rightarrow (1/16) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i2, i3] (i0=0)
n^2	0.0157	ramp	$27 \rightarrow (1/16) \cdot n^2 + (-3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0)
$n^{1.5}$	0.291	ramp	$6 \rightarrow (1/4) \cdot n$	$(7/8) \cdot n - 15$	$1.31 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=0)
$n^{1.5}$	0.187	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(3/4) \cdot n - 12$	$1.12 \cdot n^{-2}$	read A[i1, i4] (i0=0)
$n^{1.5}$	0.0418	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=0)
$n^{1.5}$	0.0312	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0)
$n^{1.5}$	0.0308	ramp	$6 \rightarrow (1/8) \cdot n + 2$	$(1/8) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0)
$n^{1.5}$	0.0296	ramp	$3 \rightarrow (1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	14	level	4	$7 \cdot n - 98$	$10.5 \cdot n^{-2}$	read A[i1, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	9.9	level	2	$7 \cdot n - 35$	$10.5 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^1	5.25	level	1	$(21/4) \cdot n - 21$	$7.88 \cdot n^{-2}$	write A[i1, i1] (i0=0)
n^1	4.95	level	2	$(7/2) \cdot n - 28$	$5.25 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	4.95	level	2	$(7/2) \cdot n - 28$	$5.25 \cdot n^{-2}$	read A[i1, i1] (i0=0)
n^1	3.5	level	1	$(7/2) \cdot n - 7$	$5.25 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	3.5	level	1	$(7/2) \cdot n - 7$	$5.25 \cdot n^{-2}$	read A[i1, i1] (i0=0)
n^1	2	level	4	$n - 8$	$1.5 \cdot n^{-2}$	read A[i2, i3] (i0=0, i1=3, i2=1, i3=0); read A[i1, i4] (i0=0, i1=9) (+1)
n^1	1.41	level	2	$n - 9$	$1.5 \cdot n^{-2}$	read A[i1, i1] (i0=0)
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$1.31 \cdot n^{-2}$	read A[i1, i4] (i0=0)
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$1.31 \cdot n^{-2}$	read A[i1, i1] (i0=0, i4=0)
n^1	0.875	level	1	$(7/8) \cdot n - 7$	$1.31 \cdot n^{-2}$	write A[i1, i1] (i0=0)
n^1	0.75	level	1	$(3/4) \cdot n$	$1.12 \cdot n^{-2}$	write A[i1, i1] (i0=0)
n^1	0.125	level	1	$(1/8) \cdot n$	$0.188 \cdot n^{-2}$	write A[i1, i1] (i0=0)
n^0	14	level	4	7	$10.5 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0)
n^0	7	level	1	7	$10.5 \cdot n^{-3}$	read A[i1, i1] (i0=0, i4=0)
n^0	6.63	level	11	2	$3 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=9, i2=7, i3=0); read A[i2, i3] (i0=0, i1=9, i2=8, i3=0)
n^0	5.39	level	29	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=17, i2=15, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	5.29	level	28	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=17, i2=8, i3=0)
n^0	5.1	level	26	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	5.1	level	26	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	5	level	25	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.9	level	24	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.9	level	24	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.8	level	23	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.69	level	22	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.69	level	22	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.58	level	21	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.47	level	20	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.47	level	20	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.36	level	19	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4.24	level	18	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.24	level	18	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	4.12	level	17	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	4	level	16	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	4	level	16	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	3.87	level	15	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i2=8, i3=0)
n^0	3.74	level	14	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	3.61	level	13	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i1=10, i2=8, i3=0)
n^0	3.16	level	10	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=9, i2=0)
n^0	3	level	9	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	3	level	9	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.83	level	8	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.83	level	8	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.83	level	8	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2.65	level	7	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.65	level	7	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.65	level	7	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2.45	level	6	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.45	level	6	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.45	level	6	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2.24	level	5	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	2.24	level	5	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2.24	level	5	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	2	level	4	1	$1.5 \cdot n^{-3}$	read A[i2, i3] (i0=0, i3=0)
n^0	2	level	4	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
n^0	1.73	level	3	1	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)

The factorization re-reads the trailing triangle: read A[i2,i3] ramps from tens of lines up to $(1/16)n^2 + (1/2)n$ (the triangular footprint), population $\sim n^3/60$, coefficient $0.0027 +$ smaller siblings — headroom $+1.0$ with the *smallest* leading coefficients in the suite. The term list quantifies exactly how latent cholesky’s tiling payoff is: the n^4 term exists, but its boundary crossover against the n^3 bulk sits far beyond the sizes any flat-miss-curve sweep reaches.

convolution — infinite-repeat [exact]

Accesses $A(n) = 163 \cdot n^2 - 2934 \cdot n + 13203$ (exact on $n \equiv 0 \pmod{16}$); DMD order n^3 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=512$, 1 at $n=528$.

DMD spectrum: $0.125 \cdot n^3 + 1.12 \cdot n^{2.5} + 399 \cdot n^2 + 35.8 \cdot n^{1.5} + 199 \cdot n^1 + 48 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0625	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 34$	$(1/8) \cdot n^2 + (-51/8) \cdot n + 76$	0.000767	read B[i1 + i3, i2 + i4] (i0=0)
n^3	0.0625	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 27$	$(1/8) \cdot n^2 + (-33/8) \cdot n + 27$	0.000767	write C[i1, i2] (i0=0, i1=0); write C[i1, i2] (i0=0)
$n^{2.5}$	1.12	level	$(5/4) \cdot n + 15$	$n^2 - 42 \cdot n + 320$	0.00613	read B[i1 + i3, i2 + i4] (i0=0, i3=0); read B[i1 + i3, i2 + i4] (i0=0)
n^2	89.1	level	2	$63 \cdot n^2 - 1134 \cdot n + 5103$	0.387	read B[i1 + i3, i2 + i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	89.1	level	2	$63 \cdot n^2 - 1134 \cdot n + 5103$	0.387	read A[i3, i4] (i0=0)
n^2	37.3	level	37	$(49/8) \cdot n^2 + (-889/8) \cdot n + 504$	0.0376	read A[i3, i4] (i0=0, i1=0); read A[i3, i4] (i0=0)
n^2	36.8	level	36	$(49/8) \cdot n^2 + (-945/8) \cdot n + 567$	0.0376	read A[i3, i4] (i0=0, i1=0, i4=0); read A[i3, i4] (i0=0, i4=0)
n^2	36.8	level	36	$(49/8) \cdot n^2 + (-889/8) \cdot n + 504$	0.0376	read B[i1 + i3, i2 + i4] (i0=0, i1=0); read B[i1 + i3, i2 + i4] (i0=0)
n^2	36.5	level	37	$6 \cdot n^2 - 110 \cdot n + 504$	0.0368	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=8, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i1=0, i4=0)
n^2	6.08	level	37	$n^2 - 17 \cdot n + 72$	0.00613	(+2) read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=8, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i1=0, i4=0)
n^2	5.92	level	35	$n^2 - 25 \cdot n + 144$	0.00613	(+2) read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=8, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i1=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-119/8) \cdot n + 63$	0.00537	read A[i3, i4] (i0=0, i1=0, i4=0); read A[i3, i4] (i0=0, i4=0)
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read A[i3, i4] (i0=0, i1=0); read A[i3, i4] (i0=0)
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read A[i3, i4] (i0=0, i1=0, i3=0); read A[i3, i4] (i0=0, i3=0)
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	write C[i1, i2] (i0=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-135/8) \cdot n + 81$	0.00537	read A[i3, i4] (i0=0, i1=0, i4=0); read A[i3, i4] (i0=0, i4=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-135/8) \cdot n + 81$	0.00537	read A[i3, i4] (i0=0, i1=0, i3=0, i4=0); read A[i3, i4] (i0=0, i3=0, i4=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0); read B[i1 + i3, i2 + i4] (i0=0, i3=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read B[i1 + i3, i2 + i4] (i0=0, i1=0); read B[i1 + i3, i2 + i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-175/8) \cdot n + 126$	0.00537	read A[i3, i4] (i0=0, i1=0); read A[i3, i4] (i0=0)
n^2	4.56	level	37	$(3/4) \cdot n^2 + (-55/4) \cdot n + 63$	0.0046	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i3=0, i4=0)
n^2	0.76	level	37	$(1/8) \cdot n^2 + (-17/8) \cdot n + 9$	0.000767	read A[i3, i4] (i0=0, i1=0, i4=0); read A[i3, i4] (i0=0, i4=0)
n^2	0.76	level	37	$(1/8) \cdot n^2 + (-17/8) \cdot n + 9$	0.000767	read A[i3, i4] (i0=0, i1=0, i3=0, i4=0); read A[i3, i4] (i0=0, i3=0, i4=0)
n^2	0.76	level	37	$(1/8) \cdot n^2 + (-17/8) \cdot n + 9$	0.000767	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i3=0, i4=0)
n^2	0.75	level	36	$(1/8) \cdot n^2 + (-25/8) \cdot n + 18$	0.000767	read A[i3, i4] (i0=0, i1=0); read A[i3, i4] (i0=0)
n^2	0.75	level	36	$(1/8) \cdot n^2 + (-25/8) \cdot n + 18$	0.000767	read A[i3, i4] (i0=0, i1=0, i3=0); read A[i3, i4] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.74	level	35	$(1/8) \cdot n^2 + (-25/8) \cdot n + 18$	0.000767	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i3=0, i4=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 34$	$n - 19$	0.00613/n	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 34$	$n - 19$	0.00613/n	read B[i1 + i3, i2 + i4] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 35$	$n - 18$	0.00613/n	read B[i1 + i3, i2 + i4] (i0=0, i1=0); read B[i1 + i3, i2 + i4] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 27$	$n - 9$	0.00613/n	write C[i1, i2] (i0=0, i1=0); write C[i1, i2] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + (39/2)$	$n - 9$	0.00613/n	write C[i1, i2] (i0=0, i1=0); write C[i1, i2] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 27$	$n - 9$	0.00613/n	write C[i1, i2] (i0=0, i1=0, i2=0); write C[i1, i2] (i0=0, i2=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 35$	$n - 19$	0.00613/n	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^2	0.411	ramp	$(1/4) \cdot n^2 - 4 \cdot n + 54 \rightarrow (1/4) \cdot n^2 + (-19/8) \cdot n + 25$	$(7/8) \cdot n - 28$	0.00537/n	read B[i1 + i3, i2 + i4] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.411	ramp	$(1/4) \cdot n^2 + (-17/4) \cdot n + 58 \rightarrow$ $(1/4) \cdot n^2 + (-5/2) \cdot n + 24$	$(7/8) \cdot n - 28$	$0.00537/n$	read B[i1 + i3, i2 + i4] (i0=0)
n^2	0.125	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 26$	$(1/4) \cdot n - 8$	$0.00153/n$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0); read B[i1 + i3, i2 + i4] (i0=0)
n^2	0.0588	ramp	$(1/4) \cdot n^2 + (-5/2) \cdot n + 30 \rightarrow$ $(1/4) \cdot n^2 + (-19/8) \cdot n + 25$	$(1/8) \cdot n - 4$	$0.000767/n$	read B[i1 + i3, i2 + i4] (i0=0)
n^2	0.0586	ramp	$(1/4) \cdot n^2 + (-17/4) \cdot n + 58 \rightarrow$ $(1/4) \cdot n^2 + (-13/4) \cdot n + 18$	$(1/8) \cdot n - 4$	$0.000767/n$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
$n^{1.5}$	7.83	level	$(5/4) \cdot n + 16$	$7 \cdot n - 70$	$0.0429/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i3=0, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
$n^{1.5}$	6.71	level	$(5/4) \cdot n + 16$	$6 \cdot n - 60$	$0.0368/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 8$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 9$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 10$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 11$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 12$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 13$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 14$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 7$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i3=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 8$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 9$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 10$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 11$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 12$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 13$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 7$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 14$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i3=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 16$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 16$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 16$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	48.7	level	37	$8 \cdot n - 80$	$0.0491/n$	read A[i3, i4] (i0=0, i2=0, i3=0, i4=0); read A[i3, i4] (i0=0, i2=0, i4=0)
n^1	36.5	level	37	$6 \cdot n - 53$	$0.0368/n$	read A[i3, i4] (i0=0, i1=0, i2=0, i4=8); read A[i3, i4] (i0=0, i2=0, i4=8)
n^1	36	level	36	$6 \cdot n - 60$	$0.0368/n$	read A[i3, i4] (i0=0, i2=0, i4=0)
n^1	6.08	level	37	$n - 1$	$0.00613/n$	read A[i3, i4] (i0=0, i1=0, i2=0, i3=0, i4=0); read A[i3, i4] (i0=0, i1=0, i2=0, i3=0, i4=8) (+5)
n^1	6.08	level	37	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i4=8)
n^1	6.08	level	37	$n - 9$	$0.00613/n$	read A[i3, i4] (i0=0, i1=0, i2=0, i4=8); read A[i3, i4] (i0=0, i2=0, i4=8)
n^1	6.08	level	37	$n - 9$	$0.00613/n$	read A[i3, i4] (i0=0, i1=0, i2=0, i3=0, i4=8); read A[i3, i4] (i0=0, i2=0, i3=0, i4=8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	6	level	36	$n - 9$	$0.00613/n$	read A[i3, i4] (i0=0, i1=0, i2=0, i4=0); read A[i3, i4] (i0=0, i1=0, i2=0, i4=8) (+2)
n^1	6	level	36	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i4=0)
n^1	6	level	36	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 4 \cdot n + 47$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-15/4) \cdot n + 44$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-7/2) \cdot n + 41$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 38$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 35$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 32$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 29$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-17/4) \cdot n + 50$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 26$	1	$0.00613 \cdot n^{-2}$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (-25/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-23/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-21/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-19/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i3=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-25/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-23/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (-21/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-19/8) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-17/4) \cdot n + 50$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 4 \cdot n + 47$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-15/4) \cdot n + 44$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-7/2) \cdot n + 41$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 38$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 35$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 32$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 29$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 26$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-25/8) \cdot n + 34$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 33$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-23/8) \cdot n + 32$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 31$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-21/8) \cdot n + 30$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 29$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-19/8) \cdot n + 28$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 27$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + (113/4)$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i3=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-17/4) \cdot n + 51$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 4 \cdot n + 48$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-15/4) \cdot n + 45$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-7/2) \cdot n + 42$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 39$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 36$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 33$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 30$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 27$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 4 \cdot n + 48$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-15/4) \cdot n + 45$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-7/2) \cdot n + 42$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 39$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 36$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 33$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 30$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-17/4) \cdot n + 51$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-13/4) \cdot n + 35$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (-25/8) \cdot n + 34$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 - 3 \cdot n + 33$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-23/8) \cdot n + 32$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-11/4) \cdot n + 31$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-21/8) \cdot n + 30$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/2) \cdot n + 29$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-19/8) \cdot n + 28$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 27$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i1=0, i2=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 27$	1	$0.00613 \cdot n^2$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^0	42	level	36	7	$0.0429 \cdot n^2$	read A[i3, i4] (i0=0, i1=0, i2=0, i4=0); read A[i3, i4] (i0=0, i1=0, i2=0, i4=8) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	6	level	36	1	$0.00613 \cdot n^{-2}$	read A[i3, i4] (i0=0, i1=0, i2=0, i3=0, i4=0); read A[i3, i4] (i0=0, i1=0, i2=0, i3=0, i4=8) (+1)

Across invocations the image itself wraps at $(1/4)n^2 + O(n)$ lines ($0.0625 + 0.0625$ for the two image arrays), giving $d = 3.0$. The filter never leaves cache at any realistic size.

convolution — single-shot [exact]

Accesses $A(n) = 163 \cdot n^2 - 2934 \cdot n + 13203$ (exact on $n \equiv 0 \pmod{8}$); DMD order $n^{2.5}$, headroom **+0.5**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $1.12 \cdot n^{2.5} + 394 \cdot n^2 + 35.8 \cdot n^{1.5} + 163 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.12	level	$(5/4) \cdot n + 15$	$n^2 - 42 \cdot n + 320$	0.00613	read B[i1 + i3, i2 + i4] (i0=0, i3=0); read B[i1 + i3, i2 + i4] (i0=0)
n^2	89.1	level	2	$63 \cdot n^2 - 1134 \cdot n + 5103$	0.387	read B[i1 + i3, i2 + i4] (i0=0)
n^2	89.1	level	2	$63 \cdot n^2 - 1134 \cdot n + 5103$	0.387	read A[i3, i4] (i0=0)
n^2	37.3	level	37	$(49/8) \cdot n^2 + (-889/8) \cdot n + 504$	0.0376	read A[i3, i4] (i0=0)
n^2	36.8	level	36	$(49/8) \cdot n^2 + (-945/8) \cdot n + 567$	0.0376	read A[i3, i4] (i0=0, i4=0)
n^2	36.8	level	36	$(49/8) \cdot n^2 + (-889/8) \cdot n + 504$	0.0376	read B[i1 + i3, i2 + i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	36.5	level	37	$6 \cdot n^2 - 110 \cdot n + 504$	0.0368	read B[i1 + i3, i2 + i4] (i0=0, i3=8, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i4=0)
n^2	6.08	level	37	$n^2 - 17 \cdot n + 72$	0.00613	read B[i1 + i3, i2 + i4] (i0=0, i3=8, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i4=0)
n^2	5.92	level	35	$n^2 - 25 \cdot n + 144$	0.00613	read B[i1 + i3, i2 + i4] (i0=0, i3=8, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i4=0)
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-119/8) \cdot n + 63$	0.00537	read A[i3, i4] (i0=0, i4=0)
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read A[i3, i4] (i0=0)
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read A[i3, i4] (i0=0, i3=0)
n^2	5.32	level	37	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	write C[i1, i2] (i0=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-135/8) \cdot n + 81$	0.00537	read A[i3, i4] (i0=0, i4=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-135/8) \cdot n + 81$	0.00537	read A[i3, i4] (i0=0, i3=0, i4=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read B[i1 + i3, i2 + i4] (i0=0)
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-127/8) \cdot n + 72$	0.00537	read B[i1 + i3, i2 + i4] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	5.25	level	36	$(7/8) \cdot n^2 + (-175/8) \cdot n + 126$	0.00537	read A[i3, i4] (i0=0)
n^2	4.56	level	37	$(3/4) \cdot n^2 + (-55/4) \cdot n + 63$	0.0046	read B[i1 + i3, i2 + i4] (i0=0, i3=0, i4=0)
n^2	0.76	level	37	$(1/8) \cdot n^2 + (-17/8) \cdot n + 9$	0.000767	read A[i3, i4] (i0=0, i4=0)
n^2	0.76	level	37	$(1/8) \cdot n^2 + (-17/8) \cdot n + 9$	0.000767	read A[i3, i4] (i0=0, i3=0, i4=0)
n^2	0.76	level	37	$(1/8) \cdot n^2 + (-17/8) \cdot n + 9$	0.000767	read B[i1 + i3, i2 + i4] (i0=0, i3=0, i4=0)
n^2	0.75	level	36	$(1/8) \cdot n^2 + (-25/8) \cdot n + 18$	0.000767	read A[i3, i4] (i0=0)
n^2	0.75	level	36	$(1/8) \cdot n^2 + (-25/8) \cdot n + 18$	0.000767	read A[i3, i4] (i0=0, i3=0)
n^2	0.74	level	35	$(1/8) \cdot n^2 + (-25/8) \cdot n + 18$	0.000767	read B[i1 + i3, i2 + i4] (i0=0, i3=0, i4=0)
$n^{1.5}$	7.83	level	$(5/4) \cdot n + 16$	$7 \cdot n - 70$	0.0429/n	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i3=0, i4=0); read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
$n^{1.5}$	6.71	level	$(5/4) \cdot n + 16$	$6 \cdot n - 60$	0.0368/n	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 8$	$n - 10$	0.00613/n	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 9$	$n - 10$	0.00613/n	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 10$	$n - 10$	0.00613/n	read B[i1 + i3, i2 + i4] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 11$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 12$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 13$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 14$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 7$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i3=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 8$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 9$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 10$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 11$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 12$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 13$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 7$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 14$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i3=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 16$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0)
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 16$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.12	level	$(5/4) \cdot n + 16$	$n - 10$	$0.00613/n$	read B[i1 + i3, i2 + i4] (i0=0, i2=0, i4=0)
n^1	48.7	level	37	$8 \cdot n - 80$	$0.0491/n$	read A[i3, i4] (i0=0, i2=0, i3=0, i4=0); read A[i3, i4] (i0=0, i2=0, i4=0)
n^1	36.5	level	37	$6 \cdot n - 60$	$0.0368/n$	read A[i3, i4] (i0=0, i2=0, i4=8)
n^1	36	level	36	$6 \cdot n - 60$	$0.0368/n$	read A[i3, i4] (i0=0, i2=0, i4=0)
n^1	6.08	level	37	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i3=0, i4=8); read A[i3, i4] (i0=0, i2=0, i4=0) (+1)
n^1	6.08	level	37	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i4=8)
n^1	6.08	level	37	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i4=8)
n^1	6.08	level	37	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i3=0, i4=8)
n^1	6	level	36	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i4=0); read A[i3, i4] (i0=0, i2=0, i4=8)
n^1	6	level	36	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i4=0)
n^1	6	level	36	$n - 10$	$0.00613/n$	read A[i3, i4] (i0=0, i2=0, i3=0, i4=0)

Two-scale kernel (image $n \times n$, filter fixed 9×9 ; all parameters except the filter extent bound to n). Single-shot: the sliding image window read B[i1+i3, i2+i4] is re-read across output

columns at $(5/4)n + 15$ lines (nine rows' working set), order $n^{2.5}$; the filter lives at constant distances (2 and 37 lines — fully resident) and carries the n^2 bulk with huge constants (81 touches per output). Headroom +0.5 over the n^2 access count.

correlation — infinite-repeat [exact]

Accesses $A(n) = 2 \cdot n^3 + (21/2) \cdot n^2 + (11/2) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma \text{mass/warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0153 \cdot n^4 + 1.33 \cdot n^{3.5} + 2.46 \cdot n^3 + 19 \cdot n^{2.5} + 45.6 \cdot n^2 + 44.1 \cdot n^{1.5} + 50 \cdot n^1 + 31.9 \cdot n^{0.5} + 1 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0115	ramp	$5 \cdot n + 32 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(3/64) \cdot n^3$ + (- $171/64) \cdot n^2$ + $(309/8) \cdot n$ - 36	0.0234	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^4	0.00193	ramp	$4 \cdot n + 22 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 14$	$(1/128) \cdot n^3$ + $(-7/16) \cdot n^2$ + $(49/8) \cdot n -$ 3	0.00391	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^4	0.00183	ramp	$6 \cdot n + 38 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 16$	$(1/128) \cdot n^3$ + (- $73/128) \cdot n^2$ + $(169/16) \cdot n -$ 10	0.00391	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.619	level	$2 \cdot n + 2$	$(7/16) \cdot n^3$ + $(-63/8) \cdot n^2$ + $35 \cdot n$	0.219	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)
$n^{3.5}$	0.475	level	$2 \cdot n + 2$	$(43/128) \cdot n^3$ + (- $99/16) \cdot n^2$ + $28 \cdot n$	0.168	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.0884	level	$2 \cdot n + 3$	$(1/16) \cdot n^3$ + $(-19/8) \cdot n^2$ + $22 \cdot n$	0.0312	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)
$n^{3.5}$	0.0663	level	$2 \cdot n + 2$	$(3/64) \cdot n^3$ + $(-3/8) \cdot n^2$	0.0234	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.0663	level	$2 \cdot n + 2$	$(3/64) \cdot n^3$ + $(-3/8) \cdot n^2$	0.0234	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.011	level	$2 \cdot n + 2$	$(1/128) \cdot n^3 + (-1/16) \cdot n^2$	0.00391	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.663	level	3	$(49/128) \cdot n^3 + (-91/16) \cdot n^2 + 21 \cdot n$	0.191	write D[i7, i8] (i0=0, i9=0); write D[i7, i8] (i0=0)
n^3	0.383	level	1	$(49/128) \cdot n^3 + (-91/16) \cdot n^2 + 21 \cdot n$	0.191	read D[i7, i8] (i0=0, i9=0); read D[i7, i8] (i0=0)
n^3	0.177	ramp	$4 \cdot n + 23 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(3/4) \cdot n^2 + (-75/4) \cdot n + 18$	0.375/n	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.15	ramp	$3 \cdot n + 14 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 9$	$(5/8) \cdot n^2 - 10 \cdot n$	0.312/n	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.148	ramp	$4 \cdot n + 23 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 9$	$(5/8) \cdot n^2 + (-125/8) \cdot n + 15$	0.312/n	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.0947	level	3	$(7/128) \cdot n^3 + (-21/16) \cdot n^2 + 7 \cdot n$	0.0273	write D[i7, i8] (i0=0, i9=0); write D[i7, i8] (i0=0)
n^3	0.0947	level	3	$(7/128) \cdot n^3 + (-7/16) \cdot n^2$	0.0273	write D[i7, i8] (i0=0, i9=0); write D[i7, i8] (i0=0)
n^3	0.0939	ramp	$n + 5 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(21/64) \cdot n^2 + (-3/2) \cdot n + 1$	0.164/n	write D[i8, i7] (i0=0)
n^3	0.0547	level	1	$(7/128) \cdot n^3 + (-21/16) \cdot n^2 + 7 \cdot n$	0.0273	read D[i7, i8] (i0=0, i9=0); read D[i7, i8] (i0=0)
n^3	0.0547	level	1	$(7/128) \cdot n^3 + (-7/16) \cdot n^2$	0.0273	read D[i7, i8] (i0=0, i9=0); read D[i7, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0451	ramp	$(1/8) \cdot n^2 + (1/8) \cdot n + 9 \rightarrow (1/4) \cdot n^2 - 4 \cdot n + 36$	$(1/8) \cdot n^2 - 4 \cdot n + 2$	0.0625/n	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
n^3	0.0409	ramp	$(1/8) \cdot n^2 + (1/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n^2 - 3 \cdot n + 1$	0.0625/n	read B[i4, i3] (i0=0, i4=0); read B[i4, i3] (i0=0)
n^3	0.036	ramp	$(5/2) \cdot n - 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0625/n	read B[i5, i6] (i0=0)
n^3	0.0347	ramp	$(9/2) \cdot n + 19 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n^2 + (11/8) \cdot n - 19$	$(1/8) \cdot n^2 + (-17/4) \cdot n + 8$	0.0625/n	read B[i9, i8] (i0=0, i7=0)
n^3	0.03	ramp	$3 \cdot n + 19 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0625/n	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i8=6)
n^3	0.0299	ramp	$3 \cdot n + 13 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 14$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0625/n	read B[i9, i8] (i0=0, i8=0, i9=0); read B[i9, i8] (i0=0, i8=0)
n^3	0.0296	ramp	$3 \cdot n + 13 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 14$	$(1/8) \cdot n^2 - 3 \cdot n + 1$	0.0625/n	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.0296	ramp	$4 \cdot n + 21 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 15$	$(1/8) \cdot n^2 - 3 \cdot n$	0.0625/n	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
n^3	0.0296	ramp	$4 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 16$	$(1/8) \cdot n^2 - 3 \cdot n$	0.0625/n	read B[i9, i8] (i0=0, i8=7, i9=0); read B[i9, i8] (i0=0, i8=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0295	ramp	$4 \cdot n + 28 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(1/8) \cdot n^2 +$ $(-25/8) \cdot n +$ 3	0.0625/n	read B[i9, i8] (i0=0, i8=14, i9=0); read B[i9, i8] (i0=0, i8=14)
n^3	0.0295	ramp	$4 \cdot n + 22 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 14$	$(1/8) \cdot n^2 +$ $(-25/8) \cdot n +$ 3	0.0625/n	read B[i9, i8] (i0=0, i8=8, i9=0); read B[i9, i8] (i0=0, i8=8)
n^3	0.0292	ramp	$5 \cdot n + 29 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 16$	$(1/8) \cdot n^2 +$ $(-33/8) \cdot n +$ 4	0.0625/n	read B[i9, i8] (i0=0, i8=15, i9=0); read B[i9, i8] (i0=0, i8=15)
n^3	0.0287	ramp	$4 \cdot n + 30 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 18$	$(1/8) \cdot n^2 +$ $(-17/4) \cdot n +$ 8	0.0625/n	read B[i9, i8] (i0=0)
n^3	0.0273	level	$(1/4) \cdot n^2 +$ $(1/4) \cdot n$	$(7/128) \cdot n^2$ $+ (-35/16) \cdot n$ $+ 21$	0.0273/n	write D[i7, i8] (i0=0)
n^3	0.0237	ramp	$(1/4) \cdot n^2 +$ $(-5/8) \cdot n +$ 21 $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 6$	$(7/128) \cdot n^2$ $+ (-37/16) \cdot n$ $+ 24$	0.0273/n	write D[i8, i7] (i0=0)
n^3	0.0154	ramp	$2 \cdot n + 11 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 7$	$(7/128) \cdot n^2$ $+ (-9/16) \cdot n$ $+ 1$	0.0273/n	write D[i8, i7] (i0=0)
n^3	0.0135	level	3	$(1/128) \cdot n^3$ $+ (-1/16) \cdot n^2$	0.00391	write A[i1] (i0=0); read B[i2, i1] (i0=0) (+9)
n^3	0.0133	ramp	$2 \cdot n + 12 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(3/64) \cdot n^2$ $+ (-3/8) \cdot n$	0.0234/n	write D[i8, i7] (i0=0)
n^3	0.0121	ramp	$4 \cdot n + 23 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(3/64) \cdot n^2$ $+ (-15/8) \cdot n$ $+ 18$	0.0234/n	read B[i9, i8] (i0=0)
n^3	0.00781	level	1	$(1/128) \cdot n^3$ $+ (-1/16) \cdot n^2$ $+ n$	0.00391	read B[i9, i8] (i0=0, i7=0, i8=0); read D[i7, i8] (i0=0, i9=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00391	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/128) \cdot n^2 + (-13/64) \cdot n + (153/128)$	$0.00391/n$	write D[i7, i8] (i0=0)
n^3	0.00391	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.00391/n$	write D[i7, i8] (i0=0)
n^3	0.00323	ramp	$(1/4) \cdot n^2 + (-5/8) \cdot n + 22 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 13$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00391/n$	write D[i8, i7] (i0=0)
n^3	0.00212	ramp	$3 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 7$	$(1/128) \cdot n^2 + (-3/16) \cdot n + 1$	$0.00391/n$	write D[i8, i7] (i0=0)
n^3	0.00193	ramp	$5 \cdot n + 29 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 16$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00391/n$	read B[i9, i8] (i0=0)
$n^{2.5}$	4.77	level	$2 \cdot n + 2$	$(27/8) \cdot n^2 - 27 \cdot n$	$1.69/n$	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)
$n^{2.5}$	2.65	level	$2 \cdot n + 2$	$(15/8) \cdot n^2 - 15 \cdot n$	$0.938/n$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{2.5}$	1.88	level	$n + 2$	$(15/8) \cdot n^2$	$0.938/n$	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i9, i7] (i0=0, i8=8, i9=0); read B[i9, i7] (i0=0, i8=8)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i9, i8] (i0=0, i8=7, i9=0); read B[i9, i8] (i0=0, i8=7)
$n^{2.5}$	1.06	level	$2 \cdot n + 3$	$(3/4) \cdot n^2 - 12 \cdot n$	$0.375/n$	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.06	level	$2 \cdot n + 2$	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i8=6)
$n^{2.5}$	0.888	ramp	$n + 5 \rightarrow 2 \cdot n + 2$	$(3/4) \cdot n^2 + (-15/2) \cdot n + 12$	$0.375/n$	read B[i9, i7] (i0=0, i8=0)
$n^{2.5}$	0.888	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$(3/4) \cdot n^2 + (-15/2) \cdot n + 12$	$0.375/n$	read B[i9, i7] (i0=0)
$n^{2.5}$	0.875	level	$n + 2$	$(7/8) \cdot n^2$	$0.438/n$	read B[i4, i3] (i0=0, i4=0); read B[i4, i3] (i0=0)
$n^{2.5}$	0.875	level	$n + 1$	$(7/8) \cdot n^2$	$0.438/n$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
$n^{2.5}$	0.75	level	$n + 2$	$(3/4) \cdot n^2$	$0.375/n$	read B[i9, i7] (i0=0, i8=1, i9=0); read B[i9, i7] (i0=0, i8=1)
$n^{2.5}$	0.177	level	$2 \cdot n + 2$	$(1/8) \cdot n^2 + (-7/8) \cdot n - 1$	$0.0625/n$	read B[i9, i7] (i0=0, i8=1, i9=0); read B[i9, i7] (i0=0, i8=1)
$n^{2.5}$	0.177	level	$2 \cdot n + 3$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0625/n$	read B[i9, i7] (i0=0, i8=8, i9=0); read B[i9, i7] (i0=0, i8=8)
$n^{2.5}$	0.177	level	$2 \cdot n + 4$	$(1/8) \cdot n^2 - n$	$0.0625/n$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.153	level	$(3/8) \cdot n + 1$	$(1/4) \cdot n^2 + (-21/4) \cdot n + 5$	$0.125/n$	read A[i6] (i0=0); read C[i6] (i0=0)
$n^{2.5}$	0.148	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$(1/8) \cdot n^2 + (-5/4) \cdot n + 2$	$0.0625/n$	read B[i9, i7] (i0=0, i8=1)
n^2	4.55	level	3	$(21/8) \cdot n^2$	$1.31/n$	read A[i6] (i0=0, i5=0); read C[i6] (i0=0, i5=0) (+3)
n^2	3.75	level	1	$(15/4) \cdot n^2 - 15 \cdot n$	$1.88/n$	read D[i7, i8] (i0=0, i9=0); read D[i7, i8] (i0=0)
n^2	3.25	level	3	$(15/8) \cdot n^2 - 15 \cdot n$	$0.938/n$	write D[i7, i8] (i0=0, i9=0); write D[i7, i8] (i0=0)
n^2	3.03	level	3	$(7/4) \cdot n^2 + (-7/8) \cdot n$	$0.875/n$	read A[i3] (i0=0); write C[i3] (i0=0)
n^2	2.65	level	2	$(15/8) \cdot n^2$	$0.938/n$	write D[i7, i8] (i0=0, i9=0); write D[i7, i8] (i0=0)
n^2	1.88	level	1	$(15/8) \cdot n^2$	$0.938/n$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2 + (21/4) \cdot n$	$0.875/n$	write A[i1] (i0=0); read A[i1] (i0=0) (+3)
n^2	1.5	level	1	$(3/2) \cdot n^2$	$0.75/n$	read D[i7, i8] (i0=0, i8=0, i9=0); read D[i7, i8] (i0=0, i8=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.3	level	3	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	write D[i7, i8] (i0=0, i8=6, i9=0); write D[i7, i8] (i0=0, i8=6)
n^2	1.24	level	2	$(7/8) \cdot n^2$	$0.438/n$	write A[i1] (i0=0)
n^2	1.08	level	3	$(5/8) \cdot n^2 - 5 \cdot n$	$0.312/n$	write D[i7, i8] (i0=0, i9=0); write D[i7, i8] (i0=0)
n^2	1.06	level	2	$(3/4) \cdot n^2$	$0.375/n$	write D[i7, i8] (i0=0, i8=0, i9=0); write D[i7, i8] (i0=0, i8=0)
n^2	0.884	level	2	$(5/8) \cdot n^2$	$0.312/n$	write D[i7, i8] (i0=0, i9=0); write D[i7, i8] (i0=0)
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.438/n$	read B[i5, i6] (i0=0)
n^2	0.75	level	1	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	read D[i7, i8] (i0=0, i8=6, i9=0); read D[i7, i8] (i0=0, i8=6)
n^2	0.625	level	1	$(5/8) \cdot n^2$	$0.312/n$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^2	0.625	level	1	$(5/8) \cdot n^2 - 5 \cdot n$	$0.312/n$	read D[i7, i8] (i0=0, i9=0); read D[i7, i8] (i0=0)
n^2	0.625	level	1	$(5/8) \cdot n^2$	$0.312/n$	read D[i7, i8] (i0=0, i9=0); read D[i7, i8] (i0=0)
n^2	0.541	level	2	$(49/128) \cdot n^2 + (-7/16) \cdot n$	$0.191/n$	write D[i7, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.5	level	$(1/4) \cdot n^2 + (-9/4) \cdot n + 22$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i1=8, i2=0); read B[i2, i1] (i0=0, i1=8)
n^2	0.5	level	$(1/4) \cdot n^2 - 4 \cdot n + 36$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i1=8, i2=0); read B[i2, i1] (i0=0, i1=8)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i1=0, i2=0); read B[i2, i1] (i0=0, i1=0)
n^2	0.5	level	$(1/4) \cdot n^2 - 2 \cdot n + 11$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i1=0, i2=0); read B[i2, i1] (i0=0, i1=0)
n^2	0.438	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(7/8) \cdot n - 21$	$0.438 \cdot n^{-2}$	write D[i7, i8] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2 - 7 \cdot n + 28$	$0.219/n$	read D[i7, i8] (i0=0)
n^2	0.433	level	3	$(1/4) \cdot n^2 + (3/4) \cdot n$	$0.125/n$	read B[i4, i3] (i0=0, i4=0); read A[i3] (i0=0, i4=0) (+3)
n^2	0.375	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 15$	$(3/4) \cdot n - 12$	$0.375 \cdot n^{-2}$	write D[i8, i7] (i0=0, i7=8)
n^2	0.375	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(3/4) \cdot n - 12$	$0.375 \cdot n^{-2}$	write D[i7, i8] (i0=0)
n^2	0.375	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 8$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	write D[i8, i7] (i0=0, i7=0)
n^2	0.375	level	1	$(3/8) \cdot n^2 - 2 \cdot n$	$0.188/n$	read D[i7, i8] (i0=0, i8=0, i9=0); read D[i7, i8] (i0=0, i8=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (11/8) \cdot n - 9$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i7=0, i9=0); read B[i9, i8] (i0=0, i7=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 2$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=8, i4=0); read B[i4, i3] (i0=0, i3=8)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	n	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0); read B[i4, i3] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 9$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n + (15/4)$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + n + (31/8)$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 2$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 1$	n	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0, i4=0); read B[i4, i3] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	n	$0.5 \cdot n^{-2}$	read B[i5, i6] (i0=0, i5=0, i6=0); read B[i5, i6] (i0=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.251	ramp	$(7/2) \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 21$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i7=0, i8=15)
n^2	0.25	level	1	$(1/4) \cdot n^2$	$0.125/n$	read D[i7, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i8=6, i9=0) (+2)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (5/8) \cdot n - 1$	$0.125/n$	read A[i1] (i0=0); write A[i1] (i0=0) (+3)
n^2	0.247	ramp	$(5/2) \cdot n + 5 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 11$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i7=0, i8=7)
n^2	0.242	ramp	$(3/2) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 2$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i7=0, i8=0)
n^2	0.242	ramp	$(3/2) \cdot n - 2 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i5, i6] (i0=0)
n^2	0.231	ramp	$(1/4) \cdot n^2 + (-3/8) \cdot n + 15 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(1/2) \cdot n - 12$	$0.25 \cdot n^{-2}$	write D[i8, i7] (i0=0)
n^2	0.217	level	3	$(1/8) \cdot n^2 - n$	$0.0625/n$	write D[i7, i8] (i0=0, i8=6, i9=0); write D[i7, i8] (i0=0, i8=6)
n^2	0.217	level	3	$(1/8) \cdot n^2 - n$	$0.0625/n$	write D[i7, i8] (i0=0, i8=0, i9=0); write D[i7, i8] (i0=0, i8=0)
n^2	0.217	level	3	$(1/8) \cdot n^2$	$0.0625/n$	write B[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.182	ramp	$n + 3 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(3/4) \cdot n$	$0.375 \cdot n^{-2}$	write D[i7, i7] (i0=0)
n^2	0.182	ramp	$n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.375 \cdot n^{-2}$	write D[i8, i7] (i0=0, i8=0)
n^2	0.182	ramp	$n + 3 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 2$	$(3/4) \cdot n - 1$	$0.375 \cdot n^{-2}$	write D[i8, i7] (i0=0)
n^2	0.18	ramp	$3 \cdot n + 14 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(3/4) \cdot n - 12$	$0.375 \cdot n^{-2}$	read B[i9, i8] (i0=0)
n^2	0.177	level	2	$(1/8) \cdot n^2$	$0.0625/n$	write D[i7, i8] (i0=0, i8=6, i9=0); write D[i7, i8] (i0=0, i8=6)
n^2	0.177	level	2	$(1/8) \cdot n^2$	$0.0625/n$	write D[i7, i8] (i0=0, i8=0, i9=0); write D[i7, i8] (i0=0, i8=0)
n^2	0.177	level	2	$(1/8) \cdot n^2$	$0.0625/n$	read B[i2, i1] (i0=0, i2=0); write A[i1] (i0=0, i2=0) (+1)
n^2	0.125	level	1	$(1/8) \cdot n^2 -$ n	$0.0625/n$	read D[i7, i8] (i0=0, i8=6, i9=0); read D[i7, i8] (i0=0, i8=6)
n^2	0.0773	level	2	$(7/128) \cdot n^2$ $+ (-7/16) \cdot n$	$0.0273/n$	write D[i7, i8] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(9/8) \cdot n +$ $(85/8)$	$(1/8) \cdot n +$ $(-25/8)$	$0.0625 \cdot n^{-2}$	write D[i8, i7] (i0=0, i7=8)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(-5/8) \cdot n +$ 15	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	write D[i8, i7] (i0=0, i7=8)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(-5/8) \cdot n +$ 14	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	write D[i8, i7] (i0=0, i7=8)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(9/8) \cdot n +$ $(29/8)$	$(1/8) \cdot n +$ $(-17/8)$	$0.0625 \cdot n^{-2}$	write D[i8, i7] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0625	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 8$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0, i7=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 7$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0, i7=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/8) \cdot n + (-25/8)$	$0.0625 \cdot n^2$	write D[i7, i7] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	write D[i7, i7] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/8) \cdot n + (-9/8)$	$0.0625 \cdot n^2$	write D[i7, i8] (i0=0, i7=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write D[i7, i8] (i0=0, i7=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write D[i7, i8] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/8) \cdot n + (-9/8)$	$0.0625 \cdot n^2$	write D[i7, i8] (i0=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2 + (-3/2) \cdot n + 8$	0.0312/n	read D[i7, i8] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/8) \cdot n + (-9/8)$	$0.0625 \cdot n^2$	write D[i7, i8] (i0=0, i8=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write D[i7, i8] (i0=0, i8=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/8) \cdot n + (-9/8)$	$0.0625 \cdot n^2$	write C[i3] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write C[i3] (i0=0)
n^2	0.0594	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write A[i1] (i0=0)
			$\rightarrow$			
n^2	0.0592	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0)
			$(1/4) \cdot n^2 + (-5/8) \cdot n + 14 \rightarrow$			
			$(1/4) \cdot n^2 + (1/4) \cdot n - 7$			
n^2	0.0578	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n + 3$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0, i8=0)
			$\rightarrow$			
			$(1/4) \cdot n^2 + (1/4) \cdot n - 1$			
n^2	0.0577	ramp	$(1/4) \cdot n^2 + (-1/2) \cdot n + 18 \rightarrow$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0, i8=5)
			$(1/4) \cdot n^2 + (1/4) \cdot n - 6$			

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0576	ramp	$(1/4) \cdot n^2 + (-5/8) \cdot n + 22 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 6$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0, i8=7)
n^2	0.0576	ramp	$(1/4) \cdot n^2 + (-5/8) \cdot n + 21 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 7$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0, i8=6)
n^2	0.055	ramp	$(11/8) \cdot n - 1 \rightarrow$ $(1/8) \cdot n^2 + (-7/4) \cdot n + 2$	$(1/4) \cdot n - 5$	$0.125 \cdot n^2$	read A[i6] (i0=0, i5=0); read C[i6] (i0=0, i5=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	read B[i5, i6] (i0=0)
n^2	0.0415	ramp	$(1/8) \cdot n^2 + (-7/8) \cdot n + 3 \rightarrow$ $(1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	read A[i3] (i0=0, i4=0)
n^2	0.0401	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 27 \rightarrow$ $(1/8) \cdot n^2 + (11/8) \cdot n - 18$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^2$	read B[i9, i8] (i0=0, i7=0, i9=0)
n^2	0.0302	ramp	$2 \cdot n + 11 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0, i8=0)
n^2	0.0302	ramp	$2 \cdot n + 10 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^2$	write D[i8, i7] (i0=0)
n^2	0.0302	ramp	$2 \cdot n + 10 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^2$	write D[i7, i7] (i0=0)
n^2	0.0293	ramp	$5 \cdot n + 28 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 17$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^2$	read B[i9, i8] (i0=0, i9=0)
n^2	0.029	ramp	$(27/8) \cdot n + 9 \rightarrow$ $(1/8) \cdot n^2 + (1/8) \cdot n - 15$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	read B[i9, i8] (i0=0, i7=0)
n^2	0.029	ramp	$(19/8) \cdot n - 2 \rightarrow$ $(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	read B[i5, i6] (i0=0, i5=0)
n^2	0.0288	ramp	$3 \cdot n + 20 \rightarrow$ $(1/8) \cdot n^2 + (1/8) \cdot n - 16$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	read B[i9, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	6	level	$n + 3$	$6 \cdot n - 12$	$3 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.87	ramp	$3 \cdot n + 21 \rightarrow$ $4 \cdot n + 18$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=15)
$n^{1.5}$	1.84	level	$(3/8) \cdot n + 1$	$3 \cdot n - 3$	$1.5 \cdot n^{-2}$	read A[i6] (i0=0, i6=0); read C[i6] (i0=0, i6=0) (+1)
$n^{1.5}$	1.73	level	$3 \cdot n + 13$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=8, i9=0); read B[i9, i8] (i0=0, i8=8)
$n^{1.5}$	1.73	level	$3 \cdot n + 14$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 15$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 16$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 17$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 18$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 19$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=14, i9=0); read B[i9, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.73	level	$3 \cdot n + 12$	n	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.58	ramp	$2 \cdot n + 12 \rightarrow$ $3 \cdot n + 9$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=7)
$n^{1.5}$	1.41	level	$2 \cdot n + 5$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 6$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 7$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 8$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 9$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 10$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i8=6)
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=0, i9=0); read B[i9, i8] (i0=0, i8=0)
$n^{1.5}$	1.22	level	$(3/8) \cdot n + 1$	$2 \cdot n - 2$	$1 \cdot n^{-2}$	read A[i6] (i0=0); read C[i6] (i0=0)
$n^{1.5}$	1.21	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.06	level	$2 \cdot n + 2$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read B[i9, i7] (i0=0)
$n^{1.5}$	1.06	level	$2 \cdot n + 3$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{1.5}$	0.75	level	$n + 3$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read B[i9, i7] (i0=0, i9=0)
$n^{1.5}$	0.75	level	$n + 4$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	0.125	level	$n + 3$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=1, i9=0)
n^1	5.25	level	1	$(21/4) \cdot n - 21$	$2.62 \cdot n^{-2}$	read D[i7, i8] (i0=0)
n^1	3.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	7	$3.5 \cdot n^{-3}$	write D[i7, i8] (i0=0)
n^1	2.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	5	$2.5 \cdot n^{-3}$	write D[i7, i8] (i0=0)
n^1	2	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	4	$2 \cdot n^{-3}$	write D[i8, i7] (i0=0)
n^1	1.75	level	1	$(7/4) \cdot n$	$0.875 \cdot n^{-2}$	write D[i7, i8] (i0=0, i8=0); read D[i7, i8] (i0=0, i8=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	2	$1 \cdot n^{-3}$	write A[i1] (i0=0); write C[i3] (i0=0, i3=0)
n^1	1	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	2	$1 \cdot n^{-3}$	write A[i1] (i0=0); write C[i3] (i0=0, i3=0)
n^1	0.875	level	1	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read D[i7, i8] (i0=0, i8=7)
n^1	0.75	level	1	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read D[i7, i8] (i0=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	2	$1 \cdot n^{-3}$	read A[i6] (i0=0, i5=0, i6=0); read C[i6] (i0=0, i5=0, i6=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (-17/4)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=5)
n^1	0.5	level	$(1/4) \cdot n^2 + (5/4) \cdot n + (17/2)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=5)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/2) \cdot n + 12$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=5)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/8) \cdot n + 10$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 8$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 6$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8)
n^1	0.5	level	$(1/4) \cdot n^2 + 4$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + (-9/2)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (9/8) \cdot n + (85/8)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 15$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + (-11/2)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=6)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (-21/4)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=6)
n^1	0.5	level	$(1/4) \cdot n^2 + (9/8) \cdot n + (77/8)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=6)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 14$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=6)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (-21/4)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (23/8)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=8, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (5/4)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + (3/2)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write D[i7, i7] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write D[i7, i7] (i0=0, i7=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i7, i7] (i0=0, i7=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + (3/2)$	1	$0.5 \cdot n^{-3}$	write D[i7, i7] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i7, i8] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write D[i7, i8] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i7, i8] (i0=0, i8=1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write D[i7, i8] (i0=0, i8=1)
n^1	0.5	level	$(1/4) \cdot n^2 + (5/4) \cdot n + (5/2)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=5)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/2) \cdot n + 6$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=5)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/8) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 4$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (15/8)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write D[i7, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i7, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i7, i7] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i3] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write C[i3] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	write A[i1] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (15/8)$	1	$0.5 \cdot n^{-3}$	write A[i1] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	write D[i7, i8] (i0=0, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write D[i7, i8] (i0=0, i8=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (9/8) \cdot n + (29/8)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 8$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (9/8) \cdot n + (21/8)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (9/8) \cdot n + (21/8)$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=6)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i7=0, i8=6)
n^1	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	1	$0.5 \cdot n^{-3}$	write D[p0 - 1, p0 - 1] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n - 9$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 18$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i7=0, i8=15, i9=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 9$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i7=0, i8=7, i9=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (31/8)$	1	$0.5 \cdot n^{-3}$	read B[i2, i1] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + n + (23/8)$	1	$0.5 \cdot n^{-3}$	read B[i2, i1] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i7=0, i8=0, i9=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i5, i6] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/4) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i6] (i0=0, i5=0, i6=8)
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read D[i7, i8] (i0=0, i8=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0)
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{0.5}$	2	level	$4 \cdot n + 19$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i8=15, i9=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 10$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i8=7, i9=0)
$n^{0.5}$	1.54	level	$(19/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i7=0, i8=7)
$n^{0.5}$	1.41	level	$2 \cdot n + 11$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i8=7)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=0)
$n^{0.5}$	1.22	level	$(3/8) \cdot n$	2	$1 \cdot n^{-3}$	read B[i5, i6] (i0=0, i5=0); read A[i6] (i0=0, i5=0) (+1)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.5 \cdot n^{-3}$	read B[i5, i6] (i0=0, i5=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 2$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i7=0, i8=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0)
$n^{0.5}$	1	level	$n + 4$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1	level	$n + 2$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=0)
n^0	1	level	1	1	$0.5 \cdot n^{-3}$	write D[p0 - 1, p0 - 1] (i0=0)

Structurally identical to covariance after normalization: same $4n \rightarrow (1/8)n^2$ ramps on read B[i9,i8], same 2n-line column levels, d = 4.0, headroom +1.0.

correlation — single-shot [exact]

Accesses $A(n) = 2 \cdot n^3 + (21/2) \cdot n^2 + (11/2) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma \text{mass/warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0161 \cdot n^4 + 1.33 \cdot n^{3.5} + 2.11 \cdot n^3 + 19.4 \cdot n^{2.5} + 25.3 \cdot n^2 + 34.5 \cdot n^{1.5} + 7.68 \cdot n^1 + 30.3 \cdot n^{0.5} + 1 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0121	ramp	$4 \cdot n + 23 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(3/64) \cdot n^3$ + $(-15/8) \cdot n^2$ + $18 \cdot n$	0.0234	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^4	0.00202	ramp	$4 \cdot n + 22 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 14$	$(1/128) \cdot n^3$ + $(-5/16) \cdot n^2$ + $3 \cdot n$	0.00391	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^4	0.00193	ramp	$5 \cdot n + 29 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 16$	$(1/128) \cdot n^3$ + $(-7/16) \cdot n^2$ + $6 \cdot n$	0.00391	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.619	level	$2 \cdot n + 2$	$(7/16) \cdot n^3$ + $(-63/8) \cdot n^2$ + $35 \cdot n$	0.219	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)
$n^{3.5}$	0.398	level	$2 \cdot n + 2$	$(9/32) \cdot n^3$ + (- $165/32) \cdot n^2$ + $(207/8) \cdot n$ - 14	0.141	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.0884	level	$2 \cdot n + 3$	$(1/16) \cdot n^3$ + $(-19/8) \cdot n^2$ + $22 \cdot n$	0.0312	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.0773	level	$2 \cdot n + 2$	$(7/128) \cdot n^3 + (-21/16) \cdot n^2 + 7 \cdot n - 7$	0.0273	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.0663	level	$2 \cdot n + 2$	$(3/64) \cdot n^3 + (-3/8) \cdot n^2$	0.0234	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.0663	level	$2 \cdot n + 2$	$(3/64) \cdot n^3 + (-3/8) \cdot n^2$	0.0234	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
$n^{3.5}$	0.011	level	$2 \cdot n + 2$	$(1/128) \cdot n^3 + (-1/16) \cdot n^2$	0.00391	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.866	level	3	$(1/2) \cdot n^3 + (-11/2) \cdot n^2 + 35 \cdot n$	0.25	read B[i4, i3] (i0=0); read A[i3] (i0=0) (+6)
n^3	0.5	level	1	$(1/2) \cdot n^3 + 2 \cdot n^2 + (7/2) \cdot n$	0.25	read A[i1] (i0=0); write A[i1] (i0=0) (+4)
n^3	0.18	ramp	$3 \cdot n + 14 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(3/4) \cdot n^2 - 12 \cdot n$	0.375/n	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.15	ramp	$3 \cdot n + 14 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(5/8) \cdot n^2 + (-79/8) \cdot n - 2$	0.312/n	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.0939	ramp	$n + 5 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(21/64) \cdot n^2 + (-3/2) \cdot n + 1$	0.164/n	write D[i8, i7] (i0=0)
n^3	0.042	ramp	$(1/8) \cdot n^2 + (1/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0625/n	read B[i4, i3] (i0=0, i4=0); read B[i4, i3] (i0=0)
n^3	0.036	ramp	$(5/2) \cdot n - 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0625/n	read B[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0355	ramp	$(7/2) \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (11/8) \cdot n - 19$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0625/n$	read B[i9, i8] (i0=0, i7=0)
n^3	0.0299	ramp	$3 \cdot n + 13 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 14$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0625/n$	read B[i9, i8] (i0=0, i8=0, i9=0); read B[i9, i8] (i0=0, i8=0)
n^3	0.0299	ramp	$3 \cdot n + 13 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 14$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0625/n$	read B[i9, i8] (i0=0, i9=0); read B[i9, i8] (i0=0)
n^3	0.0299	ramp	$3 \cdot n + 19 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0625/n$	read B[i9, i8] (i0=0, i8=6)
n^3	0.0296	ramp	$3 \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 15$	$(1/8) \cdot n^2 - 3 \cdot n + 1$	$0.0625/n$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
n^3	0.0296	ramp	$4 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 16$	$(1/8) \cdot n^2 - 3 \cdot n$	$0.0625/n$	read B[i9, i8] (i0=0, i8=7, i9=0); read B[i9, i8] (i0=0, i8=7)
n^3	0.029	ramp	$3 \cdot n + 21 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 18$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0625/n$	read B[i9, i8] (i0=0)
n^3	0.0154	ramp	$2 \cdot n + 11 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 7$	$(7/128) \cdot n^2 + (-9/16) \cdot n + 1$	$0.0273/n$	write D[i8, i7] (i0=0)
n^3	0.0133	ramp	$2 \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/64) \cdot n^2 + (-3/8) \cdot n$	$0.0234/n$	write D[i8, i7] (i0=0)
n^3	0.00212	ramp	$3 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 7$	$(1/128) \cdot n^2 + (-3/16) \cdot n + 1$	$0.00391/n$	write D[i8, i7] (i0=0)
$n^{2.5}$	4.95	level	$2 \cdot n + 2$	$(7/2) \cdot n^2 - 28 \cdot n$	$1.75/n$	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	2.65	level	$2 \cdot n + 2$	$(15/8) \cdot n^2 + (-57/4) \cdot n - 6$	$0.938/n$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)
$n^{2.5}$	2.62	level	$n + 2$	$(21/8) \cdot n^2$	$1.31/n$	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i9, i7] (i0=0, i8=8, i9=0); read B[i9, i7] (i0=0, i8=8)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i9, i8] (i0=0, i8=7, i9=0); read B[i9, i8] (i0=0, i8=7)
$n^{2.5}$	1.06	level	$2 \cdot n + 3$	$(3/4) \cdot n^2 - 12 \cdot n$	$0.375/n$	read B[i9, i7] (i0=0, i9=0); read B[i9, i7] (i0=0)
$n^{2.5}$	1.06	level	$2 \cdot n + 2$	$(3/4) \cdot n^2 + (-27/4) \cdot n + 6$	$0.375/n$	read B[i9, i8] (i0=0, i8=6)
$n^{2.5}$	1.04	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$(7/8) \cdot n^2 + (-35/4) \cdot n + 14$	$0.438/n$	read B[i9, i7] (i0=0)
$n^{2.5}$	0.888	ramp	$n + 5 \rightarrow 2 \cdot n + 2$	$(3/4) \cdot n^2 + (-15/2) \cdot n + 12$	$0.375/n$	read B[i9, i7] (i0=0, i8=0)
$n^{2.5}$	0.875	level	$n + 2$	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.438/n$	read B[i4, i3] (i0=0, i4=0); read B[i4, i3] (i0=0)
$n^{2.5}$	0.875	level	$n + 1$	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.438/n$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
$n^{2.5}$	0.398	level	$2 \cdot n + 2$	$(9/32) \cdot n^2 + (-39/8) \cdot n + 21$	$0.141/n$	read B[i9, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.177	level	$2 \cdot n + 3$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0625/n$	read B[i9, i7] (i0=0, i8=8, i9=0); read B[i9, i7] (i0=0, i8=8)
$n^{2.5}$	0.177	level	$2 \cdot n + 4$	$(1/8) \cdot n^2 - 2 \cdot n + 1$	$0.0625/n$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
$n^{2.5}$	0.153	level	$(3/8) \cdot n + 1$	$(1/4) \cdot n^2 + (-17/4) \cdot n + 4$	$0.125/n$	read A[i6] (i0=0); read C[i6] (i0=0)
n^2	7.79	level	3	$(9/2) \cdot n^2 - 36 \cdot n$	$2.25/n$	write D[i7, i8] (i0=0)
n^2	4.95	level	2	$(7/2) \cdot n^2$	$1.75/n$	write D[i7, i8] (i0=0)
n^2	2.5	level	1	$(5/2) \cdot n^2 + (1/8) \cdot n$	$1.25/n$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)
n^2	1.52	level	3	$(7/8) \cdot n^2$	$0.438/n$	read C[i6] (i0=0)
n^2	1.52	level	3	$(7/8) \cdot n^2$	$0.438/n$	read A[i6] (i0=0)
n^2	1.41	level	2	n^2	$0.5/n$	write A[i1] (i0=0); read B[i9, i7] (i0=0, i7=0, i8=0, i9=0)
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.438/n$	read B[i9, i8] (i0=0, i8=0)
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.438/n$	read B[i5, i6] (i0=0)
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.219/n$	write D[i7, i8] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (11/8) \cdot n - 9$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i7=0, i9=0); read B[i9, i8] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	n	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0); read B[i4, i3] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 1$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0, i4=0); read B[i4, i3] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i5, i6] (i0=0, i5=0, i6=0); read B[i5, i6] (i0=0, i6=0)
n^2	0.247	ramp	$(5/2) \cdot n + 5 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 11$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i7=0, i8=7)
n^2	0.242	ramp	$(3/2) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 2$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i7=0, i8=0)
n^2	0.242	ramp	$(3/2) \cdot n - 2 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i5, i6] (i0=0)
n^2	0.182	ramp	$n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/4) \cdot n$	$0.375 \cdot n^{-2}$	write D[i7, i7] (i0=0)
n^2	0.182	ramp	$n + 4 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.375 \cdot n^{-2}$	write D[i8, i7] (i0=0, i8=0)
n^2	0.182	ramp	$n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(3/4) \cdot n - 1$	$0.375 \cdot n^{-2}$	write D[i8, i7] (i0=0)
n^2	0.125	level	1	$(1/8) \cdot n^2 + (-1/8) \cdot n$	$0.0625/n$	read B[i9, i8] (i0=0, i8=6)
n^2	0.0561	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-7/4) \cdot n + 2$	$(1/4) \cdot n - 4$	$0.125 \cdot n^{-2}$	read A[i6] (i0=0, i5=0); read C[i6] (i0=0, i5=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0415	ramp	$(1/8) \cdot n^2 + (-7/8) \cdot n + 3$ $\rightarrow$ $(1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3] (i0=0, i4=0)
n^2	0.0412	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 18$ $\rightarrow$ $(1/8) \cdot n^2 + (11/8) \cdot n - 18$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i9, i8] (i0=0, i7=0, i9=0)
n^2	0.0302	ramp	$2 \cdot n + 11 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	write D[i8, i7] (i0=0, i8=0)
n^2	0.0302	ramp	$2 \cdot n + 10 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	write D[i8, i7] (i0=0)
n^2	0.0302	ramp	$2 \cdot n + 10 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	write D[i7, i7] (i0=0)
n^2	0.0296	ramp	$4 \cdot n + 19 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 17$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i9, i8] (i0=0, i9=0)
n^2	0.029	ramp	$(27/8) \cdot n + 9 \rightarrow$ $(1/8) \cdot n^2 + (1/8) \cdot n - 15$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i9, i8] (i0=0, i7=0)
n^2	0.029	ramp	$(19/8) \cdot n - 2 \rightarrow$ $(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i5, i6] (i0=0, i5=0)
n^2	0.0288	ramp	$3 \cdot n + 20 \rightarrow$ $(1/8) \cdot n^2 + (1/8) \cdot n - 16$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i9, i8] (i0=0)
$n^{1.5}$	6	level	$n + 3$	$6 \cdot n - 12$	$3 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 12$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 5$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.58	ramp	$2 \cdot n + 12 \rightarrow$ $3 \cdot n + 9$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0); read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 5$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 6$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 7$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 8$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 9$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.41	level	$2 \cdot n + 10$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=6)
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i9, i8] (i0=0, i8=0, i9=0); read B[i9, i8] (i0=0, i8=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i9, i7] (i0=0)
$n^{1.5}$	1.22	level	$(3/8) \cdot n + 1$	$2 \cdot n - 2$	$1 \cdot n^{-2}$	read A[i6] (i0=0); read C[i6] (i0=0)
$n^{1.5}$	1.22	level	$(3/8) \cdot n + 1$	$2 \cdot n - 2$	$1 \cdot n^{-2}$	read A[i6] (i0=0, i6=0); read C[i6] (i0=0, i6=0)
$n^{1.5}$	1.21	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
$n^{1.5}$	1.06	level	$2 \cdot n + 3$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{1.5}$	0.875	level	$n + 3$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i9, i7] (i0=0, i9=0)
$n^{1.5}$	0.875	level	$n + 2$	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i4, i3] (i0=0)
$n^{1.5}$	0.875	level	$n + 1$	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i2, i1] (i0=0)
$n^{1.5}$	0.75	level	$n + 4$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read B[i9, i7] (i0=0, i8=0)
n^1	1.75	level	1	$(7/4) \cdot n$	$0.875 \cdot n^{-2}$	write A[i1] (i0=0); write C[i3] (i0=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read A[i3] (i0=0, i4=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	write D[i7, i8] (i0=0, i8=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	2	$1 \cdot n^{-3}$	read A[i6] (i0=0, i5=0, i6=0); read C[i6] (i0=0, i5=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n - 9$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 9$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i7=0, i8=7, i9=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i7=0, i8=0, i9=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read B[i4, i3] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i5, i6] (i0=0, i6=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i5, i6] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i3] (i0=0, i3=0, i4=0)
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0)
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 10$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i8=7, i9=0)
$n^{0.5}$	1.54	level	$(19/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i7=0, i8=7)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0, i9=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.41	level	$2 \cdot n + 10$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i8=6, i9=0); read B[i9, i8] (i0=0, i9=0) (+1)
$n^{0.5}$	1.41	level	$2 \cdot n + 11$	1	$0.5 \cdot n^{-3}$	read B[i9, i8] (i0=0, i8=7)
$n^{0.5}$	1.22	level	$(3/8) \cdot n$	2	$1 \cdot n^{-3}$	read B[i5, i6] (i0=0, i5=0); read A[i6] (i0=0, i5=0) (+1)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 2$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i7=0, i8=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.5 \cdot n^{-3}$	read B[i5, i6] (i0=0, i5=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0)
$n^{0.5}$	1	level	$n + 2$	1	$0.5 \cdot n^{-3}$	write D[i8, i7] (i0=0, i8=0)
$n^{0.5}$	1	level	$n + 4$	1	$0.5 \cdot n^{-3}$	read B[i9, i7] (i0=0, i8=0)
n^0	1	level	1	1	$0.5 \cdot n^{-3}$	write D[p0 - 1, p0 - 1] (i0=0)

Structurally identical to covariance after normalization: same $4n \rightarrow (1/8)n^2$ ramps on read B[i9, i8], same $2n$ -line column levels, $d = 4.0$, headroom +1.0.

covariance — infinite-repeat [exact]

Accesses $A(n) = 2 \cdot n^3 + (21/2) \cdot n^2 + (11/2) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0161 \cdot n^4 + 1.33 \cdot n^{3.5} + 2.17 \cdot n^3 + 19.5 \cdot n^{2.5} + 39.3 \cdot n^2 + 28.7 \cdot n^{1.5} + 39.9 \cdot n^1 + 31.5 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0142	ramp	$4 \cdot n + 22 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(7/128) \cdot n^3$ + (- $35/16) \cdot n^2$ + $21 \cdot n$	0.0273	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^4	0.00193	ramp	$5 \cdot n + 29 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 16$	$(1/128) \cdot n^3$ + $(-7/16) \cdot n^2$ + $6 \cdot n$	0.00391	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{3.5}$	0.619	level	$2 \cdot n + 2$	$(7/16) \cdot n^3$ + $(-63/8) \cdot n^2$ + $35 \cdot n$	0.219	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{3.5}$	0.464	level	$2 \cdot n + 2$	$(21/64) \cdot n^3$ + (- $357/64) \cdot n^2$ + $(105/4) \cdot n$ - 21	0.164	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{3.5}$	0.0884	level	$2 \cdot n + 3$	$(1/16) \cdot n^3$ + $(-19/8) \cdot n^2$ + $22 \cdot n$	0.0312	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{3.5}$	0.0773	level	$2 \cdot n + 2$	$(7/128) \cdot n^3$ + $(-7/16) \cdot n^2$	0.0273	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{3.5}$	0.0773	level	$2 \cdot n + 2$	$(7/128) \cdot n^3$ + (- $21/16) \cdot n^2$ + $7 \cdot n$	0.0273	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^3	0.663	level	3	$(49/128) \cdot n^3$ + (- $91/16) \cdot n^2$ + $21 \cdot n$	0.191	write C[i5, i6] (i0=0, i7=0); write C[i5, i6] (i0=0)
n^3	0.383	level	1	$(49/128) \cdot n^3$ + (- $91/16) \cdot n^2$ + $21 \cdot n$	0.191	read C[i5, i6] (i0=0, i7=0); read C[i5, i6] (i0=0)
n^3	0.21	ramp	$3 \cdot n + 13 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(7/8) \cdot n^2 -$ $14 \cdot n$	0.438/n	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.15	ramp	$3 \cdot n + 14 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 9$	$(5/8) \cdot n^2 -$ $10 \cdot n$	$0.312/n$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^3	0.109	ramp	$n + 5 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(49/128) \cdot n^2$ $+ (-33/16) \cdot n$ $+ 2$	$0.191/n$	write C[i6, i5] (i0=0)
n^3	0.0947	level	3	$(7/128) \cdot n^3$ $+ (-$ $21/16) \cdot n^2$ $+ 7 \cdot n$	0.0273	write C[i5, i6] (i0=0, i7=0); write C[i5, i6] (i0=0)
n^3	0.0947	level	3	$(7/128) \cdot n^3$ $+$ $(-7/16) \cdot n^2$	0.0273	write C[i5, i6] (i0=0, i7=0); write C[i5, i6] (i0=0)
n^3	0.0547	level	1	$(7/128) \cdot n^3$ $+ (-$ $21/16) \cdot n^2$ $+ 7 \cdot n$	0.0273	read C[i5, i6] (i0=0, i7=0); read C[i5, i6] (i0=0)
n^3	0.0547	level	1	$(7/128) \cdot n^3$ $+$ $(-7/16) \cdot n^2$	0.0273	read C[i5, i6] (i0=0, i7=0); read C[i5, i6] (i0=0)
n^3	0.0467	ramp	$(1/8) \cdot n^2 +$ $(1/8) \cdot n +$ $32 \rightarrow$ $(1/4) \cdot n^2 -$ $4 \cdot n + 36$	$(1/8) \cdot n^2 -$ $3 \cdot n$	$0.0625/n$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
n^3	0.0359	ramp	$(19/8) \cdot n - 4$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(1/8) \cdot n - 1$	$(1/8) \cdot n^2 +$ $(-9/4) \cdot n + 4$	$0.0625/n$	read B[i3, i4] (i0=0)
n^3	0.0354	ramp	$(27/8) \cdot n +$ $12 \rightarrow$ $(1/8) \cdot n^2 +$ $(5/4) \cdot n - 19$	$(1/8) \cdot n^2 +$ $(-13/4) \cdot n +$ 6	$0.0625/n$	read B[i7, i6] (i0=0, i5=0)
n^3	0.03	ramp	$3 \cdot n + 19 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(1/8) \cdot n^2 -$ $2 \cdot n$	$0.0625/n$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i6=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0299	ramp	$3 \cdot n + 13 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 14$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0625/n$	read B[i7, i6] (i0=0, i6=1, i7=0); read B[i7, i6] (i0=0, i6=1)
n^3	0.0296	ramp	$3 \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 15$	$(1/8) \cdot n^2 - 3 \cdot n + 1$	$0.0625/n$	read B[i7, i5] (i0=0, i6=0, i7=0); read B[i7, i5] (i0=0, i6=0)
n^3	0.0296	ramp	$4 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 16$	$(1/8) \cdot n^2 - 3 \cdot n$	$0.0625/n$	read B[i7, i6] (i0=0, i6=8, i7=0); read B[i7, i6] (i0=0, i6=8)
n^3	0.029	ramp	$3 \cdot n + 21 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 18$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0625/n$	read B[i7, i6] (i0=0)
n^3	0.0273	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	$(7/128) \cdot n^2 + (-37/16) \cdot n + 24$	$0.0273/n$	write C[i5, i6] (i0=0)
n^3	0.0237	ramp	$(1/4) \cdot n^2 + (-3/4) \cdot n + 21 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n - 13$	$(7/128) \cdot n^2 + (-37/16) \cdot n + 18$	$0.0273/n$	write C[i6, i5] (i0=0)
n^3	0.0154	ramp	$2 \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(7/128) \cdot n^2 + (-9/16) \cdot n + 1$	$0.0273/n$	write C[i6, i5] (i0=0)
n^3	0.0135	level	3	$(1/128) \cdot n^3 + (-1/16) \cdot n^2$	0.00391	read B[i2, i1] (i0=0); read A[i4] (i0=0, i4=0) (+3)
n^3	0.00781	level	1	$(1/128) \cdot n^3 + (-1/16) \cdot n^2 + n$	0.00391	read B[i7, i6] (i0=0, i5=0, i6=0); read C[i5, i6] (i0=0, i7=0) (+1)
n^3	0.00391	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	$0.00391/n$	write C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00323	ramp	$(1/4) \cdot n^2 + (-3/4) \cdot n + 22 \rightarrow$ $(1/4) \cdot n^2 + (1/8) \cdot n - 13$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00391/n$	write C[i6, i5] (i0=0)
$n^{2.5}$	4.95	level	$2 \cdot n + 2$	$(7/2) \cdot n^2 - 28 \cdot n$	$1.75/n$	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{2.5}$	2.65	level	$2 \cdot n + 2$	$(15/8) \cdot n^2 - 15 \cdot n$	$0.938/n$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{2.5}$	2.62	level	$n + 2$	$(21/8) \cdot n^2$	$1.31/n$	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i7, i5] (i0=0, i6=9, i7=0); read B[i7, i5] (i0=0, i6=9)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i7, i6] (i0=0, i6=8, i7=0); read B[i7, i6] (i0=0, i6=8)
$n^{2.5}$	1.06	level	$2 \cdot n + 3$	$(3/4) \cdot n^2 - 12 \cdot n$	$0.375/n$	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{2.5}$	1.06	level	$2 \cdot n + 2$	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i6=7)
$n^{2.5}$	1.04	ramp	$n + 5 \rightarrow 2 \cdot n + 2$	$(7/8) \cdot n^2 + (-35/4) \cdot n + 14$	$0.438/n$	read B[i7, i5] (i0=0, i6=0)
$n^{2.5}$	1.04	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$(7/8) \cdot n^2 + (-35/4) \cdot n + 14$	$0.438/n$	read B[i7, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.875	level	$n + 1$	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.438/n$	read B[i7, i5] (i0=0, i6=1, i7=0); read B[i7, i5] (i0=0, i6=1)
$n^{2.5}$	0.875	level	$n + 1$	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.438/n$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
$n^{2.5}$	0.464	level	$2 \cdot n + 2$	$(21/64) \cdot n^2 + (-21/4) \cdot n + 21$	$0.164/n$	read B[i7, i6] (i0=0)
$n^{2.5}$	0.177	level	$2 \cdot n + 3$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0625/n$	read B[i7, i5] (i0=0, i6=9, i7=0); read B[i7, i5] (i0=0, i6=9)
$n^{2.5}$	0.148	ramp	$n + 3 \rightarrow 2 \cdot n$	$(1/8) \cdot n^2 + (-5/4) \cdot n + 2$	$0.0625/n$	read B[i7, i5] (i0=0, i6=1)
$n^{2.5}$	0.0625	level	$(1/4) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0625/n$	read A[i4] (i0=0)
n^2	4.5	level	1	$(9/2) \cdot n^2 - 15 \cdot n$	$2.25/n$	read C[i5, i6] (i0=0, i7=0); read C[i5, i6] (i0=0)
n^2	3.71	level	2	$(21/8) \cdot n^2$	$1.31/n$	write C[i5, i6] (i0=0, i7=0); write C[i5, i6] (i0=0)
n^2	3.25	level	3	$(15/8) \cdot n^2 - 15 \cdot n$	$0.938/n$	write C[i5, i6] (i0=0, i7=0); write C[i5, i6] (i0=0)
n^2	2.62	level	1	$(21/8) \cdot n^2 + (49/8) \cdot n + 1$	$1.31/n$	write A[i1] (i0=0); read A[i1] (i0=0) (+5)
n^2	2.47	level	2	$(7/4) \cdot n^2$	$0.875/n$	read B[i2, i1] (i0=0); write A[i1] (i0=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.88	level	1	$(15/8) \cdot n^2$	$0.938/n$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^2	1.41	level	2	n^2	$0.5/n$	read A[i4] (i0=0, i3=0); read A[i4] (i0=0) (+1)
n^2	1.31	level	1	$(21/16) \cdot n^2 - 7 \cdot n + 28$	$0.656/n$	read C[i5, i6] (i0=0); write C[i5, i6] (i0=0)
n^2	1.3	level	3	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	write C[i5, i6] (i0=0, i7=0); write C[i5, i6] (i0=0)
n^2	1.3	level	3	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	write C[i5, i6] (i0=0, i6=7, i7=0); write C[i5, i6] (i0=0, i6=7)
n^2	1.24	level	2	$(7/8) \cdot n^2$	$0.438/n$	write B[i3, i4] (i0=0)
n^2	1.06	level	2	$(3/4) \cdot n^2$	$0.375/n$	write C[i5, i6] (i0=0, i7=0); write C[i5, i6] (i0=0)
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.438/n$	read B[i3, i4] (i0=0)
n^2	0.75	level	1	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	read C[i5, i6] (i0=0, i7=0); read C[i5, i6] (i0=0)
n^2	0.75	level	1	$(3/4) \cdot n^2$	$0.375/n$	read C[i5, i6] (i0=0, i7=0); read C[i5, i6] (i0=0)
n^2	0.75	level	1	$(3/4) \cdot n^2 - 6 \cdot n$	$0.375/n$	read C[i5, i6] (i0=0, i6=7, i7=0); read C[i5, i6] (i0=0, i6=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.75	level	1	$(3/4) \cdot n^2$	$0.375/n$	read B[i7, i6] (i0=0, i6=1)
n^2	0.625	level	1	$(5/8) \cdot n^2$	$0.312/n$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^2	0.541	level	2	$(49/128) \cdot n^2 + (-7/16) \cdot n$	$0.191/n$	write C[i5, i6] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 - 2 \cdot n + 11$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i1=0, i2=0); read B[i2, i1] (i0=0, i1=0)
n^2	0.438	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	$(7/8) \cdot n - 15$	$0.438 \cdot n^{-2}$	write C[i5, i6] (i0=0)
n^2	0.438	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	$(7/8) \cdot n - 15$	$0.438 \cdot n^{-2}$	write C[i5, i6] (i0=0)
n^2	0.375	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 15$	$(3/4) \cdot n - 12$	$0.375 \cdot n^{-2}$	write C[i6, i5] (i0=0, i5=8)
n^2	0.375	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 8$	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	write C[i6, i5] (i0=0, i5=0)
n^2	0.375	level	1	$(3/8) \cdot n^2 + (-1/4) \cdot n$	$0.188/n$	read A[i1] (i0=0); write A[i1] (i0=0) (+5)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n - 9$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i5=0, i7=0); read B[i7, i6] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 9$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 1$	n	$0.5 \cdot n^{-2}$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i3, i4] (i0=0, i3=0, i4=0); read B[i3, i4] (i0=0, i4=0)
n^2	0.354	level	2	$(1/4) \cdot n^2$	$0.125/n$	write A[i1] (i0=0); write C[i5, i6] (i0=0, i6=0, i7=0) (+1)
n^2	0.288	ramp	$(1/4) \cdot n^2 + (-5/8) \cdot n + 18 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n - 2$	$(5/8) \cdot n - 15$	$0.312 \cdot n^{-2}$	write C[i6, i5] (i0=0)
n^2	0.25	level	1	$(1/4) \cdot n^2$	$0.125/n$	read C[i5, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i6=7, i7=0) (+2)
n^2	0.246	ramp	$(19/8) \cdot n + 5 \rightarrow (1/8) \cdot n^2 + 11$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i5=0, i6=8)
n^2	0.241	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + 2$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i5=0, i6=0)
n^2	0.241	ramp	$(11/8) \cdot n - 2 \rightarrow (1/8) \cdot n^2 + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i3, i4] (i0=0)
n^2	0.217	level	3	$(1/8) \cdot n^2 - n$	$0.0625/n$	write C[i5, i6] (i0=0, i6=7, i7=0); write C[i5, i6] (i0=0, i6=7)
n^2	0.212	ramp	$n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(7/8) \cdot n - 2$	$0.438 \cdot n^{-2}$	write C[i6, i5] (i0=0)
n^2	0.188	level	1	$(3/16) \cdot n^2 + (-5/2) \cdot n + 8$	$0.0938/n$	read C[i5, i6] (i0=0); write C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.182	ramp	$n + 3 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(3/4) \cdot n$	$0.375 \cdot n^{-2}$	write C[i5, i6] (i0=0, i6=0)
n^2	0.182	ramp	$n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.375 \cdot n^{-2}$	write C[i6, i5] (i0=0, i6=1)
n^2	0.177	level	2	$(1/8) \cdot n^2$	$0.0625/n$	write C[i5, i6] (i0=0, i6=7, i7=0); write C[i5, i6] (i0=0, i6=7)
n^2	0.125	level	1	$(1/8) \cdot n^2 -$ n	$0.0625/n$	read C[i5, i6] (i0=0, i6=7, i7=0); read C[i5, i6] (i0=0, i6=7)
n^2	0.125	level	1	$(1/8) \cdot n^2 +$ $(1/8) \cdot n$	$0.0625/n$	write C[i5, i6] (i0=0, i5=0, i6=1); write C[i5, i6] (i0=0, i6=1) (+1)
n^2	0.0773	level	2	$(7/128) \cdot n^2$ $+ (-7/16) \cdot n$	$0.0273/n$	write C[i5, i6] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(-3/4) \cdot n +$ 15	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	write C[i6, i5] (i0=0, i5=8)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(-3/4) \cdot n +$ 14	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	write C[i6, i5] (i0=0, i5=8)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^{-2}$	write C[i5, i6] (i0=0, i6=8)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^{-2}$	write C[i5, i6] (i0=0, i5=8)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	write C[i5, i6] (i0=0, i6=1)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	write C[i5, i6] (i0=0, i5=0)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(-3/4) \cdot n + 8$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	write C[i6, i5] (i0=0, i5=0)
n^2	0.0625	level	$(1/4) \cdot n^2 +$ $(-3/4) \cdot n + 7$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	write C[i6, i5] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^2$	write C[i5, i6] (i0=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	write C[i5, i6] (i0=0, i6=0)
n^2	0.0625	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	write A[i1] (i0=0)
n^2	0.0577	ramp	$(1/4) \cdot n^2 + 3 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	write C[i6, i5] (i0=0, i6=1)
n^2	0.056	ramp	$(1/4) \cdot n^2 + (-3/4) \cdot n + 22 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n - 13$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^2$	write C[i6, i5] (i0=0, i6=8)
n^2	0.056	ramp	$(1/4) \cdot n^2 + (-3/4) \cdot n + 21 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n - 14$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^2$	write C[i6, i5] (i0=0, i6=7)
n^2	0.056	ramp	$(1/4) \cdot n^2 + (-3/4) \cdot n + 21 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n - 14$	$(1/8) \cdot n - 4$	$0.0625 \cdot n^2$	write C[i6, i5] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	read B[i3, i4] (i0=0)
n^2	0.0412	ramp	$(1/8) \cdot n^2 + (1/8) \cdot n + 18 \rightarrow (1/8) \cdot n^2 + (5/4) \cdot n - 18$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	read B[i7, i6] (i0=0, i5=0, i7=0)
n^2	0.0302	ramp	$2 \cdot n + 11 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^2$	write C[i6, i5] (i0=0, i6=1)
n^2	0.0302	ramp	$2 \cdot n + 10 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^2$	write C[i5, i6] (i0=0, i6=0)
n^2	0.0296	ramp	$4 \cdot n + 19 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 17$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	read B[i7, i6] (i0=0, i7=0)
n^2	0.0289	ramp	$(13/4) \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 16$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	read B[i7, i6] (i0=0, i5=0)
n^2	0.0289	ramp	$(9/4) \cdot n - 2 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^2$	read B[i3, i4] (i0=0, i3=0)
n^2	0.0288	ramp	$3 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 16$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^2$	read B[i7, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0279	ramp	$(5/4) \cdot n - 1$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 2	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i4] (i0=0, i3=0)
$n^{1.5}$	6	level	$n + 3$	$6 \cdot n - 12$	$3 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 12$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0, i7=0); read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	1.58	ramp	$2 \cdot n + 12 \rightarrow$ $3 \cdot n + 9$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=8)
$n^{1.5}$	1.41	level	$2 \cdot n + 5$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 6$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 7$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 8$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 9$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 10$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i6=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=1, i7=0); read B[i7, i6] (i0=0, i6=1)
$n^{1.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 3$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{1.5}$	1.21	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	1	level	$n + 2$	n	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0, i7=0); read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	0.875	level	$n + 3$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i7=0)
$n^{1.5}$	0.875	level	$n + 4$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	0.875	level	$n + 1$	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=1)
$n^{1.5}$	0.875	level	$n + 1$	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i2, i1] (i0=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 1$	$0.5 \cdot n^{-2}$	read A[i4] (i0=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 1$	$0.5 \cdot n^{-2}$	read A[i4] (i0=0, i4=0)
$n^{1.5}$	0.177	level	$2 \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=1)
$n^{1.5}$	0.125	level	$n + 2$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=1, i7=0)
n^1	5.25	level	1	$(21/4) \cdot n - 21$	$2.62 \cdot n^{-2}$	read C[i5, i6] (i0=0)
n^1	3.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	7	$3.5 \cdot n^{-3}$	write C[i5, i6] (i0=0)
n^1	3	level	$(1/4) \cdot n^2 + (1/8) \cdot n - 6$	6	$3 \cdot n^{-3}$	write C[i6, i5] (i0=0)
n^1	2.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	5	$2.5 \cdot n^{-3}$	write C[i6, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	2	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	4	$2 \cdot n^{-3}$	write A[i1] (i0=0); write C[i5, i6] (i0=0, i5=0, i6=8) (+2)
n^1	1	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	2	$1 \cdot n^{-3}$	write C[i5, i6] (i0=0, i5=0, i6=0); write C[i5, i6] (i0=0, i5=8, i6=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read C[i5, i6] (i0=0, i6=1)
n^1	0.875	level	1	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read C[i5, i6] (i0=0, i6=8)
n^1	0.75	level	1	$(3/4) \cdot n - 6$	$0.375 \cdot n^{-2}$	read C[i5, i6] (i0=0)
n^1	0.75	level	1	$(3/4) \cdot n$	$0.375 \cdot n^{-2}$	write C[i5, i6] (i0=0, i6=1)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 15$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8, i6=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n - 6$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=8)
n^1	0.5	level	$(1/4) \cdot n^2 + n + (35/4)$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (-49/8)$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 12$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/2) \cdot n + 10$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/8) \cdot n + 8$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 6$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 4$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 14$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8, i6=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n - 7$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=7)
n^1	0.5	level	$(1/4) \cdot n^2 + 2$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8, i6=1)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=1)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n - 7$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 14$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (3/4)$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=1)
n^1	0.5	level	$(1/4) \cdot n^2 + n + (7/4)$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i5, i6] (i0=0, i6=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i5, i6] (i0=0, i5=8)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i5, i6] (i0=0, i5=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i5, i6] (i0=0, i6=1)
n^1	0.5	level	$(1/4) \cdot n^2 + (-5/8) \cdot n + 6$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/2) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/8) \cdot n + 4$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0)
n^1	0.5	level	$(1/4) \cdot n^2 + 1$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0, i6=1)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write C[i5, i6] (i0=0, i6=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (7/8)$	1	$0.5 \cdot n^{-3}$	write C[i5, i6] (i0=0, i6=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	write A[i1] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0, i6=7)
n^1	0.5	level	$(1/4) \cdot n^2 + (-3/4) \cdot n + 8$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i5=0, i6=8)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n - 9$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 9$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i5=0, i6=8, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i5=0, i6=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i4] (i0=0, i3=0, i4=0)
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read C[i5, i6] (i0=0, i6=1)
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read C[i5, i6] (i0=0, i6=8)
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.73	level	$3 \cdot n + 10$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i6=8, i7=0)
$n^{0.5}$	1.5	level	$(9/4) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i5=0, i6=8)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=1)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 11$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i6=8)
$n^{0.5}$	1.12	level	$(5/4) \cdot n - 1$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i5=0, i6=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n - 1$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0, i3=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	1	level	$n + 2$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=1)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	1	level	$n + 4$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0, i3=0); read A[i4] (i0=0, i3=0)

The column-pair sweep re-reads data columns: read B[i7,i6] ramps $4n \rightarrow (1/8)n^2 + (9/8)n$

($0.016 \cdot n^4$ combined), and the fixed column reuse at $2n + 2$ lines carries a heavy $n^{3.5}$ band ($0.62 + 0.46$). Headroom +1.0; the miss curve steps first at the two-column boundary (128n bytes), then at the matrix boundary.

covariance — single-shot [exact]

Accesses $A(n) = 2 \cdot n^3 + (21/2) \cdot n^2 + (11/2) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0161 \cdot n^4 + 1.33 \cdot n^{3.5} + 2.07 \cdot n^3 + 19.5 \cdot n^{2.5} + 26.5 \cdot n^2 + 28.7 \cdot n^{1.5} + 3 \cdot n^1 + 30.9 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0142	ramp	$4 \cdot n + 22 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 8$	$(7/128) \cdot n^3$ + (- $35/16) \cdot n^2$ + $21 \cdot n$	0.0273	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^4	0.00193	ramp	$5 \cdot n + 29 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 16$	$(1/128) \cdot n^3$ + $(-7/16) \cdot n^2$ + $6 \cdot n$	0.00391	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{3.5}$	0.619	level	$2 \cdot n + 2$	$(7/16) \cdot n^3$ + $(-63/8) \cdot n^2$ + $35 \cdot n$	0.219	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{3.5}$	0.464	level	$2 \cdot n + 2$	$(21/64) \cdot n^3$ + (- $357/64) \cdot n^2$ + $(105/4) \cdot n$ - 21	0.164	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{3.5}$	0.0884	level	$2 \cdot n + 3$	$(1/16) \cdot n^3$ + $(-19/8) \cdot n^2$ + $22 \cdot n$	0.0312	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{3.5}$	0.0773	level	$2 \cdot n + 2$	$(7/128) \cdot n^3$ + $(-7/16) \cdot n^2$	0.0273	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
$n^{3.5}$	0.0773	level	$2 \cdot n + 2$	$(7/128) \cdot n^3$ + (- $21/16) \cdot n^2$ + $7 \cdot n$	0.0273	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^3	0.866	level	3	$(1/2) \cdot n^3 +$ $(-17/2) \cdot n^2$ + $36 \cdot n$	0.25	read A[i4] (i0=0); write C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.5	level	1	$(1/2) \cdot n^3 + 3 \cdot n^2 + (11/2) \cdot n$	0.25	read A[i1] (i0=0); write A[i1] (i0=0) (+7)
n^3	0.21	ramp	$3 \cdot n + 13 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(7/8) \cdot n^2 - 14 \cdot n$	0.438/n	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^3	0.15	ramp	$3 \cdot n + 14 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(5/8) \cdot n^2 + (-79/8) \cdot n - 2$	0.312/n	read B[i7, i6] (i0=0, i7=0); read B[i7, i6] (i0=0)
n^3	0.109	ramp	$n + 5 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(49/128) \cdot n^2 + (-33/16) \cdot n + 2$	0.191/n	write C[i6, i5] (i0=0)
n^3	0.0359	ramp	$(19/8) \cdot n - 4 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0625/n	read B[i3, i4] (i0=0)
n^3	0.0354	ramp	$(27/8) \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (5/4) \cdot n - 19$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	0.0625/n	read B[i7, i6] (i0=0, i5=0)
n^3	0.0299	ramp	$3 \cdot n + 13 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 14$	$(1/8) \cdot n^2 - 2 \cdot n$	0.0625/n	read B[i7, i6] (i0=0, i6=1, i7=0); read B[i7, i6] (i0=0, i6=1)
n^3	0.0299	ramp	$3 \cdot n + 19 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0625/n	read B[i7, i6] (i0=0, i6=7)
n^3	0.0296	ramp	$3 \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 15$	$(1/8) \cdot n^2 - 3 \cdot n + 1$	0.0625/n	read B[i7, i5] (i0=0, i6=0, i7=0); read B[i7, i5] (i0=0, i6=0)
n^3	0.0296	ramp	$4 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 16$	$(1/8) \cdot n^2 - 3 \cdot n$	0.0625/n	read B[i7, i6] (i0=0, i6=8, i7=0); read B[i7, i6] (i0=0, i6=8)
n^3	0.029	ramp	$3 \cdot n + 21 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 18$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	0.0625/n	read B[i7, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0154	ramp	$2 \cdot n + 12 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(7/128) \cdot n^2$ $+ (-9/16) \cdot n$ $+ 1$	$0.0273/n$	write C[i6, i5] (i0=0)
$n^{2.5}$	4.95	level	$2 \cdot n + 2$	$(7/2) \cdot n^2 -$ $28 \cdot n$	$1.75/n$	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{2.5}$	2.65	level	$2 \cdot n + 2$	$(15/8) \cdot n^2$ $+ (-57/4) \cdot n -$ 6	$0.938/n$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)
$n^{2.5}$	2.62	level	$n + 2$	$(21/8) \cdot n^2$	$1.31/n$	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 -$ $7 \cdot n$	$0.438/n$	read B[i7, i5] (i0=0, i6=9, i7=0); read B[i7, i5] (i0=0, i6=9)
$n^{2.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n^2 -$ $7 \cdot n$	$0.438/n$	read B[i7, i6] (i0=0, i6=8, i7=0); read B[i7, i6] (i0=0, i6=8)
$n^{2.5}$	1.06	level	$2 \cdot n + 3$	$(3/4) \cdot n^2 -$ $12 \cdot n$	$0.375/n$	read B[i7, i5] (i0=0, i7=0); read B[i7, i5] (i0=0)
$n^{2.5}$	1.06	level	$2 \cdot n + 2$	$(3/4) \cdot n^2 +$ $(-27/4) \cdot n +$ 6	$0.375/n$	read B[i7, i6] (i0=0, i6=7)
$n^{2.5}$	1.04	ramp	$n + 5 \rightarrow 2 \cdot n$ $+ 2$	$(7/8) \cdot n^2 +$ $(-35/4) \cdot n +$ 14	$0.438/n$	read B[i7, i5] (i0=0, i6=0)
$n^{2.5}$	1.04	ramp	$n + 4 \rightarrow 2 \cdot n$ $+ 1$	$(7/8) \cdot n^2 +$ $(-35/4) \cdot n +$ 14	$0.438/n$	read B[i7, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.875	level	$n + 1$	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.438/n$	read B[i7, i5] (i0=0, i6=1, i7=0); read B[i7, i5] (i0=0, i6=1)
$n^{2.5}$	0.875	level	$n + 1$	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.438/n$	read B[i2, i1] (i0=0, i2=0); read B[i2, i1] (i0=0)
$n^{2.5}$	0.464	level	$2 \cdot n + 2$	$(21/64) \cdot n^2 + (-21/4) \cdot n + 21$	$0.164/n$	read B[i7, i6] (i0=0)
$n^{2.5}$	0.177	level	$2 \cdot n + 3$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0625/n$	read B[i7, i5] (i0=0, i6=9, i7=0); read B[i7, i5] (i0=0, i6=9)
$n^{2.5}$	0.148	ramp	$n + 3 \rightarrow 2 \cdot n$	$(1/8) \cdot n^2 + (-5/4) \cdot n + 2$	$0.0625/n$	read B[i7, i5] (i0=0, i6=1)
$n^{2.5}$	0.0625	level	$(1/4) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0625/n$	read A[i4] (i0=0)
n^2	7.79	level	3	$(9/2) \cdot n^2 - 36 \cdot n$	$2.25/n$	write C[i5, i6] (i0=0)
n^2	4.95	level	2	$(7/2) \cdot n^2$	$1.75/n$	write C[i5, i6] (i0=0)
n^2	4.24	level	2	$3 \cdot n^2$	$1.5/n$	write A[i1] (i0=0); read B[i3, i4] (i0=0) (+4)
n^2	2.5	level	1	$(5/2) \cdot n^2 + (1/8) \cdot n$	$1.25/n$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)
n^2	1.24	level	2	$(7/8) \cdot n^2$	$0.438/n$	read A[i4] (i0=0)
n^2	0.875	level	1	$(7/8) \cdot n^2 + (7/8) \cdot n + 1$	$0.438/n$	write A[i1] (i0=0); write C[i5, i6] (i0=0, i6=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.438/n$	read B[i7, i6] (i0=0, i6=1)
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.438/n$	read B[i3, i4] (i0=0)
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.219/n$	write C[i5, i6] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n - 9$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i5=0, i7=0); read B[i7, i6] (i0=0, i5=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i3, i4] (i0=0, i3=0, i4=0); read B[i3, i4] (i0=0, i4=0)
n^2	0.246	ramp	$(19/8) \cdot n + 5 \rightarrow (1/8) \cdot n^2 + 11$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i5=0, i6=8)
n^2	0.241	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + 2$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i5=0, i6=0)
n^2	0.241	ramp	$(11/8) \cdot n - 2 \rightarrow (1/8) \cdot n^2 + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i3, i4] (i0=0)
n^2	0.212	ramp	$n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(7/8) \cdot n - 2$	$0.438 \cdot n^{-2}$	write C[i6, i5] (i0=0)
n^2	0.182	ramp	$n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/4) \cdot n$	$0.375 \cdot n^{-2}$	write C[i5, i6] (i0=0, i6=0)
n^2	0.182	ramp	$n + 4 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.375 \cdot n^{-2}$	write C[i6, i5] (i0=0, i6=1)
n^2	0.125	level	1	$(1/8) \cdot n^2 + (-1/8) \cdot n$	$0.0625/n$	read B[i7, i6] (i0=0, i6=7)
n^2	0.125	level	1	$(1/8) \cdot n^2 - n$	$0.0625/n$	read B[i7, i6] (i0=0, i6=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0412	ramp	$(1/8) \cdot n^2 + (1/8) \cdot n + 18 \rightarrow (1/8) \cdot n^2 + (5/4) \cdot n - 18$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i7, i6] (i0=0, i5=0, i7=0)
n^2	0.0302	ramp	$2 \cdot n + 11 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	write C[i6, i5] (i0=0, i6=1)
n^2	0.0302	ramp	$2 \cdot n + 10 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 8$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	write C[i5, i6] (i0=0, i6=0)
n^2	0.0296	ramp	$4 \cdot n + 19 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 17$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i7, i6] (i0=0, i7=0)
n^2	0.0289	ramp	$(13/4) \cdot n + 12 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 16$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i7, i6] (i0=0, i5=0)
n^2	0.0289	ramp	$(9/4) \cdot n - 2 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i3, i4] (i0=0, i3=0)
n^2	0.0288	ramp	$3 \cdot n + 20 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 16$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i7, i6] (i0=0)
n^2	0.0279	ramp	$(5/4) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-15/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i4] (i0=0, i3=0)
$n^{1.5}$	6	level	$n + 3$	$6 \cdot n - 12$	$3 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 12$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0, i7=0); read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	1.58	ramp	$2 \cdot n + 12 \rightarrow 3 \cdot n + 9$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=8)
$n^{1.5}$	1.41	level	$2 \cdot n + 5$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.41	level	$2 \cdot n + 6$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 7$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 8$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 9$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)
$n^{1.5}$	1.41	level	$2 \cdot n + 10$	$n - 1$	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=7)
$n^{1.5}$	1.41	level	$2 \cdot n + 4$	n	$0.5 \cdot n^{-2}$	read B[i7, i6] (i0=0, i6=1, i7=0); read B[i7, i6] (i0=0, i6=1)
$n^{1.5}$	1.24	level	$2 \cdot n + 2$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 3$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{1.5}$	1.21	ramp	$n + 4 \rightarrow 2 \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	1	level	$n + 2$	n	$0.5 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0, i7=0); read B[i7, i5] (i0=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.875	level	$n + 1$	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=1)
$n^{1.5}$	0.875	level	$n + 3$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i7=0)
$n^{1.5}$	0.875	level	$n + 4$	$(7/8) \cdot n - 7$	$0.438 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=0)
$n^{1.5}$	0.875	level	$n + 1$	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i2, i1] (i0=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 1$	$0.5 \cdot n^{-2}$	read A[i4] (i0=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 1$	$0.5 \cdot n^{-2}$	read A[i4] (i0=0, i4=0)
$n^{1.5}$	0.177	level	$2 \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=1)
$n^{1.5}$	0.125	level	$n + 2$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i7, i5] (i0=0, i6=1, i7=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	write C[i5, i6] (i0=0, i6=1)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n - 9$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 9$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i5=0, i6=8, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i5=0, i6=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i4] (i0=0, i3=0, i4=0)
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	6	level	$n + 3$	6	$3 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 10$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i6=8, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.5	level	$(9/4) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i5=0, i6=8)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=1)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0, i7=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 10$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i6=7, i7=0); read B[i7, i6] (i0=0, i7=0) (+1)
$n^{0.5}$	1.41	level	$2 \cdot n + 11$	1	$0.5 \cdot n^{-3}$	read B[i7, i6] (i0=0, i6=8)
$n^{0.5}$	1.12	level	$(5/4) \cdot n - 1$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i5=0, i6=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n - 1$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0, i3=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.5 \cdot n^{-3}$	read B[i7, i5] (i0=0, i6=0)
$n^{0.5}$	1	level	$n + 2$	1	$0.5 \cdot n^{-3}$	write C[i6, i5] (i0=0, i6=1)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i4] (i0=0, i3=0); read A[i4] (i0=0, i3=0)

The column-pair sweep re-reads data columns: read B[i7,i6] ramps $4n \rightarrow (1/8)n^2 + (9/8)n$ ($0.016 \cdot n^4$ combined), and the fixed column reuse at $2n + 2$ lines carries a heavy $n^{3.5}$ band ($0.62 + 0.46$). Headroom +1.0; the miss curve steps first at the two-column boundary (128n

bytes), then at the matrix boundary.

doitgen — infinite-repeat [exact]

Accesses $A(n) = 4 \cdot n^4 + 3 \cdot n^3$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^5 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^5 + 1.1 \cdot n^{4.5} + 5.14 \cdot n^4 + 4.78 \cdot n^{3.5} + 7.07 \cdot n^3 + 1.5 \cdot n^{2.5} + 1.73 \cdot n^2$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^5	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(7/64) \cdot n^4 + (-15/8) \cdot n^3 + 2 \cdot n^2$	0.0273	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
n^5	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/64) \cdot n^4 + (-3/8) \cdot n^3 + 2 \cdot n^2$	0.00391	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
$n^{4.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^4 + (-7/8) \cdot n^3$	0.191	read C[i4, i3] (i0=0)
$n^{4.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^4 + (-7/8) \cdot n^3$	0.0273	read C[i4, i3] (i0=0)
$n^{4.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^4 + (-7/4) \cdot n^3$	0.0273	read B[i1, i2, i4] (i0=0)
$n^{4.5}$	0.0442	level	$(1/8) \cdot n^3 + (1/8) \cdot n^2 + (1/8) \cdot n$	$(1/8) \cdot n^3 - 2 \cdot n^2$	0.0312/n	read B[i1, i2, i4] (i0=0, i1=0, i2=0, i3=0); read B[i1, i2, i4] (i0=0, i1=0, i3=0) (+2)
$n^{4.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^4 + (-3/8) \cdot n^3 + 2 \cdot n^2$	0.00391	read B[i1, i2, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	1.73	level	3	$n^4 - n^2$	0.25	read C[i4, i3] (i0=0, i1=0, i2=0, i3=0, i4=0); read A[i3] (i0=0, i1=0, i4=0) (+5)
n^4	1.52	level	3	$(7/8) \cdot n^4$	0.219	read B[i1, i2, i4] (i0=0)
n^4	0.875	level	1	$(7/8) \cdot n^4 +$ $(7/8) \cdot n^3$	0.219	write A[i3] (i0=0, i1=0, i2=0); write A[i3] (i0=0, i1=0) (+1)
n^4	0.309	ramp	$(1/8) \cdot n^2 +$ $(1/4) \cdot n + 1$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(3/8) \cdot n$	$(7/8) \cdot n^3 -$ n^2	0.219/n	read C[i4, i3] (i0=0, i1=0, i2=0, i3=0); read C[i4, i3] (i0=0, i1=0, i3=0) (+2)
n^4	0.309	level	$(1/8) \cdot n^2 +$ $(3/8) \cdot n$	$(7/8) \cdot n^3 -$ $7 \cdot n^2$	0.219/n	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
n^4	0.125	level	1	$(1/8) \cdot n^4$	0.0312	write A[i3] (i0=0, i1=0, i2=0, i4=0); write A[i3] (i0=0, i1=0, i2=0) (+5)
n^4	0.0442	level	$(1/8) \cdot n^2 +$ $(5/4) \cdot n +$ $(21/8)$	$(1/8) \cdot n^3 +$ $(-9/8) \cdot n^2$	0.0312/n	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - 2 \cdot n^2$	$0.0312/n$	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
n^4	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - n^2$	$0.0312/n$	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
n^4	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - 2 \cdot n^2$	$0.0312/n$	read C[i4, i3] (i0=0, i1=0, i2=0, i4=0); read C[i4, i3] (i0=0, i1=0, i4=0) (+2)
n^4	0.0432	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - n^2$	$0.0312/n$	read C[i4, i3] (i0=0, i1=0, i2=0, i3=0); read C[i4, i3] (i0=0, i1=0, i3=0) (+2)
n^4	0.0281	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-7/4) \cdot n + 2$	$(1/8) \cdot n^3 - 2 \cdot n^2$	$0.0312/n$	write A[i3] (i0=0, i1=0, i2=0); write A[i3] (i0=0, i1=0) (+3)
n^4	0.028	ramp	$(5/4) \cdot n \rightarrow (1/8) \cdot n^2 + (-7/4) \cdot n$	$(1/8) \cdot n^3 - 2 \cdot n^2$	$0.0312/n$	read A[i5] (i0=0, i1=0, i2=0); read A[i5] (i0=0, i1=0) (+1)
$n^{3.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^3$	$0.219/n$	read C[i4, i3] (i0=0)
$n^{3.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^3$	$0.219/n$	read C[i4, i3] (i0=0, i4=0)
$n^{3.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^3$	$0.219/n$	read B[i1, i2, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^3$	$0.219/n$	read B[i1, i2, i4] (i0=0, i4=0)
$n^{3.5}$	0.354	level	$(1/8) \cdot n^3 +$ $(1/8) \cdot n^2 +$ $(1/8) \cdot n$	n^2	$0.25 \cdot n^{-2}$	read B[i1, i2, i4] (i0=0, i1=0, i2=0, i3=0, i4=0); read B[i1, i2, i4] (i0=0, i1=0, i3=0, i4=0) (+2)
$n^{3.5}$	0.354	level	$(1/8) \cdot n^3 +$ $(1/8) \cdot n^2 +$ $(1/8) \cdot n$	n^2	$0.25 \cdot n^{-2}$	read B[i1, i2, i4] (i0=0, i1=0, i2=0, i3=0); read B[i1, i2, i4] (i0=0, i1=0, i3=0) (+2)
$n^{3.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^3 -$ n^2	$0.0312/n$	read B[i1, i2, i4] (i0=0)
$n^{3.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^3$	$0.0312/n$	read B[i1, i2, i4] (i0=0, i4=0); write B[i1, i2, i5] (i0=0)
$n^{3.5}$	0.0964	ramp	$(1/4) \cdot n + 9$ $\rightarrow (9/8) \cdot n -$ 12	$(1/8) \cdot n^3 -$ $2 \cdot n^2$	$0.0312/n$	write B[i1, i2, i5] (i0=0)
n^3	2.12	level	$(1/8) \cdot n^2 +$ $(3/8) \cdot n$	$6 \cdot n^2$	$1.5 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
n^3	1.24	level	2	$(7/8) \cdot n^3$	$0.219/n$	write B[i1, i2, i5] (i0=0)
n^3	1.24	level	2	$(7/8) \cdot n^3$	$0.219/n$	read A[i5] (i0=0, i1=0, i2=0); read A[i5] (i0=0, i1=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (21/8)$	n^2	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	n^2	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=0, i2=0); read C[i4, i3] (i0=0, i1=0) (+2)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	n^2	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=0, i2=0, i4=0); read C[i4, i3] (i0=0, i1=0, i4=0) (+2)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	n^2	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=0, i2=0, i3=0); read C[i4, i3] (i0=0, i1=0, i3=0) (+2)
n^3	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	n^2	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=0, i2=0, i3=0, i4=0); read C[i4, i3] (i0=0, i1=0, i3=0, i4=0) (+2)
n^3	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	n^2	$0.25 \cdot n^{-2}$	write A[i3] (i0=0, i1=0, i2=0); write A[i3] (i0=0, i1=0) (+3)
n^3	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n$	n^2	$0.25 \cdot n^{-2}$	read A[i5] (i0=0, i1=0, i2=0, i5=0); read A[i5] (i0=0, i1=0, i5=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.5	level	$(1/4) \cdot n$	n^2	$0.25 \cdot n^{-2}$	write A[i3] (i0=0, i1=0, i2=0, i3=0); read B[i1, i2, i4] (i0=0, i1=0, i2=0, i3=0, i4=0) (+4)
$n^{2.5}$	0.5	level	$(1/4) \cdot n + 1$	n^2	$0.25 \cdot n^{-2}$	write B[i1, i2, i5] (i0=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n - 1$	n^2	$0.25 \cdot n^{-2}$	read A[i5] (i0=0, i1=0, i2=0); read A[i5] (i0=0, i1=0) (+1)
n^2	1.52	level	3	$(7/8) \cdot n^2$	$0.219 \cdot n^{-2}$	read A[i3] (i0=0, i1=0, i2=0, i4=0); read A[i3] (i0=0, i1=0, i2=0)
n^2	0.217	level	3	$(1/8) \cdot n^2$	$0.0312 \cdot n^{-2}$	read A[i3] (i0=0, i1=0, i2=0, i4=0); read A[i3] (i0=0, i1=0, i2=0)

The contraction buffer read $C[i4, i3]$ is re-read per (r, q) pair at $(1/8)n^2 + (3/8)n$ lines — gemm’s boundary — but the access count is n^4 , so the term order is n^5 ($0.0387 + 0.0055 \approx 0.044$, gemm’s constant one dimension up). The $n^{4.5}$ band is the row-window reuse at $(9/8)n + 1$ lines.

doitgen — single-shot [exact]

Accesses $A(n) = 4 \cdot n^4 + 3 \cdot n^3$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^5 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^5 + 1.06 \cdot n^{4.5} + 5.14 \cdot n^4 + 4.07 \cdot n^{3.5} + 7.95 \cdot n^3 + 1.5 \cdot n^{2.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^5	0.0387	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(7/64) \cdot n^4 + (-15/8) \cdot n^3 + (121/64) \cdot n^2 + (15/8) \cdot n - 2$	0.0273	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^5	0.00552	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/64) \cdot n^4 + (-3/8) \cdot n^3 + (127/64) \cdot n^2 + (3/8) \cdot n - 2$	0.00391	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)
$n^{4.5}$	0.812	level	$(9/8) \cdot n + 1$	$(49/64) \cdot n^4 + (-7/8) \cdot n^3$	0.191	read C[i4, i3] (i0=0)
$n^{4.5}$	0.116	level	$(9/8) \cdot n + 1$	$(7/64) \cdot n^4 + (-7/8) \cdot n^3$	0.0273	read C[i4, i3] (i0=0)
$n^{4.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^4 + (-7/4) \cdot n^3$	0.0273	read B[i1, i2, i4] (i0=0)
$n^{4.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^4 + (-3/8) \cdot n^3 + 2 \cdot n^2$	0.00391	read B[i1, i2, i4] (i0=0)
n^4	1.73	level	3	n^4	0.25	read B[i1, i2, i4] (i0=0, i4=0); read A[i3] (i0=0, i4=0) (+2)
n^4	1.52	level	3	$(7/8) \cdot n^4$	0.219	read B[i1, i2, i4] (i0=0)
n^4	1	level	1	n^4	0.25	write A[i3] (i0=0); read A[i5] (i0=0)
n^4	0.309	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 1 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(7/8) \cdot n^3 - n^2 + (-7/8) \cdot n + 1$	0.219/n	read C[i4, i3] (i0=0, i2=0, i3=0); read C[i4, i3] (i0=0, i3=0)
n^4	0.309	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(7/8) \cdot n^3 - 7 \cdot n^2 + (-7/8) \cdot n + 7$	0.219/n	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)
n^4	0.0442	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (21/8)$	$(1/8) \cdot n^3 + (-9/8) \cdot n^2 + (-1/8) \cdot n + (9/8)$	0.0312/n	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - 2 \cdot n^2 + (-1/8) \cdot n + 2$	$0.0312/n$	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)
n^4	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - n^2 + (-1/8) \cdot n + 1$	$0.0312/n$	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)
n^4	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - 2 \cdot n^2 + (-1/8) \cdot n + 2$	$0.0312/n$	read C[i4, i3] (i0=0, i2=0, i4=0); read C[i4, i3] (i0=0, i4=0)
n^4	0.0432	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^3 - n^2 + (-1/8) \cdot n + 1$	$0.0312/n$	read C[i4, i3] (i0=0, i2=0, i3=0); read C[i4, i3] (i0=0, i3=0)
n^4	0.0281	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-7/4) \cdot n + 2$	$(1/8) \cdot n^3 - 2 \cdot n^2 + (-1/8) \cdot n + 2$	$0.0312/n$	write A[i3] (i0=0, i2=0); write A[i3] (i0=0)
n^4	0.028	ramp	$(5/4) \cdot n \rightarrow (1/8) \cdot n^2 + (-7/4) \cdot n$	$(1/8) \cdot n^3 - 2 \cdot n^2$	$0.0312/n$	read A[i5] (i0=0)
$n^{3.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^3$	$0.219/n$	read C[i4, i3] (i0=0)
$n^{3.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^3$	$0.219/n$	read C[i4, i3] (i0=0, i4=0)
$n^{3.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^3$	$0.219/n$	read B[i1, i2, i4] (i0=0)
$n^{3.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n^3$	$0.219/n$	read B[i1, i2, i4] (i0=0, i4=0)
$n^{3.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^3 - n^2$	$0.0312/n$	read B[i1, i2, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n^3$	$0.0312/n$	read B[i1, i2, i4] (i0=0, i4=0); write B[i1, i2, i5] (i0=0)
$n^{3.5}$	0.0964	ramp	$(1/4) \cdot n + 9$ $\rightarrow (9/8) \cdot n - 12$	$(1/8) \cdot n^3 - 2 \cdot n^2$	$0.0312/n$	write B[i1, i2, i5] (i0=0)
n^3	2.12	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$6 \cdot n^2 - 6$	$1.5 \cdot n^{-2}$	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)
n^3	1.24	level	2	$(7/8) \cdot n^3$	$0.219/n$	write B[i1, i2, i5] (i0=0)
n^3	1.24	level	2	$(7/8) \cdot n^3$	$0.219/n$	read A[i5] (i0=0)
n^3	0.875	level	1	$(7/8) \cdot n^3$	$0.219/n$	write A[i3] (i0=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (21/8)$	$n^2 - 1$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$n^2 - 1$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i2=0); read C[i4, i3] (i0=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$n^2 - 1$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i2=0, i4=0); read C[i4, i3] (i0=0, i4=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$n^2 - 1$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i2=0, i3=0); read C[i4, i3] (i0=0, i3=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n^2 - 1$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i2=0, i3=0, i4=0); read C[i4, i3] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 1$	$n^2 - 1$	$0.25 \cdot n^{-2}$	write A[i3] (i0=0, i2=0); write A[i3] (i0=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n$	n^2	$0.25 \cdot n^{-2}$	read A[i5] (i0=0, i5=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n$	$n^2 - 1$	$0.25 \cdot n^{-2}$	write A[i3] (i0=0, i2=0, i3=0); write A[i3] (i0=0, i3=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n + 1$	n^2	$0.25 \cdot n^{-2}$	write B[i1, i2, i5] (i0=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n - 1$	n^2	$0.25 \cdot n^{-2}$	read A[i5] (i0=0)

The contraction buffer read C[i4, i3] is re-read per (r,q) pair at $(1/8)n^2 + (3/8)n$ lines — gemm’s boundary — but the access count is n^4 , so the term order is n^5 ($0.0387 + 0.0055 \approx 0.044$, gemm’s constant one dimension up). The $n^{4.5}$ band is the row-window reuse at $(9/8)n + 1$ lines.

fdtd — infinite-repeat [exact]

Accesses $A(n) = 14 \cdot n^3 - 18 \cdot n^2 + 6 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.617 \cdot n^4 + 0.253 \cdot n^{3.5} + 34.1 \cdot n^3 + 3.35 \cdot n^{2.5} + 41.5 \cdot n^2 + 4.34 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0765	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (1/8)$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + (17/8) \cdot n$	0.00893	read D[i5, i6] (i0=0, i1=0, i5=0); read D[i5, i6] (i0=0, i1=0) (+2)
n^4	0.0765	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	$(1/8) \cdot n^3 + (-17/8) \cdot n^2 + 2 \cdot n$	0.00893	read D[i5, i6] (i0=0, i1=0, i5=0); read D[i5, i6] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0765	level	$(3/8) \cdot n^2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read B[i7 + 1, i8] (i0=0, i1=0); read B[i7 + 1, i8] (i0=0)
n^4	0.0658	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n + 1 \rightarrow (3/8) \cdot n^2 - 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read C[i7, i8] (i0=0, i1=0); read C[i7, i8] (i0=0)
n^4	0.0658	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n + 1 \rightarrow (3/8) \cdot n^2 - 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read D[i7, i8 + 1] (i0=0, i1=0); read D[i7, i8 + 1] (i0=0)
n^4	0.0653	ramp	$(1/4) \cdot n^2 + 2 \rightarrow (3/8) \cdot n^2 + (-1/2) \cdot n - 1$	$(1/8) \cdot n^3 + (-5/2) \cdot n^2 + 8 \cdot n$	0.00893	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^4	0.0653	ramp	$(1/4) \cdot n^2 + 2 \rightarrow (3/8) \cdot n^2 + (-1/2) \cdot n - 1$	$(1/8) \cdot n^3 + (-5/2) \cdot n^2 + 8 \cdot n$	0.00893	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^4	0.0625	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (1/8)$	$(1/8) \cdot n^3 + (-3/2) \cdot n^2 + (27/8) \cdot n$	0.00893	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^4	0.0625	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	$(1/8) \cdot n^3 + (-19/8) \cdot n^2 + 6 \cdot n$	0.00893	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
$n^{3.5}$	0.0884	level	$(1/2) \cdot n + (5/2)$	$(1/8) \cdot n^3 + (-19/8) \cdot n^2 + (17/4) \cdot n$	0.00893	read B[i7, i8] (i0=0, i1=0); read B[i7, i8] (i0=0)
$n^{3.5}$	0.0884	level	$(1/2) \cdot n + 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read B[i7, i8] (i0=0, i1=0); read B[i7, i8] (i0=0)
$n^{3.5}$	0.0765	level	$(3/8) \cdot n + 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read C[i3 - 1, i4] (i0=0, i1=0); read C[i3 - 1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	3.03	level	3	$(7/4) \cdot n^3 + (-7/4) \cdot n^2$	0.125	read C[i3 - 1, i4] (i0=0, i1=0); write B[i3, i4] (i0=0, i1=0) (+2)
n^3	3	level	4	$(3/2) \cdot n^3 + (-3/2) \cdot n^2$	0.107	read B[i7, i8] (i0=0, i1=0); write C[i7, i8] (i0=0, i1=0) (+2)
n^3	1.75	level	1	$(7/4) \cdot n^3 + (-11/4) \cdot n^2 + n$	0.125	read B[i3, i4] (i0=0); read C[i7, i8] (i0=0)
n^3	1.75	level	1	$(7/4) \cdot n^3 + (-15/8) \cdot n^2 - n + 1$	0.125	read D[i7, i8] (i0=0, i1=0); read D[i5, i6] (i0=0) (+1)
n^3	1.75	level	4	$(7/8) \cdot n^3 + (-7/8) \cdot n^2$	0.0625	read B[i7 + 1, i8] (i0=0, i1=0); write C[i7, i8] (i0=0, i1=0) (+2)
n^3	1.75	level	4	$(7/8) \cdot n^3 + (-7/4) \cdot n^2$	0.0625	read C[i5, i6 - 1] (i0=0, i1=0); read D[i7, i8 + 1] (i0=0, i1=0) (+2)
n^3	1.52	level	3	$(7/8) \cdot n^3 + (-21/8) \cdot n^2 + (21/8) \cdot n + (-7/8)$	0.0625	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)
n^3	1.52	level	3	$(7/8) \cdot n^3 + (-7/4) \cdot n^2 + (7/8) \cdot n$	0.0625	read C[i3, i4] (i0=0)
n^3	1.24	level	2	$(7/8) \cdot n^3 - n^2$	0.0625	write D[i5, i6] (i0=0, i1=0); write D[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1.1	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n \rightarrow (3/8) \cdot n^2 + (-1/2) \cdot n$	$2 \cdot n^2 - 8 \cdot n$	$0.143/n$	read B[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i1=0, i4=0) (+4)
n^3	1.06	level	2	$(3/4) \cdot n^3$	0.0536	read C[i5, i6] (i0=0, i1=0, i5=0); read C[i5, i6] (i0=0, i1=0) (+2)
n^3	0.839	level	5	$(3/8) \cdot n^3 + (-27/8) \cdot n^2 + 3 \cdot n$	0.0268	read B[i7 + 1, i8] (i0=0, i1=0); read B[i7, i8] (i0=0, i1=0) (+4)
n^3	0.75	level	1	$(3/4) \cdot n^3$	0.0536	read C[i5, i6 - 1] (i0=0, i1=0); read C[i5, i6 - 1] (i0=0)
n^3	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (1/8)$	$n^2 - n$	$0.0714/n$	read D[i5, i6] (i0=0, i1=0, i5=0, i6=0); read D[i5, i6] (i0=0, i1=0, i6=0) (+2)
n^3	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	$n^2 - n$	$0.0714/n$	read D[i5, i6] (i0=0, i1=0, i5=0, i6=0); read D[i5, i6] (i0=0, i1=0, i6=0) (+2)
n^3	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (-7/8)$	$n^2 - n$	$0.0714/n$	read D[i5, i6] (i0=0, i1=0, i5=0); read D[i5, i6] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	$n^2 - n$	0.0714/n	read D[i5, i6] (i0=0, i1=0, i5=0); read D[i5, i6] (i0=0, i1=0) (+2)
n^3	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (1/8)$	$n^2 - n$	0.0714/n	read D[i5, i6] (i0=0, i1=0, i5=0); read D[i5, i6] (i0=0, i1=0) (+4)
n^3	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	n^2	0.0714/n	write B[0, i2] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=0, i3=0) (+7)
n^3	0.612	level	$(3/8) \cdot n^2$	$n^2 - n$	0.0714/n	read B[i7, i8] (i0=0, i1=0, i7=0); read B[i7 + 1, i8] (i0=0, i1=0) (+2)
n^3	0.612	level	$(3/8) \cdot n^2$	$n^2 - 2 \cdot n$	0.0714/n	read B[i7 + 1, i8] (i0=0, i1=0, i8=0); read B[i7 + 1, i8] (i0=0, i8=0)
n^3	0.555	ramp	$(1/4) \cdot n^2 + (3/8) \cdot n - 1$ → $(3/8) \cdot n^2 - 1$	$n^2 - 2 \cdot n$	0.0714/n	read C[i7, i8] (i0=0, i1=0); read C[i7, i8] (i0=0)
n^3	0.555	ramp	$(1/4) \cdot n^2 + (3/8) \cdot n - 2$ → $(3/8) \cdot n^2 - 2$	$n^2 - 2 \cdot n$	0.0714/n	read D[i7, i8 + 1] (i0=0, i1=0); read D[i7, i8 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.555	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n \rightarrow (3/8) \cdot n^2 + (-1/8) \cdot n$	$n^2 - 2 \cdot n$	0.0714/n	read D[i7, i8 + 1] (i0=0, i1=0, i8=0); read D[i7, i8 + 1] (i0=0, i8=0)
n^3	0.555	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n - 1 \rightarrow (3/8) \cdot n^2 + (-1/8) \cdot n - 1$	$n^2 - 2 \cdot n$	0.0714/n	read C[i7, i8] (i0=0, i1=0, i8=0); read C[i7, i8] (i0=0, i8=0)
n^3	0.55	ramp	$(1/4) \cdot n^2 + 1 \rightarrow (3/8) \cdot n^2 + (-5/8) \cdot n + 1$	$n^2 - 4 \cdot n$	0.0714/n	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^3	0.55	ramp	$(1/4) \cdot n^2 + 1 \rightarrow (3/8) \cdot n^2 + (-5/8) \cdot n + 1$	$n^2 - 4 \cdot n$	0.0714/n	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^3	0.5	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (1/8)$	$n^2 - 3 \cdot n$	0.0714/n	read C[i5, i6] (i0=0, i1=0, i6=0); read C[i5, i6] (i0=0, i6=0)
n^3	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	$n^2 - 3 \cdot n$	0.0714/n	read C[i5, i6] (i0=0, i1=0, i6=0); read C[i5, i6] (i0=0, i6=0)
n^3	0.5	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (1/8)$	$n^2 - 3 \cdot n$	0.0714/n	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^3	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	$n^2 - 3 \cdot n$	0.0714/n	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.306	level	6	$(1/8) \cdot n^3 + (-9/8) \cdot n^2 + n$	0.00893	read D[i7, i8 + 1] (i0=0, i1=0); read D[i7, i8 + 1] (i0=0)
n^3	0.28	level	5	$(1/8) \cdot n^3 + (-9/8) \cdot n^2 + n$	0.00893	read D[i7, i8] (i0=0, i1=0); read D[i7, i8] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 - n^2$	0.00893	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 - n^2$	0.00893	write D[i5, i6] (i0=0, i1=0); write D[i5, i6] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 + (-1/8) \cdot n^2$	0.00893	write B[i3, i4] (i0=0, i1=0); write B[i3, i4] (i0=0)
n^3	0.125	level	1	$(1/8) \cdot n^3 - n$	0.00893	read C[i5, i6 - 1] (i0=0, i1=0); read C[i5, i6 - 1] (i0=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-1/2)$	$(1/8) \cdot n^2 + (-17/8) \cdot n$	0.00893/n	read C[i3, i4] (i0=0, i1=0, i3=0); read C[i3, i4] (i0=0, i1=1, i3=0) (+1)
n^3	0.0765	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$(1/8) \cdot n^2 - 2 \cdot n$	0.00893/n	read C[i3, i4] (i0=0, i1=0, i3=0); read C[i3, i4] (i0=0, i1=1, i3=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0765	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-1/2)$	$(1/8) \cdot n^2 + (-17/8) \cdot n$	$0.00893/n$	read B[i3, i4] (i0=0, i1=0, i3=0); read B[i3, i4] (i0=0, i1=1, i3=0) (+1)
n^3	0.0765	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read B[i3, i4] (i0=0, i1=0, i3=0); read B[i3, i4] (i0=0, i1=1, i3=0) (+1)
n^3	0.0765	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n$	$0.00893/n$	read D[i5, i6] (i0=0, i1=0); read D[i5, i6] (i0=0, i1=1) (+1)
n^3	0.0765	level	$(3/8) \cdot n^2 + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read D[i5, i6] (i0=0, i1=0); read D[i5, i6] (i0=0, i1=1) (+1)
n^3	0.0765	level	$(3/8) \cdot n^2 + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read B[i7, i8] (i0=0, i1=0, i7=0); read B[i7, i8] (i0=0, i7=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n$	$0.00893/n$	read B[i7 + 1, i8] (i0=0, i1=0); read B[i7 + 1, i8] (i0=0)
n^3	0.0726	ramp	$(3/8) \cdot n^2 + (-1/8) \cdot n + 2 \rightarrow (3/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^3	0.0726	ramp	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1 \rightarrow (3/8) \cdot n^2 - 2$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read B[i7 + 1, i8] (i0=0, i1=0, i7=0); read B[i7 + 1, i8] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0725	ramp	$(3/8) \cdot n^2 + (-3/8) \cdot n + 2$ $\rightarrow$ $(3/8) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n^2 - 2 \cdot n$	0.00893/n	read C[i3 - 1, i4] (i0=0, i1=0, i3=0); read C[i3 - 1, i4] (i0=0, i1=1, i3=0) (+1)
n^3	0.0725	ramp	$(3/8) \cdot n^2 + (-1/2) \cdot n + 4$ $\rightarrow$ $(3/8) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/8) \cdot n^2 - 2 \cdot n$	0.00893/n	write B[0, i2] (i0=0, i1=0); write B[0, i2] (i0=0, i1=1) (+1)
n^3	0.0625	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n$	0.00893/n	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^3	0.0625	level	$(1/4) \cdot n^2$	$(1/8) \cdot n^2 - 2 \cdot n$	0.00893/n	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^3	0.0625	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	0.00893/n	read C[i5, i6] (i0=0, i1=0, i5=0); read C[i5, i6] (i0=0, i5=0)
n^3	0.0625	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n$	0.00893/n	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^3	0.0625	level	$(1/4) \cdot n^2 + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	0.00893/n	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^3	0.0593	ramp	$(1/4) \cdot n^2 + 2 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 4$	$(1/8) \cdot n^2 - 2 \cdot n$	0.00893/n	read C[i7, i8] (i0=0, i1=0, i7=0); read C[i7, i8] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0593	ramp	$(1/4) \cdot n^2 + 2 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 4$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read D[i7, i8 + 1] (i0=0, i1=0, i7=0); read D[i7, i8 + 1] (i0=0, i7=0)
n^3	0.0592	ramp	$(1/4) \cdot n^2 + (-1/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 - 3$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^3	0.0592	ramp	$(1/4) \cdot n^2 + (-1/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 - 3$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^3	0.0592	ramp	$(1/4) \cdot n^2 + (-1/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + (-1/8) \cdot n - 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
$n^{2.5}$	0.707	level	$(1/2) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read B[i7, i8] (i0=0, i1=0); read B[i7, i8] (i0=0)
$n^{2.5}$	0.707	level	$(1/2) \cdot n + (5/2)$	$n^2 - 2 \cdot n$	$0.0714/n$	read B[i7, i8] (i0=0, i1=0, i8=0); read B[i7, i8] (i0=0, i8=0)
$n^{2.5}$	0.707	level	$(1/2) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read B[i7, i8] (i0=0, i1=0, i8=0); read B[i7, i8] (i0=0, i8=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read C[i3 - 1, i4] (i0=0, i1=0); read C[i3 - 1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.612	level	$(3/8) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read C[i3 - 1, i4] (i0=0, i1=0, i4=0); read C[i3 - 1, i4] (i0=0, i4=0)
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.0625/n$	read C[i3, i4] (i0=0, i1=0)
n^2	1.41	level	2	$n^2 - n$	$0.0714/n$	read A[i1] (i0=0, i2=0); read A[i1] (i0=0)
n^2	1.41	level	2	n^2	$0.0714/n$	write B[0, i2] (i0=0, i1=0, i2=0); write D[i5, i6] (i0=0, i1=0) (+2)
n^2	1.24	level	2	$(7/8) \cdot n^2$	$0.0625/n$	write B[0, i2] (i0=0, i1=0); write B[0, i2] (i0=0)
n^2	1.22	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-1/2)$	$2 \cdot n$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i1=0, i3=0, i4=0); read C[i3, i4] (i0=0, i1=0, i3=0, i4=0) (+4)
n^2	1.22	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$2 \cdot n$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i1=0, i3=0, i4=0); read C[i3, i4] (i0=0, i1=0, i3=0, i4=0) (+4)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 4$	$2 \cdot n$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i1=0, i4=0) (+4)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1	level	$(1/4) \cdot n^2 - 1$	$2 \cdot n$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i1=0, i4=0); read C[i7, i8] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$2 \cdot n$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i1=0); read C[i7, i8] (i0=0, i1=0, i7=0, i8=0) (+3)
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	n	$0.0714 \cdot n^{-2}$	read D[i5, i6] (i0=0, i1=0); read D[i5, i6] (i0=0, i1=1) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + 1$	n	$0.0714 \cdot n^{-2}$	read D[i5, i6] (i0=0, i1=0); read D[i5, i6] (i0=0, i1=1) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-3/2)$	n	$0.0714 \cdot n^{-2}$	read B[i3, i4] (i0=0, i1=0, i3=0); read B[i3, i4] (i0=0, i1=1, i3=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	n	$0.0714 \cdot n^{-2}$	read B[i3, i4] (i0=0, i1=0, i3=0); read B[i3, i4] (i0=0, i1=1, i3=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-3/2)$	n	$0.0714 \cdot n^{-2}$	read C[i3, i4] (i0=0, i1=0, i3=0); read C[i3, i4] (i0=0, i1=1, i3=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0, i3=0); read C[i3, i4] (i0=0, i1=1, i3=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read A[i1] (i0=0, i1=0); read D[i5, i6] (i0=0, i1=0, i6=0) (+2)
n^2	0.612	level	$(3/8) \cdot n^2 + 1$	n	$0.0714 \cdot n^2$	read A[i1] (i0=0, i1=0); read D[i5, i6] (i0=0, i1=0, i6=0) (+2)
n^2	0.612	level	$(3/8) \cdot n^2 - 1$	n	$0.0714 \cdot n^2$	read B[i7 + 1, i8] (i0=0, i1=0, i7=0); read B[i7 + 1, i8] (i0=0, i7=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/4) \cdot n + (3/8)$	n	$0.0714 \cdot n^2$	read C[i3 - 1, i4] (i0=0, i1=0, i3=0); read C[i3 - 1, i4] (i0=0, i1=1, i3=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (-3/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i3 - 1, i4] (i0=0, i1=0, i3=0); read C[i3 - 1, i4] (i0=0, i1=1, i3=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 3$	n	$0.0714 \cdot n^2$	read C[i3 - 1, i4] (i0=0, i1=0, i3=0); read C[i3 - 1, i4] (i0=0, i1=1, i3=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (-7/8)$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 2$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (9/8)$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 2$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0, i3=0); read C[i3, i4] (i0=0, i1=1, i3=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (-7/8)$	n	$0.0714 \cdot n^2$	read B[i7 + 1, i8] (i0=0, i1=0, i7=0, i8=0); read B[i7 + 1, i8] (i0=0, i7=0, i8=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n$	n	$0.0714 \cdot n^2$	read B[i7 + 1, i8] (i0=0, i1=0, i7=0, i8=0); read B[i7 + 1, i8] (i0=0, i7=0, i8=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n - 1$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i1=1, i4=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i1=1, i4=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (3/8) \cdot n + (1/4)$	n	$0.0714 \cdot n^2$	read C[i3 - 1, i4] (i0=0, i1=0, i3=0, i4=0); read C[i3 - 1, i4] (i0=0, i1=1, i3=0, i4=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/4) \cdot n$	n	$0.0714 \cdot n^2$	read C[i3 - 1, i4] (i0=0, i1=0, i3=0, i4=0); read C[i3 - 1, i4] (i0=0, i1=1, i3=0, i4=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read B[i7 + 1, i8] (i0=0, i1=0, i8=0); read B[i7 + 1, i8] (i0=0, i1=0, i8=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (1/2)$	n	$0.0714 \cdot n^2$	write B[0, i2] (i0=0, i1=0); write B[0, i2] (i0=0, i1=1) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/2) \cdot n + 2$	n	$0.0714 \cdot n^2$	write B[0, i2] (i0=0, i1=0); write B[0, i2] (i0=0, i1=1) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.612	level	$(3/8) \cdot n^2 + (3/8) \cdot n + (-3/4)$	n	$0.0714 \cdot n^2$	write B[0, i2] (i0=0, i1=0, i2=0); write B[0, i2] (i0=0, i1=1, i2=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/4) \cdot n$	n	$0.0714 \cdot n^2$	write B[0, i2] (i0=0, i1=0, i2=0); write B[0, i2] (i0=0, i1=1, i2=0) (+1)
n^2	0.612	level	$(3/8) \cdot n^2 + 1$	n	$0.0714 \cdot n^2$	read B[i7, i8] (i0=0, i1=0, i7=0, i8=0); read B[i7, i8] (i0=0, i7=0, i8=0)
n^2	0.536	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	$(7/8) \cdot n + (-7/8)$	$0.0625 \cdot n^2$	read A[i1] (i0=0, i2=0)
n^2	0.536	level	$(3/8) \cdot n^2 + 1$	$(7/8) \cdot n$	$0.0625 \cdot n^2$	read A[i1] (i0=0, i2=0)
n^2	0.5	level	$(1/4) \cdot n^2 - 1$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i1=1, i4=0) (+1)
n^2	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (-13/4)$	n	$0.0714 \cdot n^2$	read C[i7, i8] (i0=0, i1=0, i7=0); read C[i7, i8] (i0=0, i7=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n - 2$	n	$0.0714 \cdot n^2$	read C[i7, i8] (i0=0, i1=0, i7=0); read C[i7, i8] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 1$	n	$0.0714 \cdot n^2$	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-11/4)$	n	$0.0714 \cdot n^2$	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^2	0.5	level	$(1/4) \cdot n^2$	n	$0.0714 \cdot n^2$	read B[i3, i4] (i0=0, i1=0); read B[i3, i4] (i0=0, i1=1) (+1)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-11/4)$	n	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 1$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0, i1=1) (+1)
n^2	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i1=0, i7=0, i8=0); read D[i7, i8 + 1] (i0=0, i7=0, i8=0)
n^2	0.5	level	$(1/4) \cdot n^2$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i1=0, i7=0, i8=0); read D[i7, i8 + 1] (i0=0, i7=0, i8=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0, i5=0, i6=0); read C[i5, i6] (i0=0, i5=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.5	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (-7/8)$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0, i6=0); read C[i5, i6] (i0=0, i6=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0, i6=0); read C[i5, i6] (i0=0, i6=0)
n^2	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (-5/4)$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i1=0, i7=0); read D[i7, i8 + 1] (i0=0, i7=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n - 3$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i1=0, i7=0); read D[i7, i8 + 1] (i0=0, i7=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0, i5=0); read C[i5, i6] (i0=0, i5=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-3/4)$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.5	level	$(1/4) \cdot n^2 + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0); read C[i5, i6] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2$	n	$0.0714 \cdot n^2$	read B[i3, i4] (i0=0, i1=0, i4=0); read B[i3, i4] (i0=0, i1=1, i4=0) (+1)
n^2	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0, i6=0); read C[i5, i6] (i0=0, i6=0)
n^2	0.5	level	$(1/4) \cdot n^2 + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i1=0, i6=0); read C[i5, i6] (i0=0, i6=0)
n^2	0.0765	level	$(3/8) \cdot n^2 + (11/4) \cdot n + (7/8)$	$(1/8) \cdot n + (-9/8)$	$0.00893 \cdot n^2$	read A[i1] (i0=0, i2=0)
n^2	0.0765	level	$(3/8) \cdot n^2 + (1/8) \cdot n$	$(1/8) \cdot n - 2$	$0.00893 \cdot n^2$	read A[i1] (i0=0, i2=0)
n^1	2	level	1	$2 \cdot n - 1$	$0.143 \cdot n^2$	read C[i5, i6 - 1] (i0=0, i1=0); read D[i7, i8] (i0=0, i1=0, i8=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (11/4) \cdot n + (7/8)$	1	$0.0714 \cdot n^3$	read A[i1] (i0=0, i1=0, i2=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n$	1	$0.0714 \cdot n^3$	read A[i1] (i0=0, i1=0, i2=0)
n^1	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n$	1	$0.0714 \cdot n^3$	read A[i1] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 1$	1	$0.0714 \cdot n^3$	read C[i3, i4] (i0=0, i1=1)

Multi-field 2-D stencil (B, C, D fields): cross-sweep re-reads at $(3/8)n^2 + O(n)$ lines — three field planes — with 0.077-coefficient families per field pair and a $(1/4)n^2 \rightarrow (3/8)n^2$ ramp on read $D[i7, i8+1]$. Headroom +1.0; the boundary is 3× a single plane, so fdtd crosses the cache one third the size of a same-n single-field stencil.

fdtd — single-shot [exact]

Accesses $A(n) = 14 \cdot n^3 - 18 \cdot n^2 + 6 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma\text{mass}/\text{warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.616 \cdot n^4 + 0.253 \cdot n^{3.5} + 32.6 \cdot n^3 + 3.35 \cdot n^{2.5} + 36.5 \cdot n^2$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0765	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (1/8)$	$(1/8) \cdot n^3 + (-19/8) \cdot n^2 + (35/8) \cdot n + (-17/8)$	0.00893	read D[i5, i6] (i0=0, i5=0); read D[i5, i6] (i0=0)
n^4	0.0765	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + (33/8) \cdot n - 2$	0.00893	read D[i5, i6] (i0=0, i5=0); read D[i5, i6] (i0=0)
n^4	0.0765	level	$(3/8) \cdot n^2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read B[i7 + 1, i8] (i0=0)
n^4	0.0658	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n + 1 \rightarrow (3/8) \cdot n^2 - 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read D[i7, i8 + 1] (i0=0)
n^4	0.0658	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n + 1 \rightarrow (3/8) \cdot n^2 - 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read C[i7, i8] (i0=0)
n^4	0.065	ramp	$(1/4) \cdot n^2 + 2 \rightarrow (3/8) \cdot n^2 + (-1/2) \cdot n - 1$	$(1/8) \cdot n^3 + (-21/8) \cdot n^2 + (21/2) \cdot n - 8$	0.00893	read C[i3, i4] (i0=0)
n^4	0.065	ramp	$(1/4) \cdot n^2 + 2 \rightarrow (3/8) \cdot n^2 + (-1/2) \cdot n - 1$	$(1/8) \cdot n^3 + (-21/8) \cdot n^2 + (21/2) \cdot n - 8$	0.00893	read B[i3, i4] (i0=0)
n^4	0.0625	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (1/8)$	$(1/8) \cdot n^3 + (-3/2) \cdot n^2 + (27/8) \cdot n$	0.00893	read C[i5, i6] (i0=0)
n^4	0.0625	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	$(1/8) \cdot n^3 + (-19/8) \cdot n^2 + 6 \cdot n$	0.00893	read C[i5, i6] (i0=0)
$n^{3.5}$	0.0884	level	$(1/2) \cdot n + (5/2)$	$(1/8) \cdot n^3 + (-19/8) \cdot n^2 + (17/4) \cdot n$	0.00893	read B[i7, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.0884	level	$(1/2) \cdot n + 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read B[i7, i8] (i0=0)
$n^{3.5}$	0.0765	level	$(3/8) \cdot n + 2$	$(1/8) \cdot n^3 + (-9/4) \cdot n^2 + 4 \cdot n$	0.00893	read C[i3 - 1, i4] (i0=0)
n^3	2.62	level	1	$(21/8) \cdot n^3 + (-15/8) \cdot n^2$	0.188	read D[i5, i6] (i0=0); read C[i5, i6 - 1] (i0=0) (+1)
n^3	1.95	level	3	$(9/8) \cdot n^3 - 2 \cdot n^2$	0.0804	write B[0, i2] (i0=0); read B[i3, i4] (i0=0) (+4)
n^3	1.75	level	1	$(7/4) \cdot n^3 + (-11/4) \cdot n^2 + n$	0.125	read B[i3, i4] (i0=0); read C[i7, i8] (i0=0)
n^3	1.75	level	4	$(7/8) \cdot n^3 + (-7/8) \cdot n^2$	0.0625	write C[i7, i8] (i0=0)
n^3	1.52	level	3	$(7/8) \cdot n^3 + (-7/8) \cdot n^2$	0.0625	read C[i3 - 1, i4] (i0=0)
n^3	1.52	level	3	$(7/8) \cdot n^3 + (-7/8) \cdot n^2$	0.0625	read C[i3, i4] (i0=0)
n^3	1.5	level	4	$(3/4) \cdot n^3 + (-3/4) \cdot n^2$	0.0536	read B[i7, i8] (i0=0)
n^3	1.5	level	4	$(3/4) \cdot n^3 + (-3/4) \cdot n^2$	0.0536	read B[i7 + 1, i8] (i0=0)
n^3	1.5	level	4	$(3/4) \cdot n^3 + (-3/4) \cdot n^2$	0.0536	read D[i7, i8 + 1] (i0=0)
n^3	1.24	level	2	$(7/8) \cdot n^3 + n^2 - n$	0.0625	write B[0, i2] (i0=0, i2=0); read A[i1] (i0=0) (+1)
n^3	1.1	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n \rightarrow (3/8) \cdot n^2 + (-1/2) \cdot n$	$2 \cdot n^2 - 10 \cdot n + 8$	0.143/n	read B[i3, i4] (i0=0, i4=0); read C[i3, i4] (i0=0, i4=0)
n^3	1.06	level	2	$(3/4) \cdot n^3$	0.0536	read C[i5, i6] (i0=0, i5=0); read C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (1/8)$	$n^2 - 2 \cdot n + 1$	$0.0714/n$	read D[i5, i6] (i0=0, i5=0, i6=0); read D[i5, i6] (i0=0, i6=0)
n^3	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	$n^2 - 2 \cdot n + 1$	$0.0714/n$	read D[i5, i6] (i0=0, i5=0, i6=0); read D[i5, i6] (i0=0, i6=0)
n^3	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (-7/8)$	$n^2 - 2 \cdot n + 1$	$0.0714/n$	read D[i5, i6] (i0=0, i5=0); read D[i5, i6] (i0=0)
n^3	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	$n^2 - 2 \cdot n + 1$	$0.0714/n$	read D[i5, i6] (i0=0, i5=0); read D[i5, i6] (i0=0)
n^3	0.612	level	$(3/8) \cdot n^2$	$n^2 - n$	$0.0714/n$	read B[i7, i8] (i0=0, i7=0); read B[i7 + 1, i8] (i0=0)
n^3	0.612	level	$(3/8) \cdot n^2$	$n^2 - 2 \cdot n$	$0.0714/n$	read B[i7 + 1, i8] (i0=0, i8=0)
n^3	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (1/8)$	$n^2 - 2 \cdot n + 1$	$0.0714/n$	read D[i5, i6] (i0=0, i5=0); read D[i5, i6] (i0=0)
n^3	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	$n^2 - n$	$0.0714/n$	write B[0, i2] (i0=0); read B[i3, i4] (i0=0, i3=0) (+1)
n^3	0.559	level	5	$(1/4) \cdot n^3 + (-9/4) \cdot n^2 + 2 \cdot n$	0.0179	read B[i7 + 1, i8] (i0=0); read B[i7, i8] (i0=0)
n^3	0.555	ramp	$(1/4) \cdot n^2 + (3/8) \cdot n - 1$ → $(3/8) \cdot n^2 - 1$	$n^2 - 2 \cdot n$	$0.0714/n$	read C[i7, i8] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.555	ramp	$(1/4) \cdot n^2 + (3/8) \cdot n - 2$ $\rightarrow$ $(3/8) \cdot n^2 - 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read D[i7, i8 + 1] (i0=0)
n^3	0.555	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n \rightarrow$ $(3/8) \cdot n^2 + (-1/8) \cdot n$	$n^2 - 2 \cdot n$	$0.0714/n$	read D[i7, i8 + 1] (i0=0, i8=0)
n^3	0.555	ramp	$(1/4) \cdot n^2 + (1/4) \cdot n - 1$ $\rightarrow$ $(3/8) \cdot n^2 + (-1/8) \cdot n - 1$	$n^2 - 2 \cdot n$	$0.0714/n$	read C[i7, i8] (i0=0, i8=0)
n^3	0.548	ramp	$(1/4) \cdot n^2 + 1 \rightarrow$ $(3/8) \cdot n^2 + (-5/8) \cdot n + 1$	$n^2 - 5 \cdot n + 4$	$0.0714/n$	read C[i3, i4] (i0=0)
n^3	0.548	ramp	$(1/4) \cdot n^2 + 1 \rightarrow$ $(3/8) \cdot n^2 + (-5/8) \cdot n + 1$	$n^2 - 5 \cdot n + 4$	$0.0714/n$	read B[i3, i4] (i0=0)
n^3	0.5	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (1/8)$	$n^2 - 3 \cdot n$	$0.0714/n$	read C[i5, i6] (i0=0, i6=0)
n^3	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	$n^2 - 3 \cdot n$	$0.0714/n$	read C[i5, i6] (i0=0, i6=0)
n^3	0.5	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (1/8)$	$n^2 - 3 \cdot n$	$0.0714/n$	read C[i5, i6] (i0=0)
n^3	0.306	level	6	$(1/8) \cdot n^3 + (-9/8) \cdot n^2 + n$	0.00893	read D[i7, i8 + 1] (i0=0)
n^3	0.28	level	5	$(1/8) \cdot n^3 + (-9/8) \cdot n^2 + n$	0.00893	read B[i3, i4] (i0=0, i3=0, i4=0); read C[i3, i4] (i0=0, i3=0, i4=0) (+3)
n^3	0.28	level	5	$(1/8) \cdot n^3 + (-9/8) \cdot n^2 + n$	0.00893	read D[i7, i8] (i0=0)
n^3	0.25	level	4	$(1/8) \cdot n^3 - n^2$	0.00893	read C[i5, i6 - 1] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 - n^2$	0.00893	read C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0765	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-1/2)$	$(1/8) \cdot n^2 + (-9/4) \cdot n + (17/8)$	$0.00893/n$	read C[i3, i4] (i0=0, i3=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read C[i3, i4] (i0=0, i3=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-1/2)$	$(1/8) \cdot n^2 + (-9/4) \cdot n + (17/8)$	$0.00893/n$	read B[i3, i4] (i0=0, i3=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read B[i3, i4] (i0=0, i3=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read B[i7, i8] (i0=0, i7=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n$	$0.00893/n$	read B[i7 + 1, i8] (i0=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	$0.00893/n$	read D[i5, i6] (i0=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read D[i5, i6] (i0=0)
n^3	0.0726	ramp	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1 \rightarrow (3/8) \cdot n^2 - 2$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read B[i7 + 1, i8] (i0=0, i7=0)
n^3	0.0724	ramp	$(3/8) \cdot n^2 + (-1/8) \cdot n + 2 \rightarrow (3/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read C[i3, i4] (i0=0)
n^3	0.0723	ramp	$(3/8) \cdot n^2 + (-3/8) \cdot n + 2 \rightarrow (3/8) \cdot n^2 + (-1/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read C[i3 - 1, i4] (i0=0, i3=0)
n^3	0.0723	ramp	$(3/8) \cdot n^2 + (-1/2) \cdot n + 4 \rightarrow (3/8) \cdot n^2 + (-1/4) \cdot n - 2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	write B[0, i2] (i0=0)
n^3	0.0625	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read C[i5, i6] (i0=0, i5=0)
n^3	0.0625	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n$	$0.00893/n$	read C[i5, i6] (i0=0)
n^3	0.0625	level	$(1/4) \cdot n^2 + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0625	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + (17/8)$	$0.00893/n$	read B[i3, i4] (i0=0)
n^3	0.0625	level	$(1/4) \cdot n^2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read B[i3, i4] (i0=0)
n^3	0.0593	ramp	$(1/4) \cdot n^2 + 2 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 4$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read D[i7, i8 + 1] (i0=0, i7=0)
n^3	0.0593	ramp	$(1/4) \cdot n^2 + 2 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 4$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read C[i7, i8] (i0=0, i7=0)
n^3	0.0592	ramp	$(1/4) \cdot n^2 + (-1/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + (-1/8) \cdot n - 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.00893/n$	read C[i5, i6] (i0=0)
n^3	0.0591	ramp	$(1/4) \cdot n^2 + (-1/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 - 3$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read C[i3, i4] (i0=0)
n^3	0.0591	ramp	$(1/4) \cdot n^2 + (-1/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 - 3$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.00893/n$	read B[i3, i4] (i0=0)
$n^{2.5}$	0.707	level	$(1/2) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read B[i7, i8] (i0=0)
$n^{2.5}$	0.707	level	$(1/2) \cdot n + (5/2)$	$n^2 - 2 \cdot n$	$0.0714/n$	read B[i7, i8] (i0=0, i8=0)
$n^{2.5}$	0.707	level	$(1/2) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read B[i7, i8] (i0=0, i8=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read C[i3 - 1, i4] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n + 2$	$n^2 - 2 \cdot n$	$0.0714/n$	read C[i3 - 1, i4] (i0=0, i4=0)
n^2	1.24	level	2	$(7/8) \cdot n^2$	$0.0625/n$	write B[0, i2] (i0=0)
n^2	1.22	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-1/2)$	$2 \cdot n - 2$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i3=0, i4=0); read C[i3, i4] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.22	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$2 \cdot n - 2$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i3=0, i4=0); read C[i3, i4] (i0=0, i3=0, i4=0)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 4$	$2 \cdot n - 2$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i4=0); read C[i3, i4] (i0=0, i4=0)
n^2	1	level	$(1/4) \cdot n^2 - 1$	$2 \cdot n - 1$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0, i4=0); read C[i7, i8] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$2 \cdot n - 1$	$0.143 \cdot n^{-2}$	read B[i3, i4] (i0=0); read C[i7, i8] (i0=0, i7=0, i8=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-3/2)$	$n - 1$	$0.0714 \cdot n^{-2}$	read C[i3, i4] (i0=0, i3=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$n - 1$	$0.0714 \cdot n^{-2}$	read C[i3, i4] (i0=0, i3=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (-3/2)$	$n - 1$	$0.0714 \cdot n^{-2}$	read B[i3, i4] (i0=0, i3=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/2) \cdot n$	$n - 1$	$0.0714 \cdot n^{-2}$	read B[i3, i4] (i0=0, i3=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	$n - 1$	$0.0714 \cdot n^{-2}$	read D[i5, i6] (i0=0, i6=0)
n^2	0.612	level	$(3/8) \cdot n^2 + 1$	$n - 1$	$0.0714 \cdot n^{-2}$	read D[i5, i6] (i0=0, i6=0)
n^2	0.612	level	$(3/8) \cdot n^2 - 1$	n	$0.0714 \cdot n^{-2}$	read B[i7 + 1, i8] (i0=0, i7=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/4) \cdot n + (3/8)$	$n - 1$	$0.0714 \cdot n^{-2}$	read C[i3 - 1, i4] (i0=0, i3=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-3/8) \cdot n + 1$	$n - 1$	$0.0714 \cdot n^{-2}$	read C[i3 - 1, i4] (i0=0, i3=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 3$	$n - 1$	$0.0714 \cdot n^{-2}$	read C[i3 - 1, i4] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (-7/8)$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 2$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (9/8)$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 2$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i3=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (5/2) \cdot n + (-7/8)$	n	$0.0714 \cdot n^2$	read B[i7 + 1, i8] (i0=0, i7=0, i8=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/8) \cdot n$	n	$0.0714 \cdot n^2$	read B[i7 + 1, i8] (i0=0, i7=0, i8=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n - 1$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i4=0)
n^2	0.612	level	$(3/8) \cdot n^2$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i4=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (3/8) \cdot n + (1/4)$	$n - 1$	$0.0714 \cdot n^2$	read C[i3 - 1, i4] (i0=0, i3=0, i4=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/4) \cdot n$	$n - 1$	$0.0714 \cdot n^2$	read C[i3 - 1, i4] (i0=0, i3=0, i4=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (21/8) \cdot n + 1$	$n - 1$	$0.0714 \cdot n^2$	read D[i5, i6] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + 1$	$n - 1$	$0.0714 \cdot n^2$	read D[i5, i6] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (13/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read B[i7 + 1, i8] (i0=0, i8=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (1/8) \cdot n + (1/2)$	$n - 1$	$0.0714 \cdot n^2$	write B[0, i2] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/2) \cdot n + 2$	$n - 1$	$0.0714 \cdot n^2$	write B[0, i2] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (3/8) \cdot n + (-3/4)$	$n - 1$	$0.0714 \cdot n^2$	write B[0, i2] (i0=0, i2=0)
n^2	0.612	level	$(3/8) \cdot n^2 + (-1/4) \cdot n$	$n - 1$	$0.0714 \cdot n^2$	write B[0, i2] (i0=0, i2=0)
n^2	0.612	level	$(3/8) \cdot n^2 + 1$	n	$0.0714 \cdot n^2$	read B[i7, i8] (i0=0, i7=0, i8=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.536	level	$(3/8) \cdot n^2 + 1$	$(7/8) \cdot n$	$0.0625 \cdot n^2$	read A[i1] (i0=0, i2=0)
n^2	0.5	level	$(1/4) \cdot n^2 - 1$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0, i4=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-11/4)$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 1$	$n - 1$	$0.0714 \cdot n^2$	read C[i3, i4] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 1$	$n - 1$	$0.0714 \cdot n^2$	read B[i3, i4] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-11/4)$	$n - 1$	$0.0714 \cdot n^2$	read B[i3, i4] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i7=0, i8=0)
n^2	0.5	level	$(1/4) \cdot n^2$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i7=0, i8=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i5=0, i6=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (13/8) \cdot n + (-7/8)$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i6=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i6=0)
n^2	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (-5/4)$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i7=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n - 3$	n	$0.0714 \cdot n^2$	read D[i7, i8 + 1] (i0=0, i7=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/8) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0, i5=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-3/4)$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (-1/4) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + 1$	n	$0.0714 \cdot n^2$	read C[i5, i6] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.5	level	$(1/4) \cdot n^2 + 2 \cdot n + (-13/4)$	n	$0.0714 \cdot n^{-2}$	read C[i7, i8] (i0=0, i7=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (1/4) \cdot n - 2$	n	$0.0714 \cdot n^{-2}$	read C[i7, i8] (i0=0, i7=0)
n^2	0.5	level	$(1/4) \cdot n^2$	$n - 1$	$0.0714 \cdot n^{-2}$	read B[i3, i4] (i0=0, i4=0)
n^2	0.5	level	$(1/4) \cdot n^2$	$n - 1$	$0.0714 \cdot n^{-2}$	read B[i3, i4] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	n	$0.0714 \cdot n^{-2}$	read C[i5, i6] (i0=0, i6=0)
n^2	0.5	level	$(1/4) \cdot n^2 + 1$	n	$0.0714 \cdot n^{-2}$	read C[i5, i6] (i0=0, i6=0)

Multi-field 2-D stencil (B, C, D fields): cross-sweep re-reads at $(3/8)n^2 + O(n)$ lines — three field planes — with 0.077-coefficient families per field pair and a $(1/4)n^2 \rightarrow (3/8)n^2$ ramp on read D[i7, i8+1]. Headroom +1.0; the boundary is $3 \times$ a single plane, so fdtd crosses the cache one third the size of a same-n single-field stencil.

floyd_warshall — infinite-repeat [exact]

Accesses $A(n) = 4 \cdot n^3$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^4 + 0.0625 \cdot n^{3.5} + 7.39 \cdot n^3 + 1.91 \cdot n^{2.5} + 82 \cdot n^2 + 11 \cdot n^{1.5} + 90.8 \cdot n^1 + 48.2 \cdot n^{0.5} + 89.8 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0221	level	$(1/8) \cdot n^2$	$(1/16) \cdot n^3 + (-11/4) \cdot n^2 + (129/4) \cdot n - 54$	0.0156	read A[i1, i2] (i1=0); read A[i1, i2]
n^4	0.0221	level	$(1/8) \cdot n^2$	$(1/16) \cdot n^3 + (-13/8) \cdot n^2 + (23/2) \cdot n - 26$	0.0156	read A[i1, i2] (i0=0, i1=0, i2=8); read A[i1, i2] (i0=0, i1=0) (+4)
$n^{3.5}$	0.0417	level	$(1/4) \cdot n + 2$	$(1/12) \cdot n^3 + (-91/32) \cdot n^2 + (847/24) \cdot n - 192$	0.0208	read A[i0, i2] (i0=0); read A[i0, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.0182	level	$(1/4) \cdot n + 2$	$(7/192) \cdot n^3 + (-57/64) \cdot n^2 + (61/24) \cdot n + 38$	0.00911	read A[i0, i2]
$n^{3.5}$	0.0026	level	$(1/4) \cdot n + 2$	$(1/192) \cdot n^3 + (-17/64) \cdot n^2 + (61/24) \cdot n + 20$	0.0013	read A[i0, i2]
n^3	1.77	level	3	$(49/48) \cdot n^3 + (-1983/128) \cdot n^2 + (2861/48) \cdot n - 8$	0.255	read A[i0, i2]; write A[i1, i2]
n^3	1.01	level	3	$(7/12) \cdot n^3 + (-287/32) \cdot n^2 + (259/6) \cdot n - 70$	0.146	read A[i0, i2] (i0=0); write A[i1, i2] (i0=0) (+1)
n^3	0.875	level	1	$(7/8) \cdot n^3 + (7/16) \cdot n^2 - 7$	0.219	write A[i1, i2] (i0=0, i1=0); read A[i1, i2] (+1)
n^3	0.785	level	3	$(29/64) \cdot n^3 + (-509/64) \cdot n^2 + (83/2) \cdot n - 26$	0.113	read A[i0, i2] (i0=0, i1=1, i2=0); read A[i0, i2] (i0=0, i2=0) (+9)
n^3	0.758	level	3	$(7/16) \cdot n^3 + (-115/16) \cdot n^2 + 31 \cdot n - 6$	0.109	read A[i1, i0] (i0=0, i1=0); write A[i1, i2] (i0=0) (+3)
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 9 \cdot n + 17$	0.25/n	read A[i1, i2] (i0=0, i1=0, i2=0); read A[i1, i2] (i0=0, i2=0) (+4)
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 11 \cdot n + 27$	0.25/n	read A[i1, i2] (i0=0, i1=0, i2=8); read A[i1, i2] (i0=0, i1=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.309	level	$(1/8) \cdot n^2$	$(7/8) \cdot n^2 + (-67/4) \cdot n + 30$	$0.219/n$	read A[i1, i2] (i1=0); read A[i1, i2]
n^3	0.309	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 3$	$(7/8) \cdot n^2 + (-63/4) \cdot n + 28$	$0.219/n$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^3	0.253	level	3	$(7/48) \cdot n^3 + (-469/128) \cdot n^2 + (1085/48) \cdot n - 21$	0.0365	read A[i0, i2]; write A[i1, i2]
n^3	0.125	level	4	$(1/16) \cdot n^3 + (-25/16) \cdot n^2 + (19/2) \cdot n - 8$	0.0156	read A[i0, i2] (i0=0, i1=1, i2=8); read A[i1, i0] (i0=0) (+2)
n^3	0.125	level	4	$(1/16) \cdot n^3 + (-27/16) \cdot n^2 + (93/8) \cdot n - 10$	0.0156	read A[i1, i0]
n^3	0.0947	level	3	$(7/128) \cdot n^3 + (-175/128) \cdot n^2 + (133/16) \cdot n - 7$	0.0137	write A[i1, i2]
n^3	0.0947	level	3	$(7/128) \cdot n^3 + (-159/128) \cdot n^2 + (115/16) \cdot n - 6$	0.0137	write A[i1, i2]
n^3	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0312/n$	read A[i1, i2] (i1=0); read A[i1, i2]
n^3	0.0405	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 4 \rightarrow (1/8) \cdot n^2$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0312/n$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^3	0.0394	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 4 \rightarrow (1/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 + (-17/4) \cdot n + 8$	$0.0312/n$	read A[i1, i2] (i1=0); read A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0164	ramp	$(19/8) \cdot n - 1$ → $(1/8) \cdot n^2 - 1$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0156/n	read A[i1, i2]
n^3	0.0157	ramp	$(13/4) \cdot n - 1$ → $(1/8) \cdot n^2 - 1$	$(1/16) \cdot n^2 + (-21/8) \cdot n + 27$	0.0156/n	read A[i0, i2] (i1=0)
n^3	0.0107	ramp	$(3/8) \cdot n + 3$ → $(1/8) \cdot n^2 - 2 \cdot n + 3$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0156/n	read A[i1, i2]
n^3	0.0107	ramp	$(3/8) \cdot n + 3$ → $(1/8) \cdot n^2 - 2 \cdot n + 3$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0156/n	read A[i0, i2] (i1=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n + 2$	$n^2 - 10 \cdot n + 23$	0.25/n	read A[i0, i2] (i0=0, i2=8); read A[i0, i2] (i2=0)
$n^{2.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n^2 + (-133/8) \cdot n + 42$	0.219/n	read A[i0, i2]
$n^{2.5}$	0.25	level	$(1/4) \cdot n + 1$	$(1/2) \cdot n^2 + (-31/2) \cdot n + 134$	0.125/n	read A[i0, i2] (i0=0); read A[i0, i2]
$n^{2.5}$	0.219	level	$(1/4) \cdot n + 1$	$(7/16) \cdot n^2 + (-7/4) \cdot n - 34$	0.109/n	read A[i0, i2]
$n^{2.5}$	0.156	level	$(1/4) \cdot n + 2$	$(5/16) \cdot n^2 + (-85/8) \cdot n + 90$	0.0781/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (25/4) \cdot n - 24$	0.0156/n	read A[i0, i2] (i1=0, i2=0); read A[i0, i2] (i2=0) (+1)
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	0.0156/n	read A[i0, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (1/8) \cdot n - 18$	$0.0156/n$	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-3/4) \cdot n - 18$	$0.0156/n$	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (-3/4) \cdot n - 18$	$0.0156/n$	read A[i0, i2]
$n^{2.5}$	0.0261	ramp	$(1/8) \cdot n + 2 \rightarrow (1/4) \cdot n - 1$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0156/n$	read A[i1, i2]
$n^{2.5}$	0.0238	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0156/n$	read A[i1, i2] (i2=8); read A[i1, i2]
$n^{2.5}$	0.0238	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0156/n$	read A[i0, i2]
$n^{2.5}$	0.0219	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n - 1$	$(7/128) \cdot n^2 + (-35/16) \cdot n + 21$	$0.0137/n$	read A[i0, i2]
$n^{2.5}$	0.00296	ramp	$(1/8) \cdot n + 4 \rightarrow (1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00195/n$	read A[i0, i2]
n^2	6.06	level	3	$(7/2) \cdot n^2 + (-63/2) \cdot n + 28$	$0.875/n$	read A[i0, i2]
n^2	6.06	level	3	$(7/2) \cdot n^2 + (-63/2) \cdot n + 28$	$0.875/n$	read A[i1, i0]
n^2	4.55	level	3	$(21/8) \cdot n^2 + (-189/8) \cdot n + 21$	$0.656/n$	read A[i1, i0]
n^2	3.71	level	2	$(21/8) \cdot n^2 + (-21/8) \cdot n$	$0.656/n$	read A[i0, i2]
n^2	3.71	level	2	$(21/8) \cdot n^2 + (-21/8) \cdot n$	$0.656/n$	write A[i1, i2]
n^2	3.5	level	1	$(7/2) \cdot n^2 + (-21/2) \cdot n + 7$	$0.875/n$	read A[i1, i0]
n^2	3.25	level	3	$(15/8) \cdot n^2 + (-135/8) \cdot n + 15$	$0.469/n$	write A[i1, i2]
n^2	3.25	level	3	$(15/8) \cdot n^2 + (-135/8) \cdot n + 15$	$0.469/n$	write A[i1, i2]
n^2	3.09	level	2	$(35/16) \cdot n^2 - 14 \cdot n + 49$	$0.547/n$	read A[i0, i2]; write A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	2.83	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$8 \cdot n - 16$	$2 \cdot n^{-2}$	read A[i1, i2] (i0=1, i2=0); read A[i1, i2] (i1=0, i2=0) (+1)
n^2	2.81	level	3	$(13/8) \cdot n^2 + (-117/8) \cdot n + 13$	$0.406/n$	write A[i1, i2]
n^2	2.65	level	2	$(15/8) \cdot n^2 + (-15/8) \cdot n$	$0.469/n$	write A[i1, i2]
n^2	2.62	level	1	$(21/8) \cdot n^2 + (-69/8) \cdot n + 6$	$0.656/n$	read A[i1, i0]
n^2	2.39	level	2	$(27/16) \cdot n^2 + (-21/2) \cdot n + 11$	$0.422/n$	read A[i0, i2]; write A[i1, i2]
n^2	2.12	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$6 \cdot n - 12$	$1.5 \cdot n^{-2}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^2	1.88	level	1	$(15/8) \cdot n^2 + (-23/4) \cdot n + 4$	$0.469/n$	read A[i1, i0]
n^2	1.77	level	2	$(5/4) \cdot n^2 + (-45/8) \cdot n$	$0.312/n$	read A[i0, i2]
n^2	1.62	level	3	$(15/16) \cdot n^2 + (-95/8) \cdot n + 35$	$0.234/n$	read A[i0, i2]
n^2	1.62	level	3	$(15/16) \cdot n^2 - 5 \cdot n - 20$	$0.234/n$	read A[i0, i2]
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-31/4) \cdot n + 5$	$0.219/n$	read A[i1, i0]
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-63/8) \cdot n + 7$	$0.219/n$	write A[i1, i2]
n^2	1.33	level	2	$(15/16) \cdot n^2 + (5/4) \cdot n$	$0.234/n$	read A[i0, i2]
n^2	1.3	level	3	$(3/4) \cdot n^2 + (-27/4) \cdot n + 6$	$0.188/n$	write A[i1, i2]
n^2	1.24	level	2	$(7/8) \cdot n^2 - 14 \cdot n + 56$	$0.219/n$	read A[i1, i0] (i0=0, i1=0); read A[i0, i2] (i0=0, i1=0) (+1)
n^2	1.08	level	3	$(5/8) \cdot n^2 + (-45/8) \cdot n + 5$	$0.156/n$	write A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.06	level	2	$(3/4) \cdot n^2 - 6 \cdot n$	$0.188/n$	read A[i1, i0]
n^2	1.06	level	2	$(3/4) \cdot n^2 + (-3/4) \cdot n$	$0.188/n$	write A[i1, i2]
n^2	0.758	level	3	$(7/16) \cdot n^2 - 7 \cdot n + 28$	$0.109/n$	read A[i1, i0]
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-7/8) \cdot n - 21$	$0.109/n$	read A[i1, i0]
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-45/8) \cdot n + 21$	$0.0938/n$	read A[i0, i2]
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-9/8) \cdot n - 15$	$0.0938/n$	read A[i0, i2]
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.109/n$	read A[i0, i2]
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.109/n$	read A[i0, i2]
n^2	0.53	level	2	$(3/8) \cdot n^2 + (-3/8) \cdot n$	$0.0938/n$	read A[i0, i2]
n^2	0.383	level	1	$(49/128) \cdot n^2 + (-79/16) \cdot n + 15$	$0.0957/n$	write A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 3$	$0.25 \cdot n^{-2}$	read A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 2$	$0.25 \cdot n^{-2}$	read A[i1, i2] (i1=0); read A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 9$	$0.25 \cdot n^{-2}$	read A[i1, i2] (i1=0); read A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 9$	$0.25 \cdot n^{-2}$	read A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	$n - 3$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$n - 1$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (17/8)$	$n - 2$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	2	$(1/4) \cdot n^2 + (-33/8) \cdot n + 17$	$0.0625/n$	read A[i0, i2] (i2=0); read A[i0, i2] (+1)
n^2	0.265	level	2	$(3/16) \cdot n^2 + (-7/4) \cdot n + 2$	$0.0469/n$	read A[i0, i2]; write A[i1, i2]
n^2	0.25	level	4	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0312/n$	read A[i1, i0]
n^2	0.233	ramp	$(9/4) \cdot n \rightarrow (1/8) \cdot n^2$	$n - 17$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.227	ramp	$(3/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 - n + 2$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i1=0, i2=0)
n^2	0.227	ramp	$(3/8) \cdot n + 1 \rightarrow (1/8) \cdot n^2 - n + 1$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i1, i2]
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0312/n$	read A[i1, i2] (i1=0); read A[i1, i2] (i2=8) (+1)
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0312/n$	write A[i1, i2] (i2=0)
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0312/n$	write A[i1, i2]
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	$0.0312/n$	write A[i1, i2]
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	$0.0312/n$	write A[i1, i2]
n^2	0.207	ramp	$(1/2) \cdot n \rightarrow (1/8) \cdot n^2$	$(7/8) \cdot n - 2$	$0.219 \cdot n^{-2}$	read A[i1, i2] (i2=0)
n^2	0.199	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-9/8) \cdot n + 5$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.199	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-9/8) \cdot n + 5$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^2	0.177	level	2	$(1/8) \cdot n^2 + (-1/8) \cdot n$	$0.0312/n$	write A[i1, i2]
n^2	0.177	level	2	$(1/8) \cdot n^2 - n$	$0.0312/n$	read A[i1, i0]
n^2	0.177	level	2	$(1/8) \cdot n^2 + (-1/8) \cdot n$	$0.0312/n$	write A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.146	ramp	$(5/2) \cdot n - 1$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(-1/8) \cdot n + 2$	$(5/8) \cdot n - 10$	$0.156 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (-1/2) \cdot n$	$0.0156/n$	read A[i0, i2]
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (-3/2) \cdot n +$ 8	$0.0156/n$	read A[i1, i0]
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (-1/2) \cdot n -$ 8	$0.0156/n$	read A[i1, i0]
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (-5/8) \cdot n +$ 1	$0.0156/n$	read A[i0, i2]
n^2	0.0625	level	1	$(1/16) \cdot n^2$ $+ (1/2) \cdot n - 1$	$0.0156/n$	write A[i1, i2] (i0=0, i1=0); write A[i1, i2]
n^2	0.0547	level	1	$(7/128) \cdot n^2$ $+ (5/16) \cdot n -$ 6	$0.0137/n$	write A[i1, i2]
n^2	0.0547	level	1	$(7/128) \cdot n^2$ $+ (-21/16) \cdot n$ $+ 7$	$0.0137/n$	write A[i1, i2]
n^2	0.029	ramp	$(19/8) \cdot n - 1$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(-3/4) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0289	ramp	$(9/4) \cdot n \rightarrow$ $(1/8) \cdot n^2 +$ $(-3/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i1, i2] (i2=0)
n^2	0.0289	ramp	$(9/4) \cdot n - 1$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i1, i2]
n^2	0.0289	ramp	$(9/4) \cdot n - 1$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0285	ramp	$(25/8) \cdot n - 1$ $\rightarrow$ $(1/8) \cdot n^2 -$ $n + 3$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0279	ramp	$(5/4) \cdot n \rightarrow$ $(1/8) \cdot n^2 -$ $2 \cdot n + 6$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0277	ramp	$(9/4) \cdot n - 2$ $\rightarrow$ $(1/8) \cdot n^2 -$ $2 \cdot n + 6$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i1, i0] (i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.00781	level	1	$(1/128) \cdot n^2 + (127/16) \cdot n + 30$	$0.00195/n$	read A[i1, i2] (i0=0, i1=0, i2=0); read A[i1, i0] (i0=0, i1=0, i2=0) (+9)
$n^{1.5}$	3.78	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(63/8) \cdot n - 34$	$1.97 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	2.5	level	$(1/4) \cdot n + 1$	$5 \cdot n - 65$	$1.25 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.53	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(9/8) \cdot n - 11$	$0.281 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 16$	$0.25 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 2$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i0=0, i2=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n$	n	$0.25 \cdot n^{-2}$	read A[i1, i2] (i0=1, i1=0, i2=0); read A[i1, i2] (i2=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.354	level	$(1/8) \cdot n + 1$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i1, i2]
$n^{1.5}$	0.354	level	$(1/8) \cdot n + 2$	$n - 8$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i0=0, i1=1, i2=8); read A[i0, i2] (i2=0)
$n^{1.5}$	0.288	ramp	$5 \rightarrow (1/4) \cdot n - 1$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i1, i0] (i2=0)
$n^{1.5}$	0.257	ramp	$(1/8) \cdot n + 2 \rightarrow (1/4) \cdot n - 1$	$(5/8) \cdot n - 10$	$0.156 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.125	level	$(1/4) \cdot n + 1$	$(1/4) \cdot n - 4$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=1, i1=0); read A[i0, i2] (i1=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.0514	ramp	$(1/8) \cdot n + 2 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.0514	ramp	$(1/8) \cdot n + 2 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.0502	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.0502	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i0=0, i1=1)
$n^{1.5}$	0.0502	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i1, i2]
$n^{1.5}$	0.0412	ramp	$5 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^1	22.6	level	2	$16 \cdot n - 13$	$4 \cdot n^{-2}$	read A[i1, i0] (i0=0, i1=0, i2=8); read A[i0, i2] (i0=0, i1=1, i2=0) (+10)
n^1	7	level	1	$7 \cdot n - 14$	$1.75 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^1	4.95	level	2	$(7/2) \cdot n - 28$	$0.875 \cdot n^{-2}$	read A[i1, i0]
n^1	4.5	level	1	$(9/2) \cdot n - 15$	$1.12 \cdot n^{-2}$	write A[i1, i2]
n^1	3.71	level	2	$(21/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i1, i0]
n^1	3.71	level	2	$(21/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i0, i2]
n^1	3.71	level	2	$(21/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i0, i2]
n^1	3.5	level	1	$(7/2) \cdot n - 7$	$0.875 \cdot n^{-2}$	read A[i1, i0]
n^1	2.83	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	8	$2 \cdot n^{-3}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^1	2.62	level	1	$(21/8) \cdot n - 6$	$0.656 \cdot n^{-2}$	read A[i1, i0]
n^1	2.62	level	1	$(21/8) \cdot n$	$0.656 \cdot n^{-2}$	read A[i0, i2]
n^1	2.62	level	1	$(21/8) \cdot n$	$0.656 \cdot n^{-2}$	read A[i0, i2]
n^1	2.12	level	$(1/8) \cdot n^2$	6	$1.5 \cdot n^{-3}$	read A[i1, i2]
n^1	1.88	level	1	$(15/8) \cdot n - 2$	$0.469 \cdot n^{-2}$	read A[i1, i0]
n^1	1.88	level	1	$(15/8) \cdot n - 2$	$0.469 \cdot n^{-2}$	read A[i1, i0]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.75	level	1	$(7/4) \cdot n$	$0.438 \cdot n^{-2}$	read A[i0, i2]; write A[i1, i2]
n^1	1.59	level	2	$(9/8) \cdot n - 17$	$0.281 \cdot n^{-2}$	read A[i0, i2] (i0=0, i1=0); read A[i1, i2]
n^1	1.52	level	3	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i1, i0]
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$0.219 \cdot n^{-2}$	read A[i1, i0]
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$0.219 \cdot n^{-2}$	read A[i1, i0]
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$0.219 \cdot n^{-2}$	read A[i0, i2]
n^1	1.06	level	2	$(3/4) \cdot n - 6$	$0.188 \cdot n^{-2}$	read A[i0, i2]
n^1	0.875	level	1	$(7/8) \cdot n$	$0.219 \cdot n^{-2}$	read A[i0, i2]
n^1	0.75	level	1	$(3/4) \cdot n - 6$	$0.188 \cdot n^{-2}$	read A[i0, i2]
n^1	0.707	level	$(1/8) \cdot n^2$	2	$0.5 \cdot n^{-3}$	read A[i1, i2] (i1=0); read A[i1, i2]
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i1, i2] (i1=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.25 \cdot n^{-3}$	read A[i1, i2]
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=1, i1=0, i2=8)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=1, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n - 2$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 - n + 4$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=8, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=1, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.25	level	1	$(1/4) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i0, i2]
n^1	0.177	level	2	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
n^1	0.177	level	2	$(1/8) \cdot n - 1$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0312 \cdot n^{-2}$	write A[i1, i2]
$n^{0.5}$	7	level	$(1/4) \cdot n + 1$	14	$3.5 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	6.5	level	$(1/4) \cdot n + 1$	13	$3.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	3.5	level	$(1/4) \cdot n$	7	$1.75 \cdot n^{-3}$	read A[i0, i2] (i1=0); read A[i0, i2]
$n^{0.5}$	3.5	level	$(1/4) \cdot n + 1$	7	$1.75 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	3	level	$(1/4) \cdot n + 1$	6	$1.5 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	2.5	level	$(1/4) \cdot n$	5	$1.25 \cdot n^{-3}$	read A[i0, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.46	level	$(17/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i0=8, i1=7, i2=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i1, i2] (i0=8, i1=7, i2=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1	level	$(1/4) \cdot n$	2	$0.5 \cdot n^{-3}$	read A[i0, i2] (i0=0, i1=1); read A[i1, i2] (i0=1, i1=0)
$n^{0.5}$	1	level	n	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0); read A[i0, i2]
$n^{0.5}$	0.5	level	$(1/4) \cdot n +$ $(11/4)$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.354	level	$(1/8) \cdot n + 2$	1	$0.25 \cdot n^{-3}$	read A[i1, i2]
$n^{0.5}$	0.354	level	$(1/8) \cdot n + 1$	1	$0.25 \cdot n^{-3}$	read A[i1, i2] (i0=8, i1=8)
n^0	48.5	level	3	28	$7 \cdot n^{-3}$	read A[i1, i2]
n^0	12.1	level	3	7	$1.75 \cdot n^{-3}$	read A[i1, i2]
n^0	10.4	level	3	6	$1.5 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	8.49	level	2	6	$1.5 \cdot n^{-3}$	read A[i1, i2]
n^0	2.24	level	5	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.73	level	3	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.73	level	3	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.73	level	3	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.41	level	2	1	$0.25 \cdot n^{-3}$	read A[i1, i2]
n^0	1.41	level	2	1	$0.25 \cdot n^{-3}$	read A[i1, i2]

Each k-sweep re-reads the whole distance matrix: two n^4 level terms of read A[i1,i2]/read A[i0,i2] at exactly $(1/8)n^2$ lines (no row additive term — the k-row is already resident), combined coefficient 0.0442, identical to gemm’s constant. The $n^{3.5}$ terms are the k-row reuses at $(1/4)n + 2$ lines.

floyd_warshall — single-shot [exact]

Accesses $A(n) = 4 \cdot n^3$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^4 + 0.0625 \cdot n^{3.5} + 7.39 \cdot n^3 + 1.91 \cdot n^{2.5} + 81.6 \cdot n^2 + 10.2 \cdot n^{1.5} + 66.2 \cdot n^1 + 47.8 \cdot n^{0.5} + 88.3 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0221	level	$(1/8) \cdot n^2$	$(1/16) \cdot n^3 + (-11/4) \cdot n^2 + (129/4) \cdot n - 54$	0.0156	read A[i1, i2] (i1=0); read A[i1, i2]
n^4	0.0221	level	$(1/8) \cdot n^2$	$(1/16) \cdot n^3 + (-7/4) \cdot n^2 + (27/2) \cdot n - 26$	0.0156	read A[i0, i2] (i0=1, i1=0); read A[i1, i2] (i1=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.0417	level	$(1/4) \cdot n + 2$	$(1/12) \cdot n^3 + (-95/32) \cdot n^2 + (925/24) \cdot n - 198$	0.0208	read A[i0, i2]
$n^{3.5}$	0.0182	level	$(1/4) \cdot n + 2$	$(7/192) \cdot n^3 + (-57/64) \cdot n^2 + (61/24) \cdot n + 38$	0.00911	read A[i0, i2]
$n^{3.5}$	0.0026	level	$(1/4) \cdot n + 2$	$(1/192) \cdot n^3 + (-9/64) \cdot n^2 + (-17/24) \cdot n + 26$	0.0013	read A[i0, i2]
n^3	1.62	level	3	$(15/16) \cdot n^3 + (-263/16) \cdot n^2 + (161/2) \cdot n - 57$	0.234	read A[i0, i2] (i0=0, i1=1, i2=0); read A[i0, i2] (i0=0, i2=0) (+8)
n^3	1.01	level	3	$(7/12) \cdot n^3 + (-343/32) \cdot n^2 + (707/12) \cdot n - 84$	0.146	read A[i0, i2]
n^3	0.875	level	1	$(7/8) \cdot n^3$	0.219	read A[i1, i2]
n^3	0.758	level	3	$(7/16) \cdot n^3 + (-117/16) \cdot n^2 + (257/8) \cdot n$	0.109	read A[i1, i2] (i1=0); read A[i1, i2] (+1)
n^3	0.758	level	3	$(7/16) \cdot n^3 + (-105/16) \cdot n^2 + (217/8) \cdot n - 21$	0.109	write A[i1, i2]
n^3	0.442	level	3	$(49/192) \cdot n^3 + (-469/128) \cdot n^2 + (287/48) \cdot n + 56$	0.0638	read A[i0, i2]
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 11 \cdot n + 26$	0.25/n	read A[i1, i0] (i0=8, i1=0, i2=0); read A[i1, i0] (i0=8, i2=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 12 \cdot n + 27$	$0.25/n$	read A[i1, i2] (i1=0); read A[i1, i2]
n^3	0.309	level	$(1/8) \cdot n^2$	$(7/8) \cdot n^2 + (-67/4) \cdot n + 30$	$0.219/n$	read A[i1, i2] (i1=0); read A[i1, i2]
n^3	0.309	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 3$	$(7/8) \cdot n^2 + (-63/4) \cdot n + 28$	$0.219/n$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^3	0.125	level	4	$(1/16) \cdot n^3 + (-25/16) \cdot n^2 + (19/2) \cdot n - 8$	0.0156	read A[i0, i2] (i0=0, i1=1, i2=8); read A[i1, i0]
n^3	0.125	level	4	$(1/16) \cdot n^3 + (-27/16) \cdot n^2 + (93/8) \cdot n - 10$	0.0156	read A[i1, i0]
n^3	0.108	level	3	$(1/16) \cdot n^3 + (-7/16) \cdot n^2 + (-11/4) \cdot n + 6$	0.0156	read A[i1, i2] (i1=0); read A[i1, i2] (+1)
n^3	0.0631	level	3	$(7/192) \cdot n^3 + (-7/128) \cdot n^2 + (-91/48) \cdot n$	0.00911	read A[i0, i2]
n^3	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0312/n$	read A[i1, i2] (i1=0); read A[i1, i2]
n^3	0.0405	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 4 \rightarrow (1/8) \cdot n^2$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0312/n$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^3	0.0394	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 4 \rightarrow (1/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 + (-17/4) \cdot n + 8$	$0.0312/n$	read A[i1, i2] (i1=0); read A[i1, i2]
n^3	0.0164	ramp	$(19/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 - 1$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	$0.0156/n$	read A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0157	ramp	$(13/4) \cdot n - 1$ → $(1/8) \cdot n^2 - 1$	$(1/16) \cdot n^2 + (-21/8) \cdot n + 27$	0.0156/n	read A[i0, i2] (i1=0)
n^3	0.0107	ramp	$(3/8) \cdot n + 3$ → $(1/8) \cdot n^2 - 2 \cdot n + 3$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0156/n	read A[i1, i2]
n^3	0.0107	ramp	$(3/8) \cdot n + 3$ → $(1/8) \cdot n^2 - 2 \cdot n + 3$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0156/n	read A[i0, i2] (i1=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n + 2$	$n^2 - 11 \cdot n + 25$	0.25/n	read A[i0, i2] (i2=0)
$n^{2.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n^2 + (-133/8) \cdot n + 42$	0.219/n	read A[i0, i2]
$n^{2.5}$	0.25	level	$(1/4) \cdot n + 1$	$(1/2) \cdot n^2 + (-33/2) \cdot n + 136$	0.125/n	read A[i0, i2]
$n^{2.5}$	0.219	level	$(1/4) \cdot n + 1$	$(7/16) \cdot n^2 + (-7/4) \cdot n - 34$	0.109/n	read A[i0, i2]
$n^{2.5}$	0.156	level	$(1/4) \cdot n + 2$	$(5/16) \cdot n^2 + (-85/8) \cdot n + 90$	0.0781/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (29/4) \cdot n - 26$	0.0156/n	read A[i0, i2] (i0=0, i2=0); read A[i0, i2] (i1=0, i2=0) (+2)
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (1/8) \cdot n - 18$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (1/4) \cdot n - 20$	0.0156/n	read A[i0, i2]
$n^{2.5}$	0.0312	level	$(1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (1/4) \cdot n - 20$	0.0156/n	read A[i0, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.0261	ramp	$(1/8) \cdot n + 2$ $\rightarrow (1/4) \cdot n - 1$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0156/n$	read A[i1, i2]
$n^{2.5}$	0.0238	ramp	$(1/8) \cdot n + 3$ $\rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0156/n$	read A[i1, i2] (i2=8); read A[i1, i2]
$n^{2.5}$	0.0238	ramp	$(1/8) \cdot n + 3$ $\rightarrow (1/4) \cdot n$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0156/n$	read A[i0, i2]
$n^{2.5}$	0.0227	ramp	$(1/8) \cdot n + 2$ $\rightarrow (1/4) \cdot n - 1$	$(7/128) \cdot n^2 + (-25/16) \cdot n + 11$	$0.0137/n$	read A[i0, i2]
$n^{2.5}$	0.00296	ramp	$(1/8) \cdot n + 4$ $\rightarrow (1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00195/n$	read A[i0, i2]
n^2	6.06	level	3	$(7/2) \cdot n^2 + (-63/2) \cdot n + 28$	$0.875/n$	read A[i0, i2]
n^2	5.85	level	3	$(27/8) \cdot n^2 + (-243/8) \cdot n + 27$	$0.844/n$	write A[i1, i2]
n^2	5.3	level	3	$(49/16) \cdot n^2 + (-245/8) \cdot n + 49$	$0.766/n$	read A[i1, i0]
n^2	4.55	level	3	$(21/8) \cdot n^2 + (-189/8) \cdot n + 21$	$0.656/n$	read A[i1, i0]
n^2	4.55	level	3	$(21/8) \cdot n^2 + (-189/8) \cdot n + 21$	$0.656/n$	write A[i1, i2]
n^2	3.71	level	2	$(21/8) \cdot n^2 + (-21/8) \cdot n$	$0.656/n$	read A[i0, i2]
n^2	3.71	level	2	$(21/8) \cdot n^2 + (-21/8) \cdot n$	$0.656/n$	write A[i1, i2]
n^2	3.71	level	2	$(21/8) \cdot n^2 + (-21/8) \cdot n$	$0.656/n$	write A[i1, i2]
n^2	3.44	level	1	$(55/16) \cdot n^2 + (-87/8) \cdot n + 14$	$0.859/n$	read A[i1, i0]
n^2	3.09	level	2	$(35/16) \cdot n^2 + (-63/8) \cdot n + 21$	$0.547/n$	read A[i0, i2]; write A[i1, i2]
n^2	2.83	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$8 \cdot n - 16$	$2 \cdot n^{-2}$	read A[i1, i2] (i1=0, i2=0); read A[i1, i2] (i2=0)
n^2	2.62	level	1	$(21/8) \cdot n^2 + (-21/8) \cdot n$	$0.656/n$	read A[i1, i0]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	2.12	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$6 \cdot n - 12$	$1.5 \cdot n^{-2}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^2	1.77	level	2	$(5/4) \cdot n^2 + (-45/8) \cdot n$	$0.312/n$	read A[i0, i2]
n^2	1.73	level	3	$n^2 - 9 \cdot n + 8$	$0.25/n$	write A[i1, i2]
n^2	1.62	level	3	$(15/16) \cdot n^2 + (-95/8) \cdot n + 35$	$0.234/n$	read A[i0, i2]
n^2	1.62	level	3	$(15/16) \cdot n^2 - 5 \cdot n - 20$	$0.234/n$	read A[i0, i2]
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-63/8) \cdot n + 7$	$0.219/n$	read A[i1, i0]
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-31/4) \cdot n + 5$	$0.219/n$	read A[i1, i0]
n^2	1.41	level	3	$(13/16) \cdot n^2 - 2 \cdot n - 36$	$0.203/n$	read A[i0, i2]
n^2	1.33	level	2	$(15/16) \cdot n^2 + (5/4) \cdot n$	$0.234/n$	read A[i0, i2]
n^2	1.24	level	2	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.219/n$	write A[i1, i2]
n^2	1.24	level	2	$(7/8) \cdot n^2 + (-119/8) \cdot n + 63$	$0.219/n$	read A[i1, i0]
n^2	1.06	level	2	$(3/4) \cdot n^2 - 6 \cdot n$	$0.188/n$	read A[i1, i0]
n^2	1.06	level	2	$(3/4) \cdot n^2 + (-3/4) \cdot n$	$0.188/n$	write A[i1, i2]
n^2	1	level	1	$n^2 + (5/8) \cdot n + 1$	$0.25/n$	read A[i1, i0]; read A[i0, i2]
n^2	0.873	ramp	$1 \rightarrow 2$	$(7/8) \cdot n^2 + (-7/8) \cdot n - 6$	$0.219/n$	read A[i1, i0]; read A[i0, i2]
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-7/8) \cdot n - 21$	$0.109/n$	read A[i1, i0]
n^2	0.707	level	2	$(1/2) \cdot n^2 + (-15/2) \cdot n + 35$	$0.125/n$	read A[i0, i2] (i0=0, i1=1, i2=0); write A[i1, i2] (i0=0, i2=0) (+6)
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-45/8) \cdot n + 21$	$0.0938/n$	read A[i0, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.109/n$	read A[i0, i2]
n^2	0.53	level	2	$(3/8) \cdot n^2 + (-3/8) \cdot n$	$0.0938/n$	read A[i0, i2]
n^2	0.5	level	1	$(1/2) \cdot n^2 + (-3/2) \cdot n + 18$	$0.125/n$	read A[i0, i2] (i0=0, i1=0, i2=0); write A[i1, i2] (i0=0, i1=0, i2=0) (+2)
n^2	0.442	level	2	$(5/16) \cdot n^2 + (15/8) \cdot n$	$0.0781/n$	read A[i0, i2]
n^2	0.442	level	2	$(5/16) \cdot n^2 + (-7/4) \cdot n + 2$	$0.0781/n$	read A[i1, i2]; read A[i0, i2] (+1)
n^2	0.438	level	1	$(7/16) \cdot n^2 + (-49/8) \cdot n + 21$	$0.109/n$	write A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 3$	$0.25 \cdot n^{-2}$	read A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	$n - 3$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$n - 1$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (17/8)$	$n - 2$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 9$	$0.25 \cdot n^{-2}$	read A[i1, i2]
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 8$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i0=1, i1=0, i2=8); read A[i1, i0] (i0=8, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 2$	$0.25 \cdot n^{-2}$	read A[i1, i2] (i1=0); read A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 9$	$0.25 \cdot n^{-2}$	read A[i1, i2] (i1=0); read A[i1, i2]
n^2	0.25	level	4	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0312/n$	read A[i1, i0]
n^2	0.233	ramp	$(9/4) \cdot n \rightarrow (1/8) \cdot n^2$	$n - 17$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.227	ramp	$(3/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 - n + 2$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i1=0, i2=0)
n^2	0.227	ramp	$(3/8) \cdot n + 1 \rightarrow (1/8) \cdot n^2 - n + 1$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i1, i2]
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	$0.0312/n$	write A[i1, i2]
n^2	0.207	ramp	$(1/2) \cdot n \rightarrow (1/8) \cdot n^2$	$(7/8) \cdot n - 2$	$0.219 \cdot n^{-2}$	read A[i1, i2] (i2=0)
n^2	0.199	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-9/8) \cdot n + 5$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.199	ramp	$(11/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-9/8) \cdot n + 5$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^2	0.177	level	2	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	$0.0312/n$	read A[i1, i0]
n^2	0.177	level	2	$(1/8) \cdot n^2 - n$	$0.0312/n$	read A[i1, i0]
n^2	0.146	ramp	$(5/2) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$(5/8) \cdot n - 10$	$0.156 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2 + (-1/2) \cdot n$	$0.0156/n$	read A[i0, i2]
n^2	0.0884	level	2	$(1/16) \cdot n^2 + (3/8) \cdot n$	$0.0156/n$	read A[i0, i2]
n^2	0.0884	level	2	$(1/16) \cdot n^2 + (3/8) \cdot n$	$0.0156/n$	read A[i0, i2]
n^2	0.0625	level	1	$(1/16) \cdot n^2 + (3/8) \cdot n - 7$	$0.0156/n$	read A[i1, i0]
n^2	0.0625	level	1	$(1/16) \cdot n^2 + (-1/8) \cdot n - 1$	$0.0156/n$	read A[i0, i2]; write A[i1, i2]
n^2	0.029	ramp	$(19/8) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0289	ramp	$(9/4) \cdot n \rightarrow$ $(1/8) \cdot n^2 +$ $(-3/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i1, i2] (i2=0)
n^2	0.0289	ramp	$(9/4) \cdot n - 1 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i1, i2]
n^2	0.0289	ramp	$(9/4) \cdot n - 1 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0285	ramp	$(25/8) \cdot n - 1 \rightarrow$ $(1/8) \cdot n^2 -$ $n + 3$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0279	ramp	$(5/4) \cdot n \rightarrow$ $(1/8) \cdot n^2 -$ $2 \cdot n + 6$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i1=0)
n^2	0.0277	ramp	$(9/4) \cdot n - 2 \rightarrow$ $(1/8) \cdot n^2 -$ $2 \cdot n + 6$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i1, i0] (i2=0)
$n^{1.5}$	3.78	ramp	$5 \rightarrow (1/4) \cdot n$ $+ 1$	$(63/8) \cdot n -$ 34	$1.97 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	2.5	level	$(1/4) \cdot n + 1$	$5 \cdot n - 65$	$1.25 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.53	ramp	$5 \rightarrow (1/4) \cdot n$ $+ 1$	$(9/8) \cdot n - 11$	$0.281 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 10$	$0.25 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.5	level	$(1/4) \cdot n + 1$	$n - 16$	$0.25 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.5	level	$(1/4) \cdot n$	n	$0.25 \cdot n^{-2}$	read A[i1, i2] (i0=1, i1=0, i2=0); read A[i1, i2] (i2=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.354	level	$(1/8) \cdot n + 2$	$n - 8$	$0.25 \cdot n^{-2}$	read A[i0, i2] (i0=0, i1=1, i2=8); read A[i0, i2] (i2=0)
$n^{1.5}$	0.354	level	$(1/8) \cdot n + 1$	$n - 9$	$0.25 \cdot n^{-2}$	read A[i1, i2]
$n^{1.5}$	0.288	ramp	$5 \rightarrow (1/4) \cdot n$ $- 1$	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i1, i0] (i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.125	level	$(1/4) \cdot n + 1$	$(1/4) \cdot n - 4$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=1, i1=0); read A[i0, i2] (i1=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.0514	ramp	$(1/8) \cdot n + 2 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.0514	ramp	$(1/8) \cdot n + 2 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.0502	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2] (i0=0, i1=1)
$n^{1.5}$	0.0502	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{1.5}$	0.0502	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i1, i2]
$n^{1.5}$	0.0412	ramp	$5 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0312 \cdot n^{-2}$	read A[i1, i0] (i2=0)
n^1	7	level	1	$7 \cdot n - 14$	$1.75 \cdot n^{-2}$	read A[i1, i0] (i0=7, i2=0); read A[i1, i0] (i2=0)
n^1	6.12	level	1	$(49/8) \cdot n - 21$	$1.53 \cdot n^{-2}$	write A[i1, i2]
n^1	4.95	level	2	$(7/2) \cdot n - 28$	$0.875 \cdot n^{-2}$	read A[i1, i0]
n^1	3.71	level	2	$(21/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i1, i0]
n^1	3.71	level	2	$(21/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i0, i2]
n^1	3.54	level	2	$(5/2) \cdot n - 20$	$0.625 \cdot n^{-2}$	read A[i1, i0]; read A[i0, i2]
n^1	3.5	level	1	$(7/2) \cdot n - 7$	$0.875 \cdot n^{-2}$	read A[i1, i0]
n^1	2.83	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	8	$2 \cdot n^{-3}$	read A[i1, i0] (i1=0, i2=0); read A[i1, i0] (i2=0)
n^1	2.62	level	1	$(21/8) \cdot n$	$0.656 \cdot n^{-2}$	read A[i1, i0]
n^1	2.62	level	1	$(21/8) \cdot n$	$0.656 \cdot n^{-2}$	read A[i0, i2]
n^1	2.12	level	$(1/8) \cdot n^2$	6	$1.5 \cdot n^{-3}$	read A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.88	level	1	$(15/8) \cdot n - 2$	$0.469 \cdot n^{-2}$	read A[i1, i0]
n^1	1.88	level	1	$(15/8) \cdot n - 1$	$0.469 \cdot n^{-2}$	read A[i0, i2]; write A[i1, i2]
n^1	1.88	level	1	$(15/8) \cdot n - 1$	$0.469 \cdot n^{-2}$	read A[i1, i0]
n^1	1.62	level	1	$(13/8) \cdot n - 1$	$0.406 \cdot n^{-2}$	read A[i0, i2]; write A[i1, i2]
n^1	1.52	level	3	$(7/8) \cdot n - 14$	$0.219 \cdot n^{-2}$	read A[i1, i0]
n^1	1.41	level	2	$n - 8$	$0.25 \cdot n^{-2}$	read A[i0, i2]
n^1	1.24	level	2	$(7/8) \cdot n - 13$	$0.219 \cdot n^{-2}$	read A[i1, i2]
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$0.219 \cdot n^{-2}$	read A[i1, i0]
n^1	1.24	level	2	$(7/8) \cdot n - 7$	$0.219 \cdot n^{-2}$	read A[i1, i0]
n^1	0.875	level	1	$(7/8) \cdot n$	$0.219 \cdot n^{-2}$	read A[i0, i2]
n^1	0.707	level	$(1/8) \cdot n^2$	2	$0.5 \cdot n^{-3}$	read A[i1, i2] (i1=0); read A[i1, i2]
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=1, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i1, i2] (i1=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.25 \cdot n^{-3}$	read A[i1, i2]
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 3$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n - 2$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 4$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=8, i1=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/8) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 3$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=1, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^1	0.177	level	2	$(1/8) \cdot n - 3$	$0.0312 \cdot n^{-2}$	read A[i1, i2]
n^1	0.177	level	2	$(1/8) \cdot n - 1$	$0.0312 \cdot n^{-2}$	read A[i1, i0]
n^1	0.177	level	2	$(1/8) \cdot n - 1$	$0.0312 \cdot n^{-2}$	read A[i0, i2]
$n^{0.5}$	7	level	$(1/4) \cdot n + 1$	14	$3.5 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	6.5	level	$(1/4) \cdot n + 1$	13	$3.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	3.5	level	$(1/4) \cdot n$	7	$1.75 \cdot n^{-3}$	read A[i0, i2] (i1=0); read A[i0, i2]
$n^{0.5}$	3.5	level	$(1/4) \cdot n + 1$	7	$1.75 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	3	level	$(1/4) \cdot n + 1$	6	$1.5 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	2.5	level	$(1/4) \cdot n$	5	$1.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	1.46	level	$(17/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.32	level	$(7/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i0=8, i1=7, i2=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i1, i2] (i0=8, i1=7, i2=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	1	level	n	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i1, i2] (i0=1, i1=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n +$ $(11/4)$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i1=0); read A[i0, i2]
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2] (i0=0, i1=1)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.25 \cdot n^{-3}$	read A[i0, i2]
$n^{0.5}$	0.354	level	$(1/8) \cdot n + 2$	1	$0.25 \cdot n^{-3}$	read A[i1, i2]
n^0	48.5	level	3	28	$7 \cdot n^{-3}$	read A[i1, i2]
n^0	12.1	level	3	7	$1.75 \cdot n^{-3}$	read A[i1, i2]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	10.4	level	3	6	$1.5 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	8.49	level	2	6	$1.5 \cdot n^{-3}$	read A[i1, i2]
n^0	2.24	level	5	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.73	level	3	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.73	level	3	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.73	level	3	1	$0.25 \cdot n^{-3}$	read A[i1, i0] (i2=0)
n^0	1.41	level	2	1	$0.25 \cdot n^{-3}$	read A[i1, i2] (i0=8, i1=8, i2=8)

Each k-sweep re-reads the whole distance matrix: two n^4 level terms of read A[i1,i2]/read A[i0,i2] at exactly $(1/8)n^2$ lines (no row additive term — the k-row is already resident), combined coefficient 0.0442, identical to gemm’s constant. The $n^{3.5}$ terms are the k-row reuses at $(1/4)n + 2$ lines.

gemm — infinite-repeat [exact]

Accesses $A(n) = 4 \cdot n^3 + 2 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^4 + 0.0625 \cdot n^{3.5} + 6.83 \cdot n^3 + 1.05 \cdot n^{2.5} + 7.84 \cdot n^2 + 0.354 \cdot n^{1.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0331	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(3/32) \cdot n^3 + (-3/2) \cdot n^2$	0.0234	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)
n^4	0.011	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/32) \cdot n^3 + (-3/4) \cdot n^2 + 4 \cdot n$	0.00781	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)
$n^{3.5}$	0.0547	level	$(1/4) \cdot n$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.0273	read A[i1, i4] (i0=0)
$n^{3.5}$	0.00781	level	$(1/4) \cdot n + 1$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.00391	read C[i3, i4] (i0=0, i1=0, i3=0); read A[i1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	3.03	level	3	$(7/4) \cdot n^3$	0.438	write A[i1, i4] (i0=0, i3=0); read C[i3, i4] (i0=0) (+1)
n^3	1.52	level	3	$(7/8) \cdot n^3$	0.219	read B[i1, i3] (i0=0, i1=0, i3=0, i4=0); read C[i3, i4] (i0=0, i1=0, i3=0, i4=0) (+1)
n^3	0.875	level	1	$(7/8) \cdot n^3$	0.219	read A[i1, i4] (i0=0)
n^3	0.265	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(3/4) \cdot n^2$	0.188/n	read C[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i4=0)
n^3	0.265	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(3/4) \cdot n^2$	0.188/n	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)
n^3	0.25	level	4	$(1/8) \cdot n^3 - n^2$	0.0312	read C[i3, i4] (i0=0, i1=0, i3=0, i4=0); read C[i3, i4] (i0=0, i3=0, i4=0) (+1)
n^3	0.217	level	3	$(1/8) \cdot n^3$	0.0312	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^3	0.0884	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/4) \cdot n^2 - 2 \cdot n$	0.0625/n	read C[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0765	level	$(3/8) \cdot n^2$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read B[i1, i3] (i0=0, i1=0, i4=0); read B[i1, i3] (i0=0, i4=0)
n^3	0.0765	level	$(3/8) \cdot n^2 + (-1/8) \cdot n + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read A[i1, i2] (i0=0, i1=0); read A[i1, i2] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read C[i3, i4] (i0=0, i1=0, i3=0); read C[i3, i4] (i0=0, i3=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 - n$	$0.0312/n$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 - n$	$0.0312/n$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)
n^3	0.0421	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n$	n^2	$0.25/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n$	$n^2 - n$	$0.25/n$	read A[i1, i2] (i0=0, i1=0); read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.0514	ramp	$(1/8) \cdot n + 2 \rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read A[i1, i4] (i0=0, i3=0)
n^2	1.75	level	4	$(7/8) \cdot n^2$	$0.219/n$	read B[i1, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.75	level	1	$(7/4) \cdot n^2$	$0.438/n$	read A[i1, i2] (i0=0); write A[i1, i2] (i0=0)
n^2	0.612	level	$(3/8) \cdot n^2$	n	$0.25 \cdot n^{-2}$	read B[i1, i3] (i0=0, i1=0, i4=0); read B[i1, i3] (i0=0, i4=0)
n^2	0.612	level	$(3/8) \cdot n^2$	n	$0.25 \cdot n^{-2}$	read B[i1, i3] (i0=0, i1=0, i3=0, i4=0); read B[i1, i3] (i0=0, i3=0, i4=0)
n^2	0.612	level	$(3/8) \cdot n^2 +$ $(-1/8) \cdot n + 1$	n	$0.25 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0, i2=0); read A[i1, i2] (i0=0, i2=0)
n^2	0.612	level	$(3/8) \cdot n^2 +$ $(-1/8) \cdot n$	n	$0.25 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); read A[i1, i2] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 +$ $(5/4) \cdot n +$ $(29/8)$	n	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i4=0)
n^2	0.354	level	$(1/8) \cdot n^2 +$ $(3/8) \cdot n + 1$	n	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i1=0, i4=0); read C[i3, i4] (i0=0, i4=0)
n^2	0.354	level	$(1/8) \cdot n^2 +$ $(1/4) \cdot n + 1$	n	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i1=0); read C[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	n	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i1=0, i3=0); read C[i3, i4] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	n	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i1=0, i3=0, i4=0); read C[i3, i4] (i0=0, i3=0, i4=0)
n^2	0.125	level	1	$(1/8) \cdot n^2$	$0.0312/n$	write A[i1, i2] (i0=0)
$n^{1.5}$	0.354	level	$(1/8) \cdot n$	n	$0.25 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); read A[i1, i4] (i0=0, i3=0, i4=0)

The n^4 order is carried by two bins of read C[i3, i4]: each C element is re-touched on the next outer iteration at distance $(1/8)n^2 + (3/8)n + 1$ lines — the streamed matrix plus one row — with combined population $n^{3/8}$ (one reuse per line element per sweep). Their coefficients sum to $0.0442 = (1/8) \cdot \sqrt{1/8}$, the constant the earlier study measured. Below it: A-row reuse at $(1/4)n$ lines (order $n^{3.5}$) and the line-reuse bulk at distance 1-3 (order n^3 , >90% of accesses). Under infinite repeat the same terms persist (the wraparound consumers are the first C reads of the next pass, annotated i1=0); no new order arises because all matrices are already re-touched within a pass. Miss reading: the miss ratio steps at $64 \cdot ((1/4)n)$ bytes (row) and $64 \cdot ((1/8)n^2)$ bytes (matrix) — the second is the tiling cliff.

gemm — single-shot [exact]

Accesses $A(n) = 4 \cdot n^3 + 2 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0442 \cdot n^4 + 0.0625 \cdot n^{3.5} + 6.68 \cdot n^3 + 1.05 \cdot n^{2.5} + 5.39 \cdot n^2 + 0.354 \cdot n^{1.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0331	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(3/32) \cdot n^3 + (-51/32) \cdot n^2 + (3/2) \cdot n$	0.0234	read C[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.011	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/32) \cdot n^3 + (-25/32) \cdot n^2 + (19/4) \cdot n - 4$	0.00781	read C[i3, i4] (i0=0)
$n^{3.5}$	0.0547	level	$(1/4) \cdot n$	$(7/64) \cdot n^3 + (-7/4) \cdot n^2$	0.0273	read A[i1, i4] (i0=0)
$n^{3.5}$	0.00781	level	$(1/4) \cdot n + 1$	$(1/64) \cdot n^3 + (-3/8) \cdot n^2 + 2 \cdot n$	0.00391	read A[i1, i4] (i0=0)
n^3	3.25	level	3	$(15/8) \cdot n^3$	0.469	read B[i1, i3] (i0=0); write A[i1, i4] (i0=0)
n^3	1.52	level	3	$(7/8) \cdot n^3$	0.219	read C[i3, i4] (i0=0)
n^3	0.875	level	1	$(7/8) \cdot n^3$	0.219	read A[i1, i4] (i0=0)
n^3	0.265	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(3/4) \cdot n^2 + (-3/4) \cdot n$	0.188/n	read C[i3, i4] (i0=0, i4=0)
n^3	0.265	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(3/4) \cdot n^2 + (-3/4) \cdot n$	0.188/n	read C[i3, i4] (i0=0)
n^3	0.25	level	4	$(1/8) \cdot n^3 - n^2$	0.0312	read C[i3, i4] (i0=0, i3=0, i4=0); read B[i1, i3] (i0=0)
n^3	0.0884	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/4) \cdot n^2 + (-9/4) \cdot n + 2$	0.0625/n	read C[i3, i4] (i0=0, i4=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0312/n	read C[i3, i4] (i0=0, i3=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0312/n	read C[i3, i4] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	0.0312/n	read C[i3, i4] (i0=0)
n^3	0.042	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0312/n	read C[i3, i4] (i0=0)
$n^{2.5}$	0.5	level	$(1/4) \cdot n$	n^2	0.25/n	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.5	level	$(1/4) \cdot n$	$n^2 - n$	$0.25/n$	read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.0514	ramp	$(1/8) \cdot n + 2$ $\rightarrow (1/4) \cdot n - 1$	$(1/8) \cdot n^2 - 2 \cdot n$	$0.0312/n$	read A[i1, i4] (i0=0, i3=0)
n^2	1.75	level	4	$(7/8) \cdot n^2$	$0.219/n$	read B[i1, i3] (i0=0, i4=0)
n^2	1	level	1	n^2	$0.25/n$	write A[i1, i2] (i0=0); read A[i1, i4] (i0=0, i3=0, i4=0) (+1)
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.219/n$	read A[i1, i2] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	$n - 1$	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i4=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i4=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	$n - 1$	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	$n - 1$	$0.25 \cdot n^{-2}$	read C[i3, i4] (i0=0, i3=0, i4=0)
$n^{1.5}$	0.354	level	$(1/8) \cdot n$	n	$0.25 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=0)

The n^4 order is carried by two bins of read C[i3,i4]: each C element is re-touched on the next outer iteration at distance $(1/8)n^2 + (3/8)n + 1$ lines — the streamed matrix plus one row — with combined population $n^{3/8}$ (one reuse per line element per sweep). Their coefficients sum to $0.0442 = (1/8) \cdot \sqrt{1/8}$, the constant the earlier study measured. Below it: A-row reuse at $(1/4)n$ lines (order $n^{3.5}$) and the line-reuse bulk at distance 1-3 (order n^3 , >90% of accesses). Under infinite repeat the same terms persist (the wraparound consumers are the first C reads of the next pass, annotated i1=0); no new order arises because all matrices are already re-touched within a pass. Miss reading: the miss ratio steps at $64 \cdot ((1/4)n)$ bytes (row) and $64 \cdot ((1/8)n^2)$ bytes (matrix) — the second is the tiling cliff.

gemver — infinite-repeat [exact]

Accesses $A(n) = 14 \cdot n^2 + 3 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.114 \cdot n^3 + 1.28 \cdot n^{2.5} + 24.1 \cdot n^2 + 8.36 \cdot n^{1.5} + 21.6 \cdot n^1 + 2.95 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0317	ramp	$(21/8) \cdot n \rightarrow$	$(7/64) \cdot n^2$	0.00781	read A[i4,
			$(1/8) \cdot n^2 +$	$+ (-15/8) \cdot n$		i3] (i0=0)
			$(3/4) \cdot n - 2$	$+ 2$		
n^3	0.0317	ramp	$(1/8) \cdot n^2 +$	$(3/32) \cdot n^2$	0.0067	read A[i1,
			$(1/2) \cdot n + 2$	$+ (-3/2) \cdot n$		i2] (i0=0)
			$\rightarrow$			
			$(1/8) \cdot n^2 +$			
			$(5/8) \cdot n + 1$			
n^3	0.0317	ramp	$(21/8) \cdot n - 3$	$(7/64) \cdot n^2$	0.00781	read A[i6,
			$\rightarrow$	$+ (-15/8) \cdot n$		i7] (i0=0)
			$(1/8) \cdot n^2 +$	$+ 2$		
			$(1/2) \cdot n - 1$			
n^3	0.00514	ramp	$(1/8) \cdot n^2 +$	$(1/64) \cdot n^2$	0.00112	read A[i1,
			$(1/2) \cdot n + 3$	$+ (-3/8) \cdot n +$		i2] (i0=0)
			$\rightarrow$	2		
			$(1/8) \cdot n^2 +$			
			$(5/8) \cdot n + 1$			
n^3	0.00514	ramp	$(1/8) \cdot n^2 +$	$(1/64) \cdot n^2$	0.00112	read A[i1,
			$(1/2) \cdot n + 2$	$+ (-3/8) \cdot n +$		i2] (i0=0)
			$\rightarrow$	2		
			$(1/8) \cdot n^2 +$			
			$(5/8) \cdot n$			
n^3	0.00444	ramp	$(7/2) \cdot n - 12$	$(1/64) \cdot n^2$	0.00112	read A[i4,
			$\rightarrow$	$+ (-3/8) \cdot n +$		i3] (i0=0)
			$(1/8) \cdot n^2 +$	2		
			$(3/4) \cdot n - 8$			
n^3	0.00443	ramp	$(7/2) \cdot n - 16$	$(1/64) \cdot n^2$	0.00112	read A[i6,
			$\rightarrow$	$+ (-3/8) \cdot n +$		i7] (i0=0)
			$(1/8) \cdot n^2 +$	2		
			$(1/2) \cdot n - 7$			
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2 +$	0.0625	read A[i4,
				$(-7/4) \cdot n$		i3] (i0=0)
$n^{2.5}$	0.134	level	$(3/8) \cdot n + 3$	$(7/32) \cdot n^2$	0.0156	read C[i2]
				$+ (-7/2) \cdot n$		(i0=0);
						read E[i2]
						(i0=0)
$n^{2.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^2$	0.00781	read G[i4]
				$+ (-7/4) \cdot n$		(i0=0)
$n^{2.5}$	0.0547	level	$(1/4) \cdot n + 1$	$(7/64) \cdot n^2$	0.00781	read F[i7]
				$+ (-13/8) \cdot n -$		(i0=0,
				2		i6=0); read
						F[i7]
						(i0=0)
$n^{2.5}$	0.0191	level	$(3/8) \cdot n + 5$	$(1/32) \cdot n^2$	0.00223	read C[i2]
				$+ (-3/4) \cdot n +$		(i0=0);
				4		read E[i2]
						(i0=0)
$n^{2.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^2$	0.00112	read G[i4]
				$+ (-3/8) \cdot n +$		(i0=0)
				2		
$n^{2.5}$	0.00781	level	$(1/4) \cdot n + 2$	$(1/64) \cdot n^2$	0.00112	read F[i7]
				$+ (-3/8) \cdot n +$		(i0=0)
				2		

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	4.55	level	3	$(21/8) \cdot n^2 + (-7/4) \cdot n$	0.188	read G[i4] (i0=0); read A[i6, i7] (i0=0) (+1)
n^2	4.19	level	5	$(15/8) \cdot n^2 + (-7/8) \cdot n$	0.134	read C[i2] (i0=0, i1=0); read C[i2] (i0=0) (+2)
n^2	3.91	level	5	$(7/4) \cdot n^2$	0.125	read B[i1] (i0=0); read D[i1] (i0=0)
n^2	3.03	level	3	$(7/4) \cdot n^2$	0.125	write F[i3] (i0=0); write I[i6] (i0=0)
n^2	1.96	level	5	$(7/8) \cdot n^2 + (7/8) \cdot n$	0.0625	read E[i2] (i0=0, i1=0); write A[i1, i2] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2 + (7/8) \cdot n$	0.125	read F[i3] (i0=0); read F[i5] (i0=0) (+2)
n^2	0.875	level	1	$(7/8) \cdot n^2$	0.0625	read A[i1, i2] (i0=0)
n^2	0.612	level	6	$(1/4) \cdot n^2 - 2 \cdot n$	0.0179	read B[i1] (i0=0); read D[i1] (i0=0)
n^2	0.433	level	3	$(1/4) \cdot n^2$	0.0179	write F[i3] (i0=0); write I[i6] (i0=0, i7=0) (+1)
n^2	0.354	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/4) \cdot n$	$n - 2$	0.0714/n	read A[i4, i3] (i0=0)
n^2	0.31	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 1$ → $(1/8) \cdot n^2 + (5/8) \cdot n + 1$	$(7/8) \cdot n - 1$	0.0625/n	read A[i1, i2] (i0=0, i2=0)
n^2	0.31	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 1$	0.0625/n	read A[i6, i7] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.267	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 2$ $\rightarrow$ $(1/8) \cdot n^2 + (5/8) \cdot n + 1$	$(3/4) \cdot n$	$0.0536/n$	read A[i1, i2] (i0=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (-1/4) \cdot n$	0.0179	read F[i3] (i0=0); read I[i6] (i0=0)
n^2	0.213	ramp	$(13/8) \cdot n + 1 \rightarrow$ $(1/8) \cdot n^2 + (3/8) \cdot n + 3$	$(7/8) \cdot n - 1$	$0.0625/n$	read A[i4, i3] (i0=0, i3=0)
n^2	0.183	ramp	$(13/8) \cdot n - 1 \rightarrow$ $(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$(3/4) \cdot n$	$0.0536/n$	read A[i6, i7] (i0=0)
n^2	0.177	level	$(1/8) \cdot n^2 + n$	$(1/2) \cdot n - 8$	$0.0357/n$	read B[i1] (i0=0, i2=0); read D[i1] (i0=0, i2=0) (+2)
n^2	0.0884	level	$(1/8) \cdot n^2 + (3/4) \cdot n + 4$	$(1/4) \cdot n - 4$	$0.0179/n$	read C[i2] (i0=0, i1=0); read E[i2] (i0=0, i1=0)
n^2	0.0597	ramp	$(19/8) \cdot n - 7 \rightarrow$ $(1/8) \cdot n^2 + (-1/2) \cdot n + 8$	$(1/4) \cdot n - 2$	$0.0179/n$	read A[i6, i7] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i6, i7] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (3/4) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.00893/n$	read F[i3] (i0=0, i4=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + n$	$(1/8) \cdot n - 2$	$0.00893/n$	read G[i4] (i0=0, i3=0)
n^2	0.0434	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 3 \rightarrow$ $(1/8) \cdot n^2 + (5/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.00893/n$	read A[i1, i2] (i0=0)
n^2	0.0434	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 2 \rightarrow$ $(1/8) \cdot n^2 + (5/8) \cdot n$	$(1/8) \cdot n - 1$	$0.00893/n$	read A[i1, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0434	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 2$ → $(1/8) \cdot n^2 + (5/8) \cdot n$	$(1/8) \cdot n - 1$	$0.00893/n$	read A[i1, i2] (i0=0)
n^2	0.0433	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 1$	$0.00893/n$	read A[i6, i7] (i0=0, i7=0)
n^2	0.0423	ramp	$(1/8) \cdot n^2 + (5/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i4, i3] (i0=0, i4=0)
n^2	0.0422	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 3$ → $(1/8) \cdot n^2 + (5/8) \cdot n$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i1, i2] (i0=0)
n^2	0.0421	ramp	$(1/8) \cdot n^2 + (1/4) \cdot n + 4$ → $(1/8) \cdot n^2 + (1/2) \cdot n - 2$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i1, i2] (i0=0, i1=0)
n^2	0.03	ramp	$(5/2) \cdot n - 5$ → $(1/8) \cdot n^2 + (-3/8) \cdot n + 9$	$(1/8) \cdot n - 1$	$0.00893/n$	read A[i4, i3] (i0=0, i3=0)
n^2	0.0291	ramp	$(5/2) \cdot n - 1$ → $(1/8) \cdot n^2 + (-5/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i6, i7] (i0=0, i6=0)
n^2	0.029	ramp	$(5/2) \cdot n - 2$ → $(1/8) \cdot n^2 + (-3/4) \cdot n + 4$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i4, i3] (i0=0)
n^2	0.0281	ramp	$(11/8) \cdot n - 2$ → $(1/8) \cdot n^2 + (-7/4) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.00893/n$	read F[i5] (i0=0)
$n^{1.5}$	1.15	level	$(3/8) \cdot n + 3$	$(15/8) \cdot n - 1$	$0.134/n$	read C[i2] (i0=0); read E[i2] (i0=0)
$n^{1.5}$	1.07	level	$(3/8) \cdot n + 3$	$(7/4) \cdot n$	$0.125/n$	read C[i2] (i0=0, i2=0); read E[i2] (i0=0, i2=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + (15/8)$	$(7/8) \cdot n + (-7/8)$	$0.0625/n$	read A[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	$0.0625/n$	read A[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	$0.0625/n$	read A[i4, i3] (i0=0, i4=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	$0.0625/n$	read G[i4] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	$0.0625/n$	read G[i4] (i0=0, i4=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n + 1$	$0.0625/n$	read F[i7] (i0=0, i6=0); read F[i7] (i0=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n + 1$	$0.0625/n$	read F[i7] (i0=0, i6=0, i7=0); read F[i7] (i0=0, i7=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	$0.00893/n$	read G[i4] (i0=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	$0.00893/n$	read G[i4] (i0=0, i4=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 4$	$(1/8) \cdot n - 1$	$0.00893/n$	read C[i2] (i0=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 5$	$(1/8) \cdot n - 1$	$0.00893/n$	read E[i2] (i0=0, i2=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 4$	$(1/8) \cdot n - 1$	$0.00893/n$	read C[i2] (i0=0, i2=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.00893/n$	read F[i7] (i0=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.00893/n$	read F[i7] (i0=0, i7=0)
n^1	4.29	level	6	$(7/4) \cdot n$	$0.125/n$	read B[i1] (i0=0, i2=0); read D[i1] (i0=0, i2=0)
n^1	3.03	level	3	$(7/4) \cdot n$	$0.125/n$	read G[i4] (i0=0); read F[i7] (i0=0, i6=0)
n^1	2.47	level	2	$(7/4) \cdot n$	$0.125/n$	read H[i5] (i0=0); write F[i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.41	level	$(1/8) \cdot n^2 + n$	4	$0.286 \cdot n^{-2}$	read B[i1] (i0=0, i2=0); read D[i1] (i0=0, i2=0) (+2)
n^1	1.41	level	$(1/8) \cdot n^2 + n$	4	$0.286 \cdot n^{-2}$	read B[i1] (i0=0, i1=0, i2=0); read D[i1] (i0=0, i1=0, i2=0) (+2)
n^1	1.06	level	$(1/8) \cdot n^2 + (15/8) \cdot n + 7$	3	$0.214 \cdot n^{-2}$	read B[i1] (i0=0, i2=0); read D[i1] (i0=0, i2=0) (+1)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n + (7/4)$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + 1$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0, i6=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + 1$	1	$0.0714 \cdot n^{-2}$	read F[i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (11/8) \cdot n + (11/2)$	1	$0.0714 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + 2$	1	$0.0714 \cdot n^{-2}$	read A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + 1$	1	$0.0714 \cdot n^2$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + 2$	1	$0.0714 \cdot n^2$	read E[i2] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + n$	1	$0.0714 \cdot n^2$	read G[i4] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + 3$	1	$0.0714 \cdot n^2$	read C[i2] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n$	1	$0.0714 \cdot n^2$	read F[i5] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + n$	1	$0.0714 \cdot n^2$	read H[i5] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + 1$	1	$0.0714 \cdot n^2$	read F[i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + 4$	1	$0.0714 \cdot n^2$	read E[i2] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + 3$	1	$0.0714 \cdot n^2$	read C[i2] (i0=0, i1=0, i2=0)
n^1	0.177	level	2	$(1/8) \cdot n$	$0.00893/n$	write F[i5] (i0=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n$	1	$0.0714 \cdot n^2$	read A[i6, i7] (i0=0, i6=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n$	1	$0.0714 \cdot n^2$	read A[i4, i3] (i0=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n - 1$	1	$0.0714 \cdot n^2$	read F[i5] (i0=0)

The rank-1-update matrix A is walked twice (by rows, then transposed): ramps from $\sim(21/8)n$ up to $(1/8)n^2 + (3/4)n$ lines on both read A[i4, i3] and read A[i6, i7] give $d = 3.0$, headroom +1.0 single-shot, same mechanism as mvt.

gemver — single-shot [exact]

Accesses $A(n) = 14 \cdot n^2 + 3 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom +1; conservation $\Sigma \text{mass/warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0722 \cdot n^3 + 1.28 \cdot n^{2.5} + 22.9 \cdot n^2 + 8.36 \cdot n^{1.5} + 12.7 \cdot n^1 + 2.95 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0317	ramp	$(21/8) \cdot n \rightarrow$ $(1/8) \cdot n^2 +$ $(3/4) \cdot n - 2$	$(7/64) \cdot n^2$ $+ (-15/8) \cdot n$ $+ 2$	0.00781	read A[i4, i3] (i0=0)
n^3	0.0317	ramp	$(21/8) \cdot n - 3$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(1/2) \cdot n - 1$	$(7/64) \cdot n^2$ $+ (-15/8) \cdot n$ $+ 2$	0.00781	read A[i6, i7] (i0=0)
n^3	0.00444	ramp	$(7/2) \cdot n - 12$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(3/4) \cdot n - 8$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00112	read A[i4, i3] (i0=0)
n^3	0.00443	ramp	$(7/2) \cdot n - 16$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(1/2) \cdot n - 7$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00112	read A[i6, i7] (i0=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2 +$ $(-7/4) \cdot n$	0.0625	read A[i4, i3] (i0=0)
$n^{2.5}$	0.134	level	$(3/8) \cdot n + 3$	$(7/32) \cdot n^2$ $+ (-7/2) \cdot n$	0.0156	read C[i2] (i0=0); read E[i2] (i0=0)
$n^{2.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^2$ $+ (-7/4) \cdot n$	0.00781	read G[i4] (i0=0)
$n^{2.5}$	0.0547	level	$(1/4) \cdot n + 1$	$(7/64) \cdot n^2$ $+ (-13/8) \cdot n -$ 2	0.00781	read F[i7] (i0=0, i6=0); read F[i7] (i0=0)
$n^{2.5}$	0.0191	level	$(3/8) \cdot n + 5$	$(1/32) \cdot n^2$ $+ (-3/4) \cdot n +$ 4	0.00223	read C[i2] (i0=0); read E[i2] (i0=0)
$n^{2.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00112	read G[i4] (i0=0)
$n^{2.5}$	0.00781	level	$(1/4) \cdot n + 2$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00112	read F[i7] (i0=0)
n^2	6.15	level	5	$(11/4) \cdot n^2$	0.196	read B[i1] (i0=0); read D[i1] (i0=0) (+2)
n^2	3.91	level	5	$(7/4) \cdot n^2$	0.125	read C[i2] (i0=0); read E[i2] (i0=0)
n^2	3.46	level	3	$2 \cdot n^2$	0.143	read G[i4] (i0=0); write F[i3] (i0=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	3.03	level	3	$(7/4) \cdot n^2$	0.125	read G[i4] (i0=0); read F[i7] (i0=0)
n^2	2	level	1	$2 \cdot n^2 - 2 \cdot n$	0.143	read F[i3] (i0=0); read F[i5] (i0=0) (+1)
n^2	1.52	level	3	$(7/8) \cdot n^2$	0.0625	read A[i6, i7] (i0=0)
n^2	0.875	level	1	$(7/8) \cdot n^2$	0.0625	read A[i1, i2] (i0=0)
n^2	0.612	level	6	$(1/4) \cdot n^2 - 2 \cdot n$	0.0179	read C[i2] (i0=0, i2=0); read E[i2] (i0=0, i2=0) (+2)
n^2	0.354	ramp	$(1/8) \cdot n^2 + (1/2) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/4) \cdot n$	$n - 2$	0.0714/n	read A[i4, i3] (i0=0)
n^2	0.31	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 1$	0.0625/n	read A[i6, i7] (i0=0, i7=0)
n^2	0.213	ramp	$(13/8) \cdot n + 1$ → $(1/8) \cdot n^2 + (3/8) \cdot n + 3$	$(7/8) \cdot n - 1$	0.0625/n	read A[i4, i3] (i0=0, i3=0)
n^2	0.183	ramp	$(13/8) \cdot n - 1$ → $(1/8) \cdot n^2 + (1/4) \cdot n + 2$	$(3/4) \cdot n$	0.0536/n	read A[i6, i7] (i0=0)
n^2	0.0597	ramp	$(19/8) \cdot n - 7$ → $(1/8) \cdot n^2 + (-1/2) \cdot n + 8$	$(1/4) \cdot n - 2$	0.0179/n	read A[i6, i7] (i0=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 2$	0.00893/n	read A[i6, i7] (i0=0)
n^2	0.0433	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 1$	0.00893/n	read A[i6, i7] (i0=0, i7=0)
n^2	0.0423	ramp	$(1/8) \cdot n^2 + (5/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (3/4) \cdot n - 1$	$(1/8) \cdot n - 2$	0.00893/n	read A[i4, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.03	ramp	$(5/2) \cdot n - 5$ → $(1/8) \cdot n^2 +$ $(-3/8) \cdot n + 9$	$(1/8) \cdot n - 1$	$0.00893/n$	read A[i4, i3] (i0=0, i3=0)
n^2	0.0291	ramp	$(5/2) \cdot n - 1$ → $(1/8) \cdot n^2 +$ $(-5/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i6, i7] (i0=0, i6=0)
n^2	0.029	ramp	$(5/2) \cdot n - 2$ → $(1/8) \cdot n^2 +$ $(-3/4) \cdot n + 4$	$(1/8) \cdot n - 2$	$0.00893/n$	read A[i4, i3] (i0=0)
n^2	0.0281	ramp	$(11/8) \cdot n - 2$ → $(1/8) \cdot n^2 +$ $(-7/4) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.00893/n$	read F[i5] (i0=0)
$n^{1.5}$	1.15	level	$(3/8) \cdot n + 3$	$(15/8) \cdot n - 1$	$0.134/n$	read C[i2] (i0=0); read E[i2] (i0=0)
$n^{1.5}$	1.07	level	$(3/8) \cdot n + 3$	$(7/4) \cdot n$	$0.125/n$	read C[i2] (i0=0, i2=0); read E[i2] (i0=0, i2=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n +$ $(15/8)$	$(7/8) \cdot n +$ $(-7/8)$	$0.0625/n$	read A[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	$0.0625/n$	read A[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	$0.0625/n$	read A[i4, i3] (i0=0, i4=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	$0.0625/n$	read G[i4] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	$0.0625/n$	read G[i4] (i0=0, i4=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n + 1$	$0.0625/n$	read F[i7] (i0=0, i6=0); read F[i7] (i0=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n + 1$	$0.0625/n$	read F[i7] (i0=0, i6=0, i7=0); read F[i7] (i0=0, i7=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	$0.00893/n$	read G[i4] (i0=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	$0.00893/n$	read G[i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 4$	$(1/8) \cdot n - 1$	$0.00893/n$	read C[i2] (i0=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 5$	$(1/8) \cdot n - 1$	$0.00893/n$	read E[i2] (i0=0, i2=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 4$	$(1/8) \cdot n - 1$	$0.00893/n$	read C[i2] (i0=0, i2=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.00893/n$	read F[i7] (i0=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.00893/n$	read F[i7] (i0=0, i7=0)
n^1	4.29	level	6	$(7/4) \cdot n$	$0.125/n$	read B[i1] (i0=0, i2=0); read D[i1] (i0=0, i2=0)
n^1	2.62	level	1	$(21/8) \cdot n$	$0.188/n$	read F[i3] (i0=0); read F[i5] (i0=0) (+1)
n^1	1.41	level	2	n	$0.0714/n$	read A[i4, i3] (i0=0, i3=0, i4=0); write F[i5] (i0=0)
n^1	1.24	level	2	$(7/8) \cdot n$	$0.0625/n$	read H[i5] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n + (7/4)$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (29/8)$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + 1$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0, i7=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0, i6=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n$	1	$0.0714 \cdot n^{-2}$	read F[i5] (i0=0, i5=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i6, i7] (i0=0, i6=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n$	1	$0.0714 \cdot n^{-2}$	read A[i4, i3] (i0=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n - 1$	1	$0.0714 \cdot n^{-2}$	read F[i5] (i0=0)

The rank-1-update matrix A is walked twice (by rows, then transposed): ramps from $\sim(21/8)n$ up to $(1/8)n^2 + (3/4)n$ lines on both read A[i4,i3] and read A[i6,i7] give $d = 3.0$, headroom +1.0 single-shot, same mechanism as mvt.

gesummv — infinite-repeat [exact]

Accesses $A(n) = 8 \cdot n^2 + 5 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.125 \cdot n^3 + 0.0765 \cdot n^{2.5} + 17.6 \cdot n^2 + 1.22 \cdot n^{1.5} + 14.3 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.125	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	$(1/4) \cdot n^2 + (-9/2) \cdot n + 8$	0.0312	read C[i1, i2] (i0=0); read E[i1, i2] (i0=0)
$n^{2.5}$	0.067	level	$(3/8) \cdot n + 1$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0137	read D[i2] (i0=0, i1=1); read D[i2] (i0=0)
$n^{2.5}$	0.00957	level	$(3/8) \cdot n + 3$	$(1/64) \cdot n^2 + (-1/4) \cdot n$	0.00195	read D[i2] (i0=0, i1=0); read D[i2] (i0=0)
n^2	3.5	level	4	$(7/4) \cdot n^2 + (-7/8) \cdot n$	0.219	read D[i2] (i0=0, i1=0); read D[i2] (i0=0)
n^2	3.03	level	3	$(7/4) \cdot n^2$	0.219	write A[i1] (i0=0); write B[i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	2.25	level	4	$(9/8) \cdot n^2 + (-7/8) \cdot n$	0.141	read B[i1] (i0=0, i2=0); read A[i1] (i0=0) (+1)
n^2	1.96	level	5	$(7/8) \cdot n^2$	0.109	read E[i1, i2] (i0=0)
n^2	1.96	level	5	$(7/8) \cdot n^2$	0.109	read C[i1, i2] (i0=0)
n^2	1.75	level	4	$(7/8) \cdot n^2 + (7/8) \cdot n$	0.109	read B[i1] (i0=0, i2=0); read B[i1] (i0=0) (+1)
n^2	1	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	$2 \cdot n - 4$	$0.25/n$	read C[i1, i2] (i0=0); read E[i1, i2] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	$n - 2$	$0.125/n$	read E[i1, i2] (i0=0, i2=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	$n - 2$	$0.125/n$	read C[i1, i2] (i0=0, i2=0)
n^2	0.433	level	3	$(1/4) \cdot n^2$	0.0312	write A[i1] (i0=0, i2=0); write B[i1] (i0=0, i2=0) (+2)
n^2	0.25	level	4	$(1/8) \cdot n^2 + (7/8) \cdot n$	0.0156	read D[i2] (i0=0, i1=0); read D[i2] (i0=0)
n^2	0.125	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	$(1/4) \cdot n - 4$	$0.0312/n$	read C[i1, i2] (i0=0, i1=0); read E[i1, i2] (i0=0, i1=0)
n^2	0.125	level	$(1/4) \cdot n^2 + (17/8) \cdot n + (21/8)$	$(1/4) \cdot n + (-9/4)$	$0.0312/n$	read C[i1, i2] (i0=0); read E[i1, i2] (i0=0)
n^2	0.125	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	$(1/4) \cdot n - 4$	$0.0312/n$	read C[i1, i2] (i0=0); read E[i1, i2] (i0=0)
n^2	0.125	level	$(1/4) \cdot n^2 + (-13/8) \cdot n$	$(1/4) \cdot n - 4$	$0.0312/n$	write A[i1] (i0=0); write B[i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	$0.109/n$	read D[i2] (i0=0, i1=1); read D[i2] (i0=0)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	$0.109/n$	read D[i2] (i0=0, i1=1, i2=0); read D[i2] (i0=0, i2=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n$	$0.0156/n$	read D[i2] (i0=0, i1=0); read D[i2] (i0=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n$	$0.0156/n$	read D[i2] (i0=0, i1=0, i2=0); read D[i2] (i0=0, i2=0)
n^1	2.83	level	2	$2 \cdot n$	$0.25/n$	write A[i1] (i0=0); read A[i1] (i0=0, i2=0) (+1)
n^1	2.47	level	2	$(7/4) \cdot n$	$0.219/n$	write B[i1] (i0=0); read B[i1] (i0=0)
n^1	1	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	2	$0.25 \cdot n^{-2}$	read C[i1, i2] (i0=0, i1=0); read E[i1, i2] (i0=0, i1=0)
n^1	1	level	$(1/4) \cdot n^2 + (17/8) \cdot n + (21/8)$	2	$0.25 \cdot n^{-2}$	read C[i1, i2] (i0=0); read E[i1, i2] (i0=0)
n^1	1	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	2	$0.25 \cdot n^{-2}$	read C[i1, i2] (i0=0); read E[i1, i2] (i0=0)
n^1	1	level	$(1/4) \cdot n^2 + (-13/8) \cdot n$	2	$0.25 \cdot n^{-2}$	write A[i1] (i0=0, i1=0); write B[i1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1	level	$(1/4) \cdot n^2 + (-13/8) \cdot n$	2	$0.25 \cdot n^{-2}$	write A[i1] (i0=0); write B[i1] (i0=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.109/n$	write B[i1] (i0=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read E[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read C[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (17/8) \cdot n + (21/8)$	1	$0.125 \cdot n^{-2}$	read E[i1, i2] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read E[i1, i2] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (17/8) \cdot n + (21/8)$	1	$0.125 \cdot n^{-2}$	read C[i1, i2] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (3/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read C[i1, i2] (i0=0, i2=0)
n^1	0.125	level	1	$(1/8) \cdot n$	$0.0156/n$	write B[i1] (i0=0)

Both matrices return as wraparound terms: coefficient $0.125 = 2 \times 0.0625$ at the two-matrix footprint boundary — twice atax’s jump because gesummv streams two matrices.

gesummv — single-shot [exact]

Accesses $A(n) = 8 \cdot n^2 + 5 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order $n^{2.5}$, headroom **+0.5**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0765 \cdot n^{2.5} + 15.1 \cdot n^2 + 1.22 \cdot n^{1.5} + 6.3 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.067	level	$(3/8) \cdot n + 1$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0137	read D[i2] (i0=0)
$n^{2.5}$	0.00957	level	$(3/8) \cdot n + 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00195	read D[i2] (i0=0)
n^2	6	level	4	$3 \cdot n^2$	0.375	read D[i2] (i0=0, i2=0); read B[i1] (i0=0, i2=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	3.91	level	5	$(7/4) \cdot n^2$	0.219	read C[i1, i2] (i0=0); read E[i1, i2] (i0=0)
n^2	3.46	level	3	$2 \cdot n^2$	0.25	write A[i1] (i0=0); write B[i1] (i0=0)
n^2	1.75	level	4	$(7/8) \cdot n^2$	0.109	read D[i2] (i0=0)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read D[i2] (i0=0)
$n^{1.5}$	0.536	level	$(3/8) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read D[i2] (i0=0, i2=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n - 1$	0.0156/n	read D[i2] (i0=0)
$n^{1.5}$	0.0765	level	$(3/8) \cdot n + 3$	$(1/8) \cdot n - 1$	0.0156/n	read D[i2] (i0=0, i2=0)
n^1	2.83	level	2	$2 \cdot n$	0.25/n	read A[i1] (i0=0, i2=0); read B[i1] (i0=0)
n^1	2.47	level	2	$(7/4) \cdot n$	0.219/n	write A[i1] (i0=0); write B[i1] (i0=0)
n^1	1	level	1	n	0.125/n	write B[i1] (i0=0)

Two matrices are streamed once per pass; single-shot sees only vector reuse ($d = 2.5$).

gramschmidt — infinite-repeat [exact]

Accesses $A(n) = 3 \cdot n^3 + (1/2) \cdot n^2 + (1/2) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.016 \cdot n^4 + 2.74 \cdot n^{3.5} + 2.81 \cdot n^3 + 23.9 \cdot n^{2.5} + 26.8 \cdot n^2 + 64.4 \cdot n^{1.5} + 29.2 \cdot n^1 + 77.8 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0101	ramp	$5 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n$	$(5/128) \cdot n^3$ $+ (-$ $205/128) \cdot n^2$ $+ (265/16) \cdot n -$ 15	0.013	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00203	ramp	$5 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n$	$(1/128) \cdot n^3$ + (- $41/128) \cdot n^2$ + $(53/16) \cdot n$ - 3	0.0026	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^4	0.00193	ramp	$6 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(1/128) \cdot n^3$ + (- $57/128) \cdot n^2$ + $(103/16) \cdot n -$ 6	0.0026	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^4	0.0019	ramp	$5 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(1/8) \cdot n - 1$	$(1/128) \cdot n^3$ + (- $57/128) \cdot n^2$ + $(103/16) \cdot n -$ 6	0.0026	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{3.5}$	0.795	level	$2 \cdot n + 1$	$(9/16) \cdot n^3$ + $(-9/8) \cdot n^2$ + n	0.188	read C[i6, i1] (i0=0, i4=0); read C[i5, i1] (i0=0, i5=0) (+2)
$n^{3.5}$	0.619	level	$2 \cdot n$	$(7/16) \cdot n^3$ + (- $119/16) \cdot n^2$ + $(287/8) \cdot n$ - 35	0.146	read A[i6, i4] (i0=0, i4=6, i6=0); read A[i6, i4] (i0=0)
$n^{3.5}$	0.619	level	$2 \cdot n + 1$	$(7/16) \cdot n^3$ + $(-7/8) \cdot n^2$	0.146	read C[i6, i1] (i0=0, i6=0); read C[i6, i1] (i0=0)
$n^{3.5}$	0.475	level	$2 \cdot n + 1$	$(43/128) \cdot n^3$ + (- $547/128) \cdot n^2$ + $(223/16) \cdot n -$ 10	0.112	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{3.5}$	0.0884	level	$2 \cdot n$	$(1/16) \cdot n^3$ + $(1/16) \cdot n^2$ - $6 \cdot n + 27$	0.0208	read A[i3, i1] (i0=0); read A[i6, i4] (i0=0, i4=6, i6=0) (+3)
$n^{3.5}$	0.0663	level	$2 \cdot n + 1$	$(3/64) \cdot n^3$ + (- $75/64) \cdot n^2$ + $(57/8) \cdot n -$ 6	0.0156	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.0663	level	$2 \cdot n + 1$	$(3/64) \cdot n^3 + (-75/64) \cdot n^2 + (57/8) \cdot n - 6$	0.0156	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{3.5}$	0.011	level	$2 \cdot n + 1$	$(1/128) \cdot n^3 + (-25/128) \cdot n^2 + (19/16) \cdot n - 1$	0.0026	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	1	level	4	$(1/2) \cdot n^3 - 2 \cdot n^2 + (5/2) \cdot n - 1$	0.167	read B[i1, i4] (i0=0)
n^3	0.663	level	3	$(49/128) \cdot n^3 + (-105/16) \cdot n^2 + 28 \cdot n$	0.128	write A[i6, i4] (i0=0)
n^3	0.213	ramp	$3 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(7/8) \cdot n^2 + (-71/8) \cdot n + 8$	0.292/n	read A[i2, i1] (i0=0, i2=0); read A[i2, i1] (i0=0)
n^3	0.152	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(5/8) \cdot n^2 + (-85/8) \cdot n + 10$	0.208/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	0.152	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(5/8) \cdot n^2 + (-85/8) \cdot n + 10$	0.208/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	0.0947	level	3	$(7/128) \cdot n^3 + (-21/16) \cdot n^2 + 7 \cdot n$	0.0182	write A[i6, i4] (i0=0)
n^3	0.0947	level	3	$(7/128) \cdot n^3 + (-7/16) \cdot n^2$	0.0182	write A[i6, i4] (i0=0)
n^3	0.0645	ramp	$(5/16) \cdot n^2 + (-1/2) \cdot n + 10 \rightarrow (5/16) \cdot n^2 + (1/2) \cdot n - 22$	$(1/8) \cdot n^2 - 3 \cdot n$	0.0417/n	write C[i3, i1] (i0=0, i3=0); write C[i3, i1] (i0=0)
n^3	0.0514	ramp	$(1/8) \cdot n^2 + (33/8) \cdot n + 23 \rightarrow (5/16) \cdot n^2 + (-27/8) \cdot n + 23$	$(1/8) \cdot n^2 - 4 \cdot n$	0.0417/n	read A[i5, i4] (i0=0, i1=0, i5=0); read A[i5, i4] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0442	level	$(1/8) \cdot n^2 + (9/8) \cdot n$	$(1/8) \cdot n^2 + (-25/8) \cdot n + 3$	$0.0417/n$	read A[i5, i4] (i0=0, i1=1, i5=0); read A[i5, i4] (i0=0, i1=1)
n^3	0.0303	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0417/n$	read A[i5, i4] (i0=0, i4=0, i5=0); read A[i5, i4] (i0=0, i4=0)
n^3	0.0303	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0417/n$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	0.03	ramp	$5 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(1/8) \cdot n^2 + (-25/8) \cdot n + 3$	$0.0417/n$	read A[i5, i4] (i0=0, i4=7, i5=0); read A[i5, i4] (i0=0, i4=7)
n^3	0.0295	ramp	$3 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0417/n$	read A[i2, i1] (i0=0, i2=0); read A[i2, i1] (i0=0)
n^3	0.0295	ramp	$4 \cdot n + 4 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 3$	$(1/8) \cdot n^2 + (-13/4) \cdot n + 6$	$0.0417/n$	read A[i5, i4] (i0=0)
n^3	0.0292	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 1$	$(1/8) \cdot n^2 + (-25/8) \cdot n + 3$	$0.0417/n$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
n^3	0.0292	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 1$	$(1/8) \cdot n^2 + (-25/8) \cdot n + 3$	$0.0417/n$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	0.0262	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(3/64) \cdot n^2 + (-15/8) \cdot n + 18$	$0.0156/n$	write B[i1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0135	level	3	$(1/128) \cdot n^3 + (-3/16) \cdot n^2 + 8 \cdot n$	0.0026	write C[i3, i1] (i0=0, i1=0, i3=0); write A[i6, i4] (i0=0, i1=0) (+1)
n^3	0.0122	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/64) \cdot n^2 + (-15/8) \cdot n + 18$	0.0156/n	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0)
n^3	0.00437	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/128) \cdot n^2 + (-13/64) \cdot n + (153/128)$	0.0026/n	write B[i1, i4] (i0=0)
n^3	0.00437	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.0026/n	write B[i1, i4] (i0=0)
n^3	0.00437	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/128) \cdot n^2 + (-21/64) \cdot n + (425/128)$	0.0026/n	write B[i1, i4] (i0=0)
n^3	0.00437	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.0026/n	write B[i1, i4] (i0=0)
n^3	0.00204	ramp	$5 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.0026/n	read A[i5, i4] (i0=0)
n^3	0.00194	ramp	$6 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.0026/n	read A[i5, i4] (i0=0)
$n^{2.5}$	7.42	level	$2 \cdot n$	$(21/4) \cdot n^2 + (-105/4) \cdot n + 21$	1.75/n	read A[i6, i4] (i0=0)
$n^{2.5}$	5.3	level	$2 \cdot n + 1$	$(15/4) \cdot n^2 + (-75/4) \cdot n + 15$	1.25/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{2.5}$	1.41	level	$2 \cdot n - 1$	$n^2 - 2 \cdot n + 1$	0.333/n	read C[i5, i1] (i0=0, i4=0, i5=0); read C[i5, i1] (i0=0, i4=0)
$n^{2.5}$	1.34	ramp	$2 \cdot n + 4 \rightarrow 3 \cdot n + 1$	$(7/8) \cdot n^2 + (-35/4) \cdot n + 14$	0.292/n	write C[i3, i1] (i0=0)
$n^{2.5}$	1.24	level	$2 \cdot n$	$(7/8) \cdot n^2 + (-63/8) \cdot n + 7$	0.292/n	read A[i6, i4] (i0=0, i4=6)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.24	level	$2 \cdot n$	$(7/8) \cdot n^2 + (-15/8) \cdot n + 1$	$0.292/n$	read A[i5, i4] (i0=0, i4=0, i5=0); read A[i5, i4] (i0=0, i4=0)
$n^{2.5}$	1.24	level	$2 \cdot n + 1$	$(7/8) \cdot n^2 + (-63/8) \cdot n + 7$	$0.292/n$	read A[i5, i4] (i0=0, i1=0, i5=0); read A[i5, i4] (i0=0, i1=0)
$n^{2.5}$	1.21	ramp	$n + 2 \rightarrow 2 \cdot n - 1$	$n^2 - 3 \cdot n + 2$	$0.333/n$	read A[i3, i1] (i0=0)
$n^{2.5}$	1.06	level	$2 \cdot n + 1$	$(3/4) \cdot n^2 + (-27/4) \cdot n + 6$	$0.25/n$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	1.06	level	$2 \cdot n$	$(3/4) \cdot n^2 + (21/4) \cdot n - 21$	$0.25/n$	read A[i6, i4] (i0=0, i6=0); read A[i6, i4] (i0=0)
$n^{2.5}$	0.541	level	$2 \cdot n$	$(49/128) \cdot n^2 + (-79/16) \cdot n + 15$	$0.128/n$	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0)
$n^{2.5}$	0.541	level	$2 \cdot n + 1$	$(49/128) \cdot n^2 + (-7/16) \cdot n$	$0.128/n$	write B[i1, i4] (i0=0)
$n^{2.5}$	0.177	level	$2 \cdot n + 1$	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	$0.0417/n$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	0.0773	level	$2 \cdot n$	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	$0.0182/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.0773	level	$2 \cdot n + 1$	$(7/128) \cdot n^2 + (-7/16) \cdot n$	$0.0182/n$	write B[i1, i4] (i0=0)
n^2	7.58	level	3	$(35/8) \cdot n^2 - 20 \cdot n$	$1.46/n$	write A[i6, i4] (i0=0)
n^2	2	level	4	$n^2 - 2 \cdot n + 1$	$0.333/n$	read B[i1, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.52	level	3	$(7/8) \cdot n^2 - 7 \cdot n$	$0.292/n$	write A[i6, i4] (i0=0, i4=6); read C[i6, i1] (i0=0, i6=0)
n^2	1.52	level	3	$(7/8) \cdot n^2 - 7 \cdot n$	$0.292/n$	write A[i6, i4] (i0=0, i1=0)
n^2	1.3	level	3	$(3/4) \cdot n^2$	$0.25/n$	write A[i6, i4] (i0=0, i4=0)
n^2	1.08	level	3	$(5/8) \cdot n^2 - 5 \cdot n$	$0.208/n$	write A[i6, i4] (i0=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.146/n$	read B[i1, i4] (i0=0, i6=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (9/4) \cdot n + (-121/16)$	n	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0, i1=0, i2=0); read A[i2, i1] (i0=0, i1=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (-3/8) \cdot n - 1$	n	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0, i1=0, i2=0); read A[i2, i1] (i0=0, i1=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (5/4) \cdot n + (-105/16)$	n - 1	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=0, i4=7, i5=0); read A[i5, i4] (i0=0, i1=0, i4=7)
n^2	0.559	level	$(5/16) \cdot n^2 + (-11/8) \cdot n + 7$	n - 1	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=0, i4=7, i5=0); read A[i5, i4] (i0=0, i1=0, i4=7)
n^2	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n - 14$	n	$0.333 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0); write C[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.559	level	$(5/16) \cdot n^2 + (17/8) \cdot n + (-87/16)$	n	$0.333 \cdot n^{-2}$	write C[i3, i1] (i0=0, i1=0, i3=0); write C[i3, i1] (i0=0, i1=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (-1/2) \cdot n + 2$	n	$0.333 \cdot n^{-2}$	write C[i3, i1] (i0=0, i1=0, i3=0); write C[i3, i1] (i0=0, i1=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-231/16)$	n	$0.333 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0); write C[i3, i1] (i0=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n - 7$	n	$0.333 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0); write C[i3, i1] (i0=0)
n^2	0.489	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 14$	$0.292 \cdot n^{-2}$	write B[i1, i1] (i0=0)
n^2	0.433	level	3	$(1/4) \cdot n^2 - 2 \cdot n$	$0.0833/n$	write A[i6, i4] (i0=0, i4=0)
n^2	0.419	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(3/4) \cdot n - 12$	$0.25 \cdot n^{-2}$	write B[i1, i4] (i0=0)
n^2	0.419	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(3/4) \cdot n - 12$	$0.25 \cdot n^{-2}$	write B[i1, i4] (i0=0)
n^2	0.359	ramp	$(1/8) \cdot n^2 + (9/8) \cdot n + 1$ → $(1/8) \cdot n^2 + (17/8) \cdot n - 2$	$n - 2$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=1, i4=6, i5=0); read A[i5, i4] (i0=0, i1=1, i4=6)
n^2	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=1, i5=0); read A[i5, i4] (i0=0, i1=1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (25/8) \cdot n + 7$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=0, i5=0); read A[i5, i4] (i0=0, i1=0)
n^2	0.354	level	$(1/8) \cdot n^2 + 4 \cdot n + (7/8)$	n	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=0, i5=0); read A[i5, i4] (i0=0, i1=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0, i1=1, i2=0); read A[i2, i1] (i0=0, i1=1)
n^2	0.217	level	3	$(1/8) \cdot n^2 - n$	$0.0417/n$	read C[i6, i1] (i0=0, i4=0); write A[i6, i4] (i0=0, i4=6) (+2)
n^2	0.217	level	3	$(1/8) \cdot n^2 - n$	$0.0417/n$	write A[i6, i4] (i0=0, i1=0)
n^2	0.214	ramp	$3 \cdot n + 1 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(7/8) \cdot n - 8$	$0.292 \cdot n^{-2}$	read A[i2, i1] (i0=0)
n^2	0.182	ramp	$3 \cdot n + 1 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/4) \cdot n - 12$	$0.25 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0, i4=7) (+1)
n^2	0.152	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(5/8) \cdot n - 10$	$0.208 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	$0.0208/n$	read B[i1, i4] (i0=0, i6=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/8) \cdot n + (-17/8)$	$0.0417 \cdot n^{-2}$	write B[i1, i4] (i0=0, i1=7)
n^2	0.0699	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^{-2}$	write B[i1, i4] (i0=0, i1=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0699	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/8) \cdot n + (-9/8)$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0, i1=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0, i1=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/8) \cdot n + (-17/8)$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/8) \cdot n + (-17/8)$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/8) \cdot n + (-17/8)$	$0.0417 \cdot n^2$	write B[i1, i1] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	write B[i1, i1] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	$(1/8) \cdot n + (-17/8)$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0, i4=0)
n^2	0.0699	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	write B[i1, i4] (i0=0, i4=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	read A[i5, i4] (i0=0, i1=1)
n^2	0.0304	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	read A[i5, i4] (i0=0)
n^2	0.0304	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0301	ramp	$5 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0301	ramp	$5 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	read A[i5, i4] (i0=0, i5=0)
n^2	0.0296	ramp	$3 \cdot n + 1 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 2$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	read A[i2, i1] (i0=0)
n^2	0.0293	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	read A[i5, i4] (i0=0)
$n^{1.5}$	8.66	level	$3 \cdot n + 2$	$5 \cdot n - 5$	$1.67 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	8.49	level	$2 \cdot n + 1$	$6 \cdot n - 5$	$2 \cdot n^{-2}$	write B[i1, i1] (i0=0, i1=0); write C[i3, i1] (i0=0, i1=0) (+3)
$n^{1.5}$	7.27	ramp	$n + 2 \rightarrow 2 \cdot n$ - 1	$6 \cdot n - 12$	$2 \cdot n^{-2}$	read A[i2, i1] (i0=0)
$n^{1.5}$	7.07	level	$2 \cdot n + 2$	$5 \cdot n - 10$	$1.67 \cdot n^{-2}$	write C[i3, i1] (i0=0)
$n^{1.5}$	5.3	level	$2 \cdot n$	$(15/4) \cdot n - 15$	$1.25 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	2.83	level	$2 \cdot n + 2$	$2 \cdot n$	$0.667 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0); write C[i3, i1] (i0=0)
$n^{1.5}$	1.86	ramp	$3 \cdot n + 3 \rightarrow 4 \cdot n$	$n - 2$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
$n^{1.5}$	1.73	level	$3 \cdot n + 2$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	1.73	level	$3 \cdot n + 2$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0, i2=0); read A[i2, i1] (i0=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 2$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0, i5=0); read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	1.52	level	$3 \cdot n + 2$	$(7/8) \cdot n + (-63/8)$	$0.292 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0)
$n^{1.5}$	1.52	level	$3 \cdot n + 2$	$(7/8) \cdot n - 7$	$0.292 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0)
$n^{1.5}$	1.41	level	$2 \cdot n$	n	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0, i1=0); read A[i3, i1] (i0=0, i1=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.24	level	$2 \cdot n - 1$	$(7/8) \cdot n - 1$	$0.292 \cdot n^{-2}$	read C[i5, i1] (i0=0, i4=0)
$n^{1.5}$	1.24	level	$2 \cdot n$	$(7/8) \cdot n - 7$	$0.292 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 3$	$(7/8) \cdot n +$ $(-63/8)$	$0.292 \cdot n^{-2}$	write C[i3, i1] (i0=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 3$	$(7/8) \cdot n - 7$	$0.292 \cdot n^{-2}$	write C[i3, i1] (i0=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 1$	$(7/8) \cdot n - 8$	$0.292 \cdot n^{-2}$	write B[i1, i4] (i0=0, i4=0)
$n^{1.5}$	1.21	ramp	$n + 2 \rightarrow 2 \cdot n$ $- 1$	$n - 2$	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0)
$n^{1.5}$	1.21	ramp	$n + 2 \rightarrow 2 \cdot n$ $- 1$	$n - 2$	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0)
$n^{1.5}$	1.21	ramp	$n + 2 \rightarrow 2 \cdot n$ $- 1$	$n - 2$	$0.333 \cdot n^{-2}$	read A[i3, i1] (i0=0, i1=0)
$n^{1.5}$	1.06	level	$2 \cdot n$	$(3/4) \cdot n$	$0.25 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	1	level	$n + 1$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i3, i1] (i0=0, i3=0)
$n^{1.5}$	0.884	level	$2 \cdot n$	$(5/8) \cdot n - 5$	$0.208 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.177	level	$2 \cdot n$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	0.177	level	$2 \cdot n$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.177	level	$2 \cdot n - 1$	$(1/8) \cdot n$	$0.0417 \cdot n^{-2}$	read C[i5, i1] (i0=0, i4=0)
n^1	3.35	level	$(5/16) \cdot n^2$ $+ (1/2) \cdot n$	6	$2 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	3.35	level	$(5/16) \cdot n^2$ $+ (1/2) \cdot n$	6	$2 \cdot n^{-3}$	write B[i1, i4] (i0=0)
n^1	2.8	level	$(5/16) \cdot n^2$ $+ (25/8) \cdot n +$ $(-7/16)$	5	$1.67 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	2.8	level	$(5/16) \cdot n^2$ $+ (1/2) \cdot n$	5	$1.67 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	1.68	level	$(5/16) \cdot n^2$ $+ (25/8) \cdot n +$ $(-7/16)$	3	$1 \cdot n^{-3}$	write B[i1, i1] (i0=0, i1=0); write B[i1, i1] (i0=0, i1=6) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.52	level	3	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read B[i1, i4] (i0=0, i4=0, i6=0)
n^1	1.12	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	2	$0.667 \cdot n^{-3}$	write B[i1, i1] (i0=0, i1=0); write B[i1, i1] (i0=0, i1=6)
n^1	0.559	level	$(5/16) \cdot n^2 + (-11/8) \cdot n + 7$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=0, i4=7)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i4=7)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i1=7)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i1=7)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (11/4) \cdot n + (-3/4)$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0, i1=7)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0, i1=7)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0, i1=7)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (11/4) \cdot n + (-3/4)$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i4=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i4=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i1=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i1] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (25/8) \cdot n + (-7/16)$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i1=7, i4=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i1=7, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n - 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1)
n^1	0.354	level	$(1/8) \cdot n^2 + (25/8) \cdot n + 7$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (17/8) \cdot n - 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n - 1$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0, i1=1)
n^1	0.217	level	3	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i1, i4] (i0=0, i4=0, i6=0)
$n^{0.5}$	9.9	level	$2 \cdot n$	7	$2.33 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=0, i4=0); read A[i5, i4] (i0=0, i1=0)
$n^{0.5}$	8.66	level	$3 \cdot n + 1$	5	$1.67 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	8.49	level	$2 \cdot n$	6	$2 \cdot n^{-3}$	read A[i2, i1] (i0=0, i2=0)
$n^{0.5}$	8.49	level	$2 \cdot n + 1$	6	$2 \cdot n^{-3}$	write B[i1, i4] (i0=0, i4=0)
$n^{0.5}$	7.07	level	$2 \cdot n + 2$	5	$1.67 \cdot n^{-3}$	write C[i3, i1] (i0=0)
$n^{0.5}$	7.07	level	$2 \cdot n + 2$	5	$1.67 \cdot n^{-3}$	write C[i3, i1] (i0=0, i3=0)
$n^{0.5}$	6	level	n	6	$2 \cdot n^{-3}$	read A[i2, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	2	level	$4 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7, i5=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0, i4=7) (+1)
$n^{0.5}$	1.73	level	$3 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 2$	1	$0.333 \cdot n^{-3}$	write C[i3, i1] (i0=0, i3=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0, i2=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0, i2=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 3$	1	$0.333 \cdot n^{-3}$	write C[i3, i1] (i0=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0, i2=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0, i2=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i4=0)
$n^{0.5}$	1.41	level	$2 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	write B[i1, i4] (i0=0, i4=0)
$n^{0.5}$	1	level	n	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0)
$n^{0.5}$	1	level	n	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0)
$n^{0.5}$	1	level	n	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0)
$n^{0.5}$	1	level	n	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0)
$n^{0.5}$	1	level	$n + 1$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0, i1=0); read A[i3, i1] (i0=0, i1=0)

Column orthogonalization re-reads the factor panel: read A[i5,i4] ramps from $\sim 5n$ to $(1/8)n^2 + (9/8)n$ lines ($d = 4.0$, headroom +1.0 — corrected from the old +0.5, whose fits ran over

corrupted terms). The large $n^{3.5}$ coefficient (2.74, read $C[i5, i1]$ at $2n + 1$ lines) means the pre-asymptotic behavior is dominated by column-pair reuse until the panel term takes over.

gramschmidt — single-shot [exact]

Accesses $A(n) = 3 \cdot n^3 + (1/2) \cdot n^2 + (1/2) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0161 \cdot n^4 + 2.74 \cdot n^{3.5} + 2.61 \cdot n^3 + 22.4 \cdot n^{2.5} + 12 \cdot n^2 + 50.7 \cdot n^{1.5} + 52.1 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0122	ramp	$5 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n$	$(3/64) \cdot n^3$ + (- $123/64) \cdot n^2$ + $(159/8) \cdot n$ - 18	0.0156	read $A[i5,$ $i4]$ ($i0=0,$ $i5=0$); read $A[i5, i4]$ ($i0=0$)
n^4	0.00203	ramp	$5 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n$	$(1/128) \cdot n^3$ + (- $41/128) \cdot n^2$ + $(53/16) \cdot n$ - 3	0.0026	read $A[i5,$ $i4]$ ($i0=0,$ $i5=0$); read $A[i5, i4]$ ($i0=0$)
n^4	0.00193	ramp	$6 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(1/128) \cdot n^3$ + (- $57/128) \cdot n^2$ + $(103/16) \cdot n -$ 6	0.0026	read $A[i5,$ $i4]$ ($i0=0,$ $i5=0$); read $A[i5, i4]$ ($i0=0$)
$n^{3.5}$	1.41	level	$2 \cdot n + 1$	$n^3 -$ $2 \cdot n^2 + n$	0.333	read $C[i5,$ $i1]$ ($i0=0,$ $i5=0$); read $C[i5, i1]$ ($i0=0$) (+1)
$n^{3.5}$	0.619	level	$2 \cdot n$	$(7/16) \cdot n^3$ + (- $105/16) \cdot n^2$ + $(217/8) \cdot n$ - 21	0.146	read $A[i6,$ $i4]$ ($i0=0,$ $i4=6$); read $A[i6, i4]$ ($i0=0$)
$n^{3.5}$	0.541	level	$2 \cdot n + 1$	$(49/128) \cdot n^3$ + (- $777/128) \cdot n^2$ + $(427/16) \cdot n -$ 21	0.128	read $A[i5,$ $i4]$ ($i0=0,$ $i5=0$); read $A[i5, i4]$ ($i0=0$)
$n^{3.5}$	0.0884	level	$2 \cdot n$	$(1/16) \cdot n^3$ + $(1/16) \cdot n^2$ + $(-49/8) \cdot n$ + 21	0.0208	read $A[i6,$ $i4]$ ($i0=0,$ $i4=6,$ $i6=0$); read $A[i6, i4]$ ($i0=0,$ $i4=6$) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.0773	level	$2 \cdot n + 1$	$(7/128) \cdot n^3 + (-63/128) \cdot n^2 + (7/16) \cdot n$	0.0182	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	1	level	4	$(1/2) \cdot n^3 - n^2 + (1/2) \cdot n$	0.167	read B[i1, i4] (i0=0)
n^3	0.758	level	3	$(7/16) \cdot n^3 + (-49/8) \cdot n^2 + 21 \cdot n$	0.146	write A[i6, i4] (i0=0, i4=6); write A[i6, i4] (i0=0)
n^3	0.244	ramp	$3 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$n^2 - 11 \cdot n + 10$	0.333/n	read A[i2, i1] (i0=0, i2=0); read A[i2, i1] (i0=0)
n^3	0.182	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(3/4) \cdot n^2 + (-51/4) \cdot n + 12$	0.25/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	0.152	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(5/8) \cdot n^2 + (-85/8) \cdot n + 10$	0.208/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
n^3	0.108	level	3	$(1/16) \cdot n^3 + (-3/8) \cdot n^2$	0.0208	write A[i6, i4] (i0=0, i4=6); write A[i6, i4] (i0=0)
n^3	0.0303	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0417/n	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
n^3	0.0303	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0417/n	read A[i5, i4] (i0=0, i4=0, i5=0); read A[i5, i4] (i0=0, i4=0)
n^3	0.0303	ramp	$4 \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0417/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.03	ramp	$5 \cdot n + 3 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(1/8) \cdot n^2 +$ $(-25/8) \cdot n +$ 3	0.0417/n	read A[i5, i4] (i0=0, i4=7, i5=0); read A[i5, i4] (i0=0, i4=7)
n^3	0.0295	ramp	$4 \cdot n + 4 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 3$	$(1/8) \cdot n^2 +$ $(-13/4) \cdot n +$ 6	0.0417/n	read A[i5, i4] (i0=0)
n^3	0.0123	ramp	$4 \cdot n + 2 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(3/64) \cdot n^2$ $+ (-7/4) \cdot n +$ 16	0.0156/n	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0)
n^3	0.00204	ramp	$5 \cdot n + 3 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 1$	$(1/128) \cdot n^2$ $+ (-5/16) \cdot n$ $+ 3$	0.0026/n	read A[i5, i4] (i0=0)
n^3	0.00194	ramp	$6 \cdot n + 3 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 2$	$(1/128) \cdot n^2$ $+ (-7/16) \cdot n$ $+ 6$	0.0026/n	read A[i5, i4] (i0=0)
$n^{2.5}$	7.42	level	$2 \cdot n$	$(21/4) \cdot n^2$ $+ (-105/4) \cdot n$ $+ 21$	1.75/n	read A[i6, i4] (i0=0)
$n^{2.5}$	5.3	level	$2 \cdot n + 1$	$(15/4) \cdot n^2$ $+ (-75/4) \cdot n$ $+ 15$	1.25/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{2.5}$	1.41	level	$2 \cdot n - 1$	$n^2 - 2 \cdot n +$ 1	0.333/n	read C[i5, i1] (i0=0, i4=0, i5=0); read C[i5, i1] (i0=0, i4=0)
$n^{2.5}$	1.34	ramp	$2 \cdot n + 4 \rightarrow$ $3 \cdot n + 1$	$(7/8) \cdot n^2 +$ $(-35/4) \cdot n +$ 14	0.292/n	write C[i3, i1] (i0=0)
$n^{2.5}$	1.24	level	$2 \cdot n$	$(7/8) \cdot n^2 +$ $(-7/8) \cdot n$	0.292/n	read A[i5, i4] (i0=0, i4=0, i5=0); read A[i5, i4] (i0=0, i4=0)
$n^{2.5}$	1.21	ramp	$n + 2 \rightarrow 2 \cdot n$ - 1	$n^2 - 2 \cdot n$	0.333/n	read A[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.06	level	$2 \cdot n + 1$	$(3/4) \cdot n^2 + (-27/4) \cdot n + 6$	$0.25/n$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	1.06	level	$2 \cdot n$	$(3/4) \cdot n^2 + (21/4) \cdot n - 21$	$0.25/n$	read A[i6, i4] (i0=0, i6=0); read A[i6, i4] (i0=0)
$n^{2.5}$	0.884	level	$2 \cdot n + 1$	$(5/8) \cdot n^2 + (-5/8) \cdot n$	$0.208/n$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{2.5}$	0.619	level	$2 \cdot n + 1$	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.146/n$	write B[i1, i4] (i0=0)
$n^{2.5}$	0.541	level	$2 \cdot n$	$(49/128) \cdot n^2 + (-79/16) \cdot n + 15$	$0.128/n$	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0)
$n^{2.5}$	0.177	level	$2 \cdot n + 1$	$(1/8) \cdot n^2 + (-1/8) \cdot n$	$0.0417/n$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	0.0773	level	$2 \cdot n$	$(7/128) \cdot n^2 + (-7/16) \cdot n$	$0.0182/n$	read A[i5, i4] (i0=0)
n^2	9.09	level	3	$(21/4) \cdot n^2 - 21 \cdot n$	$1.75/n$	write A[i6, i4] (i0=0)
n^2	1.3	level	3	$(3/4) \cdot n^2$	$0.25/n$	write A[i6, i4] (i0=0)
n^2	0.866	level	3	$(1/2) \cdot n^2 + (-1/2) \cdot n$	$0.167/n$	read C[i5, i1] (i0=0, i5=0); read C[i6, i1] (i0=0) (+1)
n^2	0.245	ramp	$3 \cdot n + 1 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$n - 10$	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0)
n^2	0.183	ramp	$3 \cdot n + 1 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(3/4) \cdot n - 11$	$0.25 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0, i4=7) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.152	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(5/8) \cdot n - 10$	$0.208 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.0304	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.0304	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0301	ramp	$5 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0301	ramp	$5 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
n^2	0.0293	ramp	$4 \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	8.66	level	$3 \cdot n + 2$	$5 \cdot n - 5$	$1.67 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{1.5}$	8.49	level	$2 \cdot n + 2$	$6 \cdot n - 12$	$2 \cdot n^{-2}$	write C[i3, i1] (i0=0)
$n^{1.5}$	8.48	ramp	$n + 2 \rightarrow 2 \cdot n - 1$	$7 \cdot n - 14$	$2.33 \cdot n^{-2}$	read A[i2, i1] (i0=0)
$n^{1.5}$	5.3	level	$2 \cdot n$	$(15/4) \cdot n - 15$	$1.25 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	1.86	ramp	$3 \cdot n + 3 \rightarrow 4 \cdot n$	$n - 2$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
$n^{1.5}$	1.73	level	$3 \cdot n + 2$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	1.73	level	$3 \cdot n + 2$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0, i2=0); read A[i2, i1] (i0=0)
$n^{1.5}$	1.73	level	$3 \cdot n + 2$	$n - 1$	$0.333 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0, i5=0); read A[i5, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1.52	level	$3 \cdot n + 2$	$(7/8) \cdot n - 7$	$0.292 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0)
$n^{1.5}$	1.41	level	$2 \cdot n - 1$	$n - 1$	$0.333 \cdot n^{-2}$	read C[i5, i1] (i0=0, i4=0)
$n^{1.5}$	1.41	level	$2 \cdot n + 2$	n	$0.333 \cdot n^{-2}$	write C[i3, i1] (i0=0, i3=0); write C[i3, i1] (i0=0)
$n^{1.5}$	1.41	level	$2 \cdot n$	n	$0.333 \cdot n^{-2}$	read A[i3, i1] (i0=0)
$n^{1.5}$	1.24	level	$2 \cdot n$	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 3$	$(7/8) \cdot n - 7$	$0.292 \cdot n^{-2}$	write C[i3, i1] (i0=0)
$n^{1.5}$	1.24	level	$2 \cdot n + 1$	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	write B[i1, i4] (i0=0, i4=0)
$n^{1.5}$	1.21	ramp	$n + 2 \rightarrow 2 \cdot n - 1$	$n - 2$	$0.333 \cdot n^{-2}$	read A[i2, i1] (i0=0)
$n^{1.5}$	1	level	$n + 1$	n	$0.333 \cdot n^{-2}$	read A[i3, i1] (i0=0, i3=0)
$n^{1.5}$	0.884	level	$2 \cdot n$	$(5/8) \cdot n$	$0.208 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.177	level	$2 \cdot n$	$(1/8) \cdot n$	$0.0417 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
$n^{0.5}$	9.9	level	$2 \cdot n$	7	$2.33 \cdot n^{-3}$	read A[i2, i1] (i0=0, i2=0)
$n^{0.5}$	8.66	level	$3 \cdot n + 1$	5	$1.67 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	8.49	level	$2 \cdot n + 2$	6	$2 \cdot n^{-3}$	write C[i3, i1] (i0=0)
$n^{0.5}$	8.49	level	$2 \cdot n + 2$	6	$2 \cdot n^{-3}$	write C[i3, i1] (i0=0, i3=0)
$n^{0.5}$	7	level	n	7	$2.33 \cdot n^{-3}$	read A[i2, i1] (i0=0)
$n^{0.5}$	2	level	$4 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7, i5=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6); read A[i5, i4] (i0=0, i4=7) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.73	level	$3 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0, i2=0)
$n^{0.5}$	1	level	n	1	$0.333 \cdot n^{-3}$	read A[i2, i1] (i0=0)

Column orthogonalization re-reads the factor panel: read A[i5, i4] ramps from $\sim 5n$ to $(1/8)n^2 + (9/8)n$ lines ($d = 4.0$, headroom +1.0 — corrected from the old +0.5, whose fits ran over corrupted terms). The large $n^{3.5}$ coefficient (2.74, read C[i5, i1] at $2n + 1$ lines) means the pre-asymptotic behavior is dominated by column-pair reuse until the panel term takes over.

heat-3d — single-shot [exact]

Accesses $A(n) = 16 \cdot n^4 - 64 \cdot n^3 + 96 \cdot n^2 - 64 \cdot n + 16$ (exact on $n \equiv 0 \pmod{56}$); DMD order $n^{5.5}$, headroom +1.5; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=1792$, 1 at $n=1848$.

DMD spectrum: $0.75 \cdot n^{5.5} + 1.06 \cdot n^5 + 17.7 \cdot n^{4.5} + 49.9 \cdot n^4 + 72.7 \cdot n^{3.5} + 36.8 \cdot n^3 + 71.5 \cdot n^{2.5} + 1.41 \cdot n^2 + 0.5 \cdot n^{1.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{5.5}$	0.125	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-7/2)$	$(1/4) \cdot n^4 + (-51/8) \cdot n^3 + (167/4) \cdot n^2 + (-801/8) \cdot n + (153/2)$	0.0156	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{5.5}$	0.125	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 2$	$(1/4) \cdot n^4 + (-49/8) \cdot n^3 + (317/8) \cdot n^2 + (-189/2) \cdot n + 72$	0.0156	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{5.5}$	0.125	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 3$	$(1/4) \cdot n^4 + (-37/8) \cdot n^3 + (215/8) \cdot n^2 + (-243/4) \cdot n + 45$	0.0156	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{5.5}$	0.125	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-9/2)$	$(1/4) \cdot n^4 + (-51/8) \cdot n^3 + (167/4) \cdot n^2 + (-801/8) \cdot n + (153/2)$	0.0156	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{5.5}$	0.125	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 1$	$(1/4) \cdot n^4 + (-49/8) \cdot n^3 + (317/8) \cdot n^2 + (-189/2) \cdot n + 72$	0.0156	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{5.5}$	0.125	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 4$	$(1/4) \cdot n^4 + (-37/8) \cdot n^3 + (215/8) \cdot n^2 + (-243/4) \cdot n + 45$	0.0156	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
n^5	0.177	level	$(1/2) \cdot n^2 + 3 \cdot n + (-5/2)$	$(1/4) \cdot n^4 - 6 \cdot n^3 + (133/4) \cdot n^2 + (-123/2) \cdot n + 34$	0.0156	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^5	0.177	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 1$	$(1/4) \cdot n^4 + (-23/4) \cdot n^3 + (63/2) \cdot n^2 - 58 \cdot n + 32$	0.0156	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^5	0.177	level	$(1/2) \cdot n^2 + (5/2) \cdot n - 2$	$(1/4) \cdot n^4 + (-17/4) \cdot n^3 + 21 \cdot n^2 - 37 \cdot n + 20$	0.0156	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^5	0.177	level	$(1/2) \cdot n^2 + 3 \cdot n + (-9/2)$	$(1/4) \cdot n^4 - 6 \cdot n^3 + (133/4) \cdot n^2 + (-123/2) \cdot n + 34$	0.0156	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^5	0.177	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 3$	$(1/4) \cdot n^4 + (-23/4) \cdot n^3 + (63/2) \cdot n^2 - 58 \cdot n + 32$	0.0156	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^5	0.177	level	$(1/2) \cdot n^2 + (5/2) \cdot n - 4$	$(1/4) \cdot n^4 + (-17/4) \cdot n^3 + 21 \cdot n^2 - 37 \cdot n + 20$	0.0156	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-7/2)$	$2 \cdot n^3 - 17 \cdot n^2 + 47 \cdot n - 39$	0.125/n	write B[i2, i3, i4] (i0=0, i3=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+1)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 2$	$2 \cdot n^3 - 17 \cdot n^2 + 47 \cdot n - 39$	0.125/n	write B[i2, i3, i4] (i0=0, i3=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+1)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 3$	$2 \cdot n^3 - 17 \cdot n^2 + 47 \cdot n - 39$	0.125/n	write B[i2, i3, i4] (i0=0, i3=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+1)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-9/2)$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	read A[i2 + 1, i3, i4] (i0=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 1$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	read A[i2 + 1, i3, i4] (i0=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 4$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	read A[i2 + 1, i3, i4] (i0=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-7/2)$	$2 \cdot n^3 - 17 \cdot n^2 + 47 \cdot n - 39$	0.125/n	write B[i2, i3, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-9/2)$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-7/2)$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 2$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-9/2)$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n$	$2 \cdot n^3 - 17 \cdot n^2 + 45 \cdot n - 36$	0.125/n	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{4.5}$	1	level	$(1/4) \cdot n^3 + (5/2) \cdot n^2 + (-19/4) \cdot n + 2$	$2 \cdot n^3 - 8 \cdot n^2 + 10 \cdot n - 4$	0.125/n	read A[i2, i3, i4 + 1] (i0=0); read B[i5, i6, i7 + 1] (i0=0, i5=0, i6=0) (+3)
$n^{4.5}$	0.438	level	$(1/4) \cdot n^3 + (5/2) \cdot n^2 + (-19/4) \cdot n + 2$	$(7/8) \cdot n^3 + (-135/8) \cdot n^2 + (193/4) \cdot n - 36$	0.0547/n	read A[i2, i3 - 1, i4] (i0=0); read A[i2 - 1, i3, i4] (i0=0) (+8)
$n^{4.5}$	0.438	level	$(1/4) \cdot n^3 + (9/4) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$(7/8) \cdot n^3 + (-59/4) \cdot n^2 + (329/8) \cdot n + (-121/4)$	0.0547/n	read A[i2, i3 - 1, i4] (i0=0); read A[i2 - 1, i3, i4] (i0=0) (+8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{4.5}$	0.438	level	$(1/4) \cdot n^3 + 2 \cdot n^2 + (9/4) \cdot n + (-9/2)$	$(7/8) \cdot n^3 + (-93/8) \cdot n^2 + 31 \cdot n + (-45/2)$	0.0547/n	read A[i2, i3 - 1, i4] (i0=0); read A[i2 - 1, i3, i4] (i0=0) (+8)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-5/2)$	$(1/2) \cdot n^3 + (-45/4) \cdot n^2 + (199/4) \cdot n - 51$	0.0312/n	write B[i2, i3, i4] (i0=0, i3=0); write B[i2, i3, i4] (i0=0) (+1)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 3$	$(1/2) \cdot n^3 + (-43/4) \cdot n^2 + 47 \cdot n - 48$	0.0312/n	write B[i2, i3, i4] (i0=0, i3=0); write B[i2, i3, i4] (i0=0) (+1)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 2$	$(1/2) \cdot n^3 + (-31/4) \cdot n^2 + (61/2) \cdot n - 30$	0.0312/n	write B[i2, i3, i4] (i0=0, i3=0); write B[i2, i3, i4] (i0=0) (+1)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-11/2)$	$(1/2) \cdot n^3 + (-45/4) \cdot n^2 + (199/4) \cdot n - 51$	0.0312/n	read A[i2 + 1, i3, i4] (i0=0, i3=0); read A[i2 + 1, i3, i4] (i0=0) (+1)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n$	$(1/2) \cdot n^3 + (-43/4) \cdot n^2 + 47 \cdot n - 48$	0.0312/n	read A[i2 + 1, i3, i4] (i0=0, i3=0); read A[i2 + 1, i3, i4] (i0=0) (+1)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 5$	$(1/2) \cdot n^3 + (-31/4) \cdot n^2 + (61/2) \cdot n - 30$	0.0312/n	read A[i2 + 1, i3, i4] (i0=0, i3=0); read A[i2 + 1, i3, i4] (i0=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-21/8)$	$(1/2) \cdot n^3 + (-45/4) \cdot n^2 + (199/4) \cdot n - 51$	0.0312/n	read A[i2, i3 + 1, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0) (+2)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n$	$(1/2) \cdot n^3 + (-43/4) \cdot n^2 + 47 \cdot n - 48$	0.0312/n	read A[i2, i3 + 1, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0) (+2)
$n^{4.5}$	0.25	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-3/8) \cdot n + (-9/4)$	$(1/2) \cdot n^3 + (-31/4) \cdot n^2 + (61/2) \cdot n - 30$	0.0312/n	read A[i2, i3 + 1, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0) (+2)
$n^{4.5}$	0.217	level	$(3/4) \cdot n$	$(1/4) \cdot n^4 - 5 \cdot n^3 + (69/4) \cdot n^2 + (-41/2) \cdot n + 8$	0.0156	read A[i2, i3 - 1, i4] (i0=0); read B[i5, i6 - 1, i7] (i0=0)
$n^{4.5}$	0.217	level	$(3/4) \cdot n - 1$	$(1/4) \cdot n^4 - 5 \cdot n^3 + (69/4) \cdot n^2 + (-41/2) \cdot n + 8$	0.0156	read A[i2, i3, i4 + 1] (i0=0); read B[i5, i6, i7 + 1] (i0=0)
$n^{4.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 6 \rightarrow (1/4) \cdot n^3 + n^2 + (-29/8) \cdot n$	$(1/4) \cdot n^3 + (-41/8) \cdot n^2 + (153/8) \cdot n - 18$	0.0156/n	write B[i2, i3, i4] (i0=0, i2=0); write A[i5, i6, i7] (i0=0)
$n^{4.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 3 \rightarrow (1/4) \cdot n^3 + n^2 + (-29/8) \cdot n - 3$	$(1/4) \cdot n^3 + (-41/8) \cdot n^2 + (153/8) \cdot n - 18$	0.0156/n	read A[i2 + 1, i3, i4] (i0=0, i2=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{4.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n - 1$	$(1/4) \cdot n^3 + (-43/8) \cdot n^2 + (47/2) \cdot n - 24$	0.0156/n	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{4.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n - 3$	$(1/4) \cdot n^3 + (-43/8) \cdot n^2 + (47/2) \cdot n - 24$	0.0156/n	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{4.5}$	0.0625	level	$(1/4) \cdot n^3 + (5/2) \cdot n^2 + (-19/4) \cdot n + 2$	$(1/8) \cdot n^3 + (-17/8) \cdot n^2 + (7/4) \cdot n + 4$	0.00781/n	read A[i2 - 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0, i6=0) (+2)
$n^{4.5}$	0.0625	level	$(1/4) \cdot n^3 + (9/4) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + (-1/8) \cdot n + (17/4)$	0.00781/n	read A[i2 - 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0, i6=0) (+2)
$n^{4.5}$	0.0625	level	$(1/4) \cdot n^3 + 2 \cdot n^2 + (9/4) \cdot n + (-9/2)$	$(1/8) \cdot n^3 + (-11/8) \cdot n^2 + n + (5/2)$	0.00781/n	read A[i2 - 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0, i6=0) (+2)
n^4	18.4	level	6	$(15/2) \cdot n^4 + (-49/2) \cdot n^3 + (57/2) \cdot n^2 + (-27/2) \cdot n + 2$	0.469	read A[i2 + 1, i3, i4] (i0=0); read A[i2 - 1, i3, i4] (i0=0) (+8)
n^4	3.91	level	5	$(7/4) \cdot n^4 + (-29/4) \cdot n^3 + (45/4) \cdot n^2 + (-31/4) \cdot n + 2$	0.109	read A[i2, i3, i4 + 1] (i0=0); read B[i5, i6, i7 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	3.31	level	7	$(5/4) \cdot n^4 + (-47/4) \cdot n^3 + (111/4) \cdot n^2 + (-101/4) \cdot n + 8$	0.0781	read A[i2 + 1, i3, i4] (i0=0); read A[i2 - 1, i3, i4] (i0=0) (+8)
n^4	2.12	level	2	$(3/2) \cdot n^4 + (-9/2) \cdot n^3 + (9/2) \cdot n^2 + (-3/2) \cdot n$	0.0938	read A[i2, i3, i4] (i0=0); read B[i5, i6, i7] (i0=0)
n^4	1.5	level	1	$(3/2) \cdot n^4 + (-9/2) \cdot n^3 + (9/2) \cdot n^2 + (-3/2) \cdot n$	0.0938	read A[i2, i3, i4 - 1] (i0=0); read B[i5, i6, i7 - 1] (i0=0)
n^4	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-5/2)$	$2 \cdot n^3 - 12 \cdot n^2 + 22 \cdot n - 12$	0.125/n	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i7=0)
n^4	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 1$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	0.125/n	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i7=0)
n^4	1.41	level	$(1/2) \cdot n^2 + (5/2) \cdot n - 2$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	0.125/n	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i7=0)
n^4	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-9/2)$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	0.125/n	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i7=0)
n^4	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 3$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	0.125/n	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	1.41	level	$(1/2) \cdot n^2 + (5/2) \cdot n - 4$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	$0.125/n$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i7=0)
n^4	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-5/2)$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	$0.125/n$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^4	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-9/2)$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	$0.125/n$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^4	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-5/2)$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	$0.125/n$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^4	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 1$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	$0.125/n$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^4	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-9/2)$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	$0.125/n$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^4	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 3$	$2 \cdot n^3 - 14 \cdot n^2 + 28 \cdot n - 16$	$0.125/n$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^4	0.866	level	3	$(1/2) \cdot n^4 + (-11/2) \cdot n^3 + (27/2) \cdot n^2 + (-25/2) \cdot n + 4$	0.0312	read A[i2, i3, i4] (i0=0); read B[i5, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.661	level	7	$(1/4) \cdot n^4 + (-11/4) \cdot n^3 + (27/4) \cdot n^2 + (-25/4) \cdot n + 2$	0.0156	read A[i2, i3, i4 - 1] (i0=0); read B[i5, i6, i7 - 1] (i0=0)
n^4	0.612	level	6	$(1/4) \cdot n^4 + (-3/4) \cdot n^3 + (3/4) \cdot n^2 + (-1/4) \cdot n$	0.0156	read A[i2, i3, i4 - 1] (i0=0); read B[i5, i6, i7 - 1] (i0=0)
n^4	0.177	level	$(1/2) \cdot n^2 + (13/4) \cdot n + (-7/4)$	$(1/4) \cdot n^3 - 3 \cdot n^2 + (29/4) \cdot n + (-9/2)$	0.0156/n	read A[i2, i3, i4 + 1] (i0=0); read B[i5, i6, i7 + 1] (i0=0, i6=0)
n^4	0.177	level	$(1/2) \cdot n^2 + (3/4) \cdot n - 1$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (25/2) \cdot n - 8$	0.0156/n	read A[i2, i3, i4 + 1] (i0=0); read B[i5, i6, i7 + 1] (i0=0, i6=0)
n^4	0.177	level	$(1/2) \cdot n^2 + (11/4) \cdot n + (-3/2)$	$(1/4) \cdot n^3 + (-13/4) \cdot n^2 + 8 \cdot n - 5$	0.0156/n	read A[i2, i3, i4 + 1] (i0=0); read B[i5, i6, i7 + 1] (i0=0, i6=0)
n^4	0.177	level	$(1/2) \cdot n^2 + (13/4) \cdot n + (-15/4)$	$(1/4) \cdot n^3 - 5 \cdot n^2 + (53/4) \cdot n + (-17/2)$	0.0156/n	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^4	0.177	level	$(1/2) \cdot n^2 + (3/4) \cdot n - 3$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (25/2) \cdot n - 8$	0.0156/n	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^4	0.175	ramp	$(1/2) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow (1/2) \cdot n^2 + (1/2) \cdot n - 3$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (25/2) \cdot n - 8$	0.0156/n	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.175	ramp	$(1/2) \cdot n^2 + (1/4) \cdot n + 2$ $\rightarrow$ $(1/2) \cdot n^2 + (1/2) \cdot n - 4$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (25/2) \cdot n - 8$	$0.0156/n$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i6=0)
n^4	0.175	ramp	$(1/2) \cdot n^2 + (1/4) \cdot n \rightarrow$ $(1/2) \cdot n^2 + (1/2) \cdot n - 6$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (25/2) \cdot n - 8$	$0.0156/n$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^4	0.175	ramp	$(1/2) \cdot n^2 + (1/4) \cdot n \rightarrow$ $(1/2) \cdot n^2 + (1/2) \cdot n - 6$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (25/2) \cdot n - 8$	$0.0156/n$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i6=0)
$n^{3.5}$	4	level	$(1/4) \cdot n^3 + (5/2) \cdot n^2 + (-19/4) \cdot n + 2$	$8 \cdot n^2 - 24 \cdot n + 16$	$0.5 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read A[i2, i3 - 1, i4] (i0=0) (+10)
$n^{3.5}$	4	level	$(1/4) \cdot n^3 + (5/2) \cdot n^2 + (-19/4) \cdot n + 2$	$8 \cdot n^2 - 24 \cdot n + 16$	$0.5 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read A[i2, i3 - 1, i4] (i0=0) (+10)
$n^{3.5}$	4	level	$(1/4) \cdot n^3 + (9/4) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$8 \cdot n^2 - 24 \cdot n + 16$	$0.5 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read A[i2, i3 - 1, i4] (i0=0) (+10)
$n^{3.5}$	3	level	$(1/4) \cdot n^3 + (9/4) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$6 \cdot n^2 - 18 \cdot n + 12$	$0.375 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read A[i2, i3 - 1, i4] (i0=0) (+6)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-5/2)$	$4 \cdot n^2 - 20 \cdot n + 21$	$0.25 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i3=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+1)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 3$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i3=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+1)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 2$	$4 \cdot n^2 - 20 \cdot n + 21$	$0.25 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i3=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+1)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-11/2)$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i3=0, i4=0); read A[i2 + 1, i3, i4] (i0=0, i4=0) (+1)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i3=0, i4=0); read A[i2 + 1, i3, i4] (i0=0, i4=0) (+1)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 5$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i3=0, i4=0); read A[i2 + 1, i3, i4] (i0=0, i4=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-5/2)$	$4 \cdot n^2 - 20 \cdot n + 21$	$0.25 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0, i3=0) (+3)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-11/2)$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i3=0); read A[i2 + 1, i3, i4] (i0=0) (+1)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-5/2)$	$4 \cdot n^2 - 20 \cdot n + 21$	$0.25 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0, i3=0) (+2)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 3$	$4 \cdot n^2 - 20 \cdot n + 21$	$0.25 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0, i3=0) (+2)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-11/2)$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i3=0); read A[i2 + 1, i3, i4] (i0=0) (+1)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n - 1$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i3=0); read A[i2 + 1, i3, i4] (i0=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+2)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-21/8)$	$4 \cdot n^2 - 22 \cdot n + 24$	$0.25 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0) (+2)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (9/4) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$4 \cdot n^2 - 14 \cdot n + 12$	$0.25 \cdot n^{-2}$	read A[i2, i3 - 1, i4] (i0=0, i2=0, i3=0); read A[i2 - 1, i3, i4] (i0=0, i2=0) (+5)
$n^{3.5}$	2	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-21/8)$	$4 \cdot n^2 - 20 \cdot n + 21$	$0.25 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0, i2=0); write B[i2, i3, i4] (i0=0, i3=0) (+3)
$n^{3.5}$	1.73	level	$(3/4) \cdot n$	$2 \cdot n^3 - 8 \cdot n^2 + 10 \cdot n - 4$	$0.125/n$	read A[i2, i3 - 1, i4] (i0=0); read B[i5, i6 - 1, i7] (i0=0, i7=0)
$n^{3.5}$	1.73	level	$(3/4) \cdot n - 1$	$2 \cdot n^3 - 8 \cdot n^2 + 10 \cdot n - 4$	$0.125/n$	read A[i2, i3, i4] (i0=0); read B[i5, i6, i7] (i0=0, i7=0)
$n^{3.5}$	1.73	level	$(3/4) \cdot n$	$2 \cdot n^3 - 8 \cdot n^2 + 10 \cdot n - 4$	$0.125/n$	read A[i2, i3 - 1, i4] (i0=0); read B[i5, i6 - 1, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	1.73	level	$(3/4) \cdot n + (21/4)$	$2 \cdot n^3 - 8 \cdot n^2 + 10 \cdot n - 4$	$0.125/n$	read A[i2, i3 - 1, i4] (i0=0); read B[i5, i6 - 1, i7] (i0=0)
$n^{3.5}$	1.73	level	$(3/4) \cdot n - 1$	$2 \cdot n^3 - 8 \cdot n^2 + 10 \cdot n - 4$	$0.125/n$	read A[i2, i3, i4 + 1] (i0=0, i1=0); read B[i5, i6, i7 + 1] (i0=0, i1=0) (+5)
$n^{3.5}$	1.73	level	$(3/4) \cdot n + (13/4)$	$2 \cdot n^3 - 8 \cdot n^2 + 10 \cdot n - 4$	$0.125/n$	read A[i2, i3, i4 + 1] (i0=0, i1=0); read B[i5, i6, i7 + 1] (i0=0, i1=0) (+5)
$n^{3.5}$	1	level	$(1/4) \cdot n^3 + (9/4) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read A[i2, i3 - 1, i4] (i0=0) (+10)
$n^{3.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-21/8)$	$2 \cdot n^2 - 11 \cdot n + 12$	$0.125 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-21/8)$	$2 \cdot n^2 - 11 \cdot n + 12$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i4=0); write B[i2, i3, i4] (i0=0, i4=0) (+2)
$n^{3.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n$	$2 \cdot n^2 - 9 \cdot n + 9$	$0.125 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i3=0); write B[i2, i3, i4] (i0=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 1$	$2 \cdot n^2 - 11 \cdot n + 12$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0, i2=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{3.5}$	0.999	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n + 2 \rightarrow (1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 2$	$2 \cdot n^2 - 9 \cdot n + 9$	$0.125 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i2=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.999	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n - 1 \rightarrow (1/4) \cdot n^3 + n^2 + (-29/8) \cdot n - 1$	$2 \cdot n^2 - 9 \cdot n + 9$	$0.125 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{3.5}$	0.999	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n - 2 \rightarrow (1/4) \cdot n^3 + n^2 + (-29/8) \cdot n - 2$	$2 \cdot n^2 - 9 \cdot n + 9$	$0.125 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i2=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{3.5}$	0.999	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 4 \rightarrow (1/4) \cdot n^3 + n^2 + (-31/8) \cdot n + 4$	$2 \cdot n^2 - 9 \cdot n + 9$	$0.125 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i2=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.999	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 1 \rightarrow (1/4) \cdot n^3 + n^2 + (-31/8) \cdot n + 1$	$2 \cdot n^2 - 9 \cdot n + 9$	$0.125 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{3.5}$	0.998	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-19/8) \cdot n + 1 \rightarrow (1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 1$	$2 \cdot n^2 - 11 \cdot n + 12$	$0.125 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.998	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^3 + n^2 + (-31/8) \cdot n + 3$	$2 \cdot n^2 - 11 \cdot n + 12$	$0.125 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.998	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n \rightarrow$ $(1/4) \cdot n^3 + n^2 + (-31/8) \cdot n$	$2 \cdot n^2 - 11 \cdot n + 12$	$0.125 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{3.5}$	0.125	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/4) \cdot n + (-7/4)$	$(1/4) \cdot n^2 + (-37/8) \cdot n + (51/8)$	$0.0156 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.125	level	$(1/4) \cdot n^3 + n^2 + (-9/4) \cdot n + 1$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.125	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/4) \cdot n + (-15/4)$	$(1/4) \cdot n^2 + (-21/8) \cdot n + (27/8)$	$0.0156 \cdot n^{-2}$	read A[i2, i3, i4 + 1] (i0=0, i2=0, i3=0); read B[i5, i6, i7 + 1] (i0=0)
$n^{3.5}$	0.125	level	$(1/4) \cdot n^3 + n^2 + (-9/4) \cdot n - 1$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	read A[i2, i3, i4 + 1] (i0=0, i2=0, i3=0); read B[i5, i6, i7 + 1] (i0=0)
$n^{3.5}$	0.125	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/4) \cdot n + (-7/2)$	$(1/4) \cdot n^2 + (-23/8) \cdot n + (15/4)$	$0.0156 \cdot n^{-2}$	read A[i2, i3, i4 + 1] (i0=0, i2=0, i3=0); read B[i5, i6, i7 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^3.5$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-5/2) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 1$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^2$	write B[i2, i3, i4] (i0=0, i3=0); write A[i5, i6, i7] (i0=0)
$n^3.5$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-5/2) \cdot n + 1$ $\rightarrow$ $(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 2$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^2$	read A[i2, i3 + 1, i4] (i0=0, i2=0); read B[i5, i6 + 1, i7] (i0=0)
$n^3.5$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-21/8) \cdot n + 3$ $\rightarrow$ $(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 3$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^2$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i3=0); read B[i5, i6 + 1, i7] (i0=0)
$n^3.5$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-21/8) \cdot n + 3$ $\rightarrow$ $(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 3$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^2$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^3.5$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 2$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^2$	write B[i2, i3, i4] (i0=0, i2=0); write A[i5, i6, i7] (i0=0)
$n^3.5$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 4$ $\rightarrow$ $(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n + 1$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^2$	write B[i2, i3, i4] (i0=0, i3=0); write A[i5, i6, i7] (i0=0)
$n^3.5$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 1$ $\rightarrow$ $(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n - 2$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^2$	read A[i2 + 1, i3, i4] (i0=0, i2=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.124	ramp	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 1 \rightarrow$ $(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n - 2$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i3=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{3.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n - 1$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n - 3$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{3.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n + 9 \rightarrow$ $(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0, i2=0, i3=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n + 8 \rightarrow$ $(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n - 1$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{3.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n - 5$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{3.5}$	0.124	ramp	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n - 5$	$(1/4) \cdot n^2 + (-35/8) \cdot n + 6$	$0.0156 \cdot n^{-2}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i3=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 1$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-11/2)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 4$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (11/4) \cdot n + (-13/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i6=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (1/4) \cdot n$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i6=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (9/4) \cdot n + (-5/2)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i6=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i6=0, i7=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1.41	level	$(1/2) \cdot n^2 + (1/4) \cdot n - 2$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i6=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (9/4) \cdot n + (-9/2)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i6=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (11/4) \cdot n + (-1/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (11/4) \cdot n + (-13/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (11/4) \cdot n + (-5/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (1/4) \cdot n + 1$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 2$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i6=0)
n^3	1.41	level	$(1/2) \cdot n^2 + 3 \cdot n + (-7/2)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3 + 1, i4] (i0=0); read B[i5, i6 + 1, i7] (i0=0, i6=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1.41	level	$(1/2) \cdot n^2 + (11/4) \cdot n + (-17/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (1/4) \cdot n - 2$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (1/2) \cdot n - 4$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i6=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (3/4) \cdot n - 3$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (3/4) \cdot n - 1$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3, i4] (i0=0); read B[i5, i6, i7] (i0=0, i6=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (13/4) \cdot n + (-7/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3, i4] (i0=0); read B[i5, i6, i7] (i0=0, i6=0, i7=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (13/4) \cdot n + (-27/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (3/4) \cdot n - 3$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1.41	level	$(1/2) \cdot n^2 + (13/4) \cdot n + (-11/4)$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2 - 1, i3, i4] (i0=0); read B[i5 - 1, i6, i7] (i0=0)
n^3	1.41	level	$(1/2) \cdot n^2 + (3/4) \cdot n - 1$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.125 \cdot n^{-2}$	read A[i2, i3, i4 + 1] (i0=0, i1=0); read B[i5, i6, i7 + 1] (i0=0, i1=0, i6=0) (+3)
n^3	1.41	level	$(1/2) \cdot n^2 + (13/4) \cdot n + (-7/4)$	$2 \cdot n^2 - 8 \cdot n + 8$	$0.125 \cdot n^{-2}$	read A[i2, i3, i4 + 1] (i0=0); read B[i5, i6, i7 + 1] (i0=0, i5=1, i6=0) (+1)
$n^{2.5}$	2	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-13/8) \cdot n + (15/8)$	$4 \cdot n - 6$	$0.25 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i3=0); write B[i2, i3, i4] (i0=0) (+1)
$n^{2.5}$	2	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 3$	$4 \cdot n - 6$	$0.25 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i3=0); write B[i2, i3, i4] (i0=0) (+1)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-11/8) \cdot n + (13/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-13/8) \cdot n + (7/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-21/8) \cdot n + 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (5/4) \cdot n^2 + (-9/8) \cdot n + (5/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n + 2$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (5/4) \cdot n^2 + (-11/8) \cdot n + (3/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-19/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 4$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-5/8) \cdot n + (-7/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i4=0); write A[i5, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 - 2 \cdot n + (9/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i3=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n + 6$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i3=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (5/4) \cdot n^2 + (-7/4) \cdot n + (5/2)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i2=0, i3=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-13/8) \cdot n + (-25/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n - 2$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (5/4) \cdot n^2 + (-11/8) \cdot n + (-13/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-13/2)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i3=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n - 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i3=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-1/2) \cdot n - 6$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i3=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-51/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-5/8) \cdot n + (-23/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 - 2 \cdot n + (-11/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i3=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n + 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i3=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (5/4) \cdot n^2 + (-7/4) \cdot n + (-5/2)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i3=0, i4=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-13/8) \cdot n + (31/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 - 2 \cdot n + (17/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-19/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i3=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-13/8) \cdot n + (15/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 - 2 \cdot n + (1/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-43/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i3=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-15/2)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 - 2 \cdot n + (9/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n + 5$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-27/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i3=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n + 3$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i3=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-13/8) \cdot n + (-1/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 - 2 \cdot n + (-15/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-13/4) \cdot n - 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0); read B[i5 + 1, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-51/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i3=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-29/8) \cdot n - 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i3=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/2) \cdot n + (-13/2)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-7/2) \cdot n - 2$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/2) \cdot n^2 + (-13/8) \cdot n + (-25/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i3=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (3/4) \cdot n^2 + (-23/8) \cdot n - 3$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2 + 1, i3, i4] (i0=0, i2=0, i3=0); read B[i5 + 1, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-29/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-5/2) \cdot n + 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i3=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i4=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-29/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i4=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-21/8) \cdot n + 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i3=0, i4=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-27/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i3=0, i4=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-11/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-29/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i3=0); write A[i5, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-5/8) \cdot n + (-19/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-21/8) \cdot n + 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-5/2) \cdot n - 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3 + 1, i4] (i0=0, i2=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-3/8) \cdot n + (-29/8)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i3=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-19/8) \cdot n - 2$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3 + 1, i4] (i0=0, i2=0, i3=0); read B[i5, i6 + 1, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-9/4) \cdot n + 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0, i4=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-9/4) \cdot n - 1$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3, i4] (i0=0, i2=0, i3=0, i4=0); read B[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/4) \cdot n + (-15/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read A[i2, i3, i4] (i0=0, i2=0, i3=0, i4=0); read B[i5, i6, i7] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/4) \cdot n + (-19/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-9/4) \cdot n$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/4) \cdot n + (-7/4)$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	write B[i2, i3, i4] (i0=0); write A[i5, i6, i7] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/4) \cdot n + (-15/4)$	$2 \cdot n - 4$	$0.125 \cdot n^{-3}$	read A[i2, i3, i4 + 1] (i0=0, i2=0, i3=0); read B[i5, i6, i7 + 1] (i0=0)
$n^{2.5}$	1	level	$(1/4) \cdot n^3 + n^2 + (-9/4) \cdot n - 2$	$2 \cdot n - 3$	$0.125 \cdot n^{-3}$	read B[i5, i6, i7 + 1] (i0=0, i1=0); read A[i2, i3, i4 + 1] (i0=0, i2=0, i3=0) (+1)
$n^{2.5}$	0.5	level	$(1/4) \cdot n^3 + (9/4) \cdot n^2 + (11/4) \cdot n + (-21/4)$	$n - 2$	$0.0625 \cdot n^{-3}$	read A[i2 - 1, i3, i4] (i0=0, i2=0, i3=0)
n^2	1.41	level	$(1/2) \cdot n^2 + (13/4) \cdot n + (-7/4)$	$2 \cdot n - 4$	$0.125 \cdot n^{-3}$	read A[i2, i3, i4 + 1] (i0=0, i1=0); read B[i5, i6, i7 + 1] (i0=0, i1=0, i6=0)
$n^{1.5}$	0.5	level	$(1/4) \cdot n^3 + (7/4) \cdot n^2 + (-1/4) \cdot n + (-15/4)$	1	$0.0625 \cdot n^{-4}$	read B[i5, i6, i7 + 1] (i0=0, i1=0, i5=0, i6=0)

3-D two-array stencil, the suite's only headroom +1.5 kernel: accesses grow as n^4 (time $\times$ volume) while the cross-sweep reuses of write B[i2,i3,i4] and the neighbor reads sit at the

two-array volume footprint $(1/4)n^3 + O(n^2)$ lines — reuse-distance growth $\rho = 3$, so $d = a + \rho/2 = 5.5$. Eight symmetric families of coefficient 0.125 sum to $0.75 \cdot n^{5.5}$. The time-tiling cliff is at $64 \cdot ((1/4)n^3)$ bytes; below it every sweep refetches both volumes.

jacobi-1d — infinite-repeat [exact]

Accesses $A(n) = 8 \cdot n^2 - 8 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order $n^{2.5}$, headroom **+0.5**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.25 \cdot n^{2.5} + 9.4 \cdot n^2 + 6.12 \cdot n^{1.5} + 4.25 \cdot n^1 + 2 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.125	level	$(1/4) \cdot n + 1$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0312	write B[i2] (i0=0, i1=0); write A[i3] (i0=0, i1=0) (+2)
$n^{2.5}$	0.125	level	$(1/4) \cdot n + 1$	$(1/4) \cdot n^2 + (-17/4) \cdot n + 4$	0.0312	read A[i2 + 1] (i0=0, i1=1); read A[i2 + 1] (i0=0) (+1)
n^2	2.12	level	2	$(3/2) \cdot n^2 - 2 \cdot n$	0.188	write B[i2] (i0=0, i1=0); write A[i3] (i0=0, i1=0) (+2)
n^2	2.12	level	2	$(3/2) \cdot n^2 + (-3/2) \cdot n$	0.188	read A[i2 - 1] (i0=0); read B[i3 - 1] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2 + (-15/4) \cdot n + 2$	0.219	read A[i2 + 1] (i0=0); read B[i3 + 1] (i0=0)
n^2	1.5	level	1	$(3/2) \cdot n^2$	0.188	read A[i2] (i0=0, i1=0); read B[i3] (i0=0, i1=0) (+2)
n^2	0.866	level	3	$(1/2) \cdot n^2 - 4 \cdot n$	0.0625	read A[i2 - 1] (i0=0, i1=0); read A[i2] (i0=0, i1=0) (+6)
n^2	0.433	level	3	$(1/4) \cdot n^2$	0.0312	write B[i2] (i0=0, i1=0); write A[i3] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	2	$(1/4) \cdot n^2 + (-1/2) \cdot n$	0.0312	read A[i2 - 1] (i0=0, i1=0); read B[i3 - 1] (i0=0, i1=0) (+2)
n^2	0.25	level	1	$(1/4) \cdot n^2 - 2 \cdot n + 2$	0.0312	read A[i2] (i0=0, i1=0); read B[i3] (i0=0, i1=0) (+2)
$n^{1.5}$	2	level	$(1/4) \cdot n + 1$	$4 \cdot n$	$0.5/n$	read A[i2 - 1] (i0=0, i1=0, i2=0); write B[i2] (i0=0, i1=0, i2=0) (+7)
$n^{1.5}$	1	level	$(1/4) \cdot n + 1$	$2 \cdot n$	$0.25/n$	write B[i2] (i0=0, i1=0); write A[i3] (i0=0, i1=0) (+2)
$n^{1.5}$	1	level	$(1/4) \cdot n + (7/4)$	$2 \cdot n$	$0.25/n$	write B[i2] (i0=0, i1=0); write A[i3] (i0=0, i1=0) (+2)
$n^{1.5}$	1	level	$(1/4) \cdot n - 1$	$2 \cdot n - 2$	$0.25/n$	read A[i2 + 1] (i0=0, i1=1); read A[i2 + 1] (i0=0) (+1)
$n^{1.5}$	1	level	$(1/4) \cdot n + 2$	$2 \cdot n - 2$	$0.25/n$	read A[i2 + 1] (i0=0, i1=1); read A[i2 + 1] (i0=0) (+1)
$n^{1.5}$	0.125	level	$(1/4) \cdot n + 1$	$(1/4) \cdot n - 4$	$0.0312/n$	read A[i2 + 1] (i0=0, i1=0); read B[i3 + 1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	2	level	1	$2 \cdot n$	$0.25/n$	read A[i2 - 1] (i0=0, i1=0, i2=0); read A[i2 + 1] (i0=0, i1=0, i2=0) (+7)
n^1	1.25	level	1	$(5/4) \cdot n$	$0.156/n$	read A[i2 + 1] (i0=0, i1=0); read B[i3 + 1] (i0=0, i1=0)
n^1	0.5	level	1	$(1/2) \cdot n - 4$	$0.0625/n$	read A[i2 + 1] (i0=0, i1=0); read B[i3 + 1] (i0=0, i1=0)
n^1	0.5	level	1	$(1/2) \cdot n + (-5/2)$	$0.0625/n$	read A[i2 + 1] (i0=0, i1=0); read B[i3 + 1] (i0=0, i1=0)
$n^{0.5}$	1	level	$(1/4) \cdot n - 1$	2	$0.25 \cdot n^{-2}$	read A[i2 + 1] (i0=0, i1=0); read B[i3 + 1] (i0=0, i1=0)
$n^{0.5}$	1	level	$(1/4) \cdot n + 2$	2	$0.25 \cdot n^{-2}$	read A[i2 + 1] (i0=0, i1=0); read B[i3 + 1] (i0=0, i1=0)

Two-array 1-D stencil: the cross-sweep reuses (write B[i2] consumed by the next sweep, right-neighbor reads) sit at the array footprint $(1/4)n + 1$ lines — order $n^{2.5}$, headroom +0.5, coefficient 0.25 (two arrays $\times$ two sweep directions). The bulk (distance ≤ 2) is neighbor line reuse. The correct transformation is time-tiling/fusion of the two sweeps; repetition adds nothing (the arrays are re-touched every time step already).

jacobi-1d — single-shot [exact]

Accesses $A(n) = 8 \cdot n^2 - 8 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order $n^{2.5}$, headroom **+0.5**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.25 \cdot n^{2.5} + 9.4 \cdot n^2 + 6 \cdot n^{1.5} + 2 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.125	level	$(1/4) \cdot n + 1$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0312	write B[i2] (i0=0); write A[i3] (i0=0)
$n^{2.5}$	0.125	level	$(1/4) \cdot n + 1$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0312	read A[i2 + 1] (i0=0); read B[i3 + 1] (i0=0)
n^2	2.12	level	2	$(3/2) \cdot n^2 - 2 \cdot n$	0.188	write B[i2] (i0=0); write A[i3] (i0=0)
n^2	2.12	level	2	$(3/2) \cdot n^2$	0.188	read A[i2 - 1] (i0=0); read B[i3 - 1] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2 - 2 \cdot n$	0.219	read A[i2 + 1] (i0=0); read B[i3 + 1] (i0=0)
n^2	1.5	level	1	$(3/2) \cdot n^2$	0.188	read A[i2] (i0=0); read B[i3] (i0=0)
n^2	0.866	level	3	$(1/2) \cdot n^2 - 4 \cdot n$	0.0625	read A[i2 - 1] (i0=0); read A[i2] (i0=0) (+2)
n^2	0.433	level	3	$(1/4) \cdot n^2$	0.0312	write B[i2] (i0=0); write A[i3] (i0=0)
n^2	0.354	level	2	$(1/4) \cdot n^2 - 2 \cdot n$	0.0312	read A[i2 - 1] (i0=0); read B[i3 - 1] (i0=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 - 2 \cdot n$	0.0312	read A[i2] (i0=0); read B[i3] (i0=0)
$n^{1.5}$	2	level	$(1/4) \cdot n + 1$	$4 \cdot n - 2$	0.5/n	read A[i2 - 1] (i0=0, i2=0); write B[i2] (i0=0, i2=0) (+2)
$n^{1.5}$	1	level	$(1/4) \cdot n + 1$	$2 \cdot n - 1$	0.25/n	write B[i2] (i0=0); write A[i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	1	level	$(1/4) \cdot n + (7/4)$	$2 \cdot n - 1$	$0.25/n$	write B[i2] (i0=0); write A[i3] (i0=0)
$n^{1.5}$	1	level	$(1/4) \cdot n - 1$	$2 \cdot n - 1$	$0.25/n$	read A[i2 + 1] (i0=0); read B[i3 + 1] (i0=0)
$n^{1.5}$	1	level	$(1/4) \cdot n + 2$	$2 \cdot n - 2$	$0.25/n$	read A[i2 + 1] (i0=0); read B[i3 + 1] (i0=0)
n^1	2	level	1	$2 \cdot n$	$0.25/n$	read A[i2 - 1] (i0=0, i2=0); read A[i2] (i0=0, i2=0) (+2)

Two-array 1-D stencil: the cross-sweep reuses (write B[i2] consumed by the next sweep, right-neighbor reads) sit at the array footprint $(1/4)n + 1$ lines — order $n^{2.5}$, headroom +0.5, coefficient 0.25 (two arrays $\times$ two sweep directions). The bulk (distance ≤ 2) is neighbor line reuse. The correct transformation is time-tiling/fusion of the two sweeps; repetition adds nothing (the arrays are re-touched every time step already).

jacobi-2d — infinite-repeat [exact]

Accesses $A(n) = 12 \cdot n^3 - 24 \cdot n^2 + 12 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.5 \cdot n^4 + 0.707 \cdot n^{3.5} + 30.4 \cdot n^3 + 8.84 \cdot n^{2.5} + 33.4 \cdot n^2 + 1.41 \cdot n^{1.5} + 1.5 \cdot n^1 + 7.41 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.125	level	$(1/4) \cdot n^2 + n - 2$	$(1/4) \cdot n^3 + (-21/4) \cdot n^2 + 20 \cdot n$	0.0208	write B[i2, i3] (i0=0, i1=0, i2=1); write B[i2, i3] (i0=0, i1=0) (+4)
n^4	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$(1/4) \cdot n^3 + (-7/2) \cdot n^2 + (45/4) \cdot n$	0.0208	write B[i2, i3] (i0=0, i1=0, i2=1); write B[i2, i3] (i0=0, i1=0) (+4)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.125	level	$(1/4) \cdot n^2 + n - 3$	$(1/4) \cdot n^3 + (-21/4) \cdot n^2 + 20 \cdot n$	0.0208	read A[i2 + 1, i3] (i0=0, i1=0, i2=1); read A[i2 + 1, i3] (i0=0, i1=0) (+6)
n^4	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/4) \cdot n^3 + (-7/2) \cdot n^2 + (45/4) \cdot n$	0.0208	read A[i2 + 1, i3] (i0=0, i1=0, i2=1); read A[i2 + 1, i3] (i0=0, i1=0) (+6)
$n^{3.5}$	0.177	level	$(1/2) \cdot n + (5/2)$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (17/2) \cdot n$	0.0208	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+2)
$n^{3.5}$	0.177	level	$(1/2) \cdot n$	$(1/4) \cdot n^3 + (-9/2) \cdot n^2 + 8 \cdot n$	0.0208	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+2)
$n^{3.5}$	0.177	level	$(1/2) \cdot n + 1$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (25/2) \cdot n - 8$	0.0208	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)
$n^{3.5}$	0.177	level	$(1/2) \cdot n + (7/2)$	$(1/4) \cdot n^3 - 3 \cdot n^2 + (29/4) \cdot n + (-9/2)$	0.0208	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)
n^3	6	level	4	$3 \cdot n^3 + (-17/2) \cdot n^2 + (15/2) \cdot n - 2$	0.25	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+4)
n^3	4	level	4	$2 \cdot n^3 - 6 \cdot n^2 + 4 \cdot n$	0.167	read A[i2, i3] (i0=0); read B[i4, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1.75	level	1	$(7/4) \cdot n^3 + (-7/4) \cdot n^2 - 2 \cdot n + 2$	0.146	read A[i2, i3 - 1] (i0=0, i1=0); read B[i4, i5 - 1] (i0=0, i1=0) (+2)
n^3	1.68	level	5	$(3/4) \cdot n^3 + (-11/4) \cdot n^2 + 2 \cdot n$	0.0625	read A[i2, i3 - 1] (i0=0, i1=0); read A[i2 + 1, i3] (i0=0, i1=0) (+10)
n^3	1.5	level	4	$(3/4) \cdot n^3 + (-7/4) \cdot n^2 + (-3/4) \cdot n + (7/4)$	0.0625	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+1)
n^3	1.5	level	4	$(3/4) \cdot n^3 - n^2 + (-3/4) \cdot n + 1$	0.0625	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+1)
n^3	1.5	level	4	$(3/4) \cdot n^3 + (-5/2) \cdot n^2 + (11/4) \cdot n - 1$	0.0625	read A[i2 - 1, i3] (i0=0)
n^3	1.5	level	1	$(3/2) \cdot n^3 - 3 \cdot n^2 + (3/2) \cdot n$	0.125	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)
n^3	1	level	$(1/4) \cdot n^2 + n - 2$	$2 \cdot n^2 - 8 \cdot n$	0.167/n	read A[i2, i3] (i0=0, i1=0, i2=0, i3=0); write B[i2, i3] (i0=0, i1=0, i2=2, i3=0) (+9)
n^3	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n^2 - 8 \cdot n$	0.167/n	read A[i2 + 1, i3] (i0=0, i1=0, i2=2, i3=0); read A[i2 + 1, i3] (i0=0, i1=0, i3=0) (+10)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$2 \cdot n^2 - 10 \cdot n$	$0.167/n$	write B[i2, i3] (i0=0, i1=0, i2=2, i3=0); write B[i2, i3] (i0=0, i1=0, i3=0) (+4)
n^3	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$2 \cdot n^2 - 6 \cdot n$	$0.167/n$	read A[i2, i3] (i0=0, i1=0, i2=0, i3=0); read A[i2 + 1, i3] (i0=0, i1=0, i2=2, i3=0) (+15)
n^3	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n^2 - 8 \cdot n$	$0.167/n$	write B[i2, i3] (i0=0, i1=0, i2=0); write B[i2, i3] (i0=0, i1=0) (+4)
n^3	1	level	$(1/4) \cdot n^2 + n - 4$	$2 \cdot n^2 - 10 \cdot n$	$0.167/n$	read A[i2 + 1, i3] (i0=0, i1=0, i2=1); read A[i2 + 1, i3] (i0=0, i1=0) (+6)
n^3	1	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$2 \cdot n^2 - 4 \cdot n + 2$	$0.167/n$	read A[i2, i3 + 1] (i0=0, i1=1); read B[i4, i5 + 1] (i0=0, i1=1, i4=0) (+6)
n^3	0.559	level	5	$(1/4) \cdot n^3 + (-1/2) \cdot n^2 + (1/4) \cdot n$	0.0208	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+2)
n^3	0.559	level	5	$(1/4) \cdot n^3 + (-1/4) \cdot n^2$	0.0208	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.354	level	2	$(1/4) \cdot n^3 + (-3/4) \cdot n^2 + (3/4) \cdot n + (-1/4)$	0.0208	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)
n^3	0.354	level	2	$(1/4) \cdot n^3 + (-9/4) \cdot n^2 + 2 \cdot n - 1$	0.0208	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0, i4=0) (+4)
n^3	0.25	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$(1/2) \cdot n^2 + (-13/2) \cdot n$	0.0417/n	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 2$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 3$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^3	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/4) \cdot n^2 + (-9/4) \cdot n$	0.0208/n	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 1$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^3	0.125	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0, i4=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.125	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/4) \cdot n^2 + (-9/4) \cdot n$	0.0208/n	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0, i4=0) (+3)
n^3	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n + (3/2)$	$(1/4) \cdot n^2 + (-5/2) \cdot n$	0.0208/n	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0, i4=0) (+3)
n^3	0.125	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+4)
n^3	0.125	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/4) \cdot n^2 + (-9/4) \cdot n$	0.0208/n	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+4)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 2$	$(1/4) \cdot n^2 + (-17/4) \cdot n + 4$	0.0208/n	read A[i2, i3 + 1] (i0=0, i1=1, i2=0); read A[i2, i3 + 1] (i0=0, i2=0) (+1)
n^3	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/4) \cdot n^2 + (-5/2) \cdot n + (9/4)$	0.0208/n	read A[i2, i3 + 1] (i0=0, i1=1, i2=0); read A[i2, i3 + 1] (i0=0, i2=0) (+1)
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + n - 4$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	write B[i2, i3] (i0=0, i1=0, i2=0); write A[i4, i5] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n + 1$ → $(1/4) \cdot n^2 + n - 5$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n$ → $(1/4) \cdot n^2 + n - 6$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	read A[i2 + 1, i3] (i0=0, i1=0, i2=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n$ → $(1/4) \cdot n^2 + n - 6$	$(1/4) \cdot n^2 - 4 \cdot n$	0.0208/n	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
$n^{2.5}$	1.41	level	$(1/2) \cdot n$	$2 \cdot n^2 - 4 \cdot n$	0.167/n	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0, i5=0) (+2)
$n^{2.5}$	1.41	level	$(1/2) \cdot n + 1$	$2 \cdot n^2 - 4 \cdot n$	0.167/n	read A[i2, i3] (i0=0, i1=0); read B[i4, i5] (i0=0, i1=0, i5=0) (+2)
$n^{2.5}$	1.41	level	$(1/2) \cdot n + (5/2)$	$2 \cdot n^2 - 4 \cdot n$	0.167/n	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+2)
$n^{2.5}$	1.41	level	$(1/2) \cdot n + 2$	$2 \cdot n^2 - 4 \cdot n$	0.167/n	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+2)
$n^{2.5}$	1.41	level	$(1/2) \cdot n + (9/2)$	$2 \cdot n^2 - 4 \cdot n$	0.167/n	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.41	level	$(1/2) \cdot n - 1$	$2 \cdot n^2 - 6 \cdot n + 4$	$0.167/n$	read A[i2, i3 + 1] (i0=0, i1=1); read B[i4, i5 + 1] (i0=0, i1=1) (+2)
$n^{2.5}$	0.177	level	$(1/2) \cdot n + 1$	$(1/4) \cdot n^2 + (-9/2) \cdot n + 8$	$0.0208/n$	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0)
$n^{2.5}$	0.177	level	$(1/2) \cdot n + (7/2)$	$(1/4) \cdot n^2 + (-11/4) \cdot n + (9/2)$	$0.0208/n$	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0)
n^2	3	level	4	$(3/2) \cdot n^2 + (-7/2) \cdot n + 2$	$0.125/n$	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0)
n^2	2	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$4 \cdot n$	$0.333 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0, i2=1, i3=0); write B[i2, i3] (i0=0, i1=0, i3=0) (+4)
n^2	2	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$4 \cdot n$	$0.333 \cdot n^{-2}$	read A[i2 - 1, i3] (i0=0, i1=0); read A[i2 + 1, i3] (i0=0, i1=0) (+9)
n^2	2	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$4 \cdot n$	$0.333 \cdot n^{-2}$	read A[i2 - 1, i3] (i0=0, i1=0); read A[i2 + 1, i3] (i0=0, i1=0) (+9)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.25	level	1	$(5/4) \cdot n^2 + (-5/4) \cdot n - 5$	$0.104/n$	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0, i4=0) (+1)
n^2	1	level	$(1/4) \cdot n^2 + n - 2$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0, i2=1, i3=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + (3/4) \cdot n$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0, i2=0, i3=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (1/4)$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0, i2=0, i3=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + n - 4$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0, i3=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 2$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0, i3=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0, i2=1, i3=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0, i2=1, i3=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 2$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0, i2=0, i3=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-7/4)$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0, i2=0, i3=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + n - 4$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 4$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + n - 5$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0, i2=0); read B[i4 + 1, i5] (i0=0, i1=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + n - 1$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0, i3=0); write A[i4, i5] (i0=0, i1=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + n$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	$2 \cdot n$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0); write A[i4, i5] (i0=0, i1=0) (+2)
n^2	1	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 3$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 - 1, i3] (i0=0, i1=0); read B[i4 - 1, i5] (i0=0, i1=0, i4=0) (+3)
n^2	1	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 3$	$2 \cdot n$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i1=0); read B[i4 + 1, i5] (i0=0, i1=0) (+4)
n^2	1	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$2 \cdot n - 2$	$0.167 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0, i4=0) (+2)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 3$	n	$0.0833 \cdot n^{-2}$	write B[i2, i3] (i0=0, i1=0); write B[i2, i3] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 3$	n	$0.0833 \cdot n^{-2}$	write A[i4, i5] (i0=0, i1=0); write A[i4, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.5	level	$(1/4) \cdot n^2 + n - 3$	$n - 1$	$0.0833 \cdot n^2$	read A[i2, i3 + 1] (i0=0, i1=1, i2=0); read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.5	level	$(1/4) \cdot n^2 + n - 3$	$n - 1$	$0.0833 \cdot n^2$	read B[i4, i5 + 1] (i0=0, i1=1, i4=0); read B[i4, i5 + 1] (i0=0, i4=0)
n^2	0.354	level	2	$(1/4) \cdot n^2 + (-1/2) \cdot n + (1/4)$	$0.0208/n$	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (7/4) \cdot n - 3$	$0.0208/n$	read A[i2, i3] (i0=0, i1=0, i2=0, i3=0); read A[i2, i3 - 1] (i0=0, i1=0, i3=0) (+8)
n^2	0.125	level	$(1/4) \cdot n^2 + n - 2$	$(1/4) \cdot n - 4$	$0.0208 \cdot n^2$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0); read B[i4, i5 + 1] (i0=0, i1=0)
n^2	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/4) \cdot n + (-9/4)$	$0.0208 \cdot n^2$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0); read B[i4, i5 + 1] (i0=0, i1=0)
n^2	0.125	level	1	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	$0.0104/n$	read B[i4, i5 + 1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.125	level	1	$(1/8) \cdot n^2 + (-5/4) \cdot n + (9/8)$	$0.0104/n$	read A[i2, i3 + 1] (i0=0, i1=0)
$n^{1.5}$	1.41	level	$(1/2) \cdot n - 1$	$2 \cdot n - 4$	$0.167 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 4$	1	$0.0833 \cdot n^{-3}$	read A[i2 + 1, i3] (i0=0, i1=1)
n^1	0.5	level	$(1/4) \cdot n^2 + n - 3$	1	$0.0833 \cdot n^{-3}$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + n - 3$	1	$0.0833 \cdot n^{-3}$	read B[i4, i5 + 1] (i0=0, i1=0, i4=0)
n^0	5	level	1	5	$0.417 \cdot n^{-3}$	read B[i4, i5 + 1] (i0=0, i1=0, i4=0)
n^0	1.41	level	2	1	$0.0833 \cdot n^{-3}$	read B[i4, i5 + 1] (i0=0, i1=0, i4=0)
n^0	1	level	1	1	$0.0833 \cdot n^{-3}$	read B[i4, i5 + 1] (i0=0, i1=0, i4=0)

Two-array 2-D stencil: cross-sweep plane reuses at $(1/4)n^2 + O(n)$ lines (four symmetric write/read families, combined $0.5 \cdot n^4$, headroom +1.0), row-window reuses at $(3/8)n$ lines at order $n^{3.5}$. The plane boundary $64 \cdot ((1/4)n^2)$ bytes = both arrays is the time-tiling cliff.

jacobi-2d — single-shot [exact]

Accesses $A(n) = 12 \cdot n^3 - 24 \cdot n^2 + 12 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.5 \cdot n^4 + 0.707 \cdot n^{3.5} + 28.9 \cdot n^3 + 8.49 \cdot n^{2.5} + 27 \cdot n^2$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.125	level	$(1/4) \cdot n^2 + n - 2$	$(1/4) \cdot n^3 + (-43/8) \cdot n^2 + (181/8) \cdot n - 10$	0.0208	write B[i2, i3] (i0=0, i2=1); write B[i2, i3] (i0=0) (+1)
n^4	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$(1/4) \cdot n^3 + (-29/8) \cdot n^2 + 13 \cdot n + (-45/8)$	0.0208	write B[i2, i3] (i0=0, i2=1); write B[i2, i3] (i0=0) (+1)
n^4	0.125	level	$(1/4) \cdot n^2 + n - 3$	$(1/4) \cdot n^3 + (-43/8) \cdot n^2 + (181/8) \cdot n - 10$	0.0208	read A[i2 + 1, i3] (i0=0, i2=1); read A[i2 + 1, i3] (i0=0) (+1)
n^4	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/4) \cdot n^3 + (-29/8) \cdot n^2 + 13 \cdot n + (-45/8)$	0.0208	read A[i2 + 1, i3] (i0=0, i2=1); read A[i2 + 1, i3] (i0=0) (+1)
$n^{3.5}$	0.177	level	$(1/2) \cdot n + (5/2)$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + (17/2) \cdot n$	0.0208	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
$n^{3.5}$	0.177	level	$(1/2) \cdot n$	$(1/4) \cdot n^3 + (-9/2) \cdot n^2 + 8 \cdot n$	0.0208	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
$n^{3.5}$	0.177	level	$(1/2) \cdot n + 1$	$(1/4) \cdot n^3 + (-9/2) \cdot n^2 + 8 \cdot n$	0.0208	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)
$n^{3.5}$	0.177	level	$(1/2) \cdot n + (7/2)$	$(1/4) \cdot n^3 + (-11/4) \cdot n^2 + (9/2) \cdot n$	0.0208	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	6	level	4	$3 \cdot n^3 - 7 \cdot n^2 + 4 \cdot n$	0.25	read A[i2 + 1, i3] (i0=0); write B[i2, i3] (i0=0) (+2)
n^3	4	level	4	$2 \cdot n^3 - 6 \cdot n^2 + 4 \cdot n$	0.167	read A[i2, i3] (i0=0); read B[i4, i5] (i0=0)
n^3	3	level	4	$(3/2) \cdot n^3 + (-7/2) \cdot n^2 + 2 \cdot n$	0.125	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
n^3	1.75	level	1	$(7/4) \cdot n^3 + (-7/4) \cdot n^2$	0.146	read A[i2, i3 - 1] (i0=0); read B[i4, i5 - 1] (i0=0)
n^3	1.68	level	5	$(3/4) \cdot n^3 + (-11/4) \cdot n^2 + 2 \cdot n$	0.0625	read A[i2, i3 - 1] (i0=0); read A[i2 + 1, i3] (i0=0) (+4)
n^3	1.5	level	1	$(3/2) \cdot n^3 + (-3/2) \cdot n^2$	0.125	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)
n^3	1	level	$(1/4) \cdot n^2 + n - 2$	$2 \cdot n^2 - 9 \cdot n + 4$	0.167/n	read A[i2, i3] (i0=0, i2=0, i3=0); write B[i2, i3] (i0=0, i2=2, i3=0) (+3)
n^3	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n^2 - 9 \cdot n + 4$	0.167/n	read A[i2 + 1, i3] (i0=0, i2=2, i3=0); read A[i2 + 1, i3] (i0=0, i3=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$2 \cdot n^2 - 11 \cdot n + 5$	$0.167/n$	write B[i2, i3] (i0=0, i2=2, i3=0); write B[i2, i3] (i0=0, i3=0) (+1)
n^3	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$2 \cdot n^2 - 7 \cdot n + 3$	$0.167/n$	read A[i2, i3] (i0=0, i2=0, i3=0); read A[i2 + 1, i3] (i0=0, i2=2, i3=0) (+5)
n^3	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n^2 - 9 \cdot n + 4$	$0.167/n$	write B[i2, i3] (i0=0, i2=0); write B[i2, i3] (i0=0) (+1)
n^3	1	level	$(1/4) \cdot n^2 + n - 4$	$2 \cdot n^2 - 11 \cdot n + 5$	$0.167/n$	read A[i2 + 1, i3] (i0=0, i2=1); read A[i2 + 1, i3] (i0=0) (+1)
n^3	1	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$2 \cdot n^2 - 4 \cdot n + 2$	$0.167/n$	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0, i4=0) (+2)
n^3	0.559	level	5	$(1/4) \cdot n^3 + (-1/2) \cdot n^2 + (1/4) \cdot n$	0.0208	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
n^3	0.559	level	5	$(1/4) \cdot n^3 + (-1/4) \cdot n^2$	0.0208	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
n^3	0.354	level	2	$(1/4) \cdot n^3 + (-1/2) \cdot n^2 + (1/4) \cdot n$	0.0208	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.354	level	2	$(1/4) \cdot n^3 + (-9/4) \cdot n^2 + 2 \cdot n$	0.0208	read A[i2, i3 + 1] (i0=0); read B[i4, i5 + 1] (i0=0, i4=0) (+1)
n^3	0.25	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$(1/2) \cdot n^2 + (-27/4) \cdot n + (13/4)$	0.0417/n	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 2$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 3$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/4) \cdot n^2 + (-19/8) \cdot n + (9/8)$	0.0208/n	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 1$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$(1/4) \cdot n^2 + (-17/4) \cdot n + 4$	0.0208/n	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0, i4=0)
n^3	0.125	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/4) \cdot n^2 + (-5/2) \cdot n + (9/4)$	0.0208/n	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0, i4=0)
n^3	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n + (3/2)$	$(1/4) \cdot n^2 + (-11/4) \cdot n + (5/2)$	0.0208/n	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.125	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$(1/4) \cdot n^2 + (-17/4) \cdot n + 4$	0.0208/n	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$(1/4) \cdot n^2 + (-5/2) \cdot n + (9/4)$	0.0208/n	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + n - 2$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	read A[i2, i3 + 1] (i0=0, i2=0); read B[i4, i5 + 1] (i0=0)
n^3	0.125	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$(1/4) \cdot n^2 + (-19/8) \cdot n + (9/8)$	0.0208/n	read A[i2, i3 + 1] (i0=0, i2=0); read B[i4, i5 + 1] (i0=0)
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + n - 4$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	write B[i2, i3] (i0=0, i2=0); write A[i4, i5] (i0=0)
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n + 1 \rightarrow (1/4) \cdot n^2 + n - 5$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n \rightarrow (1/4) \cdot n^2 + n - 6$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	read A[i2 + 1, i3] (i0=0, i2=0); read B[i4 + 1, i5] (i0=0)
n^3	0.119	ramp	$(1/4) \cdot n^2 + (3/4) \cdot n \rightarrow (1/4) \cdot n^2 + n - 6$	$(1/4) \cdot n^2 + (-33/8) \cdot n + 2$	0.0208/n	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
$n^{2.5}$	1.41	level	$(1/2) \cdot n$	$2 \cdot n^2 - 4 \cdot n$	0.167/n	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.41	level	$(1/2) \cdot n + 1$	$2 \cdot n^2 - 4 \cdot n$	$0.167/n$	read A[i2, i3] (i0=0); read B[i4, i5] (i0=0, i5=0)
$n^{2.5}$	1.41	level	$(1/2) \cdot n + (5/2)$	$2 \cdot n^2 - 4 \cdot n$	$0.167/n$	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
$n^{2.5}$	1.41	level	$(1/2) \cdot n + 2$	$2 \cdot n^2 - 4 \cdot n$	$0.167/n$	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
$n^{2.5}$	1.41	level	$(1/2) \cdot n + (9/2)$	$2 \cdot n^2 - 4 \cdot n$	$0.167/n$	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0)
$n^{2.5}$	1.41	level	$(1/2) \cdot n - 1$	$2 \cdot n^2 - 4 \cdot n$	$0.167/n$	read A[i2, i3 + 1] (i0=0, i1=0); read B[i4, i5 + 1] (i0=0, i1=0) (+2)
n^2	2	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$4 \cdot n - 2$	$0.333 \cdot n^{-2}$	write B[i2, i3] (i0=0, i2=1, i3=0); write B[i2, i3] (i0=0, i3=0) (+1)
n^2	2	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 2$	$4 \cdot n - 4$	$0.333 \cdot n^{-2}$	read A[i2 - 1, i3] (i0=0); read A[i2 + 1, i3] (i0=0) (+2)
n^2	2	level	$(1/4) \cdot n^2 + 2 \cdot n + (7/4)$	$4 \cdot n - 4$	$0.333 \cdot n^{-2}$	read A[i2 - 1, i3] (i0=0); read A[i2 + 1, i3] (i0=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1	level	$(1/4) \cdot n^2 + n - 2$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i2=1, i3=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (3/4) \cdot n$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i2=0, i3=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (1/4)$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i2=0, i3=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + n - 4$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i3=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 2$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i3=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i2=1, i3=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n - 1$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i2=1, i3=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 2$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i2=0, i3=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (3/2) \cdot n + (-7/4)$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i2=0, i3=0); read B[i4 + 1, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1	level	$(1/4) \cdot n^2 + n - 3$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + n - 4$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 4$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + n - 5$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0, i2=0); read B[i4 + 1, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + n - 1$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0, i3=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + n$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (7/4) \cdot n + 1$	$2 \cdot n - 1$	$0.167 \cdot n^{-2}$	write B[i2, i3] (i0=0); write A[i4, i5] (i0=0)
n^2	1	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 3$	$2 \cdot n - 2$	$0.167 \cdot n^{-2}$	read A[i2 - 1, i3] (i0=0); read B[i4 - 1, i5] (i0=0, i4=0)
n^2	1	level	$(1/4) \cdot n^2 + (9/4) \cdot n - 3$	$2 \cdot n - 2$	$0.167 \cdot n^{-2}$	read A[i2 + 1, i3] (i0=0); read B[i4 + 1, i5] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 3$	$n - 1$	$0.0833 \cdot n^{-2}$	write B[i2, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.5	level	$(1/4) \cdot n^2 + (3/4) \cdot n - 3$	n	$0.0833 \cdot n^2$	write A[i4, i5] (i0=0)
n^2	0.5	level	$(1/4) \cdot n^2 + n - 3$	$n - 1$	$0.0833 \cdot n^2$	read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.5	level	$(1/4) \cdot n^2 + n - 3$	n	$0.0833 \cdot n^2$	read B[i4, i5 + 1] (i0=0, i1=0, i4=0); read B[i4, i5 + 1] (i0=0, i4=0)

Two-array 2-D stencil: cross-sweep plane reuses at $(1/4)n^2 + O(n)$ lines (four symmetric write/read families, combined $0.5 \cdot n^4$, headroom +1.0), row-window reuses at $(3/8)n$ lines at order $n^3.5$. The plane boundary $64 \cdot ((1/4)n^2)$ bytes = both arrays is the time-tiling cliff.

lu — infinite-repeat [exact]

Accesses $A(n) = (4/3) \cdot n^3 + (-1/2) \cdot n^2 + (-5/6) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma \text{mass/warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.00643 \cdot n^4 + 0.22 \cdot n^{3.5} + 2.33 \cdot n^3 + 6.59 \cdot n^{2.5} + 39.1 \cdot n^2 + 24.9 \cdot n^{1.5} + 254 \cdot n^1 + 546 \cdot n^{0.5} + 64.3 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00309	ramp	$(21/4) \cdot n - 81 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(7/384) \cdot n^3 + (-123/64) \cdot n^2 + (1619/24) \cdot n - 790$	0.0137	read A[i3, i2] (i0=0)
n^4	0.00193	ramp	$(11/8) \cdot n + 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 91$	$(7/512) \cdot n^3 + (-195/128) \cdot n^2 + (901/16) \cdot n - 690$	0.0103	read A[i5, i4] (i0=0)
n^4	0.000411	ramp	$(25/4) \cdot n - 121 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/384) \cdot n^3 + (-21/64) \cdot n^2 + (329/24) \cdot n - 190$	0.00195	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.000316	ramp	$3 \cdot n - 16 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 86	$(7/3072) \cdot n^3$ $+ (-$ $35/128) \cdot n^2$ $+ (259/24) \cdot n -$ 140	0.00171	read A[i5, i4] (i0=0)
n^4	0.00031	ramp	$(17/8) \cdot n - 2$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 121	$(7/3072) \cdot n^3$ $+ (-$ $35/128) \cdot n^2$ $+ (259/24) \cdot n -$ 140	0.00171	read A[i5, i4] (i0=0)
n^4	0.000268	ramp	$(9/4) \cdot n - 4$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 91	$(1/512) \cdot n^3$ $+ (-$ $15/64) \cdot n^2$ $+ (37/4) \cdot n -$ 120	0.00146	read A[i5, i4] (i0=0)
n^4	6.23e-05	ramp	$2 \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 86	$(1/3072) \cdot n^3$ $+ (-55/48) \cdot n$ $+ 25$	0.000244	read A[i5, i4] (i0=0)
n^4	4.1e-05	ramp	$(25/8) \cdot n -$ 17 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 121	$(1/3072) \cdot n^3$ $+ (-3/64) \cdot n^2$ $+ (107/48) \cdot n -$ 35	0.000244	read A[i5, i4] (i0=0)
$n^{3.5}$	0.0988	ramp	11 $\rightarrow$ $(9/8) \cdot n - 2$	$(7/48) \cdot n^3$ $+ (-$ $63/16) \cdot n^2$ $+ (217/6) \cdot n$ $- 112$	0.109	read A[i3, i2] (i0=0)
$n^{3.5}$	0.0847	ramp	6 $\rightarrow (9/8) \cdot n$ $- 8$	$(49/384) \cdot n^3$ $+ (-$ $231/64) \cdot n^2$ $+ (203/6) \cdot n$ $- 105$	0.0957	read A[i5, i4] (i0=0)
$n^{3.5}$	0.012	ramp	12 $\rightarrow$ $(9/8) \cdot n - 15$	$(7/384) \cdot n^3$ $+ (-$ $35/64) \cdot n^2$ $+ (119/24) \cdot n -$ 14	0.0137	read A[i5, i4] (i0=0)
$n^{3.5}$	0.011	ramp	22 $\rightarrow$ $(9/8) \cdot n - 9$	$(7/384) \cdot n^3$ $+ (-$ $147/128) \cdot n^2$ $+ (1141/48) \cdot n$ $- 161$	0.0137	read A[i1, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.0108	ramp	14 $\rightarrow$ (9/8)·n - 16	(7/384)·n ³ + (- 35/32)·n ² + (259/12)·n - 140	0.0137	read A[i1, i5] (i0=0)
$n^{3.5}$	0.00162	ramp	21 $\rightarrow$ (9/8)·n - 15	(1/384)·n ³ + (- 17/128)·n ² + (103/48)·n - 11	0.00195	read A[i1, i3] (i0=0)
$n^{3.5}$	0.00148	ramp	15 $\rightarrow$ (9/8)·n - 23	(1/384)·n ³ + (-3/16)·n ² + (13/3)·n - 32	0.00195	read A[i1, i5] (i0=0)
n^3	0.253	level	3	(7/48)·n ³ + (- 21/32)·n ² + (-581/24)·n + 161	0.109	write A[i1, i4] (i0=0)
n^3	0.253	level	3	(7/48)·n ³ + (- 23/32)·n ² + (-713/24)·n + 223	0.109	read A[i1, i5] (i0=0, i4=0); read A[i1, i5] (i0=0)
n^3	0.221	level	3	(49/384)·n ³ + (- 469/128)·n ² + (1687/48)·n - 112	0.0957	read A[i1, i3] (i0=0)
n^3	0.189	level	3	(7/64)·n ³ + (- 399/128)·n ² + (483/16)·n - 98	0.082	write A[i1, i2] (i0=0)
n^3	0.167	level	1	(1/6)·n ³ + (41/6)·n - 2	0.125	read A[i1, i2] (i0=0, i1=1); read A[i1, i3] (i0=0) (+6)
n^3	0.146	level	1	(7/48)·n ³ + (-7/16)·n ² + (-35/6)·n + 35	0.109	read A[i1, i2] (i0=0, i3=0); read A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0737	ramp	$(3/2) \cdot n - 3$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 31$	$(49/128) \cdot n^2$ $+ (-315/16) \cdot n$ $+ 253$	$0.287/n$	read A[i5, i4] (i0=0)
n^3	0.0732	ramp	$(17/4) \cdot n - 48$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(49/128) \cdot n^2$ $+ (-371/16) \cdot n$ $+ 351$	$0.287/n$	read A[i3, i2] (i0=0)
n^3	0.0731	ramp	$(13/4) \cdot n - 23$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(49/128) \cdot n^2$ $+ (-357/16) \cdot n$ $+ 322$	$0.287/n$	read A[i3, i2] (i0=0)
n^3	0.0711	ramp	$(17/4) \cdot n - 49$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(3/8) \cdot n^2 + (-47/2) \cdot n + 370$	$0.281/n$	read A[i3, i2] (i0=0)
n^3	0.0598	ramp	$(17/4) \cdot n - 49$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(5/16) \cdot n^2 + (-75/4) \cdot n + 280$	$0.234/n$	read A[i3, i2] (i0=0)
n^3	0.0595	ramp	$(11/8) \cdot n + 2$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 57$	$(21/64) \cdot n^2 + (-87/4) \cdot n + 360$	$0.246/n$	read A[i5, i4] (i0=0)
n^3	0.0538	ramp	$(7/2) \cdot n - 31$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(35/128) \cdot n^2 + (-225/16) \cdot n + 180$	$0.205/n$	read A[i2, i2] (i0=0)
n^3	0.0516	ramp	$(19/8) \cdot n - 8$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 31$	$(35/128) \cdot n^2 + (-255/16) \cdot n + 230$	$0.205/n$	read A[i5, i4] (i0=0)
n^3	0.0379	ramp	$(1/2) \cdot n + 4$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 91$	$(9/32) \cdot n^2 - 19 \cdot n + 320$	$0.211/n$	read A[i5, i4] (i0=0)
n^3	0.0361	level	3	$(1/48) \cdot n^3 + (5/32) \cdot n^2 + (-83/24) \cdot n + 14$	0.0156	write A[i1, i4] (i0=0, i5=0); write A[i1, i4] (i0=0)
n^3	0.0316	level	3	$(7/384) \cdot n^3 + (-15/128) \cdot n^2 + (-5/48) \cdot n - 1$	0.0137	read A[i1, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0316	level	3	$(7/384) \cdot n^3 + (-63/128) \cdot n^2 + (175/48) \cdot n - 7$	0.0137	write A[i1, i2] (i0=0)
n^3	0.0316	level	3	$(7/384) \cdot n^3 + (-3/64) \cdot n^2 + (-13/24) \cdot n - 2$	0.0137	write A[i1, i2] (i0=0)
n^3	0.0271	level	3	$(1/64) \cdot n^3 + (-33/128) \cdot n^2 + (17/16) \cdot n$	0.0117	write A[i1, i2] (i0=0)
n^3	0.0208	level	1	$(1/48) \cdot n^3 + (-1/8) \cdot n^2 + (-5/24) \cdot n - 1$	0.0156	read A[i1, i2] (i0=0)
n^3	0.0193	level	$(1/8) \cdot n^2$	$(7/128) \cdot n^2 + (-53/16) \cdot n + 50$	0.041/n	read A[i1, i2] (i0=0, i3=0)
n^3	0.0127	ramp	$(3/8) \cdot n + 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 91$	$(3/32) \cdot n^2 + (-49/8) \cdot n + 100$	0.0703/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^3	0.012	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/16) \cdot n^2 + (-15/4) \cdot n + 56$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.012	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/16) \cdot n^2 + (-15/4) \cdot n + 56$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.012	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/16) \cdot n^2 + (-15/4) \cdot n + 56$	0.0469/n	read A[i3, i2] (i0=0, i3=0)
n^3	0.012	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/16) \cdot n^2 + (-15/4) \cdot n + 56$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.0119	ramp	$3 \cdot n - 18 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 28$	$(1/16) \cdot n^2 + (-29/8) \cdot n + 52$	0.0469/n	read A[i5, i4] (i0=0, i4=0)
n^3	0.0113	ramp	$(21/4) \cdot n - 81 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/16) \cdot n^2 + (-19/4) \cdot n + 90$	0.0469/n	read A[i3, i2] (i0=0, i3=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0113	ramp	$(21/4) \cdot n - 81 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/16) \cdot n^2 + (-19/4) \cdot n + 90$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.0113	ramp	$(21/4) \cdot n - 81 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/16) \cdot n^2 + (-19/4) \cdot n + 90$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.0112	ramp	$(29/8) \cdot n - 37 \rightarrow (1/8) \cdot n^2 + (-15/4) \cdot n + 68$	$(7/128) \cdot n^2 + (-51/16) \cdot n + 46$	0.041/n	read A[i1, i4] (i0=0, i5=0)
n^3	0.011	ramp	$(27/8) \cdot n - 28 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(7/128) \cdot n^2 + (-39/16) \cdot n + 27$	0.041/n	read A[i2, i2] (i0=0)
n^3	0.0105	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(7/128) \cdot n^2 + (-51/16) \cdot n + 46$	0.041/n	read A[i2, i2] (i0=0)
n^3	0.0105	ramp	$(33/8) \cdot n - 44 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 9$	$(7/128) \cdot n^2 + (-51/16) \cdot n + 46$	0.041/n	read A[i3, i2] (i0=0)
n^3	0.0104	ramp	$(25/8) \cdot n - 20 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 25$	$(7/128) \cdot n^2 + (-51/16) \cdot n + 46$	0.041/n	read A[i3, i2] (i0=0)
n^3	0.0103	ramp	$(9/4) \cdot n - 9 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(7/128) \cdot n^2 + (-49/16) \cdot n + 42$	0.041/n	read A[i5, i4] (i0=0)
n^3	0.0103	ramp	$(19/8) \cdot n - 12 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 32$	$(7/128) \cdot n^2 + (-51/16) \cdot n + 46$	0.041/n	read A[i5, i4] (i0=0)
n^3	0.0103	ramp	$(9/4) \cdot n - 6 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 46$	$(7/128) \cdot n^2 + (-51/16) \cdot n + 46$	0.041/n	read A[i5, i4] (i0=0, i4=6)
n^3	0.0103	ramp	$(17/8) \cdot n - 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(7/128) \cdot n^2 + (-51/16) \cdot n + 46$	0.041/n	read A[i5, i4] (i0=0, i4=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.01	ramp	$(21/4) \cdot n - 80 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/128) \cdot n^2 + (-65/16) \cdot n + 75$	0.041/n	read A[i3, i2] (i0=0)
n^3	0.00999	ramp	$(17/4) \cdot n - 47 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(7/128) \cdot n^2 + (-63/16) \cdot n + 70$	0.041/n	read A[i3, i2] (i0=0)
n^3	0.00997	ramp	$(41/8) \cdot n - 76 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(7/128) \cdot n^2 + (-65/16) \cdot n + 75$	0.041/n	read A[i3, i2] (i0=0)
n^3	0.0098	ramp	$3 \cdot n - 17 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 53$	$(7/128) \cdot n^2 + (-63/16) \cdot n + 70$	0.041/n	read A[i5, i4] (i0=0, i4=8)
n^3	0.00969	ramp	$(17/8) \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 81$	$(7/128) \cdot n^2 + (-63/16) \cdot n + 70$	0.041/n	read A[i5, i4] (i0=0, i4=15)
n^3	0.00902	level	3	$(1/192) \cdot n^3 + (23/64) \cdot n^2 + (113/12) \cdot n - 97$	0.00391	read A[i1, i5] (i0=0, i1=1); read A[i1, i2] (i0=0, i1=2, i2=0) (+9)
n^3	0.00901	ramp	$(35/8) \cdot n - 54 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 5$	$(3/64) \cdot n^2 + (-11/4) \cdot n + 40$	0.0352/n	read A[i2, i2] (i0=0)
n^3	0.00835	ramp	$(9/4) \cdot n - 5 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(3/64) \cdot n^2 + (-27/8) \cdot n + 60$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00754	ramp	$(33/8) \cdot n - 46 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(5/128) \cdot n^2 + (-35/16) \cdot n + 30$	0.0293/n	read A[i2, i2] (i0=0)
n^3	0.00705	ramp	$(27/8) \cdot n - 25 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/128) \cdot n^2 + (-45/16) \cdot n + 50$	0.0293/n	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00671	ramp	$(5/4) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(3/64) \cdot n^2 + (-21/8) \cdot n + 36$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00641	ramp	$2 \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(3/64) \cdot n^2 + (-27/8) \cdot n + 60$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00615	ramp	$(9/8) \cdot n + 5 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 121$	$(3/64) \cdot n^2 + (-27/8) \cdot n + 60$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00561	ramp	$(11/8) \cdot n + 2 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(5/128) \cdot n^2 + (-35/16) \cdot n + 30$	0.0293/n	read A[i5, i4] (i0=0)
n^3	0.00276	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$(1/128) \cdot n^2 + (-37/64) \cdot n + (1353/128)$	0.00586/n	read A[i1, i2] (i0=0, i3=0)
n^3	0.00276	level	$(1/8) \cdot n^2$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i1, i2] (i0=0, i3=0)
n^3	0.00211	ramp	$2 \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(1/64) \cdot n^2 + (-5/4) \cdot n + 25$	0.0117/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^3	0.00203	ramp	$(9/8) \cdot n + 5 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 121$	$(1/64) \cdot n^2 + (-5/4) \cdot n + 25$	0.0117/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^3	0.00176	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 53$	$(1/128) \cdot n^2 + (1/16) \cdot n - 9$	0.00586/n	read A[i5, i4] (i0=0, i4=8)
n^3	0.00152	ramp	$(35/8) \cdot n - 55 \rightarrow$ $(1/8) \cdot n^2 + (-37/8) \cdot n + 89$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i1, i4] (i0=0, i5=0)
n^3	0.00151	ramp	$(17/4) \cdot n - 50 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00151	ramp	$(33/8) \cdot n - 46 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i2, i2] (i0=0)
n^3	0.00151	ramp	$(33/8) \cdot n - 46 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i2, i2] (i0=0)
n^3	0.00143	ramp	$(41/8) \cdot n - 75 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 9$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i3, i2] (i0=0)
n^3	0.00143	ramp	$(41/8) \cdot n - 77 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i2, i2] (i0=0)
n^3	0.00142	ramp	$(33/8) \cdot n - 43 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 25$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i3, i2] (i0=0)
n^3	0.0014	ramp	$(13/4) \cdot n - 22 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 46$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i5, i4] (i0=0, i4=6)
n^3	0.0014	ramp	$(13/4) \cdot n - 26 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 50$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i5, i4] (i0=0)
n^3	0.0014	ramp	$(25/8) \cdot n - 19 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i5, i4] (i0=0, i4=7)
n^3	0.00136	ramp	$(49/8) \cdot n - 115 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/128) \cdot n^2 + (-11/16) \cdot n + 15$	0.00586/n	read A[i3, i2] (i0=0)
n^3	0.00131	ramp	$(25/8) \cdot n - 18 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 81$	$(1/128) \cdot n^2 + (-11/16) \cdot n + 15$	0.00586/n	read A[i5, i4] (i0=0, i4=15)
n^3	0.00114	ramp	$2 \cdot n - 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 53$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00114	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 53	$(1/128) \cdot n^2$ $+ (-7/16) \cdot n$ $+ 6$	0.00586/n	read A[i5, i4] (i0=0)
n^3	0.0011	ramp	$(5/4) \cdot n + 3$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 77	$(1/128) \cdot n^2$ $+ (-7/16) \cdot n$ $+ 6$	0.00586/n	read A[i5, i4] (i0=0)
n^3	0.0011	ramp	$(9/8) \cdot n + 4$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 81	$(1/128) \cdot n^2$ $+ (-7/16) \cdot n$ $+ 6$	0.00586/n	read A[i5, i4] (i0=0)
n^3	0.00107	ramp	$(17/8) \cdot n - 3$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 81	$(1/128) \cdot n^2$ $+ (-9/16) \cdot n$ $+ 10$	0.00586/n	read A[i5, i4] (i0=0)
n^3	0.00104	ramp	$3 \cdot n - 16 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 86	$(1/128) \cdot n^2$ $+ (-11/16) \cdot n$ $+ 15$	0.00586/n	read A[i5, i4] (i0=0)
$n^{2.5}$	1.53	ramp	$5 \rightarrow (9/8) \cdot n$ $- 2$	$(15/8) \cdot n^2$ $+ (-45/2) \cdot n$ $+ 70$	1.41/n	read A[i5, i4] (i0=0)
$n^{2.5}$	1.22	ramp	$5 \rightarrow (9/8) \cdot n$ $- 2$	$(35/16) \cdot n^2$ $+ (-235/8) \cdot n$ $+ 105$	1.64/n	read A[i3, i2] (i0=0)
$n^{2.5}$	0.315	ramp	$5 \rightarrow (9/8) \cdot n$ $- 1$	$(3/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 1	0.281/n	read A[i5, i4] (i0=0, i4=0)
$n^{2.5}$	0.298	ramp	$6 \rightarrow (9/8) \cdot n$ $- 8$	$(3/8) \cdot n^2 +$ $(-57/8) \cdot n +$ 33	0.281/n	read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	0.277	ramp	$14 \rightarrow$ $(9/8) \cdot n - 10$	$(3/8) \cdot n^2 -$ $15 \cdot n + 144$	0.281/n	read A[i1, i5] (i0=0)
$n^{2.5}$	0.267	ramp	$5 \rightarrow (9/8) \cdot n$ $- 9$	$(1/2) \cdot n^2 +$ $(-21/2) \cdot n +$ 55	0.375/n	read A[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.255	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(5/16) \cdot n^2$ $+ (-15/4) \cdot n$ $+ 10$	0.234/n	read A[i5, i4] (i0=0)
$n^{2.5}$	0.246	ramp	$4 \rightarrow (9/8) \cdot n$ $- 2$	$(7/16) \cdot n^2$ $+ (-23/8) \cdot n$ $+ 5$	0.328/n	read A[i3, i2] (i0=0)
$n^{2.5}$	0.235	ramp	$12 \rightarrow$ $(9/8) \cdot n - 15$	$(7/16) \cdot n^2$ $+ (-85/8) \cdot n$ $+ 58$	0.328/n	read A[i1, i5] (i0=0)
$n^{2.5}$	0.223	ramp	$5 \rightarrow (9/8) \cdot n$ $- 16$	$(7/16) \cdot n^2$ $- 14 \cdot n +$ 112	0.328/n	read A[i1, i5] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.211	ramp	$3 \rightarrow (9/8) \cdot n - 2$	$(3/8) \cdot n^2 + (-9/8) \cdot n$	$0.281/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.211	ramp	$4 \rightarrow (9/8) \cdot n - 2$	$(3/8) \cdot n^2 + (-17/8) \cdot n + 3$	$0.281/n$	read A[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.207	ramp	$4 \rightarrow (9/8) \cdot n - 8$	$(49/128) \cdot n^2 + (-91/16) \cdot n + 21$	$0.287/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.205	ramp	$5 \rightarrow (9/8) \cdot n - 9$	$(49/128) \cdot n^2 + (-119/16) \cdot n + 36$	$0.287/n$	read A[i5, i4] (i0=0, i5=0)
$n^{2.5}$	0.198	ramp	$14 \rightarrow (9/8) \cdot n - 9$	$(3/8) \cdot n^2 + (-105/8) \cdot n + 114$	$0.281/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.177	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$(21/64) \cdot n^2 + (-21/4) \cdot n + 21$	$0.246/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.051	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0469/n$	read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	0.0508	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/16) \cdot n^2 + (-7/8) \cdot n + 3$	$0.0469/n$	read A[i5, i4] (i0=0, i4=0)
$n^{2.5}$	0.0466	ramp	$21 \rightarrow (9/8) \cdot n - 8$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	$0.0469/n$	read A[i1, i5] (i0=0, i4=0)
$n^{2.5}$	0.0465	ramp	$20 \rightarrow (9/8) \cdot n - 9$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	$0.0469/n$	read A[i1, i5] (i0=0, i4=1)
$n^{2.5}$	0.0351	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0469/n$	read A[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.0351	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0469/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.0332	ramp	$13 \rightarrow (9/8) \cdot n - 14$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	$0.0469/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0331	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0469/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0316	ramp	$13 \rightarrow (9/8) \cdot n - 23$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	$0.0469/n$	read A[i1, i5] (i0=0)
$n^{2.5}$	0.031	ramp	$6 \rightarrow (9/8) \cdot n - 23$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	$0.0469/n$	read A[i1, i5] (i0=0, i5=0)
$n^{2.5}$	0.0294	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	$0.041/n$	read A[i5, i4] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.0294	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	$0.041/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.0294	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	$0.041/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.0277	ramp	$13 \rightarrow (9/8) \cdot n - 23$	$(7/128) \cdot n^2 + (-35/16) \cdot n + 21$	$0.041/n$	read A[i5, i4] (i0=0)
n^2	2.27	level	3	$(21/16) \cdot n^2 + (-161/8) \cdot n + 77$	$0.984/n$	read A[i1, i3] (i0=0)
n^2	1.31	level	1	$(21/16) \cdot n^2 + (-97/8) \cdot n + 33$	$0.984/n$	read A[i1, i3] (i0=0)
n^2	1.23	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$ $\rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(21/4) \cdot n - 132$	$3.94 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	1.08	level	3	$(5/8) \cdot n^2 - 10 \cdot n + 40$	$0.469/n$	write A[i1, i2] (i0=0)
n^2	1.03	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 18$ $\rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 13$	$(35/8) \cdot n - 100$	$3.28 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	1.01	ramp	$(13/4) \cdot n - 24$ $\rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(21/4) \cdot n - 139$	$3.94 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.997	ramp	$(9/4) \cdot n - 7$ $\rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(21/4) \cdot n - 133$	$3.94 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.976	ramp	$(5/8) \cdot n \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 30$	$(21/4) \cdot n - 117$	$3.94 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.972	level	2	$(11/16) \cdot n^2 + (-19/8) \cdot n$	$0.516/n$	write A[i1, i2] (i0=0)
n^2	0.964	ramp	$(13/4) \cdot n - 25$ $\rightarrow (1/16) \cdot n^2 + (5/8) \cdot n$	$5 \cdot n - 130$	$3.75 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.884	level	2	$(5/8) \cdot n^2 + (-15/4) \cdot n$	$0.469/n$	write A[i1, i2] (i0=0)
n^2	0.85	ramp	$(5/2) \cdot n - 13$ $\rightarrow (1/16) \cdot n^2 + (5/8) \cdot n - 1$	$(35/8) \cdot n - 95$	$3.28 \cdot n^{-2}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.847	ramp	$(13/4) \cdot n - 24 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(35/8) \cdot n - 110$	$3.28 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.833	ramp	$(9/4) \cdot n - 7 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(35/8) \cdot n - 105$	$3.28 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.807	ramp	$(1/2) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(9/2) \cdot n - 134$	$3.38 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-63/8) \cdot n + 35$	$0.328/n$	read A[i1, i3] (i0=0)
n^2	0.729	ramp	$(21/8) \cdot n - 14 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(15/4) \cdot n - 84$	$2.81 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.695	ramp	$(11/8) \cdot n + 1 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(15/4) \cdot n - 95$	$2.81 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.663	level	3	$(49/128) \cdot n^2 + (-105/16) \cdot n + 28$	$0.287/n$	write A[i1, i4] (i0=0, i4=0)
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-57/8) \cdot n + 33$	$0.281/n$	write A[i1, i2] (i0=0)
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.328/n$	write A[i1, i2] (i0=0)
n^2	0.578	ramp	$(13/4) \cdot n - 25 \rightarrow$ $(1/16) \cdot n^2 + (5/8) \cdot n$	$3 \cdot n - 78$	$2.25 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
n^2	0.541	level	3	$(5/16) \cdot n^2 + (-45/8) \cdot n + 25$	$0.234/n$	write A[i1, i2] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2 + (-15/8) \cdot n + 2$	$0.328/n$	read A[i1, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.438	level	1	$(7/16) \cdot n^2 + (-7/4) \cdot n - 14$	$0.328/n$	read A[i1, i4] (i0=0, i1=1, i5=0); read A[i1, i4] (i0=0, i5=0)
n^2	0.38	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n + 1$	$2 \cdot n - 68$	$1.5 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
n^2	0.365	ramp	$(7/4) \cdot n - 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 5$	$(15/8) \cdot n - 30$	$1.41 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.354	level	2	$(1/4) \cdot n^2 + (-3/4) \cdot n$	$0.188/n$	write A[i1, i2] (i0=0)
n^2	0.342	ramp	$(9/4) \cdot n - 9 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n - 1$	$(7/4) \cdot n - 31$	$1.31 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
n^2	0.309	level	$(1/8) \cdot n^2$	$(7/8) \cdot n - 16$	$0.656 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.309	level	$(1/8) \cdot n^2$	$(7/8) \cdot n - 23$	$0.656 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^2	0.269	ramp	$(3/8) \cdot n + 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(3/2) \cdot n - 43$	$1.12 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^2	0.265	level	$(1/8) \cdot n^2$	$(3/4) \cdot n - 24$	$0.562 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.233	ramp	$(3/2) \cdot n \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/4) \cdot n - 30$	$0.938 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.231	ramp	$(11/8) \cdot n + 1 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/4) \cdot n - 35$	$0.938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.231	ramp	$(11/8) \cdot n + 1 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/4) \cdot n - 35$	$0.938 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.223	ramp	$(19/8) \cdot n - 12 \rightarrow$ $(1/8) \cdot n^2 + (-15/8) \cdot n + 16$	$n - 24$	$0.75 \cdot n^{-2}$	read A[i1, i4] (i0=0, i5=0)
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-11/8) \cdot n + 3$	$0.0938/n$	write A[i1, i2] (i0=0)
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-5/4) \cdot n + 2$	$0.0938/n$	write A[i1, i2] (i0=0)
n^2	0.205	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 7 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.205	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 19 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 23$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
n^2	0.205	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 25$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.205	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 5 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 9$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.204	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 24 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 28$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)
n^2	0.193	ramp	$(21/8) \cdot n - 17 \rightarrow$ $(1/8) \cdot n^2 + (-11/4) \cdot n + 37$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i1, i4] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.171	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(3/4) \cdot n - 24$	$0.562 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.169	ramp	$(13/4) \cdot n - 25$ → $(1/16) \cdot n^2 + (5/8) \cdot n$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=8)
n^2	0.167	ramp	$(17/4) \cdot n - 48$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/8) \cdot n - 29$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=7)
n^2	0.167	ramp	$(17/4) \cdot n - 48$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/8) \cdot n - 29$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.167	ramp	$(17/4) \cdot n - 48$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(7/8) \cdot n - 29$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.167	ramp	$(9/4) \cdot n - 7$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.167	ramp	$(9/4) \cdot n - 7$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.167	ramp	$(9/4) \cdot n - 7$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.167	ramp	$(9/4) \cdot n - 7$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.165	ramp	$(13/4) \cdot n - 23$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(7/8) \cdot n - 28$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=7)
n^2	0.165	ramp	$(13/4) \cdot n - 23$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(7/8) \cdot n - 28$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.165	ramp	$(13/4) \cdot n - 23$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(7/8) \cdot n - 28$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.163	ramp	$(1/2) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 30$	$(7/8) \cdot n - 16$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
n^2	0.162	ramp	$(11/8) \cdot n - 2$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 31$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=7)
n^2	0.16	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 3$ $\rightarrow$ $(1/8) \cdot n^2 - 3 \cdot n + 50$	$(5/8) \cdot n - 20$	$0.469 \cdot n^{-2}$	read A[i1, i4] (i0=0, i5=0)
n^2	0.147	ramp	$(5/2) \cdot n - 12$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(3/4) \cdot n - 14$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.147	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 18$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 14$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.145	ramp	$(27/8) \cdot n - 27$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.145	ramp	$(27/8) \cdot n - 28$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.145	ramp	$(27/8) \cdot n - 29 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=7)
n^2	0.144	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.142	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(3/4) \cdot n - 25$	$0.562 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.141	ramp	$(17/8) \cdot n - 5 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.141	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 28$	$(3/4) \cdot n - 18$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.139	ramp	$(11/8) \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 27$	$(3/4) \cdot n - 18$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.138	ramp	$(5/4) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 48$	$(3/4) \cdot n - 18$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.138	ramp	$(5/4) \cdot n + 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 46$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
n^2	0.137	ramp	$(9/8) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 49$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.136	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 53$	$(3/4) \cdot n - 24$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)
n^2	0.133	ramp	$(9/8) \cdot n + 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 81$	$(3/4) \cdot n - 24$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=15)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.129	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n + 13 \rightarrow$ $(1/8) \cdot n^2 + (-17/8) \cdot n + 29$	$(1/2) \cdot n - 16$	$0.375 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.123	ramp	$(19/8) \cdot n - 10 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.123	ramp	$(9/4) \cdot n - 8 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.123	ramp	$(5/2) \cdot n - 12 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(5/8) \cdot n - 11$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.121	ramp	$(13/8) \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 8$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.121	ramp	$(3/2) \cdot n - 1 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.12	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 9$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.12	ramp	$(25/8) \cdot n - 22 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.12	ramp	$(25/8) \cdot n - 23 \rightarrow$ $(1/16) \cdot n^2 + (5/8) \cdot n - 7$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.119	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(5/8) \cdot n - 20$	$0.469 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.118	ramp	$(17/8) \cdot n - 5 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 25$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.117	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.116	ramp	$(3/2) \cdot n - 3$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n -$ 27	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.116	ramp	$(11/8) \cdot n +$ 1 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 31	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.116	ramp	$(11/8) \cdot n +$ 1 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 31	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.115	ramp	$(19/8) \cdot n - 8$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 31	$(5/8) \cdot n - 20$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2$ $+ (-5/8) \cdot n +$ 1	$0.0469/n$	read A[i1, i3] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2$ $+ (-13/8) \cdot n$ $+ 10$	$0.0469/n$	write A[i1, i2] (i0=0, i3=0)
n^2	0.108	level	3	$(1/16) \cdot n^2$ $+ (-3/4) \cdot n +$ 2	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2$ $+ (-3/4) \cdot n +$ 2	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2$ $+ (-13/8) \cdot n$ $+ 10$	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.0966	ramp	$(13/8) \cdot n - 2$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 11	$(1/2) \cdot n - 8$	$0.375 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0947	level	3	$(7/128) \cdot n^2$ $+ (-21/16) \cdot n$ $+ 7$	$0.041/n$	write A[i1, i4] (i0=0, i4=0)
n^2	0.0947	level	3	$(7/128) \cdot n^2$ $+ (-35/16) \cdot n$ $+ 21$	$0.041/n$	write A[i1, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.094	ramp	$23 \rightarrow$ $(1/16) \cdot n^2$ $+ (-3/8) \cdot n +$ 1	$(3/4) \cdot n - 12$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (-1/2) \cdot n$	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (1/8) \cdot n$	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (-3/4) \cdot n +$ 2	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (3/8) \cdot n$	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.0716	ramp	$(21/8) \cdot n -$ 13 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 13	$(3/8) \cdot n - 9$	$0.281 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0714	ramp	$(5/2) \cdot n - 12$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 16	$(3/8) \cdot n - 9$	$0.281 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0705	ramp	$(17/8) \cdot n - 6$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 26	$(3/8) \cdot n - 9$	$0.281 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.048	ramp	$(25/8) \cdot n -$ 23 $\rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$	$(1/4) \cdot n - 6$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
n^2	0.047	ramp	$(17/8) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 25	$(1/4) \cdot n - 6$	$0.188 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0); read A[i2, i2] (i0=0)
n^2	0.0466	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/4) \cdot n - 7$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0, i5=0); read A[i5, i4] (i0=0, i4=0, i5=7)
n^2	0.0457	ramp	$(5/4) \cdot n + 2$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 46	$(1/4) \cdot n - 7$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6, i5=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0456	ramp	$(9/8) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 49$	$(1/4) \cdot n - 7$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7, i5=0); read A[i5, i4] (i0=0, i4=7, i5=7)
n^2	0.045	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 53$	$(1/4) \cdot n - 9$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8, i5=0); read A[i5, i4] (i0=0, i4=8, i5=7)
n^2	0.0442	level	$(1/8) \cdot n^2 +$ $(7/8) \cdot n$	$(1/8) \cdot n +$ $(-25/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.0442	level	$(1/8) \cdot n^2 +$ $(7/8) \cdot n$	$(1/8) \cdot n +$ $(-33/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2 +$ $(7/8) \cdot n$	$(1/8) \cdot n +$ $(-33/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2 +$ $(7/8) \cdot n$	$(1/8) \cdot n +$ $(-33/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2 +$ $(1/8) \cdot n$	$(1/8) \cdot n +$ $(-39/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2 +$ $(7/8) \cdot n$	$(1/8) \cdot n +$ $(-33/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2 +$ $(7/8) \cdot n$	$(1/8) \cdot n +$ $(-25/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0442	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$(1/8) \cdot n + (-33/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	$(1/8) \cdot n + (-17/4)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2 + (5/8) \cdot n$	$(1/8) \cdot n + (-35/8)$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.044	ramp	$(9/8) \cdot n + 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 81$	$(1/4) \cdot n - 9$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=15, i5=0); read A[i5, i4] (i0=0, i4=15, i5=7)
n^2	0.033	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n \rightarrow (1/8) \cdot n^2 + (-11/4) \cdot n + 44$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^2	0.0324	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n + 23 \rightarrow (1/8) \cdot n^2 + (-15/8) \cdot n + 23$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0324	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n + 21 \rightarrow (1/8) \cdot n^2 - 2 \cdot n + 26$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0322	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n + 11 \rightarrow (1/8) \cdot n^2 + (-21/8) \cdot n + 41$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0321	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n + 7 \rightarrow (1/8) \cdot n^2 + (-23/8) \cdot n + 47$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=1, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0319	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$ → $(1/8) \cdot n^2 + (-29/8) \cdot n + 65$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i1, i4] (i0=0, i4=7, i5=0)
n^2	0.0293	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 20$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 24$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=6)
n^2	0.0285	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0285	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 7$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 12$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0285	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 21$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 26$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0285	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 29$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=8)
n^2	0.027	ramp	$(27/8) \cdot n - 29$ → $(1/8) \cdot n^2 + (-29/8) \cdot n + 51$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i1, i4] (i0=0, i5=0)
n^2	0.0245	ramp	$(19/8) \cdot n - 10$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 7$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0244	ramp	$(9/4) \cdot n - 8$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 8$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0241	ramp	$(13/4) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 8$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0241	ramp	$(13/4) \cdot n - 25 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0241	ramp	$(13/4) \cdot n - 26 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0, i2=7)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 22 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 22 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.024	ramp	$(25/8) \cdot n - 23 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0, i2=7)
n^2	0.024	ramp	$3 \cdot n - 19 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.024	ramp	$3 \cdot n - 20 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 12$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0239	ramp	$(3/2) \cdot n - 1 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 19$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0239	ramp	$(23/8) \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 14$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0239	ramp	$(11/8) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 21$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 44 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=7)
n^2	0.0237	ramp	$(33/8) \cdot n - 44 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 44 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(5/2) \cdot n - 11 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 19$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0237	ramp	$(33/8) \cdot n - 46 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0, i2=8)
n^2	0.0237	ramp	$(19/8) \cdot n - 9 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 21$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0237	ramp	$(19/8) \cdot n - 10 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 22$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0236	ramp	$(9/4) \cdot n - 7 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 23$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 5 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 5 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 5 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 6 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 6 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 6 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0235	ramp	$(17/8) \cdot n - 6 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 26$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 8 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n - 24$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 8 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n - 24$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0)
n^2	0.0234	ramp	$2 \cdot n - 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 28$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0234	ramp	$2 \cdot n - 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 28$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0234	ramp	$2 \cdot n - 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 28$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0234	ramp	$(7/2) \cdot n - 34 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n - 19$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i1, i4] (i0=0, i1=2, i5=0)
n^2	0.0234	ramp	$(41/8) \cdot n - 76 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/8) \cdot n - 5$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=7)
n^2	0.0234	ramp	$(41/8) \cdot n - 76 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/8) \cdot n - 5$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0234	ramp	$(41/8) \cdot n - 76 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/8) \cdot n - 5$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0233	ramp	$(27/8) \cdot n - 31 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n - 21$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i1=1, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0233	ramp	$(27/8) \cdot n - 32 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 22$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i1, i4] (i0=0, i1=1, i5=0)
n^2	0.0233	ramp	$(25/8) \cdot n - 20 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=7)
n^2	0.0233	ramp	$(25/8) \cdot n - 20 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0233	ramp	$(25/8) \cdot n - 20 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0232	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0232	ramp	$3 \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 28$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=0)
n^2	0.023	ramp	$(11/8) \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 42$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0)
n^2	0.023	ramp	$(5/4) \cdot n + 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 46$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=6)
n^2	0.023	ramp	$(5/4) \cdot n + 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 46$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=6)
n^2	0.023	ramp	$(5/4) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 48$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i5=0)
n^2	0.0229	ramp	$(5/4) \cdot n - 1 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 45$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0229	ramp	$(9/8) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0228	ramp	$(9/4) \cdot n - 6 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 46$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
n^2	0.0228	ramp	$(9/4) \cdot n - 6 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 46$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
n^2	0.0228	ramp	$(9/4) \cdot n - 8 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 48$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0228	ramp	$(9/4) \cdot n - 9 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=7)
n^2	0.0227	ramp	$(9/4) \cdot n - 10 \rightarrow$ $(1/16) \cdot n^2 + (5/8) \cdot n - 45$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.0227	ramp	$(17/8) \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0227	ramp	$(17/8) \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0224	ramp	$3 \cdot n - 17 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 53$	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)
n^2	0.0156	ramp	$22 \rightarrow$ $(1/16) \cdot n^2 + (-3/8) \cdot n - 5$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i2, i2] (i0=0)
$n^{1.5}$	4.03	ramp	$4 \rightarrow (9/8) \cdot n - 10$	$6 \cdot n - 81$	$4.5 \cdot n^{-2}$	read A[i1, i5] (i0=0, i5=0)
$n^{1.5}$	2.62	ramp	$3 \rightarrow (9/8) \cdot n - 2$	$(15/4) \cdot n - 15$	$2.81 \cdot n^{-2}$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	2.62	ramp	$4 \rightarrow (9/8) \cdot n - 2$	$(15/4) \cdot n - 20$	$2.81 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
$n^{1.5}$	2.3	ramp	$13 \rightarrow (9/8) \cdot n - 8$	$(27/8) \cdot n - 54$	$2.53 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0); read A[i1, i5] (i0=0, i4=1) (+1)
$n^{1.5}$	2.18	ramp	$4 \rightarrow (9/8) \cdot n - 2$	$(25/8) \cdot n - 15$	$2.34 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.685	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$n - 10$	$0.75 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0, i5=0)
$n^{1.5}$	0.601	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$(7/8) \cdot n - 8$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7, i5=0)
$n^{1.5}$	0.592	ramp	$13 \rightarrow (9/8) \cdot n - 14$	$(7/8) \cdot n - 14$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7); read A[i5, i4] (i0=0, i4=8) (+1)
$n^{1.5}$	0.579	ramp	$n + 4 \rightarrow (9/8) \cdot n - 1$	$(5/8) \cdot n - 20$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.54	ramp	$(1/4) \cdot n + 3 \rightarrow n - 1$	$(3/4) \cdot n - 18$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.532	ramp	$4 \rightarrow (9/8) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0, i5=0)
$n^{1.5}$	0.516	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0)
$n^{1.5}$	0.516	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	0.516	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	0.514	ramp	$4 \rightarrow (9/8) \cdot n - 9$	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=1, i5=0)
$n^{1.5}$	0.512	ramp	$3 \rightarrow (9/8) \cdot n - 10$	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.503	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(3/4) \cdot n - 12$	$0.562 \cdot n^{-2}$	read A[i1, i5] (i0=0)
$n^{1.5}$	0.475	ramp	$(3/8) \cdot n + 3 \rightarrow n - 1$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.445	ramp	$4 \rightarrow (9/8) \cdot n - 1$	$(5/8) \cdot n$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.438	ramp	$12 \rightarrow$ $(9/8) \cdot n - 6$	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.436	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
$n^{1.5}$	0.436	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.436	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.171	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(1/4) \cdot n - 3$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.156	ramp	$2 \cdot n - 2 \rightarrow$ $(17/8) \cdot n - 8$	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=7)
$n^{1.5}$	0.122	ramp	$(9/8) \cdot n + 4$ $\rightarrow (5/4) \cdot n -$ 1	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=9, i5=7)
$n^{1.5}$	0.119	ramp	$n + 3 \rightarrow$ $(9/8) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=8, i5=6)
$n^{1.5}$	0.119	ramp	$n + 3 \rightarrow$ $(9/8) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=8, i5=7)
$n^{1.5}$	0.116	ramp	$n + 4 \rightarrow$ $(9/8) \cdot n - 1$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
$n^{1.5}$	0.0876	ramp	$12 \rightarrow$ $(9/8) \cdot n - 6$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.0872	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0)
$n^{1.5}$	0.0872	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0, i5=0)
$n^{1.5}$	0.0872	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	0.0872	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	0.0872	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.0845	ramp	$13 \rightarrow$ $(9/8) \cdot n - 14$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
$n^{1.5}$	0.0842	ramp	$12 \rightarrow$ $(9/8) \cdot n - 15$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.0842	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=1, i5=0)
$n^{1.5}$	0.0839	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=1)
$n^{1.5}$	0.0839	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=1, i5=0)
$n^{1.5}$	0.0839	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
$n^{1.5}$	0.0647	ramp	$(1/4) \cdot n + 3 \rightarrow (3/8) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=2, i5=0)
n^1	24.2	level	3	$14 \cdot n - 118$	$10.5 \cdot n^{-2}$	read A[i1, i5] (i0=0)
n^1	19.7	level	3	$(91/8) \cdot n - 91$	$8.53 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	10.6	level	3	$(49/8) \cdot n - 49$	$4.59 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	10.6	level	3	$(49/8) \cdot n - 56$	$4.59 \cdot n^{-2}$	read A[i1, i5] (i0=0)
n^1	9.9	level	2	$7 \cdot n - 21$	$5.25 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	7	level	1	$7 \cdot n - 36$	$5.25 \cdot n^{-2}$	read A[i1, i5] (i0=0)
n^1	6	level	1	$6 \cdot n - 33$	$4.5 \cdot n^{-2}$	read A[i1, i3] (i0=0, i3=0)
n^1	4.55	level	3	$(21/8) \cdot n - 21$	$1.97 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	4.55	level	3	$(21/8) \cdot n - 21$	$1.97 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
n^1	4.24	level	2	$3 \cdot n + 30$	$2.25 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=1, i2=0); read A[i2, i2] (i0=0, i1=1, i2=0) (+7)
n^1	4	level	4	$2 \cdot n - 25$	$1.5 \cdot n^{-2}$	read A[i1, i5] (i0=0, i1=1); read A[i1, i2] (i0=0, i1=2, i2=0) (+3)
n^1	3.25	level	3	$(15/8) \cdot n - 15$	$1.41 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	2.65	level	2	$(15/8) \cdot n$	$1.41 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	2.62	level	1	$(21/8) \cdot n - 6$	$1.97 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
n^1	2.12	level	$(1/8) \cdot n^2$	6	$4.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	2.12	level	$(1/8) \cdot n^2$	6	$4.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	1.77	level	$(1/8) \cdot n^2$	5	$3.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	1.75	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-9/16)$	7	$5.25 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=1, i2=0); read A[i1, i2] (i0=0, i1=2, i2=0) (+3)
n^1	1.75	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	7	$5.25 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=1, i2=0); read A[i1, i2] (i0=0, i1=2, i2=0) (+3)
n^1	1.75	level	4	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=2, i5=0)
n^1	1.73	level	3	$n - 8$	$0.75 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0, i5=0)
n^1	1.52	level	3	$(7/8) \cdot n - 14$	$0.656 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	1.52	level	3	$(7/8) \cdot n - 8$	$0.656 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
n^1	1.52	level	3	$(7/8) \cdot n - 14$	$0.656 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	1.52	level	3	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	1.52	level	3	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=1, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 20$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 18$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 16$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 14$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 12$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 7$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 6$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 5$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 4$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 3$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 2$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 8$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 24$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 22$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 10$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	1.3	level	3	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 6$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	5	$3.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	1.08	level	3	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	1.06	level	2	$(3/4) \cdot n$	$0.562 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	1	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 10$	4	$3 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.884	level	2	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	0.875	level	1	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=1, i5=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.656 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^1	0.75	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 12$	3	$2.25 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.75	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.75	level	1	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=15, i2=0); read A[i1, i2] (i0=0, i1=8, i2=0)
n^1	0.707	level	$(1/8) \cdot n^2$	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=15, i2=0); read A[i1, i2] (i0=0, i1=8, i2=0)
n^1	0.5	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 14$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-15/8) \cdot n + (155/4)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=7, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-21/8) \cdot n + 35$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=7, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-9/8) \cdot n + 26$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=1, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/2) \cdot n + 33$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-19/8) \cdot n + 31$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-9/4) \cdot n + 29$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-17/8) \cdot n + 27$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 - 2 \cdot n + 25$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-15/8) \cdot n + 23$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=1, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=23, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=23, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-13/8) \cdot n + 19$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/2) \cdot n + 17$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-11/8) \cdot n + 15$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/4) \cdot n + 13$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-9/8) \cdot n + 11$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + (27/4)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 9$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/4) \cdot n + 6$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/8) \cdot n + 5$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/2) \cdot n + 4$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/8) \cdot n + 3$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + (3/8)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 9$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 7$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (1/4)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/4) \cdot n + 15$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=7, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-13/8) \cdot n + 14$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-3/2) \cdot n + 13$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-11/8) \cdot n + 12$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-5/4) \cdot n + 11$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-9/8) \cdot n + 10$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + (63/4)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=7, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 8$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=1, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (1/8)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=1, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 - n + (191/8)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/4) \cdot n + 21$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 9$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=1, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-7/8) \cdot n + 7$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (1/4)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (63/8)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=16, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=16, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=16, i2=8, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=16, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 21$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 19$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 17$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 15$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 13$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 18$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 14$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 28$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 25$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 23$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 27$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 24$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 22$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (7/8) \cdot n + (-3/2)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-9/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (9/8) \cdot n + (-51/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (-21/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n +$ $(-53/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ n +$ $(-21/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n +$ $(-107/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (9/8) \cdot n +$ $(-99/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (7/8) \cdot n +$ $(-99/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n +$ $(-139/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n +$ $(-43/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/2) \cdot n +$ $(-9/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (11/8) \cdot n +$ $(-43/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-171/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + n + (-29/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-41/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 11$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 10$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 18$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 14$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 10$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 18$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 14$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 10$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n + (-171/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n + (-203/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (7/8) \cdot n + (-131/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/2) \cdot n + (-49/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 22$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 22$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-9/16)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=7, i4=1, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=7, i4=1, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-89/16)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=2, i4=6, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=2, i4=6, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-25/16)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=1, i2=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=1, i2=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 24$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-121/16)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=1, i4=7, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=1, i4=7, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (23/16)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=9, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=9, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 3$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 4$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 5$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-235/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-57/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (11/8) \cdot n + (-219/16)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-9/16)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-203/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-49/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (7/8) \cdot n + (-131/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (11/8) \cdot n + (-187/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (1/2) \cdot n + (-73/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=14, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (103/16)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=14, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 6$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1, i4=7, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-105/16)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1, i4=7, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=15, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (119/16)$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=15, i2=8, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (11/8) \cdot n + 7$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=15, i2=8, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 7$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=15, i2=8, i3=0)
n^1	0.217	level	3	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	0.217	level	3	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	0.177	level	2	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	0.177	level	2	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	write A[i1, i4] (i0=0, i4=0)
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
$n^{0.5}$	8.84	level	$(25/8) \cdot n - 22$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.66	level	$3 \cdot n - 20$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.48	level	$(23/8) \cdot n - 18$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.29	level	$(11/4) \cdot n - 16$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.1	level	$(21/8) \cdot n - 14$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.91	level	$(5/2) \cdot n - 12$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.71	level	$(19/8) \cdot n - 10$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.71	level	$(19/8) \cdot n - 11$	5	$3.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	7.5	level	$(9/4) \cdot n - 8$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.5	level	$(9/4) \cdot n - 9$	5	$3.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	7.29	level	$(17/8) \cdot n - 6$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.29	level	$(17/8) \cdot n - 7$	5	$3.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	7.07	level	$2 \cdot n - 5$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.85	level	$(15/8) \cdot n - 4$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.61	level	$(7/4) \cdot n - 3$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.37	level	$(13/8) \cdot n - 2$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.32	level	$(5/2) \cdot n - 13$	4	$3 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	6.12	level	$(3/2) \cdot n - 1$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.86	level	$(11/8) \cdot n$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.66	level	$2 \cdot n - 6$	4	$3 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	5.59	level	$(5/4) \cdot n + 1$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.3	level	$(25/8) \cdot n - 22$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=25, i2=8, i3=0); read A[i3, i2] (i0=0, i1=25, i2=8, i3=7) (+1)
$n^{0.5}$	5.3	level	$(9/8) \cdot n + 2$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.3	level	$(9/8) \cdot n + 1$	5	$3.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=9, i2=0); read A[i2, i2] (i0=0, i1=9, i2=1) (+3)
$n^{0.5}$	5.2	level	$3 \cdot n - 20$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	5.09	level	$(23/8) \cdot n - 18$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	5	level	$n + 3$	5	$3.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8)
$n^{0.5}$	5	level	$n + 2$	5	$3.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	4.97	level	$(11/4) \cdot n - 16$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.86	level	$(21/8) \cdot n - 14$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.86	level	$(21/8) \cdot n - 15$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	4.74	level	$(5/2) \cdot n - 12$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.62	level	$(19/8) \cdot n - 10$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.5	level	$(9/4) \cdot n - 8$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.37	level	$(17/8) \cdot n - 6$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=17, i2=8, i3=0); read A[i3, i2] (i0=0, i1=17, i2=8, i3=7) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	4.24	level	$(9/8) \cdot n + 2$	4	$3 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=9, i2=8, i3=0); read A[i3, i2] (i0=0, i1=9, i2=8, i3=7) (+2)
$n^{0.5}$	4.11	level	$(15/8) \cdot n - 5$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	4.06	level	$(33/8) \cdot n - 45$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=33, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	4	level	$4 \cdot n - 42$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.94	level	$(31/8) \cdot n - 39$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.87	level	$(15/4) \cdot n - 36$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.81	level	$(29/8) \cdot n - 33$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.74	level	$(7/8) \cdot n + 2$	4	$3 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	3.74	level	$(7/2) \cdot n - 30$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	3.67	level	$(27/8) \cdot n - 27$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.61	level	$(13/4) \cdot n - 24$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.54	level	$(25/8) \cdot n - 21$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=25, i2=16, i3=6); read A[i3, i2] (i0=0, i1=25, i2=16, i3=7)
$n^{0.5}$	3.52	level	$(11/8) \cdot n - 1$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	3.46	level	$3 \cdot n - 19$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.39	level	$(23/8) \cdot n - 17$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.35	level	$(5/4) \cdot n$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	3.35	level	$(5/4) \cdot n$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=10, i2=0); read A[i2, i2] (i0=0, i1=10, i2=1) (+1)
$n^{0.5}$	3.32	level	$(11/4) \cdot n - 15$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	3.32	level	$(11/4) \cdot n - 17$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	3.24	level	$(21/8) \cdot n - 13$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.18	level	$(9/8) \cdot n + 1$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=9)
$n^{0.5}$	3.16	level	$(5/2) \cdot n - 11$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.08	level	$(19/8) \cdot n - 9$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3	level	$(9/4) \cdot n - 7$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3	level	$n + 2$	3	$2.25 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=8, i4=0, i5=0); read A[i5, i4] (i0=0, i1=8, i4=0, i5=6) (+1)
$n^{0.5}$	2.92	level	$(17/8) \cdot n - 5$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=17, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	2.92	level	$(17/8) \cdot n - 7$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=17, i2=0); read A[i2, i2] (i0=0, i2=1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	2.83	level	$2 \cdot n - 5$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.74	level	$(15/8) \cdot n - 3$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.74	level	$(15/8) \cdot n - 4$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.65	level	$(7/4) \cdot n - 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.65	level	$(7/4) \cdot n - 3$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.65	level	$(7/4) \cdot n - 4$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	2.6	level	$(3/4) \cdot n + 2$	3	$2.25 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	2.55	level	$(13/8) \cdot n - 1$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.55	level	$(13/8) \cdot n - 2$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.45	level	$(3/2) \cdot n$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	2.45	level	$(3/2) \cdot n - 1$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.45	level	$(3/2) \cdot n - 2$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	2.35	level	$(11/8) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.35	level	$(11/8) \cdot n$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.24	level	$(5/4) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.24	level	$(5/4) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=10, i4=6, i5=7); read A[i5, i4] (i0=0, i4=6)
$n^{0.5}$	2.12	level	$(9/8) \cdot n + 3$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=9, i4=7, i5=7); read A[i5, i4] (i0=0, i4=7)
$n^{0.5}$	2	level	$n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	2	level	$n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.87	level	$(7/8) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.87	level	$(7/2) \cdot n +$ $(-33/2)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=2, i5=0)
$n^{0.5}$	1.84	level	$(27/8) \cdot n +$ $(-123/8)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.84	level	$(27/8) \cdot n +$ $(-115/8)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.77	level	$(25/8) \cdot n -$ 22	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=25, i2=8)
$n^{0.5}$	1.73	level	$(3/4) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.73	level	$3 \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.73	level	$3 \cdot n - 21$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.7	level	$(23/8) \cdot n -$ 18	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.7	level	$(23/8) \cdot n -$ 19	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.7	level	$(23/8) \cdot n -$ 19	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.66	level	$(11/4) \cdot n -$ 16	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.66	level	$(11/4) \cdot n -$ 17	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.62	level	$(21/8) \cdot n -$ 14	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.62	level	$(21/8) \cdot n -$ 15	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.58	level	$(5/8) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.58	level	$(5/2) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.58	level	$(5/2) \cdot n - 13$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.58	level	$(5/2) \cdot n - 15$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=2, i5=0)
$n^{0.5}$	1.58	level	$(5/2) \cdot n +$ $(-9/2)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=2, i5=0)
$n^{0.5}$	1.54	level	$(19/8) \cdot n -$ 10	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.54	level	$(19/8) \cdot n -$ 11	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.54	level	$(19/8) \cdot n -$ 14	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.54	level	$(19/8) \cdot n +$ $(-35/8)$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.54	level	$(19/8) \cdot n -$ 13	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.54	level	$(19/8) \cdot n +$ $(-27/8)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=18, i2=7)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=18, i2=8)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 11$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=8, i5=0)
$n^{0.5}$	1.46	level	$(17/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=17, i2=7)
$n^{0.5}$	1.46	level	$(17/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=17, i2=8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.46	level	$(17/8) \cdot n - 10$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=7, i5=0)
$n^{0.5}$	1.41	level	$2 \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8, i5=7)
$n^{0.5}$	1.41	level	$2 \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=16, i2=8)
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=16, i2=7)
$n^{0.5}$	1.41	level	$(1/2) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=4, i2=0); read A[i2, i2] (i0=0, i1=4, i2=1)
$n^{0.5}$	1.41	level	$2 \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=16, i4=0, i5=7)
$n^{0.5}$	1.41	level	$2 \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.41	level	$2 \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=15, i2=7)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i5=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=12, i2=8)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.22	level	$(3/8) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=3, i2=0); read A[i2, i2] (i0=0, i1=3, i2=1)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=2, i5=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=11, i2=7)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=11, i2=1)
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=11, i2=8)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n + (-3/8)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=1, i5=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.12	level	$(5/4) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=9, i5=7)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=10, i2=7)
$n^{0.5}$	1.12	level	$(5/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=10, i2=8)
$n^{0.5}$	1.06	level	$(9/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=8, i5=6)
$n^{0.5}$	1.06	level	$(9/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=8, i5=7)
$n^{0.5}$	1.06	level	$(9/8) \cdot n + (-1/4)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n + (-65/8)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1	level	$n + 3$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8, i5=0)
$n^{0.5}$	1	level	$n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1	level	$(1/4) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=2, i4=6, i5=0); read A[i5, i4] (i0=0, i1=2, i4=6, i5=1)
$n^{0.5}$	1	level	n	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	1	level	n	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	1	level	$n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1	level	$n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.935	level	$(7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=5, i2=1)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=2, i5=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=2, i2=0)
n^0	36	level	1	36	$27 \cdot n^{-3}$	read A[i1, i5] (i0=0)
n^0	8.66	level	3	5	$3.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=2, i5=0)
n^0	3.32	level	11	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	3.16	level	10	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	3	level	9	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.83	level	8	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.65	level	7	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.45	level	6	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=2, i2=1)

bin 132: finite-size slope 3.33 at $n \approx 256-312$; converges to n^3 ; bin 133: finite-size slope 3.26 at $n \approx 256-312$; converges to n^3 ; bin 136: finite-size slope 3.27 at $n \approx 256-312$; converges to n^3 .

Row-panel and trailing-submatrix re-reads (read A[i3, i2], read A[i5, i4]) ramp to $(1/16)n^2 + (3/4)n$ lines with cubic populations — $d = 4.0$, headroom +1.0. At the anchor sizes ($n \approx 256-312$) the finite-size slope of these families still reads ≈ 4.4 because the negative n^2 corrections in

their populations have not yet died out; the exact degrees settle it at 4.0.

lu — single-shot [exact]

Accesses $A(n) = (4/3) \cdot n^3 + (-1/2) \cdot n^2 + (-5/6) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma\text{mass/warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.00643 \cdot n^4 + 0.221 \cdot n^{3.5} + 2.29 \cdot n^3 + 6.37 \cdot n^{2.5} + 37.5 \cdot n^2 + 20.5 \cdot n^{1.5} + 205 \cdot n^1 + 521 \cdot n^{0.5} + 47.6 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00309	ramp	$(21/4) \cdot n - 81 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(7/384) \cdot n^3 + (-123/64) \cdot n^2 + (1619/24) \cdot n - 790$	0.0137	read A[i3, i2] (i0=0)
n^4	0.00193	ramp	$(11/8) \cdot n + 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 91$	$(7/512) \cdot n^3 + (-195/128) \cdot n^2 + (901/16) \cdot n - 690$	0.0103	read A[i5, i4] (i0=0)
n^4	0.000411	ramp	$(25/4) \cdot n - 121 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(1/384) \cdot n^3 + (-21/64) \cdot n^2 + (329/24) \cdot n - 190$	0.00195	read A[i3, i2] (i0=0)
n^4	0.000316	ramp	$3 \cdot n - 16 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(7/3072) \cdot n^3 + (-35/128) \cdot n^2 + (259/24) \cdot n - 140$	0.00171	read A[i5, i4] (i0=0)
n^4	0.00031	ramp	$(17/8) \cdot n - 2 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 121$	$(7/3072) \cdot n^3 + (-35/128) \cdot n^2 + (259/24) \cdot n - 140$	0.00171	read A[i5, i4] (i0=0)
n^4	0.000268	ramp	$(9/4) \cdot n - 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 91$	$(1/512) \cdot n^3 + (-15/64) \cdot n^2 + (37/4) \cdot n - 120$	0.00146	read A[i5, i4] (i0=0)
n^4	6.23e-05	ramp	$2 \cdot n - 2 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(1/3072) \cdot n^3 + (-55/48) \cdot n + 25$	0.000244	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	4.1e-05	ramp	$(25/8) \cdot n - 17 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 121$	$(1/3072) \cdot n^3 + (-3/64) \cdot n^2 + (107/48) \cdot n - 35$	0.000244	read A[i5, i4] (i0=0)
$n^{3.5}$	0.0988	ramp	$11 \rightarrow (9/8) \cdot n - 2$	$(7/48) \cdot n^3 + (-63/16) \cdot n^2 + (217/6) \cdot n - 112$	0.109	read A[i3, i2] (i0=0)
$n^{3.5}$	0.0854	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$(49/384) \cdot n^3 + (-413/128) \cdot n^2 + (1309/48) \cdot n - 77$	0.0957	read A[i5, i4] (i0=0)
$n^{3.5}$	0.0126	ramp	$21 \rightarrow (9/8) \cdot n - 9$	$(1/48) \cdot n^3 + (-41/32) \cdot n^2 + (311/12) \cdot n - 172$	0.0156	read A[i1, i3] (i0=0)
$n^{3.5}$	0.0121	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(7/384) \cdot n^3 + (-63/128) \cdot n^2 + (175/48) \cdot n - 7$	0.0137	read A[i5, i4] (i0=0)
$n^{3.5}$	0.0108	ramp	$14 \rightarrow (9/8) \cdot n - 16$	$(7/384) \cdot n^3 + (-35/32) \cdot n^2 + (259/12) \cdot n - 140$	0.0137	read A[i1, i5] (i0=0)
$n^{3.5}$	0.00148	ramp	$15 \rightarrow (9/8) \cdot n - 23$	$(1/384) \cdot n^3 + (-3/16) \cdot n^2 + (13/3) \cdot n - 32$	0.00195	read A[i1, i5] (i0=0)
n^3	0.758	level	3	$(7/16) \cdot n^3 + (-273/32) \cdot n^2 + (91/2) \cdot n - 42$	0.328	read A[i1, i3] (i0=0); write A[i1, i2] (i0=0) (+1)
n^3	0.333	level	1	$(1/3) \cdot n^3 - n^2 + (23/3) \cdot n - 14$	0.25	read A[i1, i2] (i0=0); read A[i1, i4] (i0=0, i5=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.289	level	3	$(1/6) \cdot n^3 + (-169/6) \cdot n + 146$	0.125	read A[i2, i2] (i0=0, i1=2, i2=1); read A[i5, i4] (i0=0, i1=2) (+3)
n^3	0.0835	ramp	$(25/8) \cdot n - 20 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(7/16) \cdot n^2 + (-51/2) \cdot n + 368$	0.328/n	read A[i3, i2] (i0=0)
n^3	0.0737	ramp	$(3/2) \cdot n - 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(49/128) \cdot n^2 + (-315/16) \cdot n + 253$	0.287/n	read A[i5, i4] (i0=0)
n^3	0.0732	ramp	$(17/4) \cdot n - 48 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(49/128) \cdot n^2 + (-371/16) \cdot n + 351$	0.287/n	read A[i3, i2] (i0=0)
n^3	0.0711	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(3/8) \cdot n^2 + (-47/2) \cdot n + 370$	0.281/n	read A[i3, i2] (i0=0)
n^3	0.0598	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(5/16) \cdot n^2 + (-75/4) \cdot n + 280$	0.234/n	read A[i3, i2] (i0=0)
n^3	0.0595	ramp	$(11/8) \cdot n + 2 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(21/64) \cdot n^2 + (-87/4) \cdot n + 360$	0.246/n	read A[i5, i4] (i0=0)
n^3	0.0538	ramp	$(7/2) \cdot n - 31 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 4$	$(35/128) \cdot n^2 + (-225/16) \cdot n + 180$	0.205/n	read A[i2, i2] (i0=0)
n^3	0.0516	ramp	$(19/8) \cdot n - 8 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(35/128) \cdot n^2 + (-255/16) \cdot n + 230$	0.205/n	read A[i5, i4] (i0=0)
n^3	0.0379	ramp	$(1/2) \cdot n + 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 91$	$(9/32) \cdot n^2 - 19 \cdot n + 320$	0.211/n	read A[i5, i4] (i0=0)
n^3	0.0361	level	3	$(1/48) \cdot n^3 + (-11/32) \cdot n^2 + (37/24) \cdot n - 1$	0.0156	write A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0127	ramp	$(3/8) \cdot n + 4$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 91$	$(3/32) \cdot n^2$ $+ (-49/8) \cdot n$ $+ 100$	0.0703/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^3	0.012	ramp	$(17/4) \cdot n - 49$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(1/16) \cdot n^2$ $+ (-15/4) \cdot n$ $+ 56$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.012	ramp	$(17/4) \cdot n - 49$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(1/16) \cdot n^2$ $+ (-15/4) \cdot n$ $+ 56$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.012	ramp	$(17/4) \cdot n - 49$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(1/16) \cdot n^2$ $+ (-15/4) \cdot n$ $+ 56$	0.0469/n	read A[i3, i2] (i0=0, i3=0)
n^3	0.012	ramp	$(17/4) \cdot n - 49$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(1/16) \cdot n^2$ $+ (-15/4) \cdot n$ $+ 56$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.0114	ramp	$(33/8) \cdot n - 43$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(1/16) \cdot n^2$ $+ (-9/2) \cdot n + 80$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.0113	ramp	$(21/4) \cdot n - 81$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(1/16) \cdot n^2$ $+ (-19/4) \cdot n$ $+ 90$	0.0469/n	read A[i3, i2] (i0=0, i3=7)
n^3	0.0113	ramp	$(21/4) \cdot n - 81$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(1/16) \cdot n^2$ $+ (-19/4) \cdot n$ $+ 90$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.0113	ramp	$(21/4) \cdot n - 81$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(1/16) \cdot n^2$ $+ (-19/4) \cdot n$ $+ 90$	0.0469/n	read A[i3, i2] (i0=0)
n^3	0.011	ramp	$(27/8) \cdot n - 28$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(7/128) \cdot n^2$ $+ (-39/16) \cdot n$ $+ 27$	0.041/n	read A[i2, i2] (i0=0)
n^3	0.0105	ramp	$(17/4) \cdot n - 49$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(7/128) \cdot n^2$ $+ (-51/16) \cdot n$ $+ 46$	0.041/n	read A[i2, i2] (i0=0)
n^3	0.0105	ramp	$(33/8) \cdot n - 44$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(7/128) \cdot n^2$ $+ (-51/16) \cdot n$ $+ 46$	0.041/n	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0104	ramp	$3 \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(7/128) \cdot n^2$ $+ (-51/16) \cdot n$ $+ 46$	0.041/n	read A[i5, i4] (i0=0, i4=0)
n^3	0.0103	ramp	$(9/4) \cdot n - 9$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 49	$(7/128) \cdot n^2$ $+ (-49/16) \cdot n$ $+ 42$	0.041/n	read A[i5, i4] (i0=0)
n^3	0.0103	ramp	$(19/8) \cdot n -$ 12 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 32	$(7/128) \cdot n^2$ $+ (-51/16) \cdot n$ $+ 46$	0.041/n	read A[i5, i4] (i0=0)
n^3	0.0103	ramp	$(9/4) \cdot n - 6$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 46	$(7/128) \cdot n^2$ $+ (-51/16) \cdot n$ $+ 46$	0.041/n	read A[i5, i4] (i0=0, i4=6)
n^3	0.0103	ramp	$(17/8) \cdot n - 4$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 49	$(7/128) \cdot n^2$ $+ (-51/16) \cdot n$ $+ 46$	0.041/n	read A[i5, i4] (i0=0, i4=7)
n^3	0.01	ramp	$(21/4) \cdot n -$ 80 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(7/128) \cdot n^2$ $+ (-65/16) \cdot n$ $+ 75$	0.041/n	read A[i3, i2] (i0=0)
n^3	0.00997	ramp	$(41/8) \cdot n -$ 76 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 10	$(7/128) \cdot n^2$ $+ (-65/16) \cdot n$ $+ 75$	0.041/n	read A[i3, i2] (i0=0)
n^3	0.0098	ramp	$3 \cdot n - 17 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 53	$(7/128) \cdot n^2$ $+ (-63/16) \cdot n$ $+ 70$	0.041/n	read A[i5, i4] (i0=0, i4=8)
n^3	0.00969	ramp	$(17/8) \cdot n - 3$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 81	$(7/128) \cdot n^2$ $+ (-63/16) \cdot n$ $+ 70$	0.041/n	read A[i5, i4] (i0=0, i4=15)
n^3	0.00901	ramp	$(35/8) \cdot n -$ 54 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 5$	$(3/64) \cdot n^2$ $+ (-11/4) \cdot n$ $+ 40$	0.0352/n	read A[i2, i2] (i0=0)
n^3	0.00835	ramp	$(9/4) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 57	$(3/64) \cdot n^2$ $+ (-27/8) \cdot n$ $+ 60$	0.0352/n	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00754	ramp	$(33/8) \cdot n - 46 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(5/128) \cdot n^2 + (-35/16) \cdot n + 30$	0.0293/n	read A[i2, i2] (i0=0)
n^3	0.00705	ramp	$(27/8) \cdot n - 25 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/128) \cdot n^2 + (-45/16) \cdot n + 50$	0.0293/n	read A[i5, i4] (i0=0)
n^3	0.00671	ramp	$(5/4) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(3/64) \cdot n^2 + (-21/8) \cdot n + 36$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00671	ramp	$(5/4) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(3/64) \cdot n^2 + (-21/8) \cdot n + 36$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00641	ramp	$2 \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(3/64) \cdot n^2 + (-27/8) \cdot n + 60$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00615	ramp	$(9/8) \cdot n + 5 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 121$	$(3/64) \cdot n^2 + (-27/8) \cdot n + 60$	0.0352/n	read A[i5, i4] (i0=0)
n^3	0.00211	ramp	$2 \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(1/64) \cdot n^2 + (-5/4) \cdot n + 25$	0.0117/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^3	0.00203	ramp	$(9/8) \cdot n + 5 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 121$	$(1/64) \cdot n^2 + (-5/4) \cdot n + 25$	0.0117/n	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^3	0.00176	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 53$	$(1/128) \cdot n^2 + (1/16) \cdot n - 9$	0.00586/n	read A[i5, i4] (i0=0, i4=8)
n^3	0.00151	ramp	$(17/4) \cdot n - 50 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00151	ramp	$(33/8) \cdot n - 46 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i2, i2] (i0=0)
n^3	0.00151	ramp	$(33/8) \cdot n - 46 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i2, i2] (i0=0)
n^3	0.00143	ramp	$(41/8) \cdot n - 75 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 9$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i3, i2] (i0=0)
n^3	0.00143	ramp	$(41/8) \cdot n - 77 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i2, i2] (i0=0)
n^3	0.00142	ramp	$4 \cdot n - 40 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 28$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i5, i4] (i0=0, i4=0)
n^3	0.0014	ramp	$(13/4) \cdot n - 22 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 46$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i5, i4] (i0=0, i4=6)
n^3	0.0014	ramp	$(13/4) \cdot n - 26 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 50$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i5, i4] (i0=0)
n^3	0.0014	ramp	$(25/8) \cdot n - 19 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 49$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	0.00586/n	read A[i5, i4] (i0=0, i4=7)
n^3	0.00136	ramp	$(49/8) \cdot n - 115 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 10$	$(1/128) \cdot n^2 + (-11/16) \cdot n + 15$	0.00586/n	read A[i3, i2] (i0=0)
n^3	0.00131	ramp	$(25/8) \cdot n - 18 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 81$	$(1/128) \cdot n^2 + (-11/16) \cdot n + 15$	0.00586/n	read A[i5, i4] (i0=0, i4=15)
n^3	0.00114	ramp	$2 \cdot n - 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 53$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0011	ramp	$(9/8) \cdot n + 4 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 81$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00586/n$	read A[i5, i4] (i0=0)
n^3	0.00107	ramp	$(17/8) \cdot n - 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 81$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	$0.00586/n$	read A[i5, i4] (i0=0)
n^3	0.00107	ramp	$2 \cdot n - 2 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(1/128) \cdot n^2 + (-9/16) \cdot n + 10$	$0.00586/n$	read A[i5, i4] (i0=0)
n^3	0.00104	ramp	$3 \cdot n - 16 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 86$	$(1/128) \cdot n^2 + (-11/16) \cdot n + 15$	$0.00586/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	1.54	ramp	$4 \rightarrow (9/8) \cdot n - 2$	$(15/8) \cdot n^2 + (-75/4) \cdot n + 50$	$1.41/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	1.22	ramp	$5 \rightarrow (9/8) \cdot n - 2$	$(35/16) \cdot n^2 + (-235/8) \cdot n + 105$	$1.64/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.366	ramp	$5 \rightarrow (9/8) \cdot n - 1$	$(7/16) \cdot n^2 + (-21/8) \cdot n + 3$	$0.328/n$	read A[i5, i4] (i0=0, i4=0)
$n^{2.5}$	0.323	ramp	$14 \rightarrow (9/8) \cdot n - 9$	$(7/16) \cdot n^2 + (-35/2) \cdot n + 168$	$0.328/n$	read A[i1, i5] (i0=0)
$n^{2.5}$	0.3	ramp	$5 \rightarrow (9/8) \cdot n - 8$	$(3/8) \cdot n^2 + (-51/8) \cdot n + 27$	$0.281/n$	read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	0.267	ramp	$5 \rightarrow (9/8) \cdot n - 9$	$(1/2) \cdot n^2 + (-21/2) \cdot n + 55$	$0.375/n$	read A[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.257	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(5/16) \cdot n^2 + (-25/8) \cdot n + 5$	$0.234/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.246	ramp	$4 \rightarrow (9/8) \cdot n - 2$	$(7/16) \cdot n^2 + (-23/8) \cdot n + 5$	$0.328/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.235	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(7/16) \cdot n^2 + (-21/2) \cdot n + 56$	$0.328/n$	read A[i1, i5] (i0=0)
$n^{2.5}$	0.223	ramp	$5 \rightarrow (9/8) \cdot n - 16$	$(7/16) \cdot n^2 - 14 \cdot n + 112$	$0.328/n$	read A[i1, i5] (i0=0, i5=0)
$n^{2.5}$	0.211	ramp	$3 \rightarrow (9/8) \cdot n - 2$	$(3/8) \cdot n^2 + (-9/8) \cdot n$	$0.281/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.211	ramp	$4 \rightarrow (9/8) \cdot n - 2$	$(3/8) \cdot n^2 + (-17/8) \cdot n + 3$	$0.281/n$	read A[i3, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.207	ramp	$4 \rightarrow (9/8) \cdot n - 8$	$(49/128) \cdot n^2 + (-91/16) \cdot n + 21$	$0.287/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.205	ramp	$4 \rightarrow (9/8) \cdot n - 9$	$(49/128) \cdot n^2 + (-105/16) \cdot n + 28$	$0.287/n$	read A[i5, i4] (i0=0, i5=0)
$n^{2.5}$	0.198	ramp	$14 \rightarrow (9/8) \cdot n - 9$	$(3/8) \cdot n^2 + (-105/8) \cdot n + 114$	$0.281/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0513	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	$0.0469/n$	read A[i5, i4] (i0=0, i4=6)
$n^{2.5}$	0.0466	ramp	$21 \rightarrow (9/8) \cdot n - 8$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	$0.0469/n$	read A[i1, i5] (i0=0, i4=0)
$n^{2.5}$	0.0351	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0469/n$	read A[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.0351	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.0469/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.0332	ramp	$13 \rightarrow (9/8) \cdot n - 14$	$(1/16) \cdot n^2 + (-7/4) \cdot n + 12$	$0.0469/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0331	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0469/n$	read A[i1, i3] (i0=0)
$n^{2.5}$	0.0316	ramp	$13 \rightarrow (9/8) \cdot n - 23$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	$0.0469/n$	read A[i1, i5] (i0=0)
$n^{2.5}$	0.031	ramp	$6 \rightarrow (9/8) \cdot n - 23$	$(1/16) \cdot n^2 + (-5/2) \cdot n + 24$	$0.0469/n$	read A[i1, i5] (i0=0, i5=0)
$n^{2.5}$	0.0294	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	$0.041/n$	read A[i5, i4] (i0=0)
$n^{2.5}$	0.0294	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	$0.041/n$	read A[i5, i4] (i0=0, i5=0)
n^2	3.03	level	3	$(7/4) \cdot n^2 + (-49/2) \cdot n + 84$	$1.31/n$	read A[i1, i3] (i0=0)
n^2	2.27	level	3	$(21/16) \cdot n^2 + (-35/2) \cdot n + 56$	$0.984/n$	write A[i1, i2] (i0=0)
n^2	1.86	level	2	$(21/16) \cdot n^2 + (-49/8) \cdot n$	$0.984/n$	write A[i1, i2] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2 - 14 \cdot n + 35$	$1.31/n$	read A[i1, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-63/4) \cdot n + 70$	$0.656/n$	read A[i1, i3] (i0=0); write A[i1, i2] (i0=0)
n^2	1.23	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(21/4) \cdot n - 132$	$3.94 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	1.14	ramp	$(17/8) \cdot n - 5$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$6 \cdot n - 152$	$4.5 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	1.03	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 18$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 13$	$(35/8) \cdot n - 100$	$3.28 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	1.01	ramp	$(13/4) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(21/4) \cdot n - 139$	$3.94 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.976	ramp	$(5/8) \cdot n$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 30$	$(21/4) \cdot n - 117$	$3.94 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.964	ramp	$(13/4) \cdot n - 25$ → $(1/16) \cdot n^2 + (5/8) \cdot n$	$5 \cdot n - 130$	$3.75 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.95	ramp	$(17/8) \cdot n - 5$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 11$	$5 \cdot n - 120$	$3.75 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.85	ramp	$(5/2) \cdot n - 13$ → $(1/16) \cdot n^2 + (5/8) \cdot n - 1$	$(35/8) \cdot n - 95$	$3.28 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.847	ramp	$(13/4) \cdot n - 24$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(35/8) \cdot n - 110$	$3.28 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.807	ramp	$(1/2) \cdot n + 3$ → $(1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(9/2) \cdot n - 134$	$3.38 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 - 7 \cdot n + 28$	$0.328/n$	write A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.729	ramp	$(21/8) \cdot n - 14 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 3$	$(15/4) \cdot n - 84$	$2.81 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.707	level	2	$(1/2) \cdot n^2 + (3/2) \cdot n + 4$	$0.375/n$	read A[i5, i4] (i0=0, i1=1, i4=0, i5=0); read A[i5, i4] (i0=0, i1=1, i4=6, i5=0) (+6)
n^2	0.695	ramp	$(11/8) \cdot n + 1 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(15/4) \cdot n - 95$	$2.81 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.578	ramp	$(13/4) \cdot n - 25 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n$	$3 \cdot n - 78$	$2.25 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
n^2	0.53	level	2	$(3/8) \cdot n^2 + (-3/8) \cdot n$	$0.281/n$	write A[i1, i2] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2 + (-63/8) \cdot n + 35$	$0.328/n$	read A[i1, i2] (i0=0, i3=0)
n^2	0.438	level	1	$(7/16) \cdot n^2 + (-7/8) \cdot n - 21$	$0.328/n$	read A[i1, i4] (i0=0, i5=0)
n^2	0.38	ramp	$(17/4) \cdot n - 49 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n + 1$	$2 \cdot n - 68$	$1.5 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
n^2	0.365	ramp	$(7/4) \cdot n - 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 5$	$(15/8) \cdot n - 30$	$1.41 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.342	ramp	$(9/4) \cdot n - 9 \rightarrow (1/16) \cdot n^2 + (5/8) \cdot n - 1$	$(7/4) \cdot n - 31$	$1.31 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.269	ramp	$(3/8) \cdot n + 3 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 57$	$(3/2) \cdot n - 43$	$1.12 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^2	0.233	ramp	$(3/2) \cdot n \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/4) \cdot n - 30$	$0.938 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.231	ramp	$(11/8) \cdot n + 1 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/4) \cdot n - 35$	$0.938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0, i5=7)
n^2	0.231	ramp	$(11/8) \cdot n + 1 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 31$	$(5/4) \cdot n - 35$	$0.938 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.205	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 19 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 23$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
n^2	0.205	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 21 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 25$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.205	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 5 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 9$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.204	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 7 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 11$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.204	ramp	$(1/16) \cdot n^2 + (5/8) \cdot n - 24 \rightarrow (1/16) \cdot n^2 + (3/4) \cdot n - 28$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.19	ramp	$(17/8) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$n - 24$	$0.75 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.19	ramp	$(17/8) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$n - 24$	$0.75 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.19	ramp	$(17/8) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$n - 24$	$0.75 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.188	ramp	$(25/8) \cdot n - 20$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$n - 32$	$0.75 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=7)
n^2	0.188	ramp	$(25/8) \cdot n - 20$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$n - 32$	$0.75 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.188	ramp	$(25/8) \cdot n - 20$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$n - 32$	$0.75 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.171	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 4$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(3/4) \cdot n - 24$	$0.562 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.169	ramp	$(13/4) \cdot n - 24$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.169	ramp	$(13/4) \cdot n - 25 \rightarrow$ $(1/16) \cdot n^2$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=8)
n^2	0.167	ramp	$(17/4) \cdot n - 48 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n$	$(7/8) \cdot n - 29$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=7)
n^2	0.167	ramp	$(17/4) \cdot n - 48 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(7/8) \cdot n - 29$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.167	ramp	$(17/4) \cdot n - 48 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(7/8) \cdot n - 29$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.167	ramp	$(9/4) \cdot n - 7 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(7/8) \cdot n - 21$	$0.656 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.162	ramp	$(11/8) \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 31$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=7)
n^2	0.162	ramp	$(11/8) \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 27$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.158	ramp	$(1/2) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 52$	$(7/8) \cdot n - 22$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
n^2	0.147	ramp	$(5/2) \cdot n - 12 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(3/4) \cdot n - 14$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.147	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 14$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.145	ramp	$(27/8) \cdot n - 27 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.145	ramp	$(27/8) \cdot n - 28 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.145	ramp	$(27/8) \cdot n - 29 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=7)
n^2	0.144	ramp	$(11/8) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(3/4) \cdot n - 12$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.144	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.142	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(3/4) \cdot n - 25$	$0.562 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.14	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 28$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.139	ramp	$(11/8) \cdot n - 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 27$	$(3/4) \cdot n - 18$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.138	ramp	$(5/4) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 48$	$(3/4) \cdot n - 18$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.138	ramp	$(5/4) \cdot n + 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 46$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
n^2	0.137	ramp	$(9/8) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 49$	$(3/4) \cdot n - 19$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.136	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 53$	$(3/4) \cdot n - 24$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)
n^2	0.133	ramp	$(9/8) \cdot n + 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 81$	$(3/4) \cdot n - 24$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=15)
n^2	0.123	ramp	$(19/8) \cdot n - 10 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.123	ramp	$(9/4) \cdot n - 8$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.123	ramp	$(5/2) \cdot n - 12$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	$(5/8) \cdot n - 11$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.121	ramp	$(13/8) \cdot n - 2$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 8$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.121	ramp	$(3/2) \cdot n - 1$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 11	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.12	ramp	$(25/8) \cdot n -$ 21 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.12	ramp	$(25/8) \cdot n -$ 22 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 10	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.12	ramp	$(25/8) \cdot n -$ 23 $\rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.119	ramp	$(33/8) \cdot n -$ 45 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 10	$(5/8) \cdot n - 20$	$0.469 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.117	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.116	ramp	$(11/8) \cdot n +$ 1 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 31	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.116	ramp	$(11/8) \cdot n +$ 1 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 31	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.115	ramp	$(19/8) \cdot n - 8$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 31$	$(5/8) \cdot n - 20$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2$ $+ (-5/8) \cdot n + 1$	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.094	ramp	$23 \rightarrow$ $(1/16) \cdot n^2$ $+ (-3/8) \cdot n + 1$	$(3/4) \cdot n - 12$	$0.562 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2$ $+ (-1/2) \cdot n$	$0.0469/n$	write A[i1, i2] (i0=0)
n^2	0.0716	ramp	$(21/8) \cdot n - 13 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 13$	$(3/8) \cdot n - 9$	$0.281 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0714	ramp	$(5/2) \cdot n - 12 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 16$	$(3/8) \cdot n - 9$	$0.281 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0705	ramp	$(17/8) \cdot n - 6 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(3/8) \cdot n - 9$	$0.281 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.048	ramp	$(25/8) \cdot n - 23 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$	$(1/4) \cdot n - 6$	$0.188 \cdot n^{-2}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
n^2	0.0457	ramp	$(5/4) \cdot n + 2 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 46$	$(1/4) \cdot n - 7$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6, i5=7)
n^2	0.0456	ramp	$(9/8) \cdot n + 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 49$	$(1/4) \cdot n - 7$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7, i5=0); read A[i5, i4] (i0=0, i4=7, i5=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.045	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 53	$(1/4) \cdot n - 9$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8, i5=0); read A[i5, i4] (i0=0, i4=8, i5=7)
n^2	0.044	ramp	$(9/8) \cdot n + 4$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 81	$(1/4) \cdot n - 9$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=15, i5=0); read A[i5, i4] (i0=0, i4=15, i5=7)
n^2	0.0293	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n -$ 20 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 24	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6)
n^2	0.0285	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 10	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.0285	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 12	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0285	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n -$ 21 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 26	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0285	ramp	$(1/16) \cdot n^2$ $+ (5/8) \cdot n -$ 24 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 29	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)
n^2	0.0245	ramp	$(19/8) \cdot n -$ 10 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 7$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i2, i2] (i0=0)
n^2	0.0244	ramp	$(9/4) \cdot n - 8$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 8$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0241	ramp	$(13/4) \cdot n - 24 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 8$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0241	ramp	$(13/4) \cdot n - 25 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0241	ramp	$(13/4) \cdot n - 26 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0, i2=7)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 21 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 22 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0241	ramp	$(25/8) \cdot n - 22 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.024	ramp	$(25/8) \cdot n - 23 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0, i2=7)
n^2	0.024	ramp	$3 \cdot n - 19 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 11$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.024	ramp	$3 \cdot n - 20 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 12$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0239	ramp	$(23/8) \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 14$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 44 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=7)
n^2	0.0237	ramp	$(33/8) \cdot n - 44 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 44 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 45 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 10$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0237	ramp	$(5/2) \cdot n - 11 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 19$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0237	ramp	$(33/8) \cdot n - 46 \rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0, i2=8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0237	ramp	$(19/8) \cdot n - 9$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 21$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0237	ramp	$(19/8) \cdot n - 10$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 22$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0236	ramp	$(9/4) \cdot n - 7$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 23$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0236	ramp	$(9/4) \cdot n - 8$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 24$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 5$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 25$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 6$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 6$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 6$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n - 26$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)
n^2	0.0235	ramp	$(17/8) \cdot n - 8$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (5/8) \cdot n - 24$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0234	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0234	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0234	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=0, i5=0)
n^2	0.0234	ramp	$2 \cdot n - 4 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0234	ramp	$(41/8) \cdot n -$ 76 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 10	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i3, i2] (i0=0, i3=7)
n^2	0.0234	ramp	$(41/8) \cdot n -$ 76 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 10	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i3, i2] (i0=0)
n^2	0.0234	ramp	$(41/8) \cdot n -$ 76 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 10	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i3, i2] (i0=0)
n^2	0.0232	ramp	$(25/8) \cdot n -$ 21 $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 26	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i2, i2] (i0=0)
n^2	0.0232	ramp	$3 \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=0, i5=7)
n^2	0.0232	ramp	$3 \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.0232	ramp	$3 \cdot n - 18 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 28	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=0)
n^2	0.023	ramp	$(5/4) \cdot n + 2$ $\rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 46	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{\frac{1}{2}}$	read A[i5, i4] (i0=0, i4=6)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.023	ramp	$(5/4) \cdot n + 2$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 46$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=6)
n^2	0.0229	ramp	$(5/4) \cdot n - 1$ → $(1/16) \cdot n^2$ + $(5/8) \cdot n - 45$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0)
n^2	0.0229	ramp	$(9/8) \cdot n + 3$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 49$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0228	ramp	$(9/4) \cdot n - 6$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 46$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=6)
n^2	0.0228	ramp	$(9/4) \cdot n - 6$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 46$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=6)
n^2	0.0228	ramp	$(9/4) \cdot n - 9$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 49$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i5=7)
n^2	0.0227	ramp	$(9/4) \cdot n - 10$ → $(1/16) \cdot n^2$ + $(5/8) \cdot n - 45$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0)
n^2	0.0227	ramp	$(17/8) \cdot n - 4$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 49$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0227	ramp	$(17/8) \cdot n - 4$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 49$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i5, i4] (i0=0, i4=7)
n^2	0.0227	ramp	$(17/8) \cdot n - 6$ → $(1/16) \cdot n^2$ + $(3/4) \cdot n - 51$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^2$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0226	ramp	$2 \cdot n - 3 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 53	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)
n^2	0.0224	ramp	$3 \cdot n - 17 \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 53	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=8)
n^2	0.0222	ramp	$(5/4) \cdot n \rightarrow$ $(1/16) \cdot n^2$ $+ (3/4) \cdot n -$ 80	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
n^2	0.0156	ramp	$22 \rightarrow$ $(1/16) \cdot n^2$ $+ (-3/8) \cdot n -$ 5	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i2, i2] (i0=0)
$n^{1.5}$	4.71	ramp	$4 \rightarrow (9/8) \cdot n$ - 9	$7 \cdot n - 91$	$5.25 \cdot n^{-2}$	read A[i1, i5] (i0=0, i5=0)
$n^{1.5}$	2.62	ramp	$3 \rightarrow (9/8) \cdot n$ - 2	$(15/4) \cdot n -$ 15	$2.81 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	2.62	ramp	$3 \rightarrow (9/8) \cdot n$ - 2	$(15/4) \cdot n -$ 15	$2.81 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
$n^{1.5}$	2.3	ramp	$13 \rightarrow$ $(9/8) \cdot n - 8$	$(27/8) \cdot n -$ 54	$2.53 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0); read A[i1, i5] (i0=0)
$n^{1.5}$	1.03	ramp	$5 \rightarrow (9/8) \cdot n$ - 8	$(3/2) \cdot n - 12$	$1.12 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	0.872	ramp	$11 \rightarrow$ $(9/8) \cdot n - 7$	$(5/4) \cdot n - 10$	$0.938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0); read A[i5, i4] (i0=0)
$n^{1.5}$	0.685	ramp	$5 \rightarrow (9/8) \cdot n$ - 8	$n - 10$	$0.75 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0, i5=0)
$n^{1.5}$	0.62	ramp	$4 \rightarrow (9/8) \cdot n$ - 1	$(7/8) \cdot n - 2$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.601	ramp	$4 \rightarrow (9/8) \cdot n$ - 8	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=7, i5=0)
$n^{1.5}$	0.587	ramp	$11 \rightarrow$ $(9/8) \cdot n - 16$	$(7/8) \cdot n - 14$	$0.656 \cdot n^{-2}$	read A[i1, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.579	ramp	$n + 4 \rightarrow (9/8) \cdot n - 1$	$(5/8) \cdot n - 20$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.54	ramp	$(1/4) \cdot n + 3 \rightarrow n - 1$	$(3/4) \cdot n - 18$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.532	ramp	$4 \rightarrow (9/8) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0, i5=0)
$n^{1.5}$	0.512	ramp	$3 \rightarrow (9/8) \cdot n - 10$	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.475	ramp	$(3/8) \cdot n + 3 \rightarrow n - 1$	$(5/8) \cdot n - 15$	$0.469 \cdot n^{-2}$	read A[i5, i4] (i0=0)
$n^{1.5}$	0.174	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/4) \cdot n - 2$	$0.188 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=6, i5=0); read A[i5, i4] (i0=0, i4=6)
$n^{1.5}$	0.156	ramp	$2 \cdot n - 2 \rightarrow (17/8) \cdot n - 8$	$(1/8) \cdot n - 5$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=7)
$n^{1.5}$	0.122	ramp	$(9/8) \cdot n + 4 \rightarrow (5/4) \cdot n - 1$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=9, i5=7)
$n^{1.5}$	0.119	ramp	$n + 3 \rightarrow (9/8) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=8, i5=7)
$n^{1.5}$	0.116	ramp	$n + 4 \rightarrow (9/8) \cdot n - 1$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=8, i5=6)
$n^{1.5}$	0.116	ramp	$n + 4 \rightarrow (9/8) \cdot n - 1$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i5=0)
$n^{1.5}$	0.0872	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0, i5=0)
$n^{1.5}$	0.0845	ramp	$13 \rightarrow (9/8) \cdot n - 14$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
$n^{1.5}$	0.0839	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i5] (i0=0, i4=0)
$n^{1.5}$	0.0839	ramp	$11 \rightarrow (9/8) \cdot n - 16$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i4=0)
$n^{1.5}$	0.0631	ramp	$(1/4) \cdot n + 4 \rightarrow (3/8) \cdot n - 1$	$(1/8) \cdot n - 4$	$0.0938 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=2, i5=0)
n^1	30.3	level	3	$(35/2) \cdot n - 140$	$13.1 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	30.3	level	3	$(35/2) \cdot n - 140$	$13.1 \cdot n^{-2}$	read A[i1, i5] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	14.8	level	2	$(21/2) \cdot n$	$7.88 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	13.9	level	3	$8 \cdot n - 64$	$6 \cdot n^{-2}$	write A[i1, i4] (i0=0)
n^1	12.1	level	3	$7 \cdot n - 56$	$5.25 \cdot n^{-2}$	read A[i1, i5] (i0=0)
n^1	10.5	level	1	$(21/2) \cdot n - 28$	$7.88 \cdot n^{-2}$	read A[i1, i5] (i0=0)
n^1	7	level	1	$7 \cdot n - 35$	$5.25 \cdot n^{-2}$	read A[i1, i3] (i0=0, i3=0)
n^1	4	level	4	$2 \cdot n - 25$	$1.5 \cdot n^{-2}$	read A[i1, i5] (i0=0, i1=9, i4=0, i5=0); read A[i1, i3] (i0=0) (+1)
n^1	1.52	level	3	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	read A[i5, i4] (i0=0, i1=1, i5=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 20$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 18$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 16$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 14$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 12$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 27$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 24$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 22$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 10$	6	$4.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 7$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 6$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 5$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 4$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 3$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 2$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 8$	6	$4.5 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	5	$3.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 6$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	1	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 10$	4	$3 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.656 \cdot n^{-2}$	read A[i1, i4] (i0=0, i4=0, i5=0)
n^1	0.75	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 12$	3	$2.25 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.75	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.5	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 14$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 21$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 19$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 17$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 15$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 13$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 18$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 14$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 10$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 18$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 14$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 18$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 14$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 28$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 25$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 22$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 24$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 22$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 23$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 48$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=15, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 45$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 42$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 39$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 36$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 33$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 30$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 27$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 24$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=7, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 22$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=6, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (7/8) \cdot n + (-3/2)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (9/8) \cdot n + (-51/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (-21/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-9/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (-53/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (-69/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + n + (-21/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-107/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + n + (-25/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (9/8) \cdot n + (-99/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-139/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (7/8) \cdot n + (-99/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (9/8) \cdot n + (-99/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-139/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (7/8) \cdot n + (-115/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + n + (-29/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-171/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (-85/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + n + (-29/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-171/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (7/8) \cdot n +$ $(-131/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n +$ $(-203/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n +$ $(-43/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/2) \cdot n +$ $(-9/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (11/8) \cdot n +$ $(-43/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n +$ $(-171/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (13/8) \cdot n +$ $(-203/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ n +$ $(-29/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/2) \cdot n +$ $(-41/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (7/8) \cdot n +$ $(-131/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/2) \cdot n +$ $(-49/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 26$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (13/8) \cdot n + (-235/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/2) \cdot n + (-57/4)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0, i5=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 11$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 10$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0, i5=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=7)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (3/2) \cdot n +$ $(-9/16)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ $+ (5/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.25	level	$(1/16) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ + $(5/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ + $(5/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ + $(5/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ + $(5/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ + $(1/2) \cdot n$ + $(-73/16)$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2$ + $(3/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	8.84	level	$(25/8) \cdot n - 22$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.66	level	$3 \cdot n - 20$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.48	level	$(23/8) \cdot n - 18$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.29	level	$(11/4) \cdot n - 16$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	8.1	level	$(21/8) \cdot n - 14$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.91	level	$(5/2) \cdot n - 12$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.71	level	$(19/8) \cdot n - 10$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.71	level	$(19/8) \cdot n - 11$	5	$3.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	7.5	level	$(9/4) \cdot n - 8$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.5	level	$(9/4) \cdot n - 9$	5	$3.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	7.29	level	$(17/8) \cdot n - 6$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	7.29	level	$(17/8) \cdot n - 7$	5	$3.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	7.07	level	$2 \cdot n - 5$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.85	level	$(15/8) \cdot n - 4$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.61	level	$(7/4) \cdot n - 3$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.37	level	$(13/8) \cdot n - 2$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	6.32	level	$(5/2) \cdot n - 13$	4	$3 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	6.12	level	$(3/2) \cdot n - 1$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.86	level	$(11/8) \cdot n$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.66	level	$2 \cdot n - 6$	4	$3 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	5.59	level	$(5/4) \cdot n + 1$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.3	level	$(25/8) \cdot n - 22$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=25, i2=8, i3=0); read A[i3, i2] (i0=0, i1=25, i2=8, i3=7) (+1)
$n^{0.5}$	5.3	level	$(9/8) \cdot n + 2$	5	$3.75 \cdot n^{-3}$	read A[i3, i2] (i0=0)
$n^{0.5}$	5.2	level	$3 \cdot n - 20$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	5.09	level	$(23/8) \cdot n - 18$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	5	level	$n + 3$	5	$3.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8)
$n^{0.5}$	5	level	$n + 2$	5	$3.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	4.97	level	$(11/4) \cdot n - 16$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.86	level	$(21/8) \cdot n - 14$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	4.86	level	$(21/8) \cdot n - 15$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	4.74	level	$(5/2) \cdot n - 12$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.62	level	$(19/8) \cdot n - 10$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.5	level	$(9/4) \cdot n - 8$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0, i2=8, i3=7) (+1)
$n^{0.5}$	4.37	level	$(17/8) \cdot n - 6$	3	$2.25 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=17, i2=8, i3=0); read A[i3, i2] (i0=0, i1=17, i2=8, i3=7) (+1)
$n^{0.5}$	4.24	level	$(9/8) \cdot n + 1$	4	$3 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=9, i2=0); read A[i2, i2] (i0=0, i1=9, i2=1) (+2)
$n^{0.5}$	4.11	level	$(15/8) \cdot n - 5$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	4.06	level	$(33/8) \cdot n - 45$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=33, i2=16, i3=7); read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	4	level	$4 \cdot n - 42$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.94	level	$(31/8) \cdot n - 39$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.87	level	$(15/4) \cdot n - 36$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.81	level	$(29/8) \cdot n - 33$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.74	level	$(7/8) \cdot n + 2$	4	$3 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	3.74	level	$(7/2) \cdot n - 30$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.67	level	$(27/8) \cdot n - 27$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.61	level	$(13/4) \cdot n - 24$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	3.54	level	$(25/8) \cdot n - 21$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=25, i2=16, i3=6); read A[i3, i2] (i0=0, i1=25, i2=16, i3=7)
$n^{0.5}$	3.52	level	$(11/8) \cdot n - 1$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	3.46	level	$3 \cdot n - 19$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.39	level	$(23/8) \cdot n - 17$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.35	level	$(5/4) \cdot n$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	3.35	level	$(5/4) \cdot n$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=10, i2=0); read A[i2, i2] (i0=0, i1=10, i2=1) (+1)
$n^{0.5}$	3.32	level	$(11/4) \cdot n - 15$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.32	level	$(11/4) \cdot n - 17$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	3.24	level	$(21/8) \cdot n - 13$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.18	level	$(9/8) \cdot n + 1$	3	$2.25 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	3.16	level	$(5/2) \cdot n - 11$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3.08	level	$(19/8) \cdot n - 9$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3	level	$(9/4) \cdot n - 7$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	3	level	$n + 2$	3	$2.25 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=8, i4=0, i5=0); read A[i5, i4] (i0=0, i1=8, i4=0, i5=6) (+1)
$n^{0.5}$	2.92	level	$(17/8) \cdot n - 5$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=16, i3=7); read A[i3, i2] (i0=0)
$n^{0.5}$	2.92	level	$(17/8) \cdot n - 7$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=17, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	2.83	level	$2 \cdot n - 5$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.83	level	$2 \cdot n - 4$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=16, i4=0, i5=7); read A[i5, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	2.74	level	$(15/8) \cdot n - 3$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.74	level	$(15/8) \cdot n - 4$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.65	level	$(7/4) \cdot n - 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.65	level	$(7/4) \cdot n - 3$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.65	level	$(7/4) \cdot n - 4$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	2.6	level	$(3/4) \cdot n + 2$	3	$2.25 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	2.55	level	$(13/8) \cdot n - 1$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.55	level	$(13/8) \cdot n - 2$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.45	level	$(3/2) \cdot n$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.45	level	$(3/2) \cdot n - 1$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.45	level	$(3/2) \cdot n - 2$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	2.35	level	$(11/8) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=7); read A[i5, i4] (i0=0)
$n^{0.5}$	2.35	level	$(11/8) \cdot n$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.24	level	$(5/4) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.24	level	$(5/4) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=10, i4=6, i5=7); read A[i5, i4] (i0=0, i4=6)
$n^{0.5}$	2.12	level	$(9/8) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=0); read A[i3, i2] (i0=0)
$n^{0.5}$	2.12	level	$(9/8) \cdot n + 3$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=9, i4=7, i5=7); read A[i5, i4] (i0=0, i4=7)
$n^{0.5}$	2.12	level	$(9/8) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=9, i2=8, i3=7); read A[i2, i2] (i0=0, i1=9, i2=8)
$n^{0.5}$	2	level	$n + 3$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=8, i4=8, i5=6); read A[i5, i4] (i0=0, i4=8, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	2	level	$n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	2	level	$n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.87	level	$(7/8) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.77	level	$(25/8) \cdot n - 22$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=25, i2=8)
$n^{0.5}$	1.73	level	$(3/4) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.73	level	$3 \cdot n - 20$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.73	level	$3 \cdot n - 21$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.7	level	$(23/8) \cdot n - 18$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.7	level	$(23/8) \cdot n - 19$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.7	level	$(23/8) \cdot n - 19$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.66	level	$(11/4) \cdot n - 16$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.66	level	$(11/4) \cdot n - 17$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.62	level	$(21/8) \cdot n - 14$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.62	level	$(21/8) \cdot n - 15$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.58	level	$(5/8) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.58	level	$(5/2) \cdot n - 12$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.58	level	$(5/2) \cdot n - 13$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.54	level	$(19/8) \cdot n - 10$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.54	level	$(19/8) \cdot n - 11$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=18, i2=7)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 9$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 8$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=18, i2=8)
$n^{0.5}$	1.46	level	$(17/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=17, i2=7)
$n^{0.5}$	1.46	level	$(17/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=17, i2=8)
$n^{0.5}$	1.41	level	$2 \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=8, i5=7)
$n^{0.5}$	1.41	level	$2 \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=16, i2=8)
$n^{0.5}$	1.41	level	$(1/2) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=4, i2=0); read A[i2, i2] (i0=0, i1=4, i2=1)
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.41	level	$2 \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=16, i2=7)
$n^{0.5}$	1.41	level	$2 \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=15, i2=7)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=7)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=12, i2=8)
$n^{0.5}$	1.22	level	$(3/8) \cdot n + 1$	2	$1.5 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=3, i2=0); read A[i2, i2] (i0=0, i1=3, i2=1)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=11, i2=8)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=11, i2=7)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=11, i2=1)
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.12	level	$(5/4) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=9, i5=7)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=10, i2=7)
$n^{0.5}$	1.12	level	$(5/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=10, i2=8)
$n^{0.5}$	1.12	level	$(5/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=8, i3=7)
$n^{0.5}$	1.06	level	$(9/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=8, i5=7)
$n^{0.5}$	1.06	level	$(9/8) \cdot n + (-1/4)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 7$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n + (-65/8)$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 6$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 5$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 4$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 3$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n - 1$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i4=0)
$n^{0.5}$	1.06	level	$(9/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=9, i2=7)
$n^{0.5}$	1	level	n	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	1	level	n	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	1	level	$(1/4) \cdot n + 2$	2	$1.5 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=2, i4=6, i5=0); read A[i5, i4] (i0=0, i1=2, i4=6, i5=1)
$n^{0.5}$	1	level	n + 1	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1	level	n + 1	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	1	level	n + 1	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.935	level	$(7/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i2=0); read A[i2, i2] (i0=0, i2=1)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=5, i2=1)
$n^{0.5}$	0.791	level	$(5/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.707	level	$(1/2) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i5=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=2, i5=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 3$	1	$0.75 \cdot n^{-3}$	read A[i5, i4] (i0=0, i1=2, i5=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=2, i2=0)
n^0	28	level	1	28	$21 \cdot n^{-3}$	read A[i1, i5] (i0=0, i5=0)
n^0	3.32	level	11	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	3.16	level	10	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	3	level	9	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.83	level	8	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.65	level	7	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.45	level	6	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i2, i2] (i0=0, i1=2, i2=1)

bin 100: finite-size slope 3.33 at $n \approx 256-312$; converges to n^3 ; bin 101: finite-size slope 3.26 at $n \approx 256-312$; converges to n^3 ; bin 104: finite-size slope 3.27 at $n \approx 256-312$; converges to n^3 .

Row-panel and trailing-submatrix re-reads (read A[i3, i2], read A[i5, i4]) ramp to $(1/16)n^2 + (3/4)n$ lines with cubic populations — $d = 4.0$, headroom +1.0. At the anchor sizes ($n \approx 256-312$) the finite-size slope of these families still reads ≈ 4.4 because the negative n^2 corrections in their populations have not yet died out; the exact degrees settle it at 4.0.

lu_decomp — infinite-repeat [exact]

Accesses $A(n) = (4/3) \cdot n^3 + (1/2) \cdot n^2 + (1/6) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma\text{mass/warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0101 \cdot n^4 + 0.0166 \cdot n^{3.5} + 2.36 \cdot n^3 + 0.605 \cdot n^{2.5} + 24.1 \cdot n^2 + 0.8 \cdot n^{1.5} + 40.6 \cdot n^1 + 867 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00766	ramp	$72 \rightarrow (1/8) \cdot n^2$	$(1/32) \cdot n^3 + (-121/128) \cdot n^2 + (39/16) \cdot n + 75$	0.0234	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^4	0.00124	ramp	$104 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/192) \cdot n^3 + (-25/128) \cdot n^2 + (71/48) \cdot n + 5$	0.00391	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^4	0.00117	ramp	$135 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 2$	$(1/192) \cdot n^3 + (-35/128) \cdot n^2 + (143/48) \cdot n + 14$	0.00391	read A[i3, i4] (i0=0)
$n^{3.5}$	0.0125	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(1/32) \cdot n^3 + (-121/128) \cdot n^2 + (41/16) \cdot n + 73$	0.0234	read A[i1, i4] (i0=0, i4=7); read A[i1, i4] (i0=0)
$n^{3.5}$	0.0021	ramp	$10 \rightarrow (1/4) \cdot n + 2$	$(1/192) \cdot n^3 + (-9/64) \cdot n^2 + (-17/24) \cdot n + 26$	0.00391	read A[i1, i4] (i0=0)
$n^{3.5}$	0.00204	ramp	$10 \rightarrow (1/4) \cdot n + 2$	$(1/192) \cdot n^3 + (-25/128) \cdot n^2 + (71/48) \cdot n + 5$	0.00391	read A[i1, i4] (i0=0)
n^3	0.442	level	3	$(49/192) \cdot n^3 + (-413/128) \cdot n^2 + (245/48) \cdot n + 35$	0.191	write A[i3, i4] (i0=0, i4=6); read A[i3, i1] (i0=0)
n^3	0.442	level	3	$(49/192) \cdot n^3 + (-469/128) \cdot n^2 + (287/48) \cdot n + 56$	0.191	write A[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.388	level	3	$(43/192) \cdot n^3 + (-193/64) \cdot n^2 + (65/12) \cdot n + 35$	0.168	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
n^3	0.255	level	1	$(49/192) \cdot n^3 + (-365/128) \cdot n^2 + (191/48) \cdot n + 20$	0.191	read A[i3, i4] (i0=0)
n^3	0.0889	ramp	$32 \rightarrow (1/8) \cdot n^2$	$(3/8) \cdot n^2 + (-9/8) \cdot n - 65$	0.281/n	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^3	0.0872	ramp	$22 \rightarrow (1/8) \cdot n^2 + (-1/4) \cdot n + 2$	$(3/8) \cdot n^2 + (-23/8) \cdot n - 5$	0.281/n	read A[i3, i1] (i0=0)
n^3	0.0833	level	4	$(1/24) \cdot n^3 + (-29/32) \cdot n^2 + (53/24) \cdot n + 26$	0.0312	read A[i3, i1] (i0=0)
n^3	0.0738	ramp	$57 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n$	$(5/16) \cdot n^2 + (-5/4) \cdot n - 60$	0.234/n	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^3	0.0631	level	3	$(7/192) \cdot n^3 + (-119/128) \cdot n^2 + (287/48) \cdot n - 7$	0.0273	read A[i3, i1] (i0=0)
n^3	0.0631	level	3	$(7/192) \cdot n^3 + (-119/128) \cdot n^2 + (287/48) \cdot n - 7$	0.0273	write A[i3, i4] (i0=0)
n^3	0.0631	level	3	$(7/192) \cdot n^3 + (-35/128) \cdot n^2 + (-7/48) \cdot n$	0.0273	write A[i3, i4] (i0=0)
n^3	0.0541	level	3	$(1/32) \cdot n^3 + (-3/64) \cdot n^2 + (-13/8) \cdot n$	0.0234	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0541	level	3	$(1/32) \cdot n^3$ + $(-27/128) \cdot n^2$ + $(-5/16) \cdot n$	0.0234	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
n^3	0.0365	level	1	$(7/192) \cdot n^3$ + $(-7/128) \cdot n^2$ + $(-91/48) \cdot n$	0.0273	read A[i3, i4] (i0=0)
n^3	0.0183	ramp	$(1/5898240) \cdot n^6$ + $(-7/24576) \cdot n^5$ + $(7345/36864) \cdot n^4$ + $(-28525/384) \cdot n^3$ + $(22417669/1440) \cdot n^2$ + $(-41731969/24) \cdot n$ + 80868862 → $(1/8) \cdot n^2 - 2$	$(1/16) \cdot n^2$ + $(-11/4) \cdot n$ + 30	0.0469/n	read A[i3, i4] (i0=0, i1=0)
n^3	0.0144	ramp	$54 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/16) \cdot n^2$ + $(-5/8) \cdot n - 6$	0.0469/n	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^3	0.0144	ramp	$54 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/16) \cdot n^2$ + $(-3/4) \cdot n - 4$	0.0469/n	read A[i3, i4] (i0=0, i4=0)
n^3	0.0143	ramp	$20 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n + 2$	$(1/16) \cdot n^2$ + $(-3/4) \cdot n + 2$	0.0469/n	read A[i3, i1] (i0=0)
n^3	0.0141	ramp	$37 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 2$	$(1/16) \cdot n^2$ + $(-7/8) \cdot n - 2$	0.0469/n	read A[i3, i1] (i0=0)
n^3	0.0141	ramp	$78 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 2$	$(1/16) \cdot n^2$ + $(-7/8) \cdot n - 15$	0.0469/n	read A[i3, i4] (i0=0, i4=7)
n^3	0.0141	ramp	$77 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 3$	$(1/16) \cdot n^2$ + $(-7/8) \cdot n - 15$	0.0469/n	read A[i3, i4] (i0=0)
n^3	0.0123	ramp	$54 \rightarrow (1/8) \cdot n^2$	$(7/128) \cdot n^2$ + $(-21/16) \cdot n$ + 7	0.041/n	read A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0121	ramp	$(-1/5898240) \cdot n^4 + (7/24576) \cdot n^5 + (-7345/36864) \cdot n^4 + (28525/384) \cdot n^3 + (-22417399/1440) \cdot n^2 + (41731987/24) \cdot n - 80868864$ → $(1/8) \cdot n^2$	$(5/128) \cdot n^2 + (-7/16) \cdot n - 12$	0.0293/n	read A[i3, i4] (i0=0, i1=0)
n^3	0.0102	ramp	$70 \rightarrow (1/8) \cdot n^2 - 2$	$(3/64) \cdot n^2 + (-7/4) \cdot n + 16$	0.0352/n	read A[i3, i4] (i0=0)
n^3	0.00902	level	3	$(1/192) \cdot n^3 + (-1/128) \cdot n^2 + (-13/48) \cdot n$	0.00391	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
n^3	0.00902	level	3	$(1/192) \cdot n^3 + (95/128) \cdot n^2 + (-289/48) \cdot n + 5$	0.00391	read A[i3, i1] (i0=0, i1=0, i3=0); read A[i3, i1] (i0=0, i1=0, i4=7) (+2)
n^3	0.00246	ramp	$(1/3932160) \cdot n^6 + (-7/16384) \cdot n^5 + (7345/24576) \cdot n^4 + (-28525/256) \cdot n^3 + (2802203/120) \cdot n^2 + (-41731977/16) \cdot n + 121303296$ → $(1/8) \cdot n^2$	$(1/128) \cdot n^2 + (-1/16) \cdot n - 3$	0.00586/n	read A[i3, i4] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00232	ramp	$ \begin{aligned} &(-1/3932160) \cdot n^6 + (-3/16) \cdot n^5 \\ &+ (7/16384) \cdot n^4 + (-7345/24576) \cdot n^3 \\ &+ (28525/256) \cdot n^2 + (-5604361/240) \cdot n \\ &+ (41731985/16) \\ &- 121303296 \rightarrow (1/8) \cdot n^2 - n + 9 \end{aligned} $	$(1/128) \cdot n^2$	0.00586/n	read A[i3, i4] (i0=0, i1=0)
n^3	0.00232	ramp	$ \begin{aligned} &(-1/2949120) \cdot n^6 + (-3/16) \cdot n^5 \\ &+ (7/12288) \cdot n^4 + (-7345/18432) \cdot n^3 \\ &+ (28525/192) \cdot n^2 + (-44834933/1440) \cdot n \\ &+ (10432996/3) \\ &- 161737727 \rightarrow (1/8) \cdot n^2 - n + 9 \end{aligned} $	$(1/128) \cdot n^2$	0.00586/n	read A[i3, i4] (i0=0, i1=0)
n^3	0.00167	ramp	$ \begin{aligned} &101 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n - 3 \end{aligned} $	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.00586/n	read A[i3, i4] (i0=0)
n^3	0.00167	ramp	$ \begin{aligned} &100 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n + 3 \end{aligned} $	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.00586/n	read A[i1, i2] (i0=0)
n^3	0.0016	ramp	$ \begin{aligned} &164 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 1 \end{aligned} $	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i3, i4] (i0=0, i3=0)
n^3	0.00157	ramp	$ \begin{aligned} &130 \rightarrow (1/8) \cdot n^2 + (-15/8) \cdot n + 5 \end{aligned} $	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00586/n	read A[i3, i4] (i0=0)
$n^{2.5}$	0.153	ramp	$ \begin{aligned} &8 \rightarrow (1/4) \cdot n + 2 \end{aligned} $	$(3/8) \cdot n^2 + (-9/8) \cdot n - 78$	0.281/n	read A[i1, i4] (i0=0, i4=7); read A[i1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.152	ramp	$3 \rightarrow (1/4) \cdot n + 1$	$(3/8) \cdot n^2 + (-9/8) \cdot n$	$0.281/n$	read A[i1, i1] (i0=0)
$n^{2.5}$	0.127	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(5/16) \cdot n^2 + (-5/4) \cdot n - 60$	$0.234/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.026	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (1/4) \cdot n - 20$	$0.0469/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.0259	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (1/8) \cdot n - 18$	$0.0469/n$	read A[i1, i4] (i0=0, i4=6)
$n^{2.5}$	0.0258	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (1/4) \cdot n - 6$	$0.0469/n$	read A[i1, i1] (i0=0)
$n^{2.5}$	0.0251	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	$0.0469/n$	read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.0251	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	$0.0469/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.0248	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	$0.0469/n$	read A[i1, i1] (i0=0, i3=1); read A[i1, i1] (i0=0)
$n^{2.5}$	0.0149	ramp	$5 \rightarrow (1/4) \cdot n - 1$	$(3/64) \cdot n^2 + (-7/4) \cdot n + 16$	$0.0352/n$	read A[i1, i4] (i0=0, i3=0, i4=7); read A[i1, i4] (i0=0, i3=0)
$n^{2.5}$	0.00247	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.00586/n$	read A[i1, i4] (i0=0, i3=0)
$n^{2.5}$	0.00247	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.00586/n$	read A[i1, i4] (i0=0, i3=0)
n^2	2.17	level	3	$(5/4) \cdot n^2 + (-45/8) \cdot n - 35$	$0.938/n$	write A[i3, i4] (i0=0)
n^2	1.88	level	1	$(15/8) \cdot n^2 + (-15/4) \cdot n - 20$	$1.41/n$	read A[i3, i4] (i0=0)
n^2	1.62	level	3	$(15/16) \cdot n^2 - 5 \cdot n - 20$	$0.703/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-9/2) \cdot n - 20$	$0.656/n$	read A[i3, i1] (i0=0)
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-63/8) \cdot n + 7$	$0.656/n$	write A[i3, i4] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.33	level	2	$(15/16) \cdot n^2 + (5/4) \cdot n$	$0.703/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
n^2	0.972	level	2	$(11/16) \cdot n^2 + (-3/8) \cdot n$	$0.516/n$	read A[i3, i1] (i0=0)
n^2	0.938	level	1	$(15/16) \cdot n^2 + (5/4) \cdot n$	$0.703/n$	write A[i3, i4] (i0=0)
n^2	0.884	level	2	$(5/8) \cdot n^2 + (5/4) \cdot n$	$0.469/n$	read A[i3, i1] (i0=0)
n^2	0.875	level	1	$(7/8) \cdot n^2 + (-63/8) \cdot n + 35$	$0.656/n$	read A[i1, i2] (i0=0); write A[i1, i2] (i0=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-7/8) \cdot n - 21$	$0.328/n$	read A[i3, i1] (i0=0, i4=7)
n^2	0.75	level	1	$(3/4) \cdot n^2 + (-3/4) \cdot n$	$0.562/n$	read A[i3, i4] (i0=0, i4=0); write A[i3, i4] (i0=0, i4=0)
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-15/8) \cdot n - 9$	$0.281/n$	read A[i1, i4] (i0=0, i4=6)
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-9/8) \cdot n - 15$	$0.281/n$	read A[i3, i1] (i0=0, i4=6)
n^2	0.619	level	2	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.328/n$	write A[i3, i1] (i0=0)
n^2	0.53	level	2	$(3/8) \cdot n^2 + (-3/8) \cdot n - 12$	$0.281/n$	read A[i1, i4] (i0=0, i3=0, i4=0); read A[i1, i4] (i0=0, i4=0)
n^2	0.442	level	2	$(5/16) \cdot n^2 + (15/8) \cdot n$	$0.234/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
n^2	0.433	level	3	$(1/4) \cdot n^2 + (-1/4) \cdot n - 14$	$0.188/n$	read A[i3, i1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 5$	$0.75 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); read A[i3, i1] (i0=0, i1=0, i3=0) (+3)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$n - 2$	$0.75 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=1); read A[i3, i1] (i0=0, i1=1, i3=0) (+1)
n^2	0.354	level	2	$(1/4) \cdot n^2 + (-5/2) \cdot n + 4$	$0.188/n$	read A[i3, i1] (i0=0)
n^2	0.331	ramp	$(1/8) \cdot n^2 - n + 18 \rightarrow (1/8) \cdot n^2 - 1$	$n - 18$	$0.75 \cdot n^{-2}$	read A[i3, i4] (i0=0, i1=0, i4=7)
n^2	0.312	level	1	$(5/16) \cdot n^2 + (-25/8) \cdot n + 5$	$0.234/n$	write A[i3, i4] (i0=0)
n^2	0.311	ramp	$20 \rightarrow (1/8) \cdot n^2 + (-1/4) \cdot n + 2$	$(7/4) \cdot n - 16$	$1.31 \cdot n^{-2}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^2	0.273	level	1	$(35/128) \cdot n^2 + (25/16) \cdot n$	$0.205/n$	read A[i3, i4] (i0=0)
n^2	0.25	level	4	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	$0.0938/n$	read A[i3, i1] (i0=0, i1=0)
n^2	0.24	ramp	$(-1/3932160) \cdot n^6 + (7/16384) \cdot n^5 + (-7345/24576) \cdot n^4 + (28525/256) \cdot n^3 + (-5604361/240) \cdot n^2 + (41731985/16) \cdot n - 121303296 \rightarrow (1/8) \cdot n^2$	$(3/4) \cdot n - 2$	$0.562 \cdot n^{-2}$	read A[i3, i4] (i0=0, i1=0)
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-7/8) \cdot n - 1$	$0.0938/n$	read A[i3, i1] (i0=0, i4=1)
n^2	0.217	level	3	$(1/8) \cdot n^2 - n$	$0.0938/n$	read A[i3, i1] (i0=0, i4=0); write A[i3, i4] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.217	level	3	$(1/8) \cdot n^2 + (-9/8) \cdot n + 1$	$0.0938/n$	write A[i3, i4] (i0=0, i1=0)
n^2	0.177	level	2	$(1/8) \cdot n^2 + (-1/4) \cdot n + 1$	$0.0938/n$	read A[i1, i4] (i0=0, i3=0, i4=6); read A[i1, i4] (i0=0, i4=6) (+1)
n^2	0.177	level	2	$(1/8) \cdot n^2 + (-5/4) \cdot n + 2$	$0.0938/n$	read A[i3, i1] (i0=0)
n^2	0.157	ramp	$20 \rightarrow (1/8) \cdot n^2$	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.134	ramp	$30 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n$	$(3/4) \cdot n - 11$	$0.562 \cdot n^{-2}$	read A[i3, i4] (i0=0)
n^2	0.133	ramp	$20 \rightarrow (1/8) \cdot n^2 + (-3/8) \cdot n + 2$	$(3/4) \cdot n - 7$	$0.562 \cdot n^{-2}$	read A[i3, i1] (i0=0)
n^2	0.125	level	1	$(1/8) \cdot n^2 - 1$	$0.0938/n$	write A[i1, i2] (i0=0); read A[i3, i4] (i0=0, i4=0) (+1)
n^2	0.112	ramp	$55 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n - 2$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i3, i4] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (1/4) \cdot n - 6$	$0.0469/n$	read A[i3, i1] (i0=0, i4=6)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	$0.0469/n$	read A[i3, i1] (i0=0, i4=7)
n^2	0.0884	level	2	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	$0.0469/n$	read A[i1, i4] (i0=0, i3=0, i4=0); read A[i1, i4] (i0=0, i4=0)
n^2	0.0884	level	2	$(1/16) \cdot n^2 + (3/8) \cdot n$	$0.0469/n$	write A[i3, i1] (i0=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2 + (-1/2) \cdot n$	$0.0469/n$	write A[i1, i2] (i0=0); write A[i3, i4] (i0=0, i4=6)
n^2	0.0625	level	1	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	$0.0469/n$	write A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0547	level	1	$(7/128) \cdot n^2 + (5/16) \cdot n$	$0.041/n$	read A[i3, i4] (i0=0, i4=6)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i1=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i1=0, i3=0)
n^2	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0)
n^2	0.0394	ramp	$(1/3932160) \cdot n^{16} + (-7/16384) \cdot n^5 + (7345/24576) \cdot n^4 + (-28525/256) \cdot n^3 + (2802203/120) \cdot n^2 + (-41731977/16) \cdot n + 121303296$ →	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i1=0)
n^2	0.0392	ramp	$(1/8) \cdot n^2 + (-1/2949120) \cdot n^6 + (7/12288) \cdot n^5 + (-7345/18432) \cdot n^4 + (28525/192) \cdot n^3 + (-44834933/1440) \cdot n^2 + (10432996/3) \cdot n - 161737727$ →	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i1=0)
n^2	0.0391	level	$(1/8) \cdot n^2 - n + 9$ 1	$(5/128) \cdot n^2 + (-5/16) \cdot n$	$0.0293/n$	read A[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0373	ramp	$(1/3932160) \cdot n^{168} \cdot n - 3$ $+ (-7/16384) \cdot n^5$ $+$ $(7345/24576) \cdot n^4$ $+ (-28525/256) \cdot n^3$ $+$ $(2802203/120) \cdot n^2$ $+ (-41731975/16) \cdot n$ $+$ 121303295 $\rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n +$ 14	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i4] (i0=0, i1=0)
n^2	0.037	ramp	$(-1/2949120) \cdot n^6$ $+$ $(7/12288) \cdot n^5$ $+ (-7345/18432) \cdot n^4$ $+$ $(28525/192) \cdot n^3$ $+ (-44834933/1440) \cdot n^2$ $+$ $(10432996/3) \cdot n$ $-$ 161737727 $\rightarrow$ $(1/8) \cdot n^2 -$ $n + 9$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^2$	read A[i3, i4] (i0=0, i1=0)
n^2	0.0222	ramp	$54 \rightarrow$ $(1/8) \cdot n^2 +$ $(-3/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i3, i4] (i0=0, i3=0, i4=0)
n^2	0.0222	ramp	$52 \rightarrow$ $(1/8) \cdot n^2 +$ $(-3/4) \cdot n - 2$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i3, i4] (i0=0, i4=0)
n^2	0.0221	ramp	$51 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i3, i4] (i0=0)
n^2	0.0221	ramp	$51 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i1, i2] (i0=0, i2=0)
n^2	0.0221	ramp	$50 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i3, i1] (i0=0, i3=0)
n^2	0.0221	ramp	$50 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^2$	read A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0221	ramp	$18 \rightarrow (1/8) \cdot n^2 - n + 2$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i3, i1] (i0=0)
n^2	0.022	ramp	$99 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i3=0, i4=7)
n^2	0.022	ramp	$98 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i3=0)
n^2	0.0216	ramp	$34 \rightarrow (1/8) \cdot n^2 + (-15/8) \cdot n + 7$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i3, i1] (i0=0)
n^2	0.0215	ramp	$74 \rightarrow (1/8) \cdot n^2 + (-15/8) \cdot n + 6$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i4=7)
n^2	0.0214	ramp	$72 \rightarrow (1/8) \cdot n^2 - 2 \cdot n + 8$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0)
n^2	0.00781	level	1	$(1/128) \cdot n^2 + (-1/16) \cdot n$	$0.00586/n$	read A[i3, i4] (i0=0, i4=6)
$n^{1.5}$	0.251	ramp	$6 \rightarrow (1/4) \cdot n$	$(3/4) \cdot n - 12$	$0.562 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=7); read A[i1, i4] (i0=0, i3=0)
$n^{1.5}$	0.247	ramp	$2 \rightarrow (1/8) \cdot n + 1$	$n - 1$	$0.75 \cdot n^{-2}$	read A[i1, i1] (i0=0, i3=0)
$n^{1.5}$	0.156	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0)
$n^{1.5}$	0.0418	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0)
$n^{1.5}$	0.0418	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0)
$n^{1.5}$	0.0312	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=6)
$n^{1.5}$	0.0312	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	11.3	level	2	$8 \cdot n - 8$	$6 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); read A[i3, i1] (i0=0, i1=0, i3=0) (+4)
n^1	7	level	1	$7 \cdot n + 1$	$5.25 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); write A[i1, i2] (i0=0, i1=0) (+2)
n^1	3.75	level	1	$(15/4) \cdot n - 15$	$2.81 \cdot n^{-2}$	write A[i1, i2] (i0=0)
n^1	2.81	ramp	$(1/8) \cdot n^2 - n + 11 \rightarrow$ $(1/8) \cdot n^2$	8	$6 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i4=7)
n^1	1.62	level	1	$(13/8) \cdot n - 13$	$1.22 \cdot n^{-2}$	write A[i1, i2] (i0=0)
n^1	1.41	level	$(1/8) \cdot n^2$	4	$3 \cdot n^{-3}$	read A[i3, i1] (i0=0, i1=0)
n^1	1.3	level	3	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=6)
n^1	0.875	level	1	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	write A[i1, i2] (i0=0, i1=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.656 \cdot n^{-2}$	write A[i1, i2] (i0=0, i2=0)
n^1	0.75	level	1	$(3/4) \cdot n$	$0.562 \cdot n^{-2}$	write A[i1, i2] (i0=0, i2=1)
n^1	0.707	level	$(1/8) \cdot n^2 - n + 9$	2	$1.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i3=6, i4=7); read A[i3, i4] (i0=0, i1=0, i3=7, i4=7)
n^1	0.707	level	$(1/8) \cdot n^2$	2	$1.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i3=0); read A[i3, i4] (i0=0, i1=0)
n^1	0.703	ramp	$(1/8) \cdot n^2 - n + 15 \rightarrow$ $(1/8) \cdot n^2$	2	$1.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i4=7)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.625	level	1	$(5/8) \cdot n$	$0.469 \cdot n^{-2}$	read A[i3, i4] (i0=0, i3=6)
n^1	0.625	level	1	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	write A[i1, i2] (i0=0)
n^1	0.625	level	1	$(5/8) \cdot n - 5$	$0.469 \cdot n^{-2}$	write A[i1, i2] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i4=7)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 10$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i4=7)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i3=0, i4=7)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 1$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 - 2 \cdot n + 33$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i3=14, i4=7)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 17$	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i1=0, i3=15, i4=7)
n^1	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 2$	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i1=1)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 - n + 9$	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i1=0, i3=6)
n^1	0.25	level	1	$(1/4) \cdot n - 3$	$0.188 \cdot n^{-2}$	write A[i1, i2] (i0=0)
n^1	0.125	level	1	$(1/8) \cdot n$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i3=6, i4=6)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	write A[i1, i2] (i0=0, i2=7)
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	write A[i1, i2] (i0=0, i1=0)
n^0	122	level	6	50	$37.5 \cdot n^{-3}$	read A[i1, i4] (i0=0)
n^0	65.7	level	30	12	$9 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	64.7	ramp	$36 \rightarrow 48$	10	$7.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	58.2	level	28	11	$8.25 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	51	level	26	10	$7.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	44.1	level	24	9	$6.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	37.5	level	22	8	$6 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	36.7	level	6	15	$11.2 \cdot n^{-3}$	read A[i1, i4] (i0=0)
n^0	34.3	level	6	14	$10.5 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=7); read A[i1, i4] (i0=0)
n^0	31.8	level	6	13	$9.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=6)
n^0	26.8	level	20	6	$4.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=0)
n^0	21.2	level	18	5	$3.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	17.1	level	6	7	$5.25 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0)
n^0	14.7	level	6	6	$4.5 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=6)
n^0	14.1	level	2	10	$7.5 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	11.3	level	8	4	$3 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	10.4	level	3	6	$4.5 \cdot n^{-3}$	read A[i1, i1] (i0=0)
n^0	10	level	4	5	$3.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0)
n^0	8.49	level	18	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	7.94	level	7	3	$2.25 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	7	level	49	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0, i4=7)
n^0	6.56	level	43	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	6.48	level	42	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	6.4	level	41	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	5.66	level	32	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	5.66	level	8	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	5.29	level	28	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	5.29	level	7	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	5.1	level	26	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.9	level	6	2	$1.5 \cdot n^{-3}$	read A[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	4.9	level	24	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.9	level	6	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	4.69	level	22	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.47	level	20	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.47	level	20	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0, i4=0)
n^0	4.47	level	5	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	4.47	level	5	2	$1.5 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0); read A[i1, i1] (i0=0, i3=1) (+1)
n^0	4.24	level	18	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=0)
n^0	4.12	level	17	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^0	4	level	16	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=6)
n^0	4	level	4	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	4	level	16	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=0)
n^0	3.87	level	15	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.74	level	14	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.61	level	13	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.46	level	12	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.32	level	11	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	2.83	level	8	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=6)
n^0	2.83	level	2	2	$1.5 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	2.65	level	7	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.45	level	6	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.45	level	6	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0, i4=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=7); read A[i1, i4] (i0=0, i3=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=6)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	1.73	level	3	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	1.73	level	3	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	1.73	level	3	1	$0.75 \cdot n^{-3}$	read A[i1, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=0)
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=0)
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i1=1, i3=0); read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	1	level	1	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^0	1	level	1	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)

Same structure as lu in a different schedule: submatrix re-reads at up to $(1/8)n^2$ lines, $0.0076 \cdot n^4$ leading, headroom +1.0.

lu_decomp — single-shot [exact]

Accesses $A(n) = (4/3) \cdot n^3 + (1/2) \cdot n^2 + (1/6) \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma \text{mass/warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.01 \cdot n^4 + 0.0166 \cdot n^{3.5} + 2.34 \cdot n^3 + 0.631 \cdot n^{2.5} + 22.6 \cdot n^2 + 0.831 \cdot n^{1.5} + 940 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00762	ramp	$108 \rightarrow$ $(1/8) \cdot n^2$	$(1/32) \cdot n^3$ + (- $129/128) \cdot n^2$ + $(37/16) \cdot n$ + 93	0.0234	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00124	ramp	$104 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/192) \cdot n^3 + (-25/128) \cdot n^2 + (71/48) \cdot n + 5$	0.00391	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^4	0.00117	ramp	$135 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 2$	$(1/192) \cdot n^3 + (-35/128) \cdot n^2 + (143/48) \cdot n + 14$	0.00391	read A[i3, i4] (i0=0)
$n^{3.5}$	0.0145	ramp	$10 \rightarrow (1/4) \cdot n + 2$	$(7/192) \cdot n^3 + (-147/128) \cdot n^2 + (77/48) \cdot n + 119$	0.0273	read A[i1, i4] (i0=0)
$n^{3.5}$	0.00204	ramp	$10 \rightarrow (1/4) \cdot n + 2$	$(1/192) \cdot n^3 + (-25/128) \cdot n^2 + (71/48) \cdot n + 5$	0.00391	read A[i1, i4] (i0=0)
n^3	0.505	level	3	$(7/24) \cdot n^3 + (-133/32) \cdot n^2 + (49/12) \cdot n + 84$	0.219	read A[i3, i1] (i0=0)
n^3	0.505	level	3	$(7/24) \cdot n^3 + (-7/2) \cdot n^2 + (119/24) \cdot n + 35$	0.219	write A[i3, i4] (i0=0, i4=6); write A[i3, i4] (i0=0)
n^3	0.442	level	3	$(49/192) \cdot n^3 + (-7/2) \cdot n^2 + (14/3) \cdot n + 56$	0.191	read A[i1, i4] (i0=0)
n^3	0.255	level	1	$(49/192) \cdot n^3 + (-413/128) \cdot n^2 + (245/48) \cdot n + 35$	0.191	read A[i3, i4] (i0=0)
n^3	0.0889	ramp	$57 \rightarrow (1/8) \cdot n^2$	$(3/8) \cdot n^2 + (-9/8) \cdot n - 78$	0.281/n	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^3	0.0883	ramp	$22 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$(3/8) \cdot n^2 + (-15/8) \cdot n - 9$	0.281/n	read A[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0833	level	4	$(1/24) \cdot n^3 + (-27/32) \cdot n^2 + (7/12) \cdot n + 36$	0.0312	read A[i3, i1] (i0=0)
n^3	0.0738	ramp	$57 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n$	$(5/16) \cdot n^2 + (-5/4) \cdot n - 60$	0.234/n	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^3	0.0722	level	3	$(1/24) \cdot n^3 + (-1/16) \cdot n^2 + (-13/6) \cdot n$	0.0312	write A[i3, i4] (i0=0)
n^3	0.0631	level	3	$(7/192) \cdot n^3 + (-7/32) \cdot n^2 + (-7/12) \cdot n$	0.0273	read A[i1, i4] (i0=0)
n^3	0.0365	level	1	$(7/192) \cdot n^3 + (-7/128) \cdot n^2 + (-91/48) \cdot n$	0.0273	read A[i3, i4] (i0=0)
n^3	0.0151	ramp	$72 \rightarrow (1/8) \cdot n^2$	$(1/16) \cdot n^2 + (1/8) \cdot n - 18$	0.0469/n	read A[i3, i4] (i0=0, i3=0, i4=6); read A[i3, i4] (i0=0, i4=6)
n^3	0.0144	ramp	$54 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	0.0469/n	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^3	0.0144	ramp	$54 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n$	$(1/16) \cdot n^2 + (-3/4) \cdot n - 4$	0.0469/n	read A[i3, i4] (i0=0, i4=0)
n^3	0.0143	ramp	$20 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n + 2$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	0.0469/n	read A[i3, i1] (i0=0)
n^3	0.0141	ramp	$37 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 2$	$(1/16) \cdot n^2 + (-7/8) \cdot n - 2$	0.0469/n	read A[i3, i1] (i0=0)
n^3	0.0141	ramp	$78 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 2$	$(1/16) \cdot n^2 + (-7/8) \cdot n - 15$	0.0469/n	read A[i3, i4] (i0=0, i4=7)
n^3	0.0141	ramp	$77 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 3$	$(1/16) \cdot n^2 + (-7/8) \cdot n - 15$	0.0469/n	read A[i3, i4] (i0=0)
n^3	0.0123	ramp	$54 \rightarrow (1/8) \cdot n^2$	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	0.041/n	read A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0101	ramp	$105 \rightarrow (1/8) \cdot n^2 - 3$	$(3/64) \cdot n^2 + (-15/8) \cdot n + 18$	$0.0352/n$	read A[i3, i4] (i0=0)
n^3	0.00167	ramp	$101 \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n - 3$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.00586/n$	read A[i3, i4] (i0=0)
n^3	0.00167	ramp	$100 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.00586/n$	read A[i1, i2] (i0=0)
n^3	0.0016	ramp	$164 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n - 1$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00586/n$	read A[i3, i4] (i0=0, i3=0)
n^3	0.00157	ramp	$130 \rightarrow (1/8) \cdot n^2 + (-15/8) \cdot n + 5$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	$0.00586/n$	read A[i3, i4] (i0=0)
$n^{2.5}$	0.179	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(7/16) \cdot n^2 + (-7/8) \cdot n - 98$	$0.328/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.179	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(7/16) \cdot n^2 + (-7/8) \cdot n - 98$	$0.328/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.178	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(7/16) \cdot n^2 + (-7/8) \cdot n - 21$	$0.328/n$	read A[i1, i1] (i0=0)
$n^{2.5}$	0.0251	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	$0.0469/n$	read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.0251	ramp	$8 \rightarrow (1/4) \cdot n + 2$	$(1/16) \cdot n^2 + (-5/8) \cdot n - 6$	$0.0469/n$	read A[i1, i4] (i0=0)
$n^{2.5}$	0.0248	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	$0.0469/n$	read A[i1, i1] (i0=0)
$n^{2.5}$	0.0173	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(7/128) \cdot n^2 + (-35/16) \cdot n + 21$	$0.041/n$	read A[i1, i4] (i0=0, i3=0)
$n^{2.5}$	0.00247	ramp	$7 \rightarrow (1/4) \cdot n - 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.00586/n$	read A[i1, i4] (i0=0, i3=0)
n^2	3.03	level	3	$(7/4) \cdot n^2 - 7 \cdot n - 56$	$1.31/n$	read A[i1, i4] (i0=0)
n^2	2.27	level	3	$(21/16) \cdot n^2 + (-49/8) \cdot n - 35$	$0.984/n$	read A[i3, i1] (i0=0)
n^2	2.27	level	3	$(21/16) \cdot n^2 + (-49/8) \cdot n - 35$	$0.984/n$	write A[i3, i4] (i0=0)
n^2	1.88	level	1	$(15/8) \cdot n^2 + (-15/4) \cdot n - 20$	$1.41/n$	read A[i3, i4] (i0=0)
n^2	1.86	level	2	$(21/16) \cdot n^2 + (7/2) \cdot n$	$0.984/n$	read A[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.86	level	2	$(21/16) \cdot n^2 + (7/2) \cdot n$	$0.984/n$	read A[i1, i4] (i0=0)
n^2	1.31	level	1	$(21/16) \cdot n^2 + (7/8) \cdot n$	$0.984/n$	write A[i3, i4] (i0=0)
n^2	1.24	level	2	$(7/8) \cdot n^2$	$0.656/n$	read A[i1, i4] (i0=0, i4=0); read A[i3, i1] (i0=0, i4=0)
n^2	0.758	level	3	$(7/16) \cdot n^2 - 28$	$0.328/n$	read A[i3, i1] (i0=0, i4=8)
n^2	0.758	level	3	$(7/16) \cdot n^2 + (-7/8) \cdot n - 21$	$0.328/n$	read A[i3, i1] (i0=0, i4=7)
n^2	0.707	level	2	$(1/2) \cdot n^2 + (-1/2) \cdot n$	$0.375/n$	write A[i3, i1] (i0=0)
n^2	0.65	level	3	$(3/8) \cdot n^2 + (-3/8) \cdot n - 21$	$0.281/n$	read A[i3, i1] (i0=0)
n^2	0.5	level	1	$(1/2) \cdot n^2 + (1/2) \cdot n$	$0.375/n$	write A[i1, i2] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2$	$0.328/n$	read A[i1, i2] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2$	$0.328/n$	read A[i3, i4] (i0=0, i4=0)
n^2	0.375	level	1	$(3/8) \cdot n^2 + (-9/8) \cdot n - 15$	$0.281/n$	read A[i3, i4] (i0=0, i4=6)
n^2	0.313	ramp	$20 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n + 2$	$(7/4) \cdot n - 14$	$1.31 \cdot n^{-2}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^2	0.312	level	1	$(5/16) \cdot n^2 + (15/8) \cdot n$	$0.234/n$	read A[i3, i4] (i0=0)
n^2	0.312	level	1	$(5/16) \cdot n^2 + (15/8) \cdot n$	$0.234/n$	write A[i3, i4] (i0=0)
n^2	0.157	ramp	$20 \rightarrow (1/8) \cdot n^2$	$(7/8) \cdot n - 7$	$0.656 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.134	ramp	$54 \rightarrow (1/8) \cdot n^2 + (-1/8) \cdot n$	$(3/4) \cdot n - 12$	$0.562 \cdot n^{-2}$	read A[i3, i4] (i0=0, i4=7); read A[i3, i4] (i0=0)
n^2	0.134	ramp	$20 \rightarrow (1/8) \cdot n^2 + (-1/4) \cdot n + 2$	$(3/4) \cdot n - 6$	$0.562 \cdot n^{-2}$	read A[i3, i1] (i0=0)
n^2	0.125	level	4	$(1/16) \cdot n^2 + (-1/2) \cdot n - 8$	$0.0469/n$	read A[i3, i1] (i0=0, i4=8)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.112	ramp	$55 \rightarrow$ $(1/8) \cdot n^2 +$ $(-1/8) \cdot n - 2$	$(5/8) \cdot n - 10$	$0.469 \cdot n^{-2}$	read A[i3, i4] (i0=0)
n^2	0.108	level	3	$(1/16) \cdot n^2 +$ $(3/8) \cdot n - 7$	$0.0469/n$	read A[i3, i1] (i0=0, i4=7)
n^2	0.108	level	3	$(1/16) \cdot n^2 +$ $(-1/2) \cdot n$	$0.0469/n$	read A[i3, i1] (i0=0, i4=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2 +$ $(3/8) \cdot n$	$0.0469/n$	read A[i3, i4] (i0=0, i4=6)
n^2	0.0625	level	1	$(1/16) \cdot n^2 +$ $(3/8) \cdot n$	$0.0469/n$	write A[i3, i4] (i0=0, i4=0)
n^2	0.0625	level	1	$(1/16) \cdot n^2 +$ $(3/8) \cdot n$	$0.0469/n$	write A[i3, i4] (i0=0, i4=6)
n^2	0.0226	ramp	$70 \rightarrow$ $(1/8) \cdot n^2 -$ 2	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i4=6)
n^2	0.0222	ramp	$54 \rightarrow$ $(1/8) \cdot n^2 +$ $(-3/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i3=0, i4=0)
n^2	0.0222	ramp	$52 \rightarrow$ $(1/8) \cdot n^2 +$ $(-3/4) \cdot n - 2$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i4=0)
n^2	0.0221	ramp	$51 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0)
n^2	0.0221	ramp	$51 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.0221	ramp	$50 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i3, i1] (i0=0, i3=0)
n^2	0.0221	ramp	$50 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.0221	ramp	$18 \rightarrow$ $(1/8) \cdot n^2 -$ $n + 2$	$(1/8) \cdot n - 1$	$0.0938 \cdot n^{-2}$	read A[i3, i1] (i0=0)
n^2	0.022	ramp	$99 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n - 1$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i3=0, i4=7)
n^2	0.022	ramp	$98 \rightarrow$ $(1/8) \cdot n^2 +$ $(-7/8) \cdot n - 2$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i3=0)
n^2	0.0216	ramp	$34 \rightarrow$ $(1/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 7	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0215	ramp	$74 \rightarrow (1/8) \cdot n^2 + (-15/8) \cdot n + 6$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0, i4=7)
n^2	0.0214	ramp	$72 \rightarrow (1/8) \cdot n^2 - 2 \cdot n + 8$	$(1/8) \cdot n - 3$	$0.0938 \cdot n^{-2}$	read A[i3, i4] (i0=0)
$n^{1.5}$	0.293	ramp	$6 \rightarrow (1/4) \cdot n$	$(7/8) \cdot n - 14$	$0.656 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0)
$n^{1.5}$	0.247	ramp	$2 \rightarrow (1/8) \cdot n + 1$	$n - 1$	$0.75 \cdot n^{-2}$	read A[i1, i1] (i0=0, i3=0)
$n^{1.5}$	0.218	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(7/8) \cdot n - 14$	$0.656 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0)
$n^{1.5}$	0.0418	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0)
$n^{1.5}$	0.0312	ramp	$5 \rightarrow (1/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0938 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	189	level	6	77	$57.8 \cdot n^{-3}$	read A[i1, i4] (i0=0)
n^0	73.5	level	32	13	$9.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0, i4=6); read A[i3, i4] (i0=0, i4=6)
n^0	65.7	level	30	12	$9 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	64.7	ramp	$36 \rightarrow 48$	10	$7.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	58.2	level	28	11	$8.25 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	51.4	level	6	21	$15.8 \cdot n^{-3}$	read A[i1, i4] (i0=0)
n^0	51	level	26	10	$7.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	44.1	level	24	9	$6.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	37.5	level	22	8	$6 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0); read A[i3, i4] (i0=0)
n^0	36.4	level	3	21	$15.8 \cdot n^{-3}$	read A[i1, i1] (i0=0)
n^0	26.8	level	20	6	$4.5 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=0)
n^0	21.2	level	18	5	$3.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	17.1	level	6	7	$5.25 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0)
n^0	14	level	4	7	$5.25 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0)
n^0	11.3	level	8	4	$3 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	8.49	level	18	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	7.94	level	7	3	$2.25 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	7	level	49	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0, i4=7)
n^0	6.56	level	43	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	6.48	level	42	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	6.4	level	41	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7)
n^0	5.66	level	32	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=7); read A[i3, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	5.66	level	8	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	5.48	level	30	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=6)
n^0	5.29	level	28	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	5.29	level	7	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	5.1	level	26	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.9	level	6	2	$1.5 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	4.9	level	24	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.9	level	6	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	4.69	level	22	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.47	level	20	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0)
n^0	4.47	level	20	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0, i4=0)
n^0	4.47	level	5	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	4.24	level	18	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i4=0)
n^0	4.12	level	17	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^0	4	level	16	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=6)
n^0	4	level	4	2	$1.5 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	4	level	16	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=0)
n^0	3.87	level	15	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.74	level	14	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.61	level	13	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.46	level	12	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	3.32	level	11	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.83	level	8	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=6)
n^0	2.65	level	7	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.45	level	6	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.45	level	6	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i4=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2.24	level	5	1	$0.75 \cdot n^{-3}$	read A[i1, i1] (i0=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i3, i4] (i0=0, i3=0, i4=0)
n^0	2	level	4	1	$0.75 \cdot n^{-3}$	read A[i1, i4] (i0=0, i3=0, i4=0)
n^0	1.73	level	3	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0)
n^0	1.73	level	3	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0); read A[i3, i1] (i0=0, i3=0)
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=0)
n^0	1.41	level	2	1	$0.75 \cdot n^{-3}$	read A[i3, i1] (i0=0, i3=0)
n^0	1	level	1	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^0	1	level	1	1	$0.75 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)

Same structure as lu in a different schedule: submatrix re-reads at up to $(1/8)n^2$ lines, $0.0076 \cdot n^4$ leading, headroom +1.0.

mvt — infinite-repeat [exact]

Accesses $A(n) = 8 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0721 \cdot n^3 + 1.12 \cdot n^{2.5} + 11.5 \cdot n^2 + 5.91 \cdot n^{1.5} + 9.4 \cdot n^1 + 2.35 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0316	ramp	$(5/2) \cdot n - 1$ → $(1/8) \cdot n^2 +$ $(1/2) \cdot n - 2$	$(7/64) \cdot n^2$ + $(-15/8) \cdot n$ + 2	0.0137	read B[i4, i3] (i0=0)
n^3	0.0316	ramp	$(5/2) \cdot n - 1$ → $(1/8) \cdot n^2 +$ $(1/2) \cdot n - 2$	$(7/64) \cdot n^2$ + $(-15/8) \cdot n$ + 2	0.0137	read B[i1, i2] (i0=0)
n^3	0.00442	ramp	$(27/8) \cdot n -$ 14 → $(1/8) \cdot n^2 +$ $(1/2) \cdot n - 8$	$(1/64) \cdot n^2$ + $(-3/8) \cdot n +$ 2	0.00195	read B[i1, i2] (i0=0)
n^3	0.00442	ramp	$(27/8) \cdot n -$ 14 → $(1/8) \cdot n^2 +$ $(1/2) \cdot n - 8$	$(1/64) \cdot n^2$ + $(-3/8) \cdot n +$ 2	0.00195	read B[i4, i3] (i0=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2 +$ $(-7/4) \cdot n$	0.109	read B[i4, i3] (i0=0)
$n^{2.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^2$ + $(-7/4) \cdot n$	0.0137	read E[i4] (i0=0)
$n^{2.5}$	0.0547	level	$(1/4) \cdot n + 1$	$(7/64) \cdot n^2$ + $(-7/4) \cdot n$	0.0137	read C[i2] (i0=0)
$n^{2.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^2$ + $(-3/8) \cdot n +$ 2	0.00195	read E[i4] (i0=0)
$n^{2.5}$	0.00781	level	$(1/4) \cdot n + 2$	$(1/64) \cdot n^2$ + $(-3/8) \cdot n +$ 2	0.00195	read C[i2] (i0=0)
n^2	4.55	level	3	$(21/8) \cdot n^2$ + $(-7/4) \cdot n$	0.328	read B[i1, i2] (i0=0); read C[i2] (i0=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	3.03	level	3	$(7/4) \cdot n^2$	0.219	write A[i1] (i0=0); write D[i3] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2$	0.219	read A[i1] (i0=0); read D[i3] (i0=0, i4=0) (+1)
n^2	0.433	level	3	$(1/4) \cdot n^2$	0.0312	write A[i1] (i0=0); write D[i3] (i0=0, i4=0) (+1)
n^2	0.31	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 1$	0.109/n	read B[i4, i3] (i0=0)
n^2	0.31	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 1$	0.109/n	read B[i1, i2] (i0=0, i2=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (-1/4) \cdot n$	0.0312	read A[i1] (i0=0); read D[i3] (i0=0)
n^2	0.212	ramp	$(3/2) \cdot n \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 3$	$(7/8) \cdot n - 1$	0.109/n	read B[i4, i3] (i0=0, i3=0)
n^2	0.183	ramp	$(3/2) \cdot n \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 3$	$(3/4) \cdot n$	0.0938/n	read B[i1, i2] (i0=0)
n^2	0.0884	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/4) \cdot n - 4$	0.0312/n	read A[i1] (i0=0); read D[i3] (i0=0, i4=0)
n^2	0.0884	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$	$(1/4) \cdot n - 4$	0.0312/n	read C[i2] (i0=0); read E[i4] (i0=0, i3=0)
n^2	0.0595	ramp	$(9/4) \cdot n - 6$ → $(1/8) \cdot n^2 + (-5/8) \cdot n + 9$	$(1/4) \cdot n - 2$	0.0312/n	read B[i1, i2] (i0=0)
n^2	0.0433	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 1$	0.0156/n	read B[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0433	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 1$	0.0156/n	read B[i1, i2] (i0=0, i2=0)
n^2	0.0422	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n - 1$	$(1/8) \cdot n - 2$	0.0156/n	read B[i4, i3] (i0=0, i4=0)
n^2	0.0422	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n - 1$	$(1/8) \cdot n - 2$	0.0156/n	read B[i1, i2] (i0=0)
n^2	0.0298	ramp	$(19/8) \cdot n - 7$ → $(1/8) \cdot n^2 + (-5/8) \cdot n + 9$	$(1/8) \cdot n - 1$	0.0156/n	read B[i4, i3] (i0=0, i3=0)
n^2	0.029	ramp	$(19/8) \cdot n \rightarrow (1/8) \cdot n^2 + (-3/4) \cdot n + 3$	$(1/8) \cdot n - 2$	0.0156/n	read B[i4, i3] (i0=0)
n^2	0.029	ramp	$(19/8) \cdot n - 1$ → $(1/8) \cdot n^2 + (-3/4) \cdot n + 2$	$(1/8) \cdot n - 2$	0.0156/n	read B[i1, i2] (i0=0, i1=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + (15/8)$	$(7/8) \cdot n + (-7/8)$	0.109/n	read B[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read B[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read B[i4, i3] (i0=0, i4=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	0.109/n	read E[i4] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	0.109/n	read E[i4] (i0=0, i4=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read C[i2] (i0=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read C[i2] (i0=0, i2=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	0.0156/n	read E[i4] (i0=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	0.0156/n	read E[i4] (i0=0, i4=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	0.0156/n	read C[i2] (i0=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	0.0156/n	read C[i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	3.03	level	3	$(7/4) \cdot n$	$0.219/n$	read C[i2] (i0=0); read E[i4] (i0=0, i3=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (9/8) \cdot n + (15/4)$	2	$0.25 \cdot n^{-2}$	read B[i1, i2] (i0=0); read B[i4, i3] (i0=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	2	$0.25 \cdot n^{-2}$	read B[i1, i2] (i0=0); read B[i4, i3] (i0=0, i3=0, i4=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	2	$0.25 \cdot n^{-2}$	read A[i1] (i0=0); read D[i3] (i0=0, i4=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$	2	$0.25 \cdot n^{-2}$	read C[i2] (i0=0); read E[i4] (i0=0, i3=0)
n^1	0.707	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	2	$0.25 \cdot n^{-2}$	read A[i1] (i0=0); read D[i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	1	$0.125 \cdot n^{-2}$	read B[i1, i2] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (11/8) \cdot n + (7/2)$	1	$0.125 \cdot n^{-2}$	read B[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.125 \cdot n^{-2}$	read B[i1, i2] (i0=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (5/4) \cdot n + (37/8)$	1	$0.125 \cdot n^{-2}$	read E[i4] (i0=0, i3=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$	1	$0.125 \cdot n^{-2}$	read C[i2] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$	1	$0.125 \cdot n^{-2}$	read E[i4] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.125 \cdot n^{-2}$	read B[i1, i2] (i0=0, i1=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n + 1$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)

Unlike atax, mvt multiplies by A and A^T in one pass, so the matrix is re-touched *within* a single invocation: the transposed walk read B[i4, i3] forms ramp families from $(5/2)n$ up to $(1/8)n^2 + (1/2)n$ lines with population $n^2/8 - d = 3.0$, headroom +1.0 already single-shot (the earlier +0.5 reading came from the corrupted iterator terms). The $n^{2.5}$ levels below it are the row/vector reuses at $(9/8)n$ and $(1/4)n$ lines. Repetition changes only constants ($0.036 \rightarrow 0.072$: the second pass doubles the far reuses).

mvt — single-shot [exact]

Accesses $A(n) = 8 \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.036 \cdot n^3 + 1.12 \cdot n^{2.5} + 10.7 \cdot n^2 + 5.91 \cdot n^{1.5} + 3.16 \cdot n^1 + 1.17 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0316	ramp	$(5/2) \cdot n - 1 \rightarrow (1/8) \cdot n^2 + (1/2) \cdot n - 2$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0137	read B[i4, i3] (i0=0)
n^3	0.00442	ramp	$(27/8) \cdot n - 14 \rightarrow (1/8) \cdot n^2 + (1/2) \cdot n - 8$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00195	read B[i4, i3] (i0=0)
$n^{2.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n^2 + (-7/4) \cdot n$	0.109	read B[i4, i3] (i0=0)
$n^{2.5}$	0.116	level	$(9/8) \cdot n - 6$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0137	read E[i4] (i0=0)
$n^{2.5}$	0.0547	level	$(1/4) \cdot n + 1$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	0.0137	read C[i2] (i0=0)
$n^{2.5}$	0.0166	level	$(9/8) \cdot n - 5$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00195	read E[i4] (i0=0)
$n^{2.5}$	0.00781	level	$(1/4) \cdot n + 2$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00195	read C[i2] (i0=0)
n^2	4.55	level	3	$(21/8) \cdot n^2$	0.328	read B[i1, i2] (i0=0); read C[i2] (i0=0) (+1)
n^2	3.46	level	3	$2 \cdot n^2$	0.25	read C[i2] (i0=0); write A[i1] (i0=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	2	level	1	$2 \cdot n^2 - 2 \cdot n$	0.25	read A[i1] (i0=0); read D[i3] (i0=0)
n^2	0.31	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 1$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 1$	0.109/n	read B[i4, i3] (i0=0)
n^2	0.212	ramp	$(3/2) \cdot n$ → $(1/8) \cdot n^2 + (1/8) \cdot n + 3$	$(7/8) \cdot n - 1$	0.109/n	read B[i4, i3] (i0=0, i3=0)
n^2	0.0433	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n$	$(1/8) \cdot n - 1$	0.0156/n	read B[i4, i3] (i0=0)
n^2	0.0422	ramp	$(1/8) \cdot n^2 + (3/8) \cdot n + 2$ → $(1/8) \cdot n^2 + (1/2) \cdot n - 1$	$(1/8) \cdot n - 2$	0.0156/n	read B[i4, i3] (i0=0, i4=0)
n^2	0.0298	ramp	$(19/8) \cdot n - 7$ → $(1/8) \cdot n^2 + (-5/8) \cdot n + 9$	$(1/8) \cdot n - 1$	0.0156/n	read B[i4, i3] (i0=0, i3=0)
n^2	0.029	ramp	$(19/8) \cdot n$ → $(1/8) \cdot n^2 + (-3/4) \cdot n + 3$	$(1/8) \cdot n - 2$	0.0156/n	read B[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + (15/8)$	$(7/8) \cdot n + (-7/8)$	0.109/n	read B[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read B[i4, i3] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read B[i4, i3] (i0=0, i4=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	0.109/n	read E[i4] (i0=0)
$n^{1.5}$	0.928	level	$(9/8) \cdot n - 6$	$(7/8) \cdot n$	0.109/n	read E[i4] (i0=0, i4=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read C[i2] (i0=0)
$n^{1.5}$	0.438	level	$(1/4) \cdot n + 1$	$(7/8) \cdot n$	0.109/n	read C[i2] (i0=0, i2=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	0.0156/n	read E[i4] (i0=0)
$n^{1.5}$	0.133	level	$(9/8) \cdot n - 5$	$(1/8) \cdot n - 1$	0.0156/n	read E[i4] (i0=0, i4=0)
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	0.0156/n	read C[i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.0625	level	$(1/4) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.0156/n$	read C[i2] (i0=0, i2=0)
n^1	1.75	level	1	$(7/4) \cdot n$	$0.219/n$	read A[i1] (i0=0); read D[i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (9/8) \cdot n + (15/4)$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 2$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/4) \cdot n + 1$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0, i4=0)
$n^{0.5}$	1.17	level	$(11/8) \cdot n + 1$	1	$0.125 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)

Unlike atax, mvt multiplies by A and A^T in one pass, so the matrix is re-touched *within* a single invocation: the transposed walk read B[i4, i3] forms ramp families from $(5/2)n$ up to $(1/8)n^2 + (1/2)n$ lines with population $n^{2/8} - d = 3.0$, headroom +1.0 already single-shot (the earlier +0.5 reading came from the corrupted iterator terms). The $n^{2.5}$ levels below it are the row/vector reuses at $(9/8)n$ and $(1/4)n$ lines. Repetition changes only constants ($0.036 \rightarrow 0.072$: the second pass doubles the far reuses).

seidel-2d — infinite-repeat [exact]

Accesses $A(n) = 10 \cdot n^3 - 40 \cdot n^2 + 40 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0884 \cdot n^4 + 0.306 \cdot n^{3.5} + 15.4 \cdot n^3 + 6.43 \cdot n^{2.5} + 25 \cdot n^2 + 4.29 \cdot n^{1.5} + 14.3 \cdot n^1$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0442	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$(1/8) \cdot n^3 + (-23/8) \cdot n^2 + (107/8) \cdot n + (-85/8)$	0.0125	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n^3 + (-11/4) \cdot n^2 + (101/8) \cdot n - 10$	0.0125	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0)
$n^{3.5}$	0.153	level	$(3/8) \cdot n + (13/8)$	$(1/4) \cdot n^3 + (-21/4) \cdot n^2 + (71/4) \cdot n + (-51/4)$	0.025	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)
$n^{3.5}$	0.153	level	$(3/8) \cdot n - 1$	$(1/4) \cdot n^3 - 5 \cdot n^2 + (67/4) \cdot n - 12$	0.025	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)
n^3	2.38	level	3	$(11/8) \cdot n^3 + (-15/4) \cdot n^2 + 2 \cdot n$	0.138	read A[i2 + 1, i3 - 1] (i0=0, i1=0); read A[i2 - 1, i3 - 1] (i0=0) (+1)
n^3	1.75	level	1	$(7/4) \cdot n^3 + (-33/4) \cdot n^2 + (25/2) \cdot n - 6$	0.175	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)
n^3	1.5	level	1	$(3/2) \cdot n^3 + (-17/4) \cdot n^2 + (1/2) \cdot n + 4$	0.15	read A[i2 - 1, i3] (i0=0, i1=0); read A[i2, i3] (i0=0, i1=0) (+3)
n^3	1.22	level	6	$(1/2) \cdot n^3 + (-39/8) \cdot n^2 + (27/4) \cdot n + 2$	0.05	read A[i2 - 1, i3 - 1] (i0=0, i1=0); read A[i2 - 1, i3] (i0=0, i1=0) (+7)
n^3	0.884	level	2	$(5/8) \cdot n^3 + (-5/4) \cdot n^2$	0.0625	read A[i2, i3 - 1] (i0=0, i1=0); read A[i2, i3 - 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.884	level	2	$(5/8) \cdot n^3 + (-5/4) \cdot n^2$	0.0625	write A[i2, i3] (i0=0, i1=0); write A[i2, i3] (i0=0)
n^3	0.875	level	1	$(7/8) \cdot n^3 + (-37/8) \cdot n^2 + (31/4) \cdot n - 4$	0.0875	read A[i2 + 1, i3 + 1] (i0=0, i2=5); read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.875	level	1	$(7/8) \cdot n^3 + (-29/8) \cdot n^2 + (19/4) \cdot n - 2$	0.0875	read A[i2, i3] (i0=0)
n^3	0.559	level	5	$(1/4) \cdot n^3 + (-5/2) \cdot n^2 + 4 \cdot n$	0.025	read A[i2 + 1, i3 - 1] (i0=0, i1=0); read A[i2 - 1, i3 - 1] (i0=0) (+1)
n^3	0.433	level	3	$(1/4) \cdot n^3 + (-5/2) \cdot n^2 + 4 \cdot n$	0.025	read A[i2, i3 - 1] (i0=0, i1=0); read A[i2, i3 - 1] (i0=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n^2 - 2 \cdot n$	0.1/n	read A[i2 - 1, i3 - 1] (i0=0, i1=0, i2=0, i3=0); read A[i2 + 1, i3 - 1] (i0=0, i1=0, i2=1, i3=0) (+5)
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 2 \cdot n$	0.1/n	read A[i2 - 1, i3 - 1] (i0=0, i1=0, i2=0, i3=0); read A[i2 + 1, i3 - 1] (i0=0, i1=0, i3=0) (+3)
n^3	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n^2 - 5 \cdot n + 4$	0.1/n	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0, i2=0) (+2)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 5 \cdot n + 4$	$0.1/n$	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0, i2=0) (+1)
n^3	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n^2 - 6 \cdot n + 5$	$0.1/n$	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0, i2=1) (+1)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	$n^2 - 6 \cdot n + 5$	$0.1/n$	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.306	level	6	$(1/8) \cdot n^3 + (-11/8) \cdot n^2 + (13/4) \cdot n - 2$	0.0125	read A[i2, i3] (i0=0)
n^3	0.25	level	1	$(1/4) \cdot n^3 + (-19/8) \cdot n^2 + (11/4) \cdot n + 2$	0.025	read A[i2 - 1, i3] (i0=0, i1=0); read A[i2, i3] (i0=0, i1=0) (+3)
n^3	0.25	level	4	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	write A[i2, i3] (i0=0, i1=0); write A[i2, i3] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	read A[i2 + 1, i3 - 1] (i0=0, i1=0); read A[i2 + 1, i3 - 1] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	write A[i2, i3] (i0=0, i1=0); write A[i2, i3] (i0=0)
n^3	0.177	level	2	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	read A[i2, i3 - 1] (i0=0, i1=0); read A[i2, i3 - 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.177	level	2	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	write A[i2, i3] (i0=0, i1=0); write A[i2, i3] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$(1/8) \cdot n^2 + (-9/4) \cdot n + (17/8)$	0.0125/n	read A[i2 - 1, i3 + 1] (i0=0, i2=0)
n^3	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 - 1, i3 + 1] (i0=0, i2=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$(1/8) \cdot n^2 + (-21/8) \cdot n + (17/2)$	0.0125/n	read A[i2 - 1, i3 + 1] (i0=0, i1=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^3	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n^2 + (-5/2) \cdot n + 8$	0.0125/n	read A[i2 - 1, i3 + 1] (i0=0, i1=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^3	0.0418	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.0418	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2, i3 + 1] (i0=0, i2=0)
n^3	0.0417	ramp	$(1/8) \cdot n^2 + (-1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 - 2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 + 1, i3 + 1] (i0=0, i2=0)
n^3	0.0417	ramp	$(1/8) \cdot n^2 + (-1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 - 2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 + 1, i3 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.22	level	$(3/8) \cdot n + (13/8)$	$2 \cdot n^2 - 6 \cdot n$	$0.2/n$	read A[i2 - 1, i3 - 1] (i0=0, i1=0, i3=0); read A[i2, i3 - 1] (i0=0, i1=0, i3=0) (+2)
$n^{2.5}$	1.22	level	$(3/8) \cdot n - 1$	$2 \cdot n^2 - 6 \cdot n$	$0.2/n$	read A[i2 - 1, i3 - 1] (i0=0, i1=0, i3=0); read A[i2, i3 - 1] (i0=0, i1=0, i3=0) (+2)
$n^{2.5}$	1.22	level	$(3/8) \cdot n + (1/4)$	$2 \cdot n^2 - 8 \cdot n + 6$	$0.2/n$	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n - 1$	$n^2 - 4 \cdot n + 3$	$0.1/n$	read A[i2, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n - 2$	$n^2 - 4 \cdot n + 3$	$0.1/n$	read A[i2 - 1, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n + (-3/8)$	$n^2 - 4 \cdot n + 3$	$0.1/n$	read A[i2, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n + (13/8)$	$n^2 - 4 \cdot n + 3$	$0.1/n$	read A[i2 - 1, i3 + 1] (i0=0)
$n^{2.5}$	0.0765	level	$(3/8) \cdot n + (13/8)$	$(1/8) \cdot n^2 + (-5/2) \cdot n + (51/8)$	$0.0125/n$	read A[i2, i3 + 1] (i0=0, i1=0)
$n^{2.5}$	0.0765	level	$(3/8) \cdot n - 1$	$(1/8) \cdot n^2 + (-19/8) \cdot n + 6$	$0.0125/n$	read A[i2, i3 + 1] (i0=0, i1=0)
$n^{2.5}$	0.0765	level	$(3/8) \cdot n + (13/8)$	$(1/8) \cdot n^2 + (-5/2) \cdot n + (51/8)$	$0.0125/n$	read A[i2 - 1, i3 + 1] (i0=0, i1=0)
$n^{2.5}$	0.0765	level	$(3/8) \cdot n - 1$	$(1/8) \cdot n^2 + (-19/8) \cdot n + 6$	$0.0125/n$	read A[i2 - 1, i3 + 1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	7	level	1	$7 \cdot n^2 - 21 \cdot n + 14$	$0.7/n$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	4	level	1	$4 \cdot n^2 - 5 \cdot n - 6$	$0.4/n$	read A[i2 - 1, i3 + 1] (i0=0, i1=0, i2=0); read A[i2 - 1, i3 - 1] (i0=0, i1=0, i3=0) (+14)
n^2	1.41	level	2	$n^2 - 2 \cdot n$	$0.1/n$	read A[i2, i3 - 1] (i0=0, i1=0, i2=0, i3=0); read A[i2, i3 - 1] (i0=0, i1=0, i3=0) (+5)
n^2	1.25	level	1	$(5/4) \cdot n^2 + (-5/2) \cdot n$	$0.125/n$	read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=5); read A[i2 - 1, i3 + 1] (i0=0, i1=0) (+1)
n^2	0.707	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$2 \cdot n - 2$	$0.2 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0); read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.707	level	$(1/8) \cdot n^2$	$2 \cdot n - 2$	$0.2 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0); read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.625	level	1	$(5/8) \cdot n^2 + (-5/4) \cdot n$	$0.0625/n$	read A[i2, i3 + 1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (1/4)$	n	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 - 1] (i0=0, i1=0, i2=0, i3=0); read A[i2 + 1, i3 - 1] (i0=0, i2=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 2$	n	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 - 1] (i0=0, i1=0, i2=0, i3=0); read A[i2 + 1, i3 - 1] (i0=0, i2=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (9/4)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + (5/2)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 2$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (1/8)$	n	$0.1 \cdot n^{-2}$	read A[i2, i3 - 1] (i0=0, i1=0, i2=0, i3=0); read A[i2, i3 - 1] (i0=0, i2=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 1$	n	$0.1 \cdot n^{-2}$	read A[i2, i3 - 1] (i0=0, i1=0, i2=0, i3=0); read A[i2, i3 - 1] (i0=0, i2=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (5/4)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (-15/8)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=0) (+2)
n^2	0.354	level	$(1/8) \cdot n^2$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=0) (+1)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (9/4)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n - 2$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + (5/2)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n - 2$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=1); read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n - 1$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n - 1$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n - 4$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=1) (+1)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	$n - 4$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (-5/2) \cdot n + 4$	$0.025/n$	read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=5); read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (-19/4) \cdot n + (17/2)$	$0.025/n$	read A[i2, i3 + 1] (i0=0, i1=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (-5/2) \cdot n + 4$	$0.025/n$	read A[i2, i3 + 1] (i0=0, i1=0)
n^2	0.25	level	1	$(1/4) \cdot n^2 + (-5/2) \cdot n + 4$	$0.025/n$	read A[i2 - 1, i3 + 1] (i0=0, i1=0)
n^2	0.0419	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$ → $(1/8) \cdot n^2 - 1$	$(1/8) \cdot n - 2$	$0.0125 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^2	0.0419	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2$ → $(1/8) \cdot n^2 - 1$	$(1/8) \cdot n - 2$	$0.0125 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0419	ramp	$(1/8) \cdot n^2 + (-1/4) \cdot n + 4$ → $(1/8) \cdot n^2 - 2$	$(1/8) \cdot n - 2$	$0.0125 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^2	0.0419	ramp	$(1/8) \cdot n^2 + (-1/4) \cdot n + 4$ → $(1/8) \cdot n^2 - 2$	$(1/8) \cdot n - 2$	$0.0125 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + (1/4)$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + (1/4)$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n - 1$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + (-19/8)$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n - 2$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + (13/8)$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i1=0)
$n^{1.5}$	0.612	level	$(3/8) \cdot n + (-3/8)$	$n - 3$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0)
n^1	5	level	1	$5 \cdot n - 10$	$0.5 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	1	level	1	$n - 2$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	1	level	1	$n - 2$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1	level	1	$n - 2$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0)
n^1	1	level	1	$n - 2$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(5/8) \cdot n +$ $(9/4)$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(1/2) \cdot n +$ $(5/2)$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(-1/4) \cdot n + 2$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(5/8) \cdot n +$ $(5/4)$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(3/4) \cdot n +$ $(-15/8)$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(-1/8) \cdot n$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(7/8) \cdot n$		$0.1 \cdot n^{-3}$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2$	1	$0.1 \cdot n^{-3}$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(5/8) \cdot n +$ $(9/4)$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(7/8) \cdot n - 2$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + 1$ $(1/2) \cdot n +$ $(5/2)$		$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n - 2$	1	$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	1	$0.1 \cdot n^{-3}$	read A[i2 + 1, i3 + 1] (i0=0, i1=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n - 1$	1	$0.1 \cdot n^{-3}$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n - 1$	1	$0.1 \cdot n^{-3}$	read A[i2, i3 + 1] (i0=0, i1=0, i2=0)

In-place 2-D sweep: the south-east neighbor read A[i2+1, i3+1] is the value written a full sweep earlier — distance $(1/8)n^2 + (7/8)n$ at the row seam), one array's footprint, $0.0884 \cdot n^4$ total. The three-row working window shows up as $n^{3.5}$ terms at $(3/8)n$ lines. Headroom +1.0 with a single-array boundary, half of jacobi-2d's.

seidel-2d — single-shot [exact]

Accesses $A(n) = 10 \cdot n^3 - 40 \cdot n^2 + 40 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass}}/\text{warm} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0884 \cdot n^4 + 0.306 \cdot n^{3.5} + 15.3 \cdot n^3 + 6.12 \cdot n^{2.5} + 17.1 \cdot n^2$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0442	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$(1/8) \cdot n^3 + (-23/8) \cdot n^2 + (107/8) \cdot n + (-85/8)$	0.0125	read A[i2 + 1, i3 + 1] (i0=0)
n^4	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n^3 + (-11/4) \cdot n^2 + (101/8) \cdot n - 10$	0.0125	read A[i2 + 1, i3 + 1] (i0=0)
$n^{3.5}$	0.153	level	$(3/8) \cdot n + (13/8)$	$(1/4) \cdot n^3 - 5 \cdot n^2 + (51/4) \cdot n$	0.025	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)
$n^{3.5}$	0.153	level	$(3/8) \cdot n - 1$	$(1/4) \cdot n^3 + (-19/4) \cdot n^2 + 12 \cdot n$	0.025	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	2.38	level	3	$(11/8) \cdot n^3 + (-15/4) \cdot n^2 + 2 \cdot n$	0.138	read A[i2 - 1, i3 - 1] (i0=0); read A[i2 + 1, i3 - 1] (i0=0)
n^3	1.75	level	1	$(7/4) \cdot n^3 + (-11/2) \cdot n^2 + 4 \cdot n$	0.175	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)
n^3	1.5	level	1	$(3/2) \cdot n^3 - 5 \cdot n^2 + 4 \cdot n$	0.15	read A[i2 - 1, i3] (i0=0); read A[i2 + 1, i3] (i0=0)
n^3	1.22	level	6	$(1/2) \cdot n^3 - 5 \cdot n^2 + 8 \cdot n$	0.05	read A[i2 + 1, i3 - 1] (i0=0, i2=0); read A[i2 - 1, i3 - 1] (i0=0) (+3)
n^3	1.06	level	2	$(3/4) \cdot n^3 + (-5/2) \cdot n^2 + 2 \cdot n$	0.075	read A[i2, i3 - 1] (i0=0, i2=0, i3=0); read A[i2, i3 - 1] (i0=0, i3=0) (+1)
n^3	0.884	level	2	$(5/8) \cdot n^3 + (-5/4) \cdot n^2$	0.0625	write A[i2, i3] (i0=0)
n^3	0.875	level	1	$(7/8) \cdot n^3 + (-11/4) \cdot n^2 + 2 \cdot n$	0.0875	read A[i2 + 1, i3 + 1] (i0=0, i2=5); read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.875	level	1	$(7/8) \cdot n^3 + (-11/4) \cdot n^2 + 2 \cdot n$	0.0875	read A[i2, i3] (i0=0)
n^3	0.559	level	5	$(1/4) \cdot n^3 + (-5/2) \cdot n^2 + 4 \cdot n$	0.025	read A[i2 - 1, i3 - 1] (i0=0); read A[i2 + 1, i3 - 1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.433	level	3	$(1/4) \cdot n^3 + (-5/2) \cdot n^2 + 4 \cdot n$	0.025	read A[i2 - 1, i3 - 1] (i0=0, i2=0, i3=0); read A[i2 + 1, i3 - 1] (i0=0, i2=0, i3=0) (+2)
n^3	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n^2 - 3 \cdot n + 2$	0.1/n	read A[i2 - 1, i3 - 1] (i0=0, i2=0, i3=0); read A[i2 + 1, i3 - 1] (i0=0, i2=1, i3=0) (+1)
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 3 \cdot n + 2$	0.1/n	read A[i2 - 1, i3 - 1] (i0=0, i2=0, i3=0); read A[i2 + 1, i3 - 1] (i0=0, i3=0) (+1)
n^3	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n^2 - 5 \cdot n + 4$	0.1/n	read A[i2 + 1, i3 + 1] (i0=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0, i2=1) (+1)
n^3	0.354	level	$(1/8) \cdot n^2$	$n^2 - 5 \cdot n + 4$	0.1/n	read A[i2 + 1, i3 + 1] (i0=0, i2=0); read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n^2 - 6 \cdot n + 5$	0.1/n	read A[i2 + 1, i3 + 1] (i0=0, i2=1); read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	$n^2 - 6 \cdot n + 5$	0.1/n	read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.306	level	6	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	read A[i2, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.25	level	4	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	write A[i2, i3] (i0=0)
n^3	0.25	level	1	$(1/4) \cdot n^3 + (-5/2) \cdot n^2 + 4 \cdot n$	0.025	read A[i2 - 1, i3] (i0=0); read A[i2 + 1, i3] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	read A[i2 + 1, i3 - 1] (i0=0)
n^3	0.217	level	3	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0125	write A[i2, i3] (i0=0)
n^3	0.177	level	2	$(1/8) \cdot n^3 + (-1/4) \cdot n^2$	0.0125	write A[i2, i3] (i0=0)
n^3	0.0442	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$(1/8) \cdot n^2 + (-9/4) \cdot n + (17/8)$	0.0125/n	read A[i2 - 1, i3 + 1] (i0=0, i2=0)
n^3	0.0442	level	$(1/8) \cdot n^2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 - 1, i3 + 1] (i0=0, i2=0)
n^3	0.0418	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 + 1, i3 + 1] (i0=0)
n^3	0.0418	ramp	$(1/8) \cdot n^2 + (-1/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 - 1$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2, i3 + 1] (i0=0, i2=0)
n^3	0.0417	ramp	$(1/8) \cdot n^2 + (-1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 - 2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 + 1, i3 + 1] (i0=0, i2=0)
n^3	0.0417	ramp	$(1/8) \cdot n^2 + (-1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 - 2$	$(1/8) \cdot n^2 + (-17/8) \cdot n + 2$	0.0125/n	read A[i2 + 1, i3 + 1] (i0=0)
$n^{2.5}$	1.22	level	$(3/8) \cdot n + (13/8)$	$2 \cdot n^2 - 6 \cdot n$	0.2/n	read A[i2 - 1, i3 - 1] (i0=0, i3=0); read A[i2, i3 - 1] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.22	level	$(3/8) \cdot n - 1$	$2 \cdot n^2 - 6 \cdot n$	$0.2/n$	read A[i2 - 1, i3 - 1] (i0=0, i3=0); read A[i2, i3 - 1] (i0=0, i3=0)
$n^{2.5}$	1.22	level	$(3/8) \cdot n + (1/4)$	$2 \cdot n^2 - 6 \cdot n$	$0.2/n$	read A[i2 - 1, i3 + 1] (i0=0); read A[i2, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n - 1$	$n^2 - 3 \cdot n$	$0.1/n$	read A[i2, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n - 2$	$n^2 - 3 \cdot n$	$0.1/n$	read A[i2 - 1, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n + (-3/8)$	$n^2 - 3 \cdot n$	$0.1/n$	read A[i2, i3 + 1] (i0=0)
$n^{2.5}$	0.612	level	$(3/8) \cdot n + (13/8)$	$n^2 - 3 \cdot n$	$0.1/n$	read A[i2 - 1, i3 + 1] (i0=0)
n^2	7	level	1	$7 \cdot n^2 - 14 \cdot n$	$0.7/n$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	2	level	1	$2 \cdot n^2 - 4 \cdot n$	$0.2/n$	read A[i2 - 1, i3 - 1] (i0=0, i3=0); read A[i2 - 1, i3] (i0=0, i3=0) (+1)
n^2	0.707	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$2 \cdot n - 2$	$0.2 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0); read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.707	level	$(1/8) \cdot n^2$	$2 \cdot n - 2$	$0.2 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0); read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (1/4)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 - 1] (i0=0, i2=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 2$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 - 1] (i0=0, i2=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (9/4)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + (5/2)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/4) \cdot n + 2$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (1/8)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2, i3 - 1] (i0=0, i2=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n + 1$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2, i3 - 1] (i0=0, i2=0, i3=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (5/4)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (-15/8)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (-1/8) \cdot n$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (5/8) \cdot n + (9/4)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n - 2$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (1/2) \cdot n + (5/2)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n - 2$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n + (9/8)$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 + 1, i3 + 1] (i0=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n - 1$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n - 1$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (7/8) \cdot n$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0)
n^2	0.354	level	$(1/8) \cdot n^2 + (3/4) \cdot n$	$n - 1$	$0.1 \cdot n^{-2}$	read A[i2 - 1, i3 + 1] (i0=0, i2=0)

In-place 2-D sweep: the south-east neighbor read A[i2+1, i3+1] is the value written a full sweep earlier — distance $(1/8)n^2 + (7/8)n$ at the row seam), one array's footprint, $0.0884 \cdot n^4$ total. The three-row working window shows up as $n^{3.5}$ terms at $(3/8)n$ lines. Headroom +1.0 with a single-array boundary, half of jacobi-2d's.

symm — infinite-repeat [exact]

Accesses $A(n) = (5/2) \cdot n^3 + (3/2) \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0466 \cdot n^4 + 1.08 \cdot n^{3.5} + 4.13 \cdot n^3 + 6.28 \cdot n^{2.5} + 15.8 \cdot n^2 + 0.333 \cdot n^{1.5} + 43.6 \cdot n^1 + 13.6 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0176	ramp	$n + 5 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(3/64) \cdot n^3 + (-63/64) \cdot n^2 + (31/8) \cdot n - 2$	0.0187	read A[i3, i2] (i0=0)
n^4	0.0155	ramp	$n + 5 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(21/512) \cdot n^3 + (-53/64) \cdot n^2 + (11/4) \cdot n$	0.0164	read B[i3, i2] (i0=0)
n^4	0.00286	ramp	$(9/4) \cdot n + 7 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 3$	$(1/128) \cdot n^3 + (-7/32) \cdot n^2 + (7/4) \cdot n - 4$	0.00313	read A[i3, i2] (i0=0)
n^4	0.00283	ramp	$2 \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 4$	$(1/128) \cdot n^3 + (-15/64) \cdot n^2 + (17/8) \cdot n - 6$	0.00313	read B[i3, i2] (i0=0)
n^4	0.00283	ramp	$2 \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 4$	$(1/128) \cdot n^3 + (-15/64) \cdot n^2 + (17/8) \cdot n - 6$	0.00313	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00251	ramp	$(9/4) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(7/1024) \cdot n^3$ $+ (-$ $23/128) \cdot n^2$ $+ (5/4) \cdot n - 2$	0.00273	read B[i3, i2] (i0=0)
n^4	0.00217	ramp	$(11/4) \cdot n +$ $7 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(3/512) \cdot n^3$ $+ (-$ $5/32) \cdot n^2$ $+ (9/8) \cdot n - 2$	0.00234	read B[i3, i2] (i0=0)
n^4	0.000345	ramp	$(17/4) \cdot n +$ $9 \rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(1/1024) \cdot n^3$ $+ (-$ $5/128) \cdot n^2$ $+ (1/2) \cdot n - 2$	0.000391	read B[i3, i2] (i0=0)
$n^{3.5}$	0.443	ramp	$9 \rightarrow$ $(17/8) \cdot n$	$(49/128) \cdot n^3$ $+ (-$ $49/32) \cdot n^2$ $+ (7/8) \cdot n$	0.153	read A[i3, i2] (i0=0)
$n^{3.5}$	0.389	ramp	$9 \rightarrow$ $(17/8) \cdot n$	$(343/1024) \cdot n^3$ $+ (-$ $133/128) \cdot n^2$	0.134	read B[i3, i2] (i0=0)
$n^{3.5}$	0.0613	ramp	$20 \rightarrow$ $(17/8) \cdot n -$ 14	$(7/128) \cdot n^3$ $+ (-$ $21/32) \cdot n^2$ $+ (7/4) \cdot n$	0.0219	read B[i3, i2] (i0=0)
$n^{3.5}$	0.0613	ramp	$20 \rightarrow$ $(17/8) \cdot n -$ 14	$(7/128) \cdot n^3$ $+ (-$ $21/32) \cdot n^2$ $+ (7/4) \cdot n$	0.0219	read A[i3, i2] (i0=0)
$n^{3.5}$	0.0544	ramp	$24 \rightarrow$ $(17/8) \cdot n$	$(49/1024) \cdot n^3$ $+ (-$ $63/128) \cdot n^2$ $+ (7/8) \cdot n$	0.0191	read B[i3, i2] (i0=0)
$n^{3.5}$	0.0516	ramp	$26 \rightarrow$ $(17/8) \cdot n -$ 13	$(49/1024) \cdot n^3$ $+ (-$ $147/128) \cdot n^2$ $+ (49/8) \cdot n$	0.0191	read C[i1, i3] (i0=0)
$n^{3.5}$	0.00719	ramp	$27 \rightarrow$ $(17/8) \cdot n -$ 12	$(7/1024) \cdot n^3$ $+ (-$ $7/32) \cdot n^2$ $+ (35/16) \cdot n$ $- 7$	0.00273	read C[i1, i3] (i0=0)
$n^{3.5}$	0.00697	ramp	$41 \rightarrow$ $(17/8) \cdot n -$ 27	$(7/1024) \cdot n^3$ $+ (-$ $35/128) \cdot n^2$ $+ (21/8) \cdot n$	0.00273	read C[i1, i3] (i0=0)
$n^{3.5}$	0.000971	ramp	$42 \rightarrow$ $(17/8) \cdot n -$ 26	$(1/1024) \cdot n^3$ $+ (-$ $3/64) \cdot n^2$ $+ (11/16) \cdot n$ $- 3$	0.000391	read C[i1, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	1.12	level	5	$(1/2) \cdot n^3 + (-3/2) \cdot n^2 + (15/8) \cdot n$	0.2	read A[i3, i2] (i0=0, i1=1, i3=0); read B[i1, i2] (i0=0)
n^3	0.978	level	5	$(7/16) \cdot n^3 + (-21/16) \cdot n^2 + (7/8) \cdot n$	0.175	read C[i1, i3] (i0=0)
n^3	0.758	level	3	$(7/16) \cdot n^3 + (7/16) \cdot n^2 + (7/8) \cdot n$	0.175	read B[i1, i2] (i0=0, i1=0); write A[i3, i2] (i0=0) (+1)
n^3	0.323	ramp	$n + 3 \rightarrow (1/4) \cdot n^2 + 1$	$(105/128) \cdot n^3 + (-77/16) \cdot n + 6$	0.328/n	read A[i3, i2] (i0=0, i2=0); read B[i3, i2] (i0=0, i2=0)
n^3	0.149	ramp	$(5/4) \cdot n + 1 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 1$	$(3/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.15/n	read A[i3, i2] (i0=0)
n^3	0.13	ramp	$(5/4) \cdot n + 1 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 1$	$(21/64) \cdot n^2 + (-11/8) \cdot n$	0.131/n	read B[i3, i2] (i0=0)
n^3	0.108	level	3	$(1/16) \cdot n^3 + (1/16) \cdot n^2$	0.025	write A[i3, i2] (i0=0); write A[i1, i2] (i0=0)
n^3	0.0465	ramp	$(39/8) \cdot n - 9 \rightarrow (5/16) \cdot n^2 + (3/8) \cdot n - 4$	$(7/64) \cdot n^2 + (-17/8) \cdot n + 6$	0.0437/n	read A[i1, i2] (i0=0)
n^3	0.0463	ramp	$(41/8) \cdot n - 13 \rightarrow (5/16) \cdot n^2 + (3/8) \cdot n - 5$	$(7/64) \cdot n^2 + (-9/4) \cdot n + 8$	0.0437/n	read B[i1, i2] (i0=0, i3=0)
n^3	0.0452	ramp	$(9/4) \cdot n + 4 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n$	$(15/128) \cdot n^2 + (-21/16) \cdot n + 3$	0.0469/n	read A[i3, i2] (i0=0, i2=0); read B[i3, i2] (i0=0, i2=0)
n^3	0.0306	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	0.0219/n	read C[i1, i3] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0298	ramp	$(1/2) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(3/32) \cdot n^2$ $+ (-3/2) \cdot n$	0.0375/n	read B[i3, i2] (i0=0)
n^3	0.0298	ramp	$(1/2) \cdot n + 4$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 4$	$(3/32) \cdot n^2$ $+ (-3/2) \cdot n$	0.0375/n	read A[i3, i2] (i0=0)
n^3	0.0298	ramp	$(3/4) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(3/32) \cdot n^2$ $+ (-13/8) \cdot n$ $+ 2$	0.0375/n	read B[i3, i2] (i0=0, i3=0)
n^3	0.0298	ramp	$(3/4) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(3/32) \cdot n^2$ $+ (-13/8) \cdot n$ $+ 2$	0.0375/n	read A[i3, i2] (i0=0)
n^3	0.0298	ramp	$(3/4) \cdot n + 4$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 4$	$(3/32) \cdot n^2$ $+ (-13/8) \cdot n$ $+ 2$	0.0375/n	read A[i3, i2] (i0=0, i3=0)
n^3	0.0249	ramp	$(3/4) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(5/64) \cdot n^2$ $+ (-5/4) \cdot n$	0.0312/n	read B[i3, i2] (i0=0)
n^3	0.0241	ramp	$(5/2) \cdot n + 3$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 1$	$(1/16) \cdot n^2$ $+ (-3/4) \cdot n$ $+ 2$	0.025/n	read A[i3, i2] (i0=0)
n^3	0.0239	ramp	$(9/4) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 2$	$(1/16) \cdot n^2$ $+ (-7/8) \cdot n$ $+ 3$	0.025/n	read A[i3, i2] (i0=0)
n^3	0.0212	ramp	$(5/2) \cdot n + 3$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 1$	$(7/128) \cdot n^2$ $+ (-9/16) \cdot n$ $+ 1$	0.0219/n	read B[i3, i2] (i0=0)
n^3	0.0212	ramp	$(11/4) \cdot n +$ $5 \rightarrow$ $(1/4) \cdot n^2 +$ 1	$(7/128) \cdot n^2$ $+ (-11/16) \cdot n$ $+ 2$	0.0219/n	read B[i3, i2] (i0=0, i2=0)
n^3	0.021	ramp	$(9/4) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 2$	$(7/128) \cdot n^2$ $+ (-11/16) \cdot n$ $+ 2$	0.0219/n	read B[i3, i2] (i0=0)
n^3	0.0183	ramp	$3 \cdot n + 2 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 2$	$(3/64) \cdot n^2$ $+ (-1/2) \cdot n$ $+ 1$	0.0187/n	read B[i3, i2] (i0=0)
n^3	0.00654	ramp	$(51/8) \cdot n -$ $26 \rightarrow$ $(5/16) \cdot n^2$ $+ (-1/4) \cdot n -$ 14	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n$ $+ 2$	0.00625/n	read A[i1, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00654	ramp	$(51/8) \cdot n - 27 \rightarrow (5/16) \cdot n^2 + (-1/4) \cdot n - 15$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i1, i2] (i0=0, i3=0)
n^3	0.0049	ramp	$(5/2) \cdot n + 7 \rightarrow (1/4) \cdot n^2 - n - 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i3, i2] (i0=0, i3=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read A[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 4$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read A[i3, i2] (i0=0, i3=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 4$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read A[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 4$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 4$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i3, i2] (i0=0, i3=0)
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 4$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read A[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 4$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 5 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 5$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read A[i3, i2] (i0=0, i3=0)
n^3	0.00486	ramp	$2 \cdot n + 5 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 5$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00625/n	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00437	level	$(5/16) \cdot n^2 + 2 \cdot n + (3/4)$	$(1/128) \cdot n^2 + (-7/32) \cdot n + (45/32)$	$0.00313/n$	read C[i1, i3] (i0=0, i2=0)
n^3	0.00437	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	$0.00313/n$	read C[i1, i3] (i0=0, i2=0)
n^3	0.00291	ramp	$(9/2) \cdot n + 4 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 2$	$(1/128) \cdot n^2 + (-3/16) \cdot n + 1$	$0.00313/n$	read B[i3, i2] (i0=0)
n^3	0.00291	ramp	$(17/4) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n$	$(1/128) \cdot n^2 + (-3/16) \cdot n + 1$	$0.00313/n$	read B[i3, i2] (i0=0, i2=0)
n^3	0.00291	ramp	$(17/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n - 3$	$(1/128) \cdot n^2 + (-3/16) \cdot n + 1$	$0.00313/n$	read B[i3, i2] (i0=0)
$n^{2.5}$	1.6	ramp	$5 \rightarrow (17/8) \cdot n$	$(105/64) \cdot n^2 + (-7/4) \cdot n$	$0.656/n$	read A[i3, i2] (i0=0, i3=0); read A[i1, i2] (i0=0)
$n^{2.5}$	0.853	ramp	$7 \rightarrow (17/8) \cdot n$	$(7/8) \cdot n^2 + (-7/4) \cdot n$	$0.35/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.837	ramp	$21 \rightarrow (17/8) \cdot n - 13$	$(7/8) \cdot n^2 + (-63/8) \cdot n + 7$	$0.35/n$	read C[i1, i3] (i0=0)
$n^{2.5}$	0.748	ramp	$7 \rightarrow (17/8) \cdot n$	$(49/64) \cdot n^2 + (-7/8) \cdot n$	$0.306/n$	read B[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.726	ramp	$9 \rightarrow (17/8) \cdot n - 13$	$(49/64) \cdot n^2 + (-49/8) \cdot n$	$0.306/n$	read C[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.643	ramp	$7 \rightarrow (17/8) \cdot n$	$(21/32) \cdot n^2$	$0.263/n$	read B[i3, i2] (i0=0)
$n^{2.5}$	0.209	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$(7/32) \cdot n^2 + (-7/4) \cdot n$	$0.0875/n$	read A[i3, i2] (i0=0, i3=0); read B[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.105	ramp	$22 \rightarrow (17/8) \cdot n - 12$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0437/n$	read B[i3, i2] (i0=0)
$n^{2.5}$	0.105	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0437/n$	read B[i3, i2] (i0=0)
$n^{2.5}$	0.105	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0437/n$	read C[i1, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.102	ramp	$24 \rightarrow (17/8) \cdot n - 27$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	$0.0437/n$	read C[i1, i3] (i0=0)
$n^{2.5}$	0.102	ramp	$24 \rightarrow (17/8) \cdot n - 27$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	$0.0437/n$	read C[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.101	ramp	$10 \rightarrow (17/8) \cdot n - 12$	$(7/64) \cdot n^2 + (-7/4) \cdot n + 7$	$0.0437/n$	read C[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.0146	ramp	$21 \rightarrow (17/8) \cdot n - 13$	$(1/64) \cdot n^2 + (-1/4) \cdot n + 1$	$0.00625/n$	read C[i1, i1] (i0=0)
$n^{2.5}$	0.0142	ramp	$25 \rightarrow (17/8) \cdot n - 26$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00625/n$	read C[i1, i3] (i0=0)
$n^{2.5}$	0.0142	ramp	$25 \rightarrow (17/8) \cdot n - 26$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00625/n$	read C[i1, i3] (i0=0, i3=0)
n^2	2.24	level	5	$n^2 - n$	$0.4/n$	read A[i1, i2] (i0=0, i1=0, i2=0); read B[i1, i2] (i0=0, i1=0, i2=0) (+3)
n^2	1.96	level	5	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.35/n$	read C[i1, i1] (i0=0)
n^2	1.75	level	4	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.35/n$	read B[i1, i2] (i0=0, i3=0)
n^2	0.978	level	5	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.175/n$	read C[i1, i3] (i0=0, i2=0)
n^2	0.978	level	5	$(7/16) \cdot n^2 + (-7/8) \cdot n$	$0.175/n$	read C[i1, i3] (i0=0, i2=0)
n^2	0.515	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n + 3 \rightarrow (5/16) \cdot n^2 + (3/8) \cdot n$	$n - 4$	$0.4 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.513	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n + 2 \rightarrow (5/16) \cdot n^2 + (1/4) \cdot n - 1$	$n - 5$	$0.4 \cdot n^{-2}$	read B[i1, i2] (i0=0, i2=0, i3=0)
n^2	0.489	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 8$	$0.35 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0, i3=0)
n^2	0.489	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(7/8) \cdot n - 8$	$0.35 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.462	ramp	$n + 1 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(11/8) \cdot n - 1$	$0.55 \cdot n^{-2}$	read A[i3, i2] (i0=0); read B[i3, i2] (i0=0)
n^2	0.318	ramp	$(23/8) \cdot n - 3 \rightarrow$ $(5/16) \cdot n^2$ $+ (-1/4) \cdot n +$ 5	$(7/8) \cdot n - 3$	$0.35 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.317	ramp	$(25/8) \cdot n - 5 \rightarrow$ $(5/16) \cdot n^2$ $+ (-1/4) \cdot n +$ 4	$(7/8) \cdot n - 4$	$0.35 \cdot n^{-2}$	read B[i1, i2] (i0=0, i3=0)
n^2	0.292	ramp	$(1/2) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 +$ 1	$(7/8) \cdot n - 1$	$0.35 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)
n^2	0.292	ramp	$(1/2) \cdot n + 2 \rightarrow$ $(1/4) \cdot n^2$	$(7/8) \cdot n - 1$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.292	ramp	$(3/4) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 +$ 1	$(7/8) \cdot n - 2$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.292	ramp	$(3/4) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 +$ $(-1/8) \cdot n + 2$	$(7/8) \cdot n - 2$	$0.35 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.292	ramp	$(3/4) \cdot n + 1 \rightarrow$ $(1/4) \cdot n^2 +$ $(-1/8) \cdot n$	$(7/8) \cdot n - 2$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.252	ramp	$n + 1 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.3 \cdot n^{-2}$	read B[i3, i2] (i0=0, i3=0)
n^2	0.252	ramp	$n \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 2$	$(3/4) \cdot n - 1$	$0.3 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.252	ramp	$(3/4) \cdot n + 1 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(3/4) \cdot n$	$0.3 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.252	ramp	$(3/4) \cdot n - 1 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/8) \cdot n - 2$	$(3/4) \cdot n$	$0.3 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.251	ramp	$(3/4) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 +$ 1	$(3/4) \cdot n - 1$	$0.3 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0824	ramp	$(5/2) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-11/8) \cdot n$	$(1/4) \cdot n - 2$	$0.1 \cdot n^{-2}$	read A[i3, i2] (i0=0); read B[i3, i2] (i0=0)
n^2	0.082	ramp	$(9/4) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-13/8) \cdot n$	$(1/4) \cdot n - 2$	$0.1 \cdot n^{-2}$	read A[i3, i2] (i0=0); read B[i3, i2] (i0=0)
n^2	0.0815	ramp	$(35/8) \cdot n - 3$ → $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 3	$(1/4) \cdot n - 4$	$0.1 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); read B[i1, i2] (i0=0, i1=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(9/4) \cdot n +$ $(7/16)$	$(1/8) \cdot n +$ $(-9/8)$	$0.05 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(9/4) \cdot n +$ $(7/16)$	$(1/8) \cdot n +$ $(-9/8)$	$0.05 \cdot n^{-2}$	read B[i1, i2] (i0=0, i3=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i1, i2] (i0=0, i3=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(1/2) \cdot n$	$(1/8) \cdot n + 4$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0, i3=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(7/4) \cdot n +$ $(9/4)$	$(1/8) \cdot n +$ $(-5/4)$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $2 \cdot n +$ $(27/16)$	$(1/8) \cdot n +$ $(-17/8)$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(1/4) \cdot n +$ 3	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $2 \cdot n +$ $(3/4)$	$(1/8) \cdot n +$ $(-9/4)$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(1/2) \cdot n$	$(1/8) \cdot n - 3$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i2=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(9/4) \cdot n +$ $(7/16)$	$(1/8) \cdot n +$ $(-9/8)$	$0.05 \cdot n^{-2}$	read C[i1, i1] (i0=0, i2=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read C[i1, i1] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0664	ramp	$(5/16) \cdot n^2 + (1/4) \cdot n + 4 \rightarrow$	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.0664	ramp	$(5/16) \cdot n^2 + (1/2) \cdot n - 2 \rightarrow$ $(5/16) \cdot n^2 + (1/4) \cdot n + 3 \rightarrow$	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i1, i2] (i0=0, i3=0)
n^2	0.0449	ramp	$(5/16) \cdot n^2 + (1/2) \cdot n - 3 \rightarrow$ $(35/8) \cdot n - 9 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.0449	ramp	$(5/16) \cdot n^2 + (-17/8) \cdot n + 15 \rightarrow$ $(35/8) \cdot n - 10 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i1, i2] (i0=0, i3=0)
n^2	0.0414	ramp	$(5/16) \cdot n^2 + (-17/8) \cdot n + 14 \rightarrow$ $(11/4) \cdot n + 2 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.0412	ramp	$(1/4) \cdot n^2 + (-9/8) \cdot n \rightarrow$ $(5/2) \cdot n + 5 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)
n^2	0.0412	ramp	$(1/4) \cdot n^2 + (-5/4) \cdot n + 1 \rightarrow$ $(5/2) \cdot n + 3 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i3=0)
n^2	0.0412	ramp	$(1/4) \cdot n^2 + (-5/4) \cdot n - 1 \rightarrow$ $(5/2) \cdot n + 2 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0412	ramp	$(1/4) \cdot n^2 + (-5/4) \cdot n - 2 \rightarrow$ $(5/2) \cdot n + 2 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.0412	ramp	$(1/4) \cdot n^2 + (-11/8) \cdot n \rightarrow$ $(5/2) \cdot n \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.0411	ramp	$(1/4) \cdot n^2 + (-11/8) \cdot n - 2 \rightarrow$ $(9/4) \cdot n + 5 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)
n^2	0.0411	ramp	$(1/4) \cdot n^2 + (-3/2) \cdot n + 1 \rightarrow$ $(9/4) \cdot n + 4 \rightarrow$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)
			$(1/4) \cdot n^2 + (-3/2) \cdot n$			

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0411	ramp	$(9/4) \cdot n + 4$ → $(1/4) \cdot n^2 +$ $(-3/2) \cdot n$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 3$ → $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 1$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 2$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i3=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 1$ → $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 3$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.041	ramp	$(9/4) \cdot n + 4$ → $(1/4) \cdot n^2 +$ $(-13/8) \cdot n +$ 2	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.041	ramp	$(9/4) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-13/8) \cdot n$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.041	ramp	$(9/4) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-13/8) \cdot n$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.041	ramp	$(9/4) \cdot n \rightarrow$ $(1/4) \cdot n^2 +$ $(-13/8) \cdot n - 2$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.0409	ramp	$(39/8) \cdot n -$ 10 → $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 8	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i1, i2] (i0=0, i1=2, i3=0)
n^2	0.0408	ramp	$(37/8) \cdot n - 6$ → $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 6	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=1)
n^2	0.0408	ramp	$(37/8) \cdot n - 7$ → $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 5	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i1, i2] (i0=0, i1=1, i3=0)
$n^{1.5}$	0.0941	ramp	$(1/2) \cdot n + 5$ → $(3/4) \cdot n -$ 1	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i1=2, i3=0)
$n^{1.5}$	0.0939	ramp	$(1/2) \cdot n + 4$ → $(3/4) \cdot n -$ 2	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=2, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.0729	ramp	$(1/4) \cdot n + 5$ $\rightarrow (1/2) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i1=1, i3=0)
$n^{1.5}$	0.0723	ramp	$(1/4) \cdot n + 3$ $\rightarrow (1/2) \cdot n - 3$	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=1, i3=0)
n^1	14.2	level	4	$(57/8) \cdot n - 8$	$2.85 \cdot n^{-2}$	read C[i1, i1] (i0=0, i1=0, i2=0); read C[i1, i1] (i0=0, i1=0) (+1)
n^1	2.32	level	7	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read B[i1, i2] (i0=0, i1=1, i2=0, i3=0); read B[i3, i2] (i0=0, i1=2, i2=0, i3=0) (+1)
n^1	1.96	level	5	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read B[i3, i2] (i0=0, i1=1, i3=0)
n^1	1.96	level	5	$(7/8) \cdot n + (-7/8)$	$0.35 \cdot n^{-2}$	read C[i1, i1] (i0=0, i2=0)
n^1	1.96	level	5	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read C[i1, i1] (i0=0, i2=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read B[i1, i2] (i0=0, i1=0, i2=0); read C[i1, i1] (i0=0, i1=0, i2=0) (+1)
n^1	1.12	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	2	$0.8 \cdot n^{-3}$	read C[i1, i3] (i0=0, i1=1, i2=0, i3=0); read C[i1, i3] (i0=0, i1=8, i2=0, i3=0)
n^1	1.12	level	$(5/16) \cdot n^2 + (9/4) \cdot n + (7/16)$	2	$0.8 \cdot n^{-3}$	read B[i1, i2] (i0=0, i2=0, i3=0); read A[i1, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.12	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	2	$0.8 \cdot n^{-3}$	read B[i1, i2] (i0=0, i2=0, i3=0); read A[i1, i2] (i0=0, i2=0)
n^1	1	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (7/8)$	2	$0.8 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i2=0); read B[i1, i2] (i0=0, i1=1, i2=0, i3=0)
n^1	1	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	2	$0.8 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i2=0); read B[i1, i2] (i0=0, i1=1, i2=0, i3=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/8) \cdot n + 3$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/8) \cdot n + 2$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i3=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i3=0)
n^1	0.559	level	$(5/16) \cdot n^2 + 2 \cdot n + (27/16)$	1	$0.4 \cdot n^{-3}$	read C[i1, i3] (i0=0, i2=0, i3=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/4) \cdot n + 3$	1	$0.4 \cdot n^{-3}$	read C[i1, i3] (i0=0, i2=0, i3=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (17/8) \cdot n + (-7/16)$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i2=0, i3=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (3/8) \cdot n - 1$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i2=0, i3=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (9/4) \cdot n + (7/16)$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (7/4) \cdot n + (5/4)$	1	$0.4 \cdot n^{-3}$	read C[i1, i3] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/4) \cdot n + 2$	1	$0.4 \cdot n^{-3}$	read C[i1, i3] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.4 \cdot n^{-3}$	read C[i1, i1] (i0=0, i1=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + 2 \cdot n + (3/4)$	1	$0.4 \cdot n^{-3}$	read C[i1, i3] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.4 \cdot n^{-3}$	read C[i1, i3] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + 2 \cdot n + (27/16)$	1	$0.4 \cdot n^{-3}$	read C[i1, i3] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.4 \cdot n^{-3}$	read C[i1, i1] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + 2 \cdot n + (27/16)$	1	$0.4 \cdot n^{-3}$	read C[i1, i1] (i0=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (15/8)$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i1=2, i2=0, i3=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i1=2, i2=0, i3=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (23/8)$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=1, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 2$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=1, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (15/8) \cdot n + (7/8)$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.354	level	8	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i1=8, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	3.08	level	$(19/8) \cdot n - 1$	2	$0.8 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0); read B[i1, i2] (i0=0, i1=0)
$n^{0.5}$	1.7	level	$(23/8) \cdot n - 4$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i1=2, i3=0)
$n^{0.5}$	1.62	level	$(21/8) \cdot n - 2$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=1)
$n^{0.5}$	1.62	level	$(21/8) \cdot n - 3$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i1=1, i3=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=2, i3=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.4 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=2, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 3$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=2, i2=0, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=1, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n - 2$	1	$0.4 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=1, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.4 \cdot n^{-3}$	read B[i1, i2] (i0=0, i1=0, i2=0); read A[i3, i2] (i0=0, i1=2, i2=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.4 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i2=0); read A[i3, i2] (i0=0, i1=1, i2=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 3$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=1, i2=0, i3=0)

Symmetric multiply: the symmetric operand and the accumulating output are re-read across output rows — ramps to $(1/4)n^2 + O(n)$ lines with population $n^3/16$, coefficient $0.0466 \cdot n^4$ (numerically identical to syr2k's two-source structure), headroom +1.0. The old headroom-0 reading was the rendering artifact.

symm — single-shot [exact]

Accesses $A(n) = (5/2) \cdot n^3 + (3/2) \cdot n^2$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0466 \cdot n^4 + 1.08 \cdot n^{3.5} + 3.99 \cdot n^3 + 6.28 \cdot n^{2.5} + 7.71 \cdot n^2 + 0.333 \cdot n^{1.5} + 23.2 \cdot n^1 + 5.56 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0176	ramp	$n + 5 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(3/64) \cdot n^3$ + (- $63/64) \cdot n^2$ + $(31/8) \cdot n -$ 2	0.0187	read A[i3, i2] (i0=0)
n^4	0.0155	ramp	$n + 5 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(21/512) \cdot n^3$ + (- $53/64) \cdot n^2$ + $(11/4) \cdot n$	0.0164	read B[i3, i2] (i0=0)
n^4	0.00286	ramp	$(9/4) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(1/128) \cdot n^3$ + $(-7/32) \cdot n^2$ + $(7/4) \cdot n - 4$	0.00313	read A[i3, i2] (i0=0)
n^4	0.00283	ramp	$2 \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 4$	$(1/128) \cdot n^3$ + (- $15/64) \cdot n^2$ + $(17/8) \cdot n -$ 6	0.00313	read B[i3, i2] (i0=0)
n^4	0.00283	ramp	$2 \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 4$	$(1/128) \cdot n^3$ + (- $15/64) \cdot n^2$ + $(17/8) \cdot n -$ 6	0.00313	read A[i3, i2] (i0=0)
n^4	0.00251	ramp	$(9/4) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(7/1024) \cdot n^3$ + (- $23/128) \cdot n^2$ + $(5/4) \cdot n - 2$	0.00273	read B[i3, i2] (i0=0)
n^4	0.00217	ramp	$(11/4) \cdot n +$ 7 $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(3/512) \cdot n^3$ + $(-5/32) \cdot n^2$ + $(9/8) \cdot n - 2$	0.00234	read B[i3, i2] (i0=0)
n^4	0.000345	ramp	$(17/4) \cdot n +$ 9 $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(1/1024) \cdot n^3$ + (- $5/128) \cdot n^2$ + $(1/2) \cdot n - 2$	0.000391	read B[i3, i2] (i0=0)
$n^{3.5}$	0.443	ramp	9 $\rightarrow$ $(17/8) \cdot n$	$(49/128) \cdot n^3$ + (- $49/32) \cdot n^2$ + $(7/8) \cdot n$	0.153	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.389	ramp	$9 \rightarrow (17/8) \cdot n$	$(343/1024) \cdot n^3 + (-133/128) \cdot n^2$	0.134	read B[i3, i2] (i0=0)
$n^{3.5}$	0.0613	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$(7/128) \cdot n^3 + (-21/32) \cdot n^2 + (7/4) \cdot n$	0.0219	read B[i3, i2] (i0=0)
$n^{3.5}$	0.0613	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$(7/128) \cdot n^3 + (-21/32) \cdot n^2 + (7/4) \cdot n$	0.0219	read A[i3, i2] (i0=0)
$n^{3.5}$	0.0544	ramp	$24 \rightarrow (17/8) \cdot n$	$(49/1024) \cdot n^3 + (-63/128) \cdot n^2 + (7/8) \cdot n$	0.0191	read B[i3, i2] (i0=0)
$n^{3.5}$	0.0516	ramp	$26 \rightarrow (17/8) \cdot n - 13$	$(49/1024) \cdot n^3 + (-147/128) \cdot n^2 + (49/8) \cdot n$	0.0191	read C[i1, i3] (i0=0)
$n^{3.5}$	0.00719	ramp	$27 \rightarrow (17/8) \cdot n - 12$	$(7/1024) \cdot n^3 + (-7/32) \cdot n^2 + (35/16) \cdot n - 7$	0.00273	read C[i1, i3] (i0=0)
$n^{3.5}$	0.00697	ramp	$41 \rightarrow (17/8) \cdot n - 27$	$(7/1024) \cdot n^3 + (-35/128) \cdot n^2 + (21/8) \cdot n$	0.00273	read C[i1, i3] (i0=0)
$n^{3.5}$	0.000971	ramp	$42 \rightarrow (17/8) \cdot n - 26$	$(1/1024) \cdot n^3 + (-3/64) \cdot n^2 + (11/16) \cdot n - 3$	0.000391	read C[i1, i3] (i0=0)
n^3	1.12	level	5	$(1/2) \cdot n^3 + (-1/2) \cdot n^2$	0.2	read B[i3, i2] (i0=0, i1=1, i2=0, i3=0); read A[i3, i2] (i0=0, i1=2, i2=0, i3=0) (+1)
n^3	0.978	level	5	$(7/16) \cdot n^3 + (-7/8) \cdot n^2$	0.175	read C[i1, i3] (i0=0)
n^3	0.866	level	3	$(1/2) \cdot n^3 + (1/2) \cdot n^2 + (7/8) \cdot n$	0.2	read C[i1, i1] (i0=0, i1=0); write A[i3, i2] (i0=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.323	ramp	$n + 3 \rightarrow$ $(1/4) \cdot n^2 + 1$	$(105/128) \cdot n^2 + (-77/16) \cdot n + 6$	$0.328/n$	read A[i3, i2] (i0=0, i2=0); read B[i3, i2] (i0=0, i2=0)
n^3	0.149	ramp	$(5/4) \cdot n + 1 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 1$	$(3/8) \cdot n^2 + (-15/8) \cdot n + 1$	$0.15/n$	read A[i3, i2] (i0=0)
n^3	0.13	ramp	$(5/4) \cdot n + 1 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 1$	$(21/64) \cdot n^2 + (-11/8) \cdot n$	$0.131/n$	read B[i3, i2] (i0=0)
n^3	0.0452	ramp	$(9/4) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 + (-3/2) \cdot n$	$(15/128) \cdot n^2 + (-21/16) \cdot n + 3$	$0.0469/n$	read A[i3, i2] (i0=0, i2=0); read B[i3, i2] (i0=0, i2=0)
n^3	0.0298	ramp	$(1/2) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(3/32) \cdot n^2 + (-3/2) \cdot n$	$0.0375/n$	read B[i3, i2] (i0=0)
n^3	0.0298	ramp	$(1/2) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 4$	$(3/32) \cdot n^2 + (-3/2) \cdot n$	$0.0375/n$	read A[i3, i2] (i0=0)
n^3	0.0298	ramp	$(3/4) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(3/32) \cdot n^2 + (-13/8) \cdot n + 2$	$0.0375/n$	read B[i3, i2] (i0=0, i3=0)
n^3	0.0298	ramp	$(3/4) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(3/32) \cdot n^2 + (-13/8) \cdot n + 2$	$0.0375/n$	read A[i3, i2] (i0=0)
n^3	0.0298	ramp	$(3/4) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 4$	$(3/32) \cdot n^2 + (-13/8) \cdot n + 2$	$0.0375/n$	read A[i3, i2] (i0=0, i3=0)
n^3	0.0249	ramp	$(3/4) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(5/64) \cdot n^2 + (-5/4) \cdot n$	$0.0312/n$	read B[i3, i2] (i0=0)
n^3	0.0241	ramp	$(5/2) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 + (-5/4) \cdot n - 1$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	$0.025/n$	read A[i3, i2] (i0=0)
n^3	0.0239	ramp	$(9/4) \cdot n + 2 \rightarrow$ $(1/4) \cdot n^2 + (-3/2) \cdot n - 2$	$(1/16) \cdot n^2 + (-7/8) \cdot n + 3$	$0.025/n$	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0212	ramp	$(5/2) \cdot n + 3$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 1$	$(7/128) \cdot n^2$ $+ (-9/16) \cdot n$ $+ 1$	0.0219/n	read B[i3, i2] (i0=0)
n^3	0.0212	ramp	$(11/4) \cdot n +$ $5 \rightarrow$ $(1/4) \cdot n^2 +$ 1	$(7/128) \cdot n^2$ $+ (-11/16) \cdot n$ $+ 2$	0.0219/n	read B[i3, i2] (i0=0, i2=0)
n^3	0.021	ramp	$(9/4) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 2$	$(7/128) \cdot n^2$ $+ (-11/16) \cdot n$ $+ 2$	0.0219/n	read B[i3, i2] (i0=0)
n^3	0.0183	ramp	$3 \cdot n + 2 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 2$	$(3/64) \cdot n^2$ $+ (-1/2) \cdot n +$ 1	0.0187/n	read B[i3, i2] (i0=0)
n^3	0.0049	ramp	$(5/2) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 -$ $n - 3$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read B[i3, i2] (i0=0, i3=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 3$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read A[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 6$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 4$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read A[i3, i2] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 6$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 4$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read A[i3, i2] (i0=0, i3=0)
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 4$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 4$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read B[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 4$	$(1/64) \cdot n^2$ $+ (-3/8) \cdot n +$ 2	0.00625/n	read B[i3, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.00486	ramp	$2 \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 4$	$(1/64) \cdot n^2 +$ $+ (-3/8) \cdot n +$ 2	$0.00625/n$	read A[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 5$	$(1/64) \cdot n^2 +$ $+ (-3/8) \cdot n +$ 2	$0.00625/n$	read A[i3, i2] (i0=0)
n^3	0.00486	ramp	$2 \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 5$	$(1/64) \cdot n^2 +$ $+ (-3/8) \cdot n +$ 2	$0.00625/n$	read A[i3, i2] (i0=0, i3=0)
n^3	0.00291	ramp	$(9/2) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 2$	$(1/128) \cdot n^2 +$ $+ (-3/16) \cdot n$ $+ 1$	$0.00313/n$	read B[i3, i2] (i0=0)
n^3	0.00291	ramp	$(17/4) \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n$	$(1/128) \cdot n^2 +$ $+ (-3/16) \cdot n$ $+ 1$	$0.00313/n$	read B[i3, i2] (i0=0, i2=0)
n^3	0.00291	ramp	$(17/4) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 3$	$(1/128) \cdot n^2 +$ $+ (-3/16) \cdot n$ $+ 1$	$0.00313/n$	read B[i3, i2] (i0=0)
$n^{2.5}$	0.853	ramp	$5 \rightarrow$ $(17/8) \cdot n$	$(7/8) \cdot n^2 +$ $(-7/8) \cdot n$	$0.35/n$	read A[i1, i2] (i0=0)
$n^{2.5}$	0.853	ramp	$7 \rightarrow$ $(17/8) \cdot n$	$(7/8) \cdot n^2 +$ $(-7/4) \cdot n$	$0.35/n$	read A[i3, i2] (i0=0)
$n^{2.5}$	0.837	ramp	$21 \rightarrow$ $(17/8) \cdot n -$ 13	$(7/8) \cdot n^2 +$ $(-63/8) \cdot n +$ 7	$0.35/n$	read C[i1, i3] (i0=0)
$n^{2.5}$	0.748	ramp	$7 \rightarrow$ $(17/8) \cdot n$	$(49/64) \cdot n^2 +$ $+ (-7/8) \cdot n$	$0.306/n$	read B[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.748	ramp	$7 \rightarrow$ $(17/8) \cdot n$	$(49/64) \cdot n^2 +$ $+ (-7/8) \cdot n$	$0.306/n$	read A[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.726	ramp	$9 \rightarrow$ $(17/8) \cdot n -$ 13	$(49/64) \cdot n^2 +$ $+ (-49/8) \cdot n$	$0.306/n$	read C[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.643	ramp	$7 \rightarrow$ $(17/8) \cdot n$	$(21/32) \cdot n^2$	$0.263/n$	read B[i3, i2] (i0=0)
$n^{2.5}$	0.209	ramp	$20 \rightarrow$ $(17/8) \cdot n -$ 14	$(7/32) \cdot n^2 +$ $+ (-7/4) \cdot n$	$0.0875/n$	read A[i3, i2] (i0=0, i3=0); read B[i3, i2] (i0=0, i3=0)
$n^{2.5}$	0.105	ramp	$22 \rightarrow$ $(17/8) \cdot n -$ 12	$(7/64) \cdot n^2 +$ $+ (-7/8) \cdot n$	$0.0437/n$	read B[i3, i2] (i0=0)
$n^{2.5}$	0.105	ramp	$20 \rightarrow$ $(17/8) \cdot n -$ 14	$(7/64) \cdot n^2 +$ $+ (-7/8) \cdot n$	$0.0437/n$	read B[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.105	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0437/n$	read C[i1, i1] (i0=0)
$n^{2.5}$	0.102	ramp	$24 \rightarrow (17/8) \cdot n - 27$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	$0.0437/n$	read C[i1, i3] (i0=0)
$n^{2.5}$	0.102	ramp	$24 \rightarrow (17/8) \cdot n - 27$	$(7/64) \cdot n^2 + (-7/4) \cdot n$	$0.0437/n$	read C[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.101	ramp	$10 \rightarrow (17/8) \cdot n - 12$	$(7/64) \cdot n^2 + (-7/4) \cdot n + 7$	$0.0437/n$	read C[i1, i3] (i0=0, i3=0)
$n^{2.5}$	0.0146	ramp	$21 \rightarrow (17/8) \cdot n - 13$	$(1/64) \cdot n^2 + (-1/4) \cdot n + 1$	$0.00625/n$	read C[i1, i1] (i0=0)
$n^{2.5}$	0.0142	ramp	$25 \rightarrow (17/8) \cdot n - 26$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00625/n$	read C[i1, i3] (i0=0)
$n^{2.5}$	0.0142	ramp	$25 \rightarrow (17/8) \cdot n - 26$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00625/n$	read C[i1, i3] (i0=0, i3=0)
n^2	1.96	level	5	$(7/8) \cdot n^2$	$0.35/n$	read C[i1, i1] (i0=0)
n^2	1.75	level	4	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.35/n$	read B[i1, i2] (i0=0, i3=0)
n^2	0.462	ramp	$n + 1 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n$	$(11/8) \cdot n - 1$	$0.55 \cdot n^{-2}$	read A[i3, i2] (i0=0); read B[i3, i2] (i0=0)
n^2	0.293	ramp	$(3/4) \cdot n - 1 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n - 2$	$(7/8) \cdot n - 1$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0)
n^2	0.292	ramp	$(1/2) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + 1$	$(7/8) \cdot n - 1$	$0.35 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)
n^2	0.292	ramp	$(1/2) \cdot n + 2 \rightarrow (1/4) \cdot n^2$	$(7/8) \cdot n - 1$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.292	ramp	$(3/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + 1$	$(7/8) \cdot n - 2$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.292	ramp	$(3/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + (-1/8) \cdot n + 2$	$(7/8) \cdot n - 2$	$0.35 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.292	ramp	$(3/4) \cdot n + 1 \rightarrow (1/4) \cdot n^2 + (-1/8) \cdot n$	$(7/8) \cdot n - 2$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.252	ramp	$n + 1 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 1$	$(3/4) \cdot n - 1$	$0.3 \cdot n^{-2}$	read B[i3, i2] (i0=0, i3=0)
n^2	0.252	ramp	$n \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 2$	$(3/4) \cdot n - 1$	$0.3 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.252	ramp	$(3/4) \cdot n + 1$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(3/4) \cdot n$	$0.3 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.251	ramp	$(3/4) \cdot n + 3$ $\rightarrow$ $(1/4) \cdot n^2 +$ 1	$(3/4) \cdot n - 1$	$0.3 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)
n^2	0.0824	ramp	$(5/2) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-11/8) \cdot n$	$(1/4) \cdot n - 2$	$0.1 \cdot n^{-2}$	read A[i3, i2] (i0=0); read B[i3, i2] (i0=0)
n^2	0.0821	ramp	$(9/4) \cdot n + 4$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n$	$(1/4) \cdot n - 2$	$0.1 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0); read B[i3, i2] (i0=0, i2=0)
n^2	0.082	ramp	$(9/4) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-13/8) \cdot n$	$(1/4) \cdot n - 2$	$0.1 \cdot n^{-2}$	read A[i3, i2] (i0=0); read B[i3, i2] (i0=0)
n^2	0.0414	ramp	$(11/4) \cdot n +$ 2 $\rightarrow$ $(1/4) \cdot n^2 +$ $(-9/8) \cdot n$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 3$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 1$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i3=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-5/4) \cdot n - 2$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-11/8) \cdot n$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0411	ramp	$(9/4) \cdot n + 3$ → $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 1$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 2$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i3=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 1$ → $(1/4) \cdot n^2 +$ $(-3/2) \cdot n - 3$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.041	ramp	$(9/4) \cdot n + 4$ → $(1/4) \cdot n^2 +$ $(-13/8) \cdot n +$ 2	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.041	ramp	$(9/4) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-13/8) \cdot n$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0)
n^2	0.041	ramp	$(9/4) \cdot n + 2$ → $(1/4) \cdot n^2 +$ $(-13/8) \cdot n$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0, i3=0)
n^2	0.041	ramp	$(9/4) \cdot n \rightarrow$ $(1/4) \cdot n^2 +$ $(-13/8) \cdot n - 2$	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0)
$n^{1.5}$	0.0941	ramp	$(1/2) \cdot n + 5$ → $(3/4) \cdot n -$ 1	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i1=2, i3=0)
$n^{1.5}$	0.0939	ramp	$(1/2) \cdot n + 4$ → $(3/4) \cdot n -$ 2	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=2, i3=0)
$n^{1.5}$	0.0729	ramp	$(1/4) \cdot n + 5$ → $(1/2) \cdot n -$ 1	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read B[i3, i2] (i0=0, i1=1, i3=0)
$n^{1.5}$	0.0723	ramp	$(1/4) \cdot n + 3$ → $(1/2) \cdot n -$ 3	$(1/8) \cdot n - 2$	$0.05 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=1, i3=0)
n^1	14.2	level	4	$(57/8) \cdot n - 8$	$2.85 \cdot n^{-2}$	read C[i1, i1] (i0=0, i1=0); read C[i1, i3] (i0=0, i3=0)
n^1	2.32	level	7	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read B[i3, i2] (i0=0, i1=2, i2=0, i3=0); read C[i1, i3] (i0=0, i1=8, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	1.96	level	5	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read B[i3, i2] (i0=0, i1=1, i3=0)
n^1	1.96	level	5	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=1, i3=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read B[i1, i2] (i0=0, i1=0)
n^1	0.875	level	1	$(7/8) \cdot n$	$0.35 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0)
n^1	0.354	level	8	$(1/8) \cdot n - 1$	$0.05 \cdot n^{-2}$	read C[i1, i3] (i0=0, i1=8, i3=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=2, i3=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.4 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=2, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=1, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 3$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=2, i2=0, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.4 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=2, i2=0, i3=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n - 2$	1	$0.4 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=1, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.4 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=1, i2=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 3$	1	$0.4 \cdot n^{-3}$	read B[i3, i2] (i0=0, i1=1, i2=0, i3=0)

Symmetric multiply: the symmetric operand and the accumulating output are re-read across output rows — ramps to $(1/4)n^2 + O(n)$ lines with population $n^3/16$, coefficient $0.0466 \cdot n^4$ (numerically identical to syr2k's two-source structure), headroom +1.0. The old headroom-0 reading was the rendering artifact.

syr2k — infinite-repeat [exact]

Accesses $A(n) = 3 \cdot n^3 + 4 \cdot n^2 + n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0466 \cdot n^4 + 1.08 \cdot n^{3.5} + 4.98 \cdot n^3 + 5.47 \cdot n^{2.5} + 20.9 \cdot n^2 + 0.572 \cdot n^{1.5} + 57.4 \cdot n^1 + 70.9 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0409	ramp	$n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 2$	$(7/64) \cdot n^3$ $+ (-$ $77/32) \cdot n^2$ $+ (45/4) \cdot n -$ 12	0.0365	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
n^4	0.00572	ramp	$(11/4) \cdot n +$ $8 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 2$	$(1/64) \cdot n^3$ $+ (-$ $15/32) \cdot n^2$ $+ (17/4) \cdot n -$ 12	0.00521	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
$n^{3.5}$	0.885	ramp	$9 \rightarrow$ $(17/8) \cdot n$	$(49/64) \cdot n^3$ $+ (-$ $49/16) \cdot n^2$ $+ (7/4) \cdot n$	0.255	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
$n^{3.5}$	0.124	ramp	$24 \rightarrow$ $(17/8) \cdot n$	$(7/64) \cdot n^3$ $+ (-$ $21/16) \cdot n^2$ $+ (7/2) \cdot n$	0.0365	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
$n^{3.5}$	0.0583	ramp	$23 \rightarrow$ $(17/8) \cdot n -$ 16	$(7/128) \cdot n^3$ $+ (-$ $91/64) \cdot n^2$ $+ (35/4) \cdot n$	0.0182	read A[i1, i4] (i0=0)
$n^{3.5}$	0.00814	ramp	$25 \rightarrow$ $(17/8) \cdot n -$ 14	$(1/128) \cdot n^3$ $+ (-$ $17/64) \cdot n^2$ $+ (23/8) \cdot n -$ 10	0.0026	read A[i1, i4] (i0=0)
n^3	1.96	level	5	$(7/8) \cdot n^3 +$ $(-13/8) \cdot n^2$ $- n$	0.292	read C[i4, i3] (i0=0, i1=1, i3=0, i4=0); read C[i1, i3] (i0=0) (+1)
n^3	0.978	level	5	$(7/16) \cdot n^3$ $+$ $(-7/8) \cdot n^2$	0.146	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^3	0.438	level	1	$(7/16) \cdot n^3$	0.146	read A[i1, i4] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.393	ramp	$n + 4 \rightarrow$ $(1/4) \cdot n^2 + 2$	$n^2 - 7 \cdot n + 12$	$0.333/n$	read B[i4, i3] (i0=0, i3=0); read C[i4, i3] (i0=0, i3=0)
n^3	0.306	level	6	$(1/8) \cdot n^3 +$ $(-5/4) \cdot n^2$ $+ 2 \cdot n$	0.0417	read B[i4, i3] (i0=0, i1=0, i3=0, i4=0); read B[i1, i3] (i0=0, i1=1, i3=0, i4=0) (+3)
n^3	0.148	ramp	$(5/4) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n$	$(3/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 1	$0.125/n$	read C[i4, i3] (i0=0)
n^3	0.148	ramp	$(5/4) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n$	$(3/8) \cdot n^2 +$ $(-15/8) \cdot n +$ 1	$0.125/n$	read B[i4, i3] (i0=0)
n^3	0.14	level	5	$(1/16) \cdot n^3$ $+ (3/8) \cdot n^2$	0.0208	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^3	0.0511	ramp	$(67/8) \cdot n -$ 41 $\rightarrow$ $(5/16) \cdot n^2$ $+ (3/8) \cdot n - 4$	$(1/8) \cdot n^2 +$ $(-17/4) \cdot n +$ 36	$0.0417/n$	read B[i1, i3] (i0=0, i4=0)
n^3	0.0511	ramp	$(67/8) \cdot n -$ 42 $\rightarrow$ $(5/16) \cdot n^2$ $+ (3/8) \cdot n - 5$	$(1/8) \cdot n^2 +$ $(-17/4) \cdot n +$ 36	$0.0417/n$	read C[i1, i3] (i0=0, i4=0)
n^3	0.0485	ramp	$(11/4) \cdot n +$ 4 $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n$	$(1/8) \cdot n^2 +$ $(-3/2) \cdot n + 4$	$0.0417/n$	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
n^3	0.0483	ramp	$3 \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 1$	$(1/8) \cdot n^2 +$ $(-7/4) \cdot n + 6$	$0.0417/n$	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
n^3	0.0396	ramp	$(3/4) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 3$	$(1/8) \cdot n^2 +$ $(-19/8) \cdot n +$ 6	$0.0417/n$	read C[i4, i3] (i0=0, i4=0)
n^3	0.0396	ramp	$(3/4) \cdot n + 4$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 4$	$(1/8) \cdot n^2 +$ $(-19/8) \cdot n +$ 6	$0.0417/n$	read B[i4, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0349	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	0.0208/n	read A[i1, i2] (i0=0)
n^3	0.0347	ramp	$(1/2) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0365/n	read C[i4, i3] (i0=0)
n^3	0.0347	ramp	$(1/2) \cdot n + 5 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0365/n	read B[i4, i3] (i0=0)
n^3	0.0347	ramp	$(3/4) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(7/64) \cdot n^2 - 2 \cdot n + 4$	0.0365/n	read C[i4, i3] (i0=0)
n^3	0.0347	ramp	$(3/4) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(7/64) \cdot n^2 - 2 \cdot n + 4$	0.0365/n	read B[i4, i3] (i0=0)
n^3	0.0049	ramp	$(5/2) \cdot n + 8 \rightarrow (1/4) \cdot n^2 - n - 2$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read C[i4, i3] (i0=0)
n^3	0.0049	ramp	$(5/2) \cdot n + 8 \rightarrow (1/4) \cdot n^2 - n - 2$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read B[i4, i3] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 8 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 2$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read C[i4, i3] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n - 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read B[i4, i3] (i0=0)
$n^{2.5}$	1.71	ramp	$7 \rightarrow (17/8) \cdot n$	$(7/4) \cdot n^2 + (-7/2) \cdot n$	0.583/n	read B[i4, i3] (i0=0, i4=0); read C[i4, i3] (i0=0, i4=0)
$n^{2.5}$	0.958	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$n^2 - 8 \cdot n$	0.333/n	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.937	ramp	$6 \rightarrow (17/8) \cdot n - 16$	$n^2 - 10 \cdot n + 9$	0.333/n	read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.747	ramp	$7 \rightarrow (17/8) \cdot n$	$(49/64) \cdot n^2 + (-7/8) \cdot n$	0.255/n	read B[i4, i3] (i0=0)
$n^{2.5}$	0.747	ramp	$7 \rightarrow (17/8) \cdot n$	$(49/64) \cdot n^2 + (-7/8) \cdot n$	0.255/n	read C[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.149	ramp	$21 \rightarrow (17/8) \cdot n - 30$	$(11/64) \cdot n^2 + (-13/4) \cdot n + 8$	$0.0573/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.105	ramp	$22 \rightarrow (17/8) \cdot n - 12$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0365/n$	read B[i4, i3] (i0=0)
$n^{2.5}$	0.105	ramp	$22 \rightarrow (17/8) \cdot n - 12$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0365/n$	read C[i4, i3] (i0=0)
$n^{2.5}$	0.0141	ramp	$23 \rightarrow (17/8) \cdot n - 28$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00521/n$	read A[i1, i4] (i0=0)
n^2	2.32	level	2	$(105/64) \cdot n^2 + (7/8) \cdot n$	$0.547/n$	read B[i4, i3] (i0=0, i1=0, i4=0); read B[i4, i3] (i0=0) (+1)
n^2	2.14	level	6	$(7/8) \cdot n^2 - 7 \cdot n$	$0.292/n$	read B[i1, i3] (i0=0, i4=0)
n^2	1.96	level	5	$(7/8) \cdot n^2 - 7 \cdot n$	$0.292/n$	read C[i1, i3] (i0=0, i4=0)
n^2	1.75	level	4	$(7/8) \cdot n^2 + n - 1$	$0.292/n$	read A[i1, i4] (i0=0, i1=8, i4=0); read C[i1, i3] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2$	$0.583/n$	read A[i1, i2] (i0=0); write A[i1, i2] (i0=0) (+1)
n^2	1.52	level	3	$(7/8) \cdot n^2$	$0.292/n$	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$n - 9$	$0.333 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	$n - 9$	$0.333 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.505	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n + 10 \rightarrow (5/16) \cdot n^2 + (3/8) \cdot n$	$n - 10$	$0.333 \cdot n^{-2}$	read B[i1, i3] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.505	ramp	$(1/4) \cdot n^2 + (1/8) \cdot n + 9$ $\rightarrow$ $(5/16) \cdot n^2 + (3/8) \cdot n - 1$	$n - 10$	$0.333 \cdot n^{-2}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^2	0.334	ramp	$(51/8) \cdot n - 26 \rightarrow$ $(1/4) \cdot n^2 + (-15/8) \cdot n + 56$	$n - 16$	$0.333 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.334	ramp	$(51/8) \cdot n - 27 \rightarrow$ $(1/4) \cdot n^2 + (-15/8) \cdot n + 55$	$n - 16$	$0.333 \cdot n^{-2}$	read C[i1, i3] (i0=0, i4=0)
n^2	0.334	ramp	$n + 1 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 1$	$n - 3$	$0.333 \cdot n^{-2}$	read C[i4, i3] (i0=0, i4=0)
n^2	0.334	ramp	$n \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 2$	$n - 3$	$0.333 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0)
n^2	0.333	ramp	$(3/4) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 + 2$	$n - 3$	$0.333 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)
n^2	0.333	ramp	$(3/4) \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 + 1$	$n - 3$	$0.333 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0, i4=0)
n^2	0.333	ramp	$(3/4) \cdot n + 2 \rightarrow$ $(1/4) \cdot n^2$	$n - 3$	$0.333 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0, i4=0)
n^2	0.331	level	2	$(15/64) \cdot n^2$	$0.0781/n$	read B[i1, i3] (i0=0, i1=0, i4=0); read B[i4, i3] (i0=0) (+1)
n^2	0.313	ramp	$(51/8) \cdot n - 9 \rightarrow$ $(5/16) \cdot n^2 + (-1/4) \cdot n + 5$	$(7/8) \cdot n - 15$	$0.292 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.313	ramp	$(51/8) \cdot n - 10 \rightarrow$ $(5/16) \cdot n^2 + (-1/4) \cdot n + 4$	$(7/8) \cdot n - 15$	$0.292 \cdot n^{-2}$	read C[i1, i3] (i0=0, i4=0)
n^2	0.293	ramp	$(3/4) \cdot n + 2 \rightarrow$ $(1/4) \cdot n^2 + (1/8) \cdot n + 1$	$(7/8) \cdot n - 1$	$0.292 \cdot n^{-2}$	read C[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.293	ramp	$(3/4) \cdot n + 1$ → $(1/4) \cdot n^2 +$ $(1/8) \cdot n$	$(7/8) \cdot n - 1$	$0.292 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.292	ramp	$(3/4) \cdot n + 4$ → $(1/4) \cdot n^2 +$ 2	$(7/8) \cdot n - 2$	$0.292 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.251	ramp	$n + 2$ → $(1/4) \cdot n^2 +$ $(1/8) \cdot n + 1$	$(3/4) \cdot n - 1$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0)
n^2	0.251	ramp	$n + 2$ → $(1/4) \cdot n^2 +$ $(1/8) \cdot n + 1$	$(3/4) \cdot n - 1$	$0.25 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.251	ramp	$(1/2) \cdot n + 4$ → $(1/4) \cdot n^2 +$ 2	$(3/4) \cdot n$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)
n^2	0.251	ramp	$(1/2) \cdot n + 3$ → $(1/4) \cdot n^2 +$ 1	$(3/4) \cdot n$	$0.25 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.247	ramp	$(39/8) \cdot n - 9$ → $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 24	$(3/4) \cdot n - 12$	$0.25 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.247	ramp	$(39/8) \cdot n -$ 10 → $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 23	$(3/4) \cdot n - 12$	$0.25 \cdot n^{-2}$	read C[i1, i3] (i0=0, i4=0)
n^2	0.217	level	3	$(1/8) \cdot n^2$	$0.0417/n$	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^2	0.189	level	3	$(7/64) \cdot n^2$ + $(-7/8) \cdot n$	$0.0365/n$	read B[i4, i3] (i0=0)
n^2	0.188	level	1	$(3/16) \cdot n^2$ + $(1/2) \cdot n$	$0.0625/n$	read C[i4, i3] (i0=0, i1=0, i4=0); write A[i1, i2] (i0=0) (+1)
n^2	0.14	level	$(5/16) \cdot n^2$ + $(1/2) \cdot n$	$(1/4) \cdot n - 4$	$0.0833 \cdot n^{-2}$	read C[i1, i3] (i0=0, i4=0); read B[i1, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0824	ramp	$(5/2) \cdot n + 4$ → $(1/4) \cdot n^2 +$ $(-11/8) \cdot n +$ 2	$(1/4) \cdot n - 2$	$0.0833 \cdot n^2$	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
n^2	0.0815	ramp	$(35/8) \cdot n - 3$ → $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 3	$(1/4) \cdot n - 4$	$0.0833 \cdot n^2$	read B[i4, i3] (i0=0, i1=0, i4=0); read C[i1, i3] (i0=0, i1=0, i4=0)
n^2	0.0699	level	$(5/16) \cdot n^2$ + $(1/4) \cdot n +$ 2	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	read A[i1, i2] (i0=0)
n^2	0.0664	ramp	$(5/16) \cdot n^2$ + $(1/4) \cdot n +$ 4 → $(5/16) \cdot n^2$ + $(1/2) \cdot n - 2$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	read B[i1, i3] (i0=0, i4=0)
n^2	0.0664	ramp	$(5/16) \cdot n^2$ + $(1/4) \cdot n +$ 3 → $(5/16) \cdot n^2$ + $(1/2) \cdot n - 3$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^2$	read C[i1, i3] (i0=0, i4=0)
n^2	0.0439	ramp	$(65/8) \cdot n - 2$ → $(5/16) \cdot n^2$ + $(-5/2) \cdot n +$ 18	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	read B[i1, i3] (i0=0, i4=0)
n^2	0.0439	ramp	$(65/8) \cdot n - 3$ → $(5/16) \cdot n^2$ + $(-5/2) \cdot n +$ 17	$(1/8) \cdot n - 3$	$0.0417 \cdot n^2$	read C[i1, i3] (i0=0, i4=0)
n^2	0.0414	ramp	$(11/4) \cdot n +$ 3 → $(1/4) \cdot n^2 +$ $(-9/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^2$	read C[i4, i3] (i0=0)
n^2	0.0414	ramp	$(11/4) \cdot n +$ 3 → $(1/4) \cdot n^2 +$ $(-9/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^2$	read B[i4, i3] (i0=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 6$ → $(1/4) \cdot n^2 +$ $(-5/4) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^2$	read B[i4, i3] (i0=0, i3=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 3$ → $(1/4) \cdot n^2 +$ $(-11/8) \cdot n +$ 1	$(1/8) \cdot n - 1$	$0.0417 \cdot n^2$	read C[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0412	ramp	$(5/2) \cdot n + 2$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-11/8) \cdot n$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 6$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 5$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-3/2) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.0409	ramp	$2 \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 +$ $(-7/4) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)
n^2	0.0409	ramp	$2 \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 +$ $(-7/4) \cdot n$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.0408	ramp	$(37/8) \cdot n - 6$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 6	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read B[i1, i3] (i0=0, i1=1, i4=0)
n^2	0.0408	ramp	$(37/8) \cdot n - 7$ $\rightarrow$ $(1/4) \cdot n^2 +$ $(-15/8) \cdot n +$ 5	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read C[i1, i3] (i0=0, i1=1, i4=0)
n^2	0.0271	level	3	$(1/64) \cdot n^2 +$ $(-1/8) \cdot n$	$0.00521/n$	read B[i4, i3] (i0=0)
$n^{1.5}$	0.238	ramp	$2 \rightarrow (1/8) \cdot n$	$n - 8$	$0.333 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=0)
$n^{1.5}$	0.0941	ramp	$(1/2) \cdot n + 5$ $\rightarrow (3/4) \cdot n -$ 1	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=2, i4=0)
$n^{1.5}$	0.0939	ramp	$(1/2) \cdot n + 4$ $\rightarrow (3/4) \cdot n -$ 2	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=2, i4=0)
$n^{1.5}$	0.0729	ramp	$(1/4) \cdot n + 5$ $\rightarrow (1/2) \cdot n -$ 1	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=1, i4=0)
$n^{1.5}$	0.0723	ramp	$(1/4) \cdot n + 3$ $\rightarrow (1/2) \cdot n -$ 3	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
n^1	13.7	level	5	$(49/8) \cdot n$	$2.04 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^1	12.2	level	4	$(49/8) \cdot n$	$2.04 \cdot n^{-2}$	read C[i1, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	8	level	1	$8 \cdot n$	$2.67 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); read A[i1, i4] (i0=0, i3=0, i4=0) (+1)
n^1	3.91	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	7	$2.33 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	1.96	level	5	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=1, i4=0)
n^1	1.96	level	5	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read C[i1, i3] (i0=0, i1=0, i4=0)
n^1	1.12	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	2	$0.667 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0); read B[i1, i3] (i0=0, i4=0)
n^1	1.12	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	2	$0.667 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0); read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	1	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 1$	2	$0.667 \cdot n^{-3}$	read C[i1, i3] (i0=0, i1=0, i3=0, i4=0); read C[i1, i3] (i0=0, i1=1, i3=0, i4=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/8) \cdot n + 2$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/4) \cdot n + 2$	1	$0.333 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/8) \cdot n + 3$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n - 1$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/4) \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i1, i2] (i0=0)
n^1	0.559	level	$(5/16) \cdot n^2 + (1/2) \cdot n$	1	$0.333 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 3$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 4$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 5$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 6$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 7$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 8$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 2$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i1=1, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 2$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 3$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 4$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 5$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 6$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 7$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.5	level	$(1/4) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	3.08	level	$(19/8) \cdot n - 1$	2	$0.667 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=0, i4=0); read C[i1, i3] (i0=0, i1=0, i4=0)
$n^{0.5}$	2.47	level	$(49/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.47	level	$(49/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.42	level	$(47/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.42	level	$(47/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.37	level	$(45/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.37	level	$(45/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.32	level	$(43/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.32	level	$(43/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.26	level	$(41/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.26	level	$(41/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.21	level	$(39/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.21	level	$(39/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.15	level	$(37/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.15	level	$(37/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.09	level	$(35/8) \cdot n - 10$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	2.09	level	$(35/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	2.03	level	$(33/8) \cdot n - 9$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i1=7, i4=0)
$n^{0.5}$	2.03	level	$(33/8) \cdot n - 8$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i1=7, i4=0)
$n^{0.5}$	1.97	level	$(31/8) \cdot n - 8$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.97	level	$(31/8) \cdot n - 7$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.9	level	$(29/8) \cdot n - 7$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.9	level	$(29/8) \cdot n - 6$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.84	level	$(27/8) \cdot n - 6$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.84	level	$(27/8) \cdot n - 5$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.77	level	$(25/8) \cdot n - 5$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.77	level	$(25/8) \cdot n - 4$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.7	level	$(23/8) \cdot n - 4$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.7	level	$(23/8) \cdot n - 3$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.62	level	$(21/8) \cdot n - 3$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	1.62	level	$(21/8) \cdot n - 2$	1	$0.333 \cdot n^{-3}$	read B[i1, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=2, i4=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=2, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n - 1$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 3$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=2, i3=0, i4=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=0, i3=0, i4=0); read B[i4, i3] (i0=0, i1=2, i3=0, i4=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 3$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=1, i3=0, i4=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read C[i1, i3] (i0=0, i1=0, i3=0, i4=0); read B[i4, i3] (i0=0, i1=1, i3=0, i4=0)

Two rank-k sources (B and C) double syrk's structure: ramps to $(1/4)n^2 + (1/4)n$ lines with combined $0.0466 \cdot n^4$ ($\approx 2.8 \times$ syrk, two matrices and a wider window), headroom +1.0.

syr2k — single-shot [exact]

Accesses $A(n) = 3 \cdot n^3 + 4 \cdot n^2 + n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0466 \cdot n^4 + 1.08 \cdot n^{3.5} + 4.84 \cdot n^3 + 5.47 \cdot n^{2.5} + 16.4 \cdot n^2 + 0.572 \cdot n^{1.5} + 18.9 \cdot n^1 + 5.56 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0409	ramp	$n + 6 \rightarrow$ $(1/4) \cdot n^2 +$ $(1/4) \cdot n - 2$	$(7/64) \cdot n^3$ $+ (-$ $77/32) \cdot n^2$ $+ (45/4) \cdot n -$ 12	0.0365	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.00572	ramp	$(11/4) \cdot n + 8 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(1/64) \cdot n^3 + (-15/32) \cdot n^2 + (17/4) \cdot n - 12$	0.00521	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
$n^{3.5}$	0.885	ramp	$9 \rightarrow (17/8) \cdot n$	$(49/64) \cdot n^3 + (-49/16) \cdot n^2 + (7/4) \cdot n$	0.255	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
$n^{3.5}$	0.124	ramp	$24 \rightarrow (17/8) \cdot n$	$(7/64) \cdot n^3 + (-21/16) \cdot n^2 + (7/2) \cdot n$	0.0365	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
$n^{3.5}$	0.0583	ramp	$23 \rightarrow (17/8) \cdot n - 16$	$(7/128) \cdot n^3 + (-91/64) \cdot n^2 + (35/4) \cdot n$	0.0182	read A[i1, i4] (i0=0)
$n^{3.5}$	0.00814	ramp	$25 \rightarrow (17/8) \cdot n - 14$	$(1/128) \cdot n^3 + (-17/64) \cdot n^2 + (23/8) \cdot n - 10$	0.0026	read A[i1, i4] (i0=0)
n^3	3.07	level	5	$(11/8) \cdot n^3 + (-17/8) \cdot n^2 - n$	0.458	read C[i4, i3] (i0=0, i1=1, i3=0, i4=0); read C[i1, i3] (i0=0) (+2)
n^3	0.438	level	1	$(7/16) \cdot n^3$	0.146	read A[i1, i4] (i0=0)
n^3	0.393	ramp	$n + 4 \rightarrow (1/4) \cdot n^2 + 2$	$n^2 - 7 \cdot n + 12$	0.333/n	read B[i4, i3] (i0=0, i3=0); read C[i4, i3] (i0=0, i3=0)
n^3	0.306	level	6	$(1/8) \cdot n^3 + (-5/4) \cdot n^2 + 2 \cdot n$	0.0417	read B[i4, i3] (i0=0, i1=2, i3=0, i4=0); read C[i1, i3] (i0=0) (+1)
n^3	0.148	ramp	$(5/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n$	$(3/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.125/n	read C[i4, i3] (i0=0)
n^3	0.148	ramp	$(5/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n$	$(3/8) \cdot n^2 + (-15/8) \cdot n + 1$	0.125/n	read B[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0485	ramp	$(11/4) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n$	$(1/8) \cdot n^2 + (-3/2) \cdot n + 4$	0.0417/n	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
n^3	0.0483	ramp	$3 \cdot n + 3 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-7/4) \cdot n + 6$	0.0417/n	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
n^3	0.0396	ramp	$(3/4) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(1/8) \cdot n^2 + (-19/8) \cdot n + 6$	0.0417/n	read C[i4, i3] (i0=0, i4=0)
n^3	0.0396	ramp	$(3/4) \cdot n + 4 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 4$	$(1/8) \cdot n^2 + (-19/8) \cdot n + 6$	0.0417/n	read B[i4, i3] (i0=0, i4=0)
n^3	0.0347	ramp	$(1/2) \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0365/n	read C[i4, i3] (i0=0)
n^3	0.0347	ramp	$(1/2) \cdot n + 5 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 3$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0365/n	read B[i4, i3] (i0=0)
n^3	0.0347	ramp	$(3/4) \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(7/64) \cdot n^2 - 2 \cdot n + 4$	0.0365/n	read C[i4, i3] (i0=0)
n^3	0.0347	ramp	$(3/4) \cdot n + 6 \rightarrow$ $(1/4) \cdot n^2 + (1/4) \cdot n - 2$	$(7/64) \cdot n^2 - 2 \cdot n + 4$	0.0365/n	read B[i4, i3] (i0=0)
n^3	0.0049	ramp	$(5/2) \cdot n + 8 \rightarrow$ $(1/4) \cdot n^2 - n - 2$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read C[i4, i3] (i0=0)
n^3	0.0049	ramp	$(5/2) \cdot n + 8 \rightarrow$ $(1/4) \cdot n^2 - n - 2$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read B[i4, i3] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 8 \rightarrow$ $(1/4) \cdot n^2 + (-5/4) \cdot n - 2$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read C[i4, i3] (i0=0)
n^3	0.00488	ramp	$(9/4) \cdot n + 7 \rightarrow$ $(1/4) \cdot n^2 + (-5/4) \cdot n - 3$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00521/n	read B[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	1.71	ramp	$7 \rightarrow (17/8) \cdot n$	$(7/4) \cdot n^2 + (-7/2) \cdot n$	$0.583/n$	read B[i4, i3] (i0=0, i4=0); read C[i4, i3] (i0=0, i4=0)
$n^{2.5}$	0.958	ramp	$20 \rightarrow (17/8) \cdot n - 14$	$n^2 - 8 \cdot n$	$0.333/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.937	ramp	$6 \rightarrow (17/8) \cdot n - 16$	$n^2 - 10 \cdot n + 9$	$0.333/n$	read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.747	ramp	$7 \rightarrow (17/8) \cdot n$	$(49/64) \cdot n^2 + (-7/8) \cdot n$	$0.255/n$	read B[i4, i3] (i0=0)
$n^{2.5}$	0.747	ramp	$7 \rightarrow (17/8) \cdot n$	$(49/64) \cdot n^2 + (-7/8) \cdot n$	$0.255/n$	read C[i4, i3] (i0=0)
$n^{2.5}$	0.149	ramp	$21 \rightarrow (17/8) \cdot n - 30$	$(11/64) \cdot n^2 + (-13/4) \cdot n + 8$	$0.0573/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.105	ramp	$22 \rightarrow (17/8) \cdot n - 12$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0365/n$	read B[i4, i3] (i0=0)
$n^{2.5}$	0.105	ramp	$22 \rightarrow (17/8) \cdot n - 12$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0365/n$	read C[i4, i3] (i0=0)
$n^{2.5}$	0.0141	ramp	$23 \rightarrow (17/8) \cdot n - 28$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00521/n$	read A[i1, i4] (i0=0)
n^2	2.65	level	2	$(15/8) \cdot n^2$	$0.625/n$	read B[i4, i3] (i0=0); read B[i1, i3] (i0=0)
n^2	2.14	level	6	$(7/8) \cdot n^2 - 7 \cdot n$	$0.292/n$	read B[i1, i3] (i0=0, i4=0)
n^2	1.96	level	5	$(7/8) \cdot n^2 + (-7/8) \cdot n$	$0.292/n$	read C[i1, i3] (i0=0, i4=0); read B[i1, i3] (i0=0, i4=0)
n^2	1.95	level	3	$(9/8) \cdot n^2 - n$	$0.375/n$	read B[i4, i3] (i0=0, i1=1, i3=0, i4=0); read B[i4, i3] (i0=0) (+1)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1.75	level	4	$(7/8) \cdot n^2 + n - 1$	$0.292/n$	read A[i1, i4] (i0=0, i1=8, i4=0); read C[i1, i3] (i0=0)
n^2	1.5	level	1	$(3/2) \cdot n^2 + (17/2) \cdot n$	$0.5/n$	write A[i1, i2] (i0=0); read A[i1, i4] (i0=0, i3=0, i4=0) (+2)
n^2	0.438	level	1	$(7/16) \cdot n^2$	$0.146/n$	read A[i1, i2] (i0=0)
n^2	0.334	ramp	$n + 1 \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 1$	$n - 3$	$0.333 \cdot n^{-2}$	read C[i4, i3] (i0=0, i4=0)
n^2	0.334	ramp	$n \rightarrow (1/4) \cdot n^2 + (1/4) \cdot n - 2$	$n - 3$	$0.333 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0)
n^2	0.333	ramp	$(3/4) \cdot n + 4 \rightarrow (1/4) \cdot n^2 + 2$	$n - 3$	$0.333 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)
n^2	0.333	ramp	$(3/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + 1$	$n - 3$	$0.333 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0, i4=0)
n^2	0.333	ramp	$(3/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2$	$n - 3$	$0.333 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0, i4=0)
n^2	0.293	ramp	$(3/4) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n + 1$	$(7/8) \cdot n - 1$	$0.292 \cdot n^{-2}$	read C[i4, i3] (i0=0)
n^2	0.293	ramp	$(3/4) \cdot n + 1 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n$	$(7/8) \cdot n - 1$	$0.292 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.292	ramp	$(3/4) \cdot n + 4 \rightarrow (1/4) \cdot n^2 + 2$	$(7/8) \cdot n - 2$	$0.292 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.251	ramp	$n + 2 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n + 1$	$(3/4) \cdot n - 1$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0)
n^2	0.251	ramp	$n + 2 \rightarrow (1/4) \cdot n^2 + (1/8) \cdot n + 1$	$(3/4) \cdot n - 1$	$0.25 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.251	ramp	$(1/2) \cdot n + 4 \rightarrow (1/4) \cdot n^2 + 2$	$(3/4) \cdot n$	$0.25 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.251	ramp	$(1/2) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + 1$	$(3/4) \cdot n$	$0.25 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.0824	ramp	$(5/2) \cdot n + 4 \rightarrow (1/4) \cdot n^2 + (-11/8) \cdot n + 2$	$(1/4) \cdot n - 2$	$0.0833 \cdot n^{-2}$	read B[i4, i3] (i0=0); read C[i4, i3] (i0=0)
n^2	0.0414	ramp	$(11/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + (-9/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0)
n^2	0.0414	ramp	$(11/4) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + (-9/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-5/4) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 3 \rightarrow (1/4) \cdot n^2 + (-11/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0)
n^2	0.0412	ramp	$(5/2) \cdot n + 2 \rightarrow (1/4) \cdot n^2 + (-11/8) \cdot n$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 6 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n + 2$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)
n^2	0.0411	ramp	$(9/4) \cdot n + 5 \rightarrow (1/4) \cdot n^2 + (-3/2) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.0409	ramp	$2 \cdot n + 5 \rightarrow (1/4) \cdot n^2 + (-7/4) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i3=0)
n^2	0.0409	ramp	$2 \cdot n + 4 \rightarrow (1/4) \cdot n^2 + (-7/4) \cdot n$	$(1/8) \cdot n - 1$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
$n^{1.5}$	0.238	ramp	$2 \rightarrow (1/8) \cdot n$	$n - 8$	$0.333 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=0)
$n^{1.5}$	0.0941	ramp	$(1/2) \cdot n + 5 \rightarrow (3/4) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=2, i4=0)
$n^{1.5}$	0.0939	ramp	$(1/2) \cdot n + 4 \rightarrow (3/4) \cdot n - 2$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=2, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.0729	ramp	$(1/4) \cdot n + 5$ $\rightarrow (1/2) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=1, i4=0)
$n^{1.5}$	0.0723	ramp	$(1/4) \cdot n + 3$ $\rightarrow (1/2) \cdot n - 3$	$(1/8) \cdot n - 2$	$0.0417 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
n^1	12.2	level	4	$(49/8) \cdot n$	$2.04 \cdot n^{-2}$	read C[i1, i3] (i0=0, i4=0)
n^1	1.96	level	5	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read C[i4, i3] (i0=0, i1=1, i4=0)
n^1	1.96	level	5	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read C[i1, i3] (i0=0, i1=0, i4=0)
n^1	1.24	level	2	$(7/8) \cdot n$	$0.292 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=0, i4=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=2, i4=0)
$n^{0.5}$	0.866	level	$(3/4) \cdot n$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=2, i4=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n - 1$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 3$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=2, i3=0, i4=0)
$n^{0.5}$	0.707	level	$(1/2) \cdot n + 2$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=2, i3=0, i4=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 3$	1	$0.333 \cdot n^{-3}$	read C[i4, i3] (i0=0, i1=1, i3=0, i4=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 1$	1	$0.333 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=1, i3=0, i4=0)

Two rank-k sources (B and C) double syrks's structure: ramps to $(1/4)n^2 + (1/4)n$ lines with combined $0.0466 \cdot n^4$ ($\approx 2.8 \times$ syrks, two matrices and a wider window), headroom +1.0.

syrk — infinite-repeat [exact]

Accesses $A(n) = 2 \cdot n^3 + 3 \cdot n^2 + n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0166 \cdot n^4 + 0.416 \cdot n^{3.5} + 2.59 \cdot n^3 + 2.74 \cdot n^{2.5} + 7.9 \cdot n^2 + 0.291 \cdot n^{1.5} + 27.4 \cdot n^1 + 24.3 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0146	ramp	$(3/8) \cdot n + 4$ → $(1/8) \cdot n^2 +$ $(1/4) \cdot n - 1$	$(7/128) \cdot n^3$ + (- $35/32) \cdot n^2$ + $(29/8) \cdot n -$ 2	0.0273	read B[i4, i3] (i0=0)
n^4	0.00204	ramp	$(5/4) \cdot n + 6$ → $(1/8) \cdot n^2 +$ $(1/4) \cdot n - 1$	$(1/128) \cdot n^3$ + $(-7/32) \cdot n^2$ + $(7/4) \cdot n - 4$	0.00391	read B[i4, i3] (i0=0)
$n^{3.5}$	0.322	ramp	5 → $(9/8) \cdot n$	$(49/128) \cdot n^3$ + (- $49/32) \cdot n^2$ + $(7/8) \cdot n$	0.191	read B[i4, i3] (i0=0)
$n^{3.5}$	0.045	ramp	13 → $(9/8) \cdot n$	$(7/128) \cdot n^3$ + (- $21/32) \cdot n^2$ + $(7/4) \cdot n$	0.0273	read B[i4, i3] (i0=0)
$n^{3.5}$	0.0425	ramp	13 → $(9/8) \cdot n - 8$	$(7/128) \cdot n^3$ + (- $91/64) \cdot n^2$ + $(35/4) \cdot n$	0.0273	read A[i1, i4] (i0=0)
$n^{3.5}$	0.00593	ramp	14 → $(9/8) \cdot n - 7$	$(1/128) \cdot n^3$ + (- $17/64) \cdot n^2$ + $(23/8) \cdot n -$ 10	0.00391	read A[i1, i4] (i0=0)
n^3	0.758	level	3	$(7/16) \cdot n^3$ + $n - 1$	0.219	read A[i1, i4] (i0=0, i1=8, i4=0); read B[i1, i3] (i0=0)
n^3	0.758	level	3	$(7/16) \cdot n^3$ + $(-7/8) \cdot n^2$	0.219	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^3	0.438	level	1	$(7/16) \cdot n^3$	0.219	read A[i1, i4] (i0=0)
n^3	0.14	ramp	$(3/8) \cdot n + 3$ → $(1/8) \cdot n^2 +$ $(1/8) \cdot n + 1$	$(1/2) \cdot n^2 +$ $(-5/2) \cdot n + 3$	0.25/n	read B[i4, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.125	level	4	$(1/16) \cdot n^3 + (-1/2) \cdot n^2$	0.0312	read B[i4, i3] (i0=0, i1=1, i3=0, i4=0); read B[i1, i3] (i0=0)
n^3	0.108	level	3	$(1/16) \cdot n^3 + (3/8) \cdot n^2$	0.0312	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^3	0.106	ramp	$(1/2) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n$	$(3/8) \cdot n^2 + (-9/8) \cdot n$	0.188/n	read B[i4, i3] (i0=0)
n^3	0.0381	ramp	$(17/4) \cdot n - 7 \rightarrow (3/16) \cdot n^2 + (1/2) \cdot n - 1$	$(1/8) \cdot n^2 + (-33/8) \cdot n + 34$	0.0625/n	read B[i1, i3] (i0=0, i4=0)
n^3	0.028	ramp	$(1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0625/n	read B[i4, i3] (i0=0, i4=0)
n^3	0.0271	level	$(3/16) \cdot n^2 + (1/2) \cdot n$	$(1/16) \cdot n^2 + (-13/8) \cdot n + 10$	0.0312/n	read A[i1, i2] (i0=0)
n^3	0.0246	ramp	$(1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0547/n	read B[i4, i3] (i0=0)
n^3	0.0173	ramp	$(5/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n$	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	0.0312/n	read B[i4, i3] (i0=0)
n^3	0.0172	ramp	$(11/8) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	0.0312/n	read B[i4, i3] (i0=0)
n^3	0.00346	ramp	$(9/8) \cdot n + 6 \rightarrow (1/8) \cdot n^2 + (-1/2) \cdot n - 1$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00781/n	read B[i4, i3] (i0=0)
$n^{2.5}$	0.698	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$n^2 - 8 \cdot n$	0.5/n	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.683	ramp	$4 \rightarrow (9/8) \cdot n - 8$	$n^2 - 10 \cdot n + 9$	0.5/n	read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.621	ramp	$4 \rightarrow (9/8) \cdot n$	$(7/8) \cdot n^2 + (-7/4) \cdot n$	0.438/n	read B[i4, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.544	ramp	$4 \rightarrow (9/8) \cdot n$	$(49/64) \cdot n^2 + (-7/8) \cdot n$	$0.383/n$	read B[i4, i3] (i0=0)
$n^{2.5}$	0.11	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(11/64) \cdot n^2 + (-13/4) \cdot n + 8$	$0.0859/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.0766	ramp	$12 \rightarrow (9/8) \cdot n - 6$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0547/n$	read B[i4, i3] (i0=0)
$n^{2.5}$	0.0103	ramp	$13 \rightarrow (9/8) \cdot n - 14$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00781/n$	read A[i1, i4] (i0=0)
n^2	1.75	level	1	$(7/4) \cdot n^2$	$0.875/n$	read A[i1, i2] (i0=0); write A[i1, i2] (i0=0) (+1)
n^2	1.52	level	3	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i1, i3] (i0=0, i4=0)
n^2	1.24	level	2	$(7/8) \cdot n^2$	$0.438/n$	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^2	0.433	level	$(3/16) \cdot n^2 + (1/2) \cdot n$	$n - 9$	$0.5 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0)
n^2	0.433	level	$(3/16) \cdot n^2 + (1/2) \cdot n$	$n - 9$	$0.5 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.372	ramp	$(1/8) \cdot n^2 + (1/8) \cdot n + 10 \rightarrow (3/16) \cdot n^2 + (1/2) \cdot n$	$n - 9$	$0.5 \cdot n^{-2}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^2	0.237	ramp	$(13/4) \cdot n + 8 \rightarrow (3/16) \cdot n^2 + (1/4) \cdot n + 2$	$(7/8) \cdot n - 14$	$0.438 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.237	ramp	$(13/4) \cdot n - 8 \rightarrow (1/8) \cdot n^2 + (-7/8) \cdot n + 40$	$n - 16$	$0.5 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.237	ramp	$(3/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0)
n^2	0.236	ramp	$(1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.207	ramp	$(3/8) \cdot n + 2$ → $(1/8) \cdot n^2 +$ $(1/8) \cdot n + 1$	$(7/8) \cdot n - 1$	$0.438 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.204	ramp	$(19/8) \cdot n - 2$ → $(1/8) \cdot n^2 +$ $(-7/8) \cdot n +$ 16	$(7/8) \cdot n - 14$	$0.438 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.188	level	1	$(3/16) \cdot n^2$ + $(1/2) \cdot n$	$0.0938/n$	read B[i4, i3] (i0=0, i1=0, i4=0); write A[i1, i2] (i0=0) (+1)
n^2	0.178	ramp	$(1/4) \cdot n + 3$ → $(1/8) \cdot n^2 +$ $(1/8) \cdot n + 1$	$(3/4) \cdot n$	$0.375 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.177	level	2	$(1/8) \cdot n^2$	$0.0625/n$	write A[i1, i4] (i0=0, i3=0); write A[i1, i4] (i0=0)
n^2	0.0541	level	$(3/16) \cdot n^2$ + $(1/2) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.0541	level	$(3/16) \cdot n^2$ + $(3/8) \cdot n +$ 1	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^2	0.033	ramp	$(33/8) \cdot n +$ 22 → $(3/16) \cdot n^2$ + $(-3/2) \cdot n +$ 10	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^2	0.0292	ramp	$(5/4) \cdot n + 3$ → $(1/8) \cdot n^2 +$ $(-5/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.0291	ramp	$(9/8) \cdot n + 5$ → $(1/8) \cdot n^2 +$ $(-5/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.029	ramp	$n + 4$ → $(1/8) \cdot n^2 +$ $(-3/4) \cdot n$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.0289	ramp	$(9/4) \cdot n - 1$ → $(1/8) \cdot n^2 +$ $(-7/8) \cdot n + 2$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i1, i3] (i0=0, i1=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{1.5}$	0.238	ramp	$2 \rightarrow (1/8) \cdot n$	$n - 8$	$0.5 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=0)
$n^{1.5}$	0.0522	ramp	$(1/8) \cdot n + 4$ $\rightarrow (1/4) \cdot n +$ 1	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
n^1	9.9	level	2	$7 \cdot n$	$3.5 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^1	8	level	1	$8 \cdot n$	$4 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0); read A[i1, i4] (i0=0, i3=0, i4=0) (+1)
n^1	3.03	level	$(3/16) \cdot n^2$ $+ (1/2) \cdot n$	7	$3.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
n^1	0.433	level	$(3/16) \cdot n^2$ $+ (1/2) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
n^1	0.433	level	$(3/16) \cdot n^2$ $+ (3/8) \cdot n +$ 1	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2$ $+ (3/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0)
n^1	0.433	level	$(3/16) \cdot n^2$ $+ (1/2) \cdot n$	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2$ $+ (1/2) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(1/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(1/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(1/8) \cdot n + 4$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(1/8) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 +$ $(1/8) \cdot n + 6$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 8$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i3=0, i4=0)
n^1	0.354	level	$(1/8) \cdot n^2 + (1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i1=0, i3=0, i4=0)
$n^{0.5}$	1.77	level	$(25/8) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.73	level	$3 \cdot n + 6$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.7	level	$(23/8) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.66	level	$(11/4) \cdot n + 4$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.62	level	$(21/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.58	level	$(5/2) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.54	level	$(19/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.5	level	$(9/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.46	level	$(17/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i1=7, i4=0)
$n^{0.5}$	1.41	level	$2 \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.37	level	$(15/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.32	level	$(7/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.27	level	$(13/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.22	level	$(3/2) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	1.17	level	$(11/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i4=0)
$n^{0.5}$	1.12	level	$(5/4) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i1, i3] (i0=0, i1=0, i4=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.354	level	$(1/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=1, i3=0, i4=0)

The rank-k source read B[i4,i3] is re-read for every output row: ramp families from $(3/8)n + 4$ up to $(1/8)n^2 + (1/4)n$ lines (the growing triangular window) with population $n^{3/16}$, giving $0.0166 \cdot n^4 - \text{headroom} + 1.0$. The old headroom-0 reading was an artifact: the ramp values collapsed to constants under the rendering bug and the scale approximation could not see the whole-matrix distances. The $n^{3.5}$ terms are row-window reuses (up to $(9/8)n$ lines) of both A and B. The cache cliff sits at $64 \cdot ((1/8)n^2)$ bytes, a constant factor past gemm's — matching the measured $0.04\% \rightarrow 1.9\%$ jump at $N \approx 560$ in the empirical section.

syrk — single-shot [exact]

Accesses $A(n) = 2 \cdot n^3 + 3 \cdot n^2 + n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0166 \cdot n^4 + 0.416 \cdot n^{3.5} + 2.52 \cdot n^3 + 2.74 \cdot n^{2.5} + 5.81 \cdot n^2 + 0.291 \cdot n^{1.5} + 11.4 \cdot n^1 + 0.854 \cdot n^{0.5}$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0146	ramp	$(3/8) \cdot n + 4$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(1/4) \cdot n - 1$	$(7/128) \cdot n^3$ $+ (-$ $35/32) \cdot n^2$ $+ (29/8) \cdot n -$ 2	0.0273	read B[i4, i3] (i0=0)
n^4	0.00204	ramp	$(5/4) \cdot n + 6$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(1/4) \cdot n - 1$	$(1/128) \cdot n^3$ $+ (-7/32) \cdot n^2$ $+ (7/4) \cdot n - 4$	0.00391	read B[i4, i3] (i0=0)
$n^{3.5}$	0.322	ramp	$5 \rightarrow (9/8) \cdot n$	$(49/128) \cdot n^3$ $+ (-$ $49/32) \cdot n^2$ $+ (7/8) \cdot n$	0.191	read B[i4, i3] (i0=0)
$n^{3.5}$	0.045	ramp	$13 \rightarrow$ $(9/8) \cdot n$	$(7/128) \cdot n^3$ $+ (-$ $21/32) \cdot n^2$ $+ (7/4) \cdot n$	0.0273	read B[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0425	ramp	$13 \rightarrow (9/8) \cdot n - 8$	$(7/128) \cdot n^3 + (-91/64) \cdot n^2 + (35/4) \cdot n$	0.0273	read A[i1, i4] (i0=0)
n^3	0.00593	ramp	$14 \rightarrow (9/8) \cdot n - 7$	$(1/128) \cdot n^3 + (-17/64) \cdot n^2 + (23/8) \cdot n - 10$	0.00391	read A[i1, i4] (i0=0)
n^3	1.62	level	3	$(15/16) \cdot n^3 + (-1/2) \cdot n^2 + n - 1$	0.469	read A[i1, i4] (i0=0, i1=8, i4=0); read B[i1, i3] (i0=0) (+1)
n^3	0.438	level	1	$(7/16) \cdot n^3$	0.219	read A[i1, i4] (i0=0)
n^3	0.14	ramp	$(3/8) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 1$	$(1/2) \cdot n^2 + (-5/2) \cdot n + 3$	0.25/n	read B[i4, i3] (i0=0, i3=0)
n^3	0.125	level	4	$(1/16) \cdot n^3 + (-1/2) \cdot n^2$	0.0312	read B[i4, i3] (i0=0, i1=1, i3=0, i4=0); read B[i1, i3] (i0=0)
n^3	0.106	ramp	$(1/2) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n$	$(3/8) \cdot n^2 + (-9/8) \cdot n$	0.188/n	read B[i4, i3] (i0=0)
n^3	0.028	ramp	$(1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/8) \cdot n^2 + (-9/4) \cdot n + 4$	0.0625/n	read B[i4, i3] (i0=0, i4=0)
n^3	0.0246	ramp	$(1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0547/n	read B[i4, i3] (i0=0)
n^3	0.0173	ramp	$(5/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n$	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	0.0312/n	read B[i4, i3] (i0=0)
n^3	0.0172	ramp	$(11/8) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n - 1$	$(1/16) \cdot n^2 + (-3/4) \cdot n + 2$	0.0312/n	read B[i4, i3] (i0=0)
n^3	0.00346	ramp	$(9/8) \cdot n + 6 \rightarrow (1/8) \cdot n^2 + (-1/2) \cdot n - 1$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00781/n	read B[i4, i3] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.698	ramp	$11 \rightarrow (9/8) \cdot n - 7$	$n^2 - 8 \cdot n$	$0.5/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.683	ramp	$4 \rightarrow (9/8) \cdot n - 8$	$n^2 - 10 \cdot n + 9$	$0.5/n$	read A[i1, i4] (i0=0, i4=0)
$n^{2.5}$	0.621	ramp	$4 \rightarrow (9/8) \cdot n$	$(7/8) \cdot n^2 + (-7/4) \cdot n$	$0.438/n$	read B[i4, i3] (i0=0, i4=0)
$n^{2.5}$	0.544	ramp	$4 \rightarrow (9/8) \cdot n$	$(49/64) \cdot n^2 + (-7/8) \cdot n$	$0.383/n$	read B[i4, i3] (i0=0)
$n^{2.5}$	0.11	ramp	$12 \rightarrow (9/8) \cdot n - 15$	$(11/64) \cdot n^2 + (-13/4) \cdot n + 8$	$0.0859/n$	read A[i1, i4] (i0=0, i3=0); read A[i1, i4] (i0=0)
$n^{2.5}$	0.0766	ramp	$12 \rightarrow (9/8) \cdot n - 6$	$(7/64) \cdot n^2 + (-7/8) \cdot n$	$0.0547/n$	read B[i4, i3] (i0=0)
$n^{2.5}$	0.0103	ramp	$13 \rightarrow (9/8) \cdot n - 14$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00781/n$	read A[i1, i4] (i0=0)
n^2	1.52	level	3	$(7/8) \cdot n^2 - 7 \cdot n$	$0.438/n$	read B[i1, i3] (i0=0, i4=0)
n^2	1.5	level	1	$(3/2) \cdot n^2 + (17/2) \cdot n$	$0.75/n$	write A[i1, i2] (i0=0); read A[i1, i4] (i0=0, i3=0, i4=0) (+2)
n^2	1.41	level	2	n^2	$0.5/n$	write A[i1, i4] (i0=0)
n^2	0.438	level	1	$(7/16) \cdot n^2$	$0.219/n$	read A[i1, i2] (i0=0)
n^2	0.237	ramp	$(3/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/4) \cdot n$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i4=0)
n^2	0.236	ramp	$(1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 1$	$n - 2$	$0.5 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0, i4=0)
n^2	0.207	ramp	$(3/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 1$	$(7/8) \cdot n - 1$	$0.438 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.178	ramp	$(1/4) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n + 1$	$(3/4) \cdot n$	$0.375 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0292	ramp	$(5/4) \cdot n + 3$ → $(1/8) \cdot n^2 +$ $(-5/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0)
n^2	0.0291	ramp	$(9/8) \cdot n + 5$ → $(1/8) \cdot n^2 +$ $(-5/8) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
n^2	0.029	ramp	$n + 4$ → $(1/8) \cdot n^2 +$ $(-3/4) \cdot n$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0, i3=0)
$n^{1.5}$	0.238	ramp	$2 \rightarrow (1/8) \cdot n$	$n - 8$	$0.5 \cdot n^{-2}$	read A[i1, i4] (i0=0, i3=0, i4=0)
$n^{1.5}$	0.0522	ramp	$(1/8) \cdot n + 4$ → $(1/4) \cdot n +$ 1	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
n^1	9.9	level	2	$7 \cdot n$	$3.5 \cdot n^{-2}$	read B[i1, i3] (i0=0, i4=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i4, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=1, i4=0)
$n^{0.5}$	0.354	level	$(1/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i4, i3] (i0=0, i1=1, i3=0, i4=0)

The rank- k source read B[i4,i3] is re-read for every output row: ramp families from $(3/8)n + 4$ up to $(1/8)n^2 + (1/4)n$ lines (the growing triangular window) with population $n^{3/16}$, giving $0.0166 \cdot n^4$ — headroom +1.0. The old headroom-0 reading was an artifact: the ramp values collapsed to constants under the rendering bug and the scale approximation could not see the whole-matrix distances. The $n^{3.5}$ terms are row-window reuses (up to $(9/8)n$ lines) of both A and B. The cache cliff sits at $64 \cdot ((1/8)n^2)$ bytes, a constant factor past gemm's — matching the measured $0.04\% \rightarrow 1.9\%$ jump at $N \approx 560$ in the empirical section.

trisolve — infinite-repeat [exact]

Accesses $A(n) = 2 \cdot n^2 + 3 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^3 , headroom +1; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0156 \cdot n^3 + 0.0226 \cdot n^{2.5} + 3.52 \cdot n^2 + 1.02 \cdot n^{1.5} + 38.7 \cdot n^1 + 1 \cdot n^{0.5} + 50.8 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0137	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(7/128) \cdot n^2 + (-23/16) \cdot n + 9$	0.0273	read C[i0, i1]
n^3	0.00195	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.00391	read C[i0, i1]
$n^{2.5}$	0.0171	ramp	$9 \rightarrow (1/4) \cdot n + 1$	$(3/64) \cdot n^2 + (-15/8) \cdot n + 18$	0.0234	read B[i1]
$n^{2.5}$	0.00285	ramp	$9 \rightarrow (1/4) \cdot n + 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.00391	read B[i1]
$n^{2.5}$	0.00269	ramp	$10 \rightarrow (1/4) \cdot n$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00391	read B[i1]
n^2	1.52	level	3	$(7/8) \cdot n^2 + (-105/8) \cdot n + 49$	0.438	read C[i0, i1]; read B[i1]
n^2	0.568	level	3	$(21/64) \cdot n^2 + (-21/4) \cdot n + 21$	0.164	write B[i0]
n^2	0.438	level	1	$(7/16) \cdot n^2 + (7/8) \cdot n - 7$	0.219	read B[i0] (i1=0);
n^2	0.219	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(7/8) \cdot n - 8$	0.438/n	read B[i0]
n^2	0.219	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(7/8) \cdot n - 8$	0.438/n	read C[i0, i1]
n^2	0.0947	level	3	$(7/128) \cdot n^2 + (-21/16) \cdot n + 7$	0.0273	read C[i0, i1] (i1=0)
n^2	0.0947	level	3	$(7/128) \cdot n^2 + (-5/16) \cdot n - 1$	0.0273	write B[i0]
n^2	0.0812	level	3	$(3/64) \cdot n^2 + (-3/8) \cdot n$	0.0234	write B[i0]
n^2	0.0625	level	1	$(1/16) \cdot n^2 + (-3/8) \cdot n - 1$	0.0312	read B[i0] (i1=0);
n^2	0.0312	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (35/16)$	$(1/8) \cdot n + (-17/8)$	0.0625/n	read B[i0]
n^2	0.0312	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(1/8) \cdot n - 2$	0.0625/n	read C[i0, i1]
n^2	0.0312	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(1/8) \cdot n - 3$	0.0625/n	read C[i0, i1]
n^2	0.0312	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(1/8) \cdot n - 2$	0.0625/n	read C[i0, i1] (i1=0)
n^2	0.0312	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	$(1/8) \cdot n - 2$	0.0625/n	read C[i0, i0]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0291	ramp	$(1/16) \cdot n^2 + (-1/8) \cdot n + 7 \rightarrow$ $(1/16) \cdot n^2 + (3/4) \cdot n - 21$	$(1/8) \cdot n - 3$	$0.0625/n$	read A[i0]
n^2	0.0271	level	3	$(1/64) \cdot n^2 + (-3/8) \cdot n + 5$	0.00781	read A[i0] (i0=0); read C[i0, i0] (i0=0) (+3)
n^2	0.0146	ramp	$(1/4) \cdot n + 8 \rightarrow$ $(1/16) \cdot n^2 + (-9/4) \cdot n + 24$	$(1/8) \cdot n - 3$	$0.0625/n$	write B[i0]
$n^{1.5}$	0.257	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(3/4) \cdot n - 6$	$0.375/n$	read A[i0]
$n^{1.5}$	0.254	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(3/4) \cdot n - 12$	$0.375/n$	read B[i1] (i1=0)
$n^{1.5}$	0.254	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(3/4) \cdot n - 12$	$0.375/n$	read B[i1]
$n^{1.5}$	0.0428	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(1/8) \cdot n - 1$	$0.0625/n$	read A[i0]
$n^{1.5}$	0.0424	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625/n$	read B[i1] (i1=0)
$n^{1.5}$	0.0424	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625/n$	read B[i1]
$n^{1.5}$	0.0418	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625/n$	read B[i1]
$n^{1.5}$	0.0414	ramp	$8 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 3$	$0.0625/n$	read B[i1]
$n^{1.5}$	0.0414	ramp	$8 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 3$	$0.0625/n$	read B[i1] (i1=0)
n^1	10.6	level	3	$(49/8) \cdot n - 49$	$3.06/n$	read C[i0, i1]; read B[i1]
n^1	7.42	level	2	$(21/4) \cdot n$	$2.62/n$	read C[i0, i1]; read B[i1]
n^1	2.47	level	2	$(7/4) \cdot n$	$0.875/n$	write B[i0]
n^1	2.17	level	3	$(5/4) \cdot n - 10$	$0.625/n$	write B[i0]
n^1	1.41	level	2	$n - 1$	$0.5/n$	read C[i0, i0]; write B[i0]
n^1	1.38	level	1	$(11/8) \cdot n$	$0.688/n$	write B[i0]
n^1	1.3	level	3	$(3/4) \cdot n - 6$	$0.375/n$	write B[i0]
n^1	1.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	5	$2.5 \cdot n^{-2}$	read C[i0, i1] (i1=0)
n^1	1.25	level	1	$(5/4) \cdot n$	$0.625/n$	write B[i0]
n^1	1.08	level	3	$(5/8) \cdot n - 5$	$0.312/n$	write B[i0]
n^1	1.06	level	2	$(3/4) \cdot n - 6$	$0.375/n$	read B[i1]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.5	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	2	$1 \cdot n^{-2}$	read C[i0, i1] (i0=7, i1=0); read C[i0, i1] (i0=8, i1=0)
n^1	0.5	level	1	$(1/2) \cdot n$	$0.25/n$	write B[i0]
n^1	0.433	level	3	$(1/4) \cdot n - 2$	$0.125/n$	write B[i0]
n^1	0.433	level	3	$(1/4) \cdot n - 2$	$0.125/n$	write B[i0]
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	1	$0.5 \cdot n^{-2}$	read C[i0, i1]
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	1	$0.5 \cdot n^{-2}$	read C[i0, i1]
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (35/16)$	1	$0.5 \cdot n^{-2}$	read C[i0, i1] (i1=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	1	$0.5 \cdot n^{-2}$	read C[i0, i1] (i1=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	1	$0.5 \cdot n^{-2}$	read C[i0, i1] (i0=1, i1=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	1	$0.5 \cdot n^{-2}$	read C[i0, i0] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (35/16)$	1	$0.5 \cdot n^{-2}$	read A[i0]
n^1	0.25	level	$(1/16) \cdot n^2 + (-1/8) \cdot n$	1	$0.5 \cdot n^{-2}$	read A[i0] (i0=8)
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n$	1	$0.5 \cdot n^{-2}$	read C[i0, i0]
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n + (35/16)$	1	$0.5 \cdot n^{-2}$	read C[i0, i0]
n^1	0.25	level	$(1/16) \cdot n^2 + (-5/4) \cdot n + 8$	1	$0.5 \cdot n^{-2}$	write B[i0]
n^1	0.25	level	$(1/16) \cdot n^2 + (-1/4) \cdot n$	1	$0.5 \cdot n^{-2}$	write B[i0]
n^1	0.25	level	$(1/16) \cdot n^2 + (3/4) \cdot n - 7$	1	$0.5 \cdot n^{-2}$	read A[i0] (i0=0)
n^1	0.25	level	$(1/16) \cdot n^2 + (-3/8) \cdot n + (37/16)$	1	$0.5 \cdot n^{-2}$	write B[i0]
n^1	0.217	level	3	$(1/8) \cdot n - 2$	$0.0625/n$	write B[i0] (i1=0)
n^1	0.217	level	3	$(1/8) \cdot n - 1$	$0.0625/n$	write B[i0]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.217	level	3	$(1/8) \cdot n - 1$	$0.0625/n$	write B[i0]
n^1	0.217	level	3	$(1/8) \cdot n - 2$	$0.0625/n$	write B[i0]
n^1	0.177	level	2	$(1/8) \cdot n - 1$	$0.0625/n$	read B[i1]
n^1	0.125	level	1	$(1/8) \cdot n$	$0.0625/n$	write B[i0]
n^1	0.125	level	1	$(1/8) \cdot n$	$0.0625/n$	write B[i0]
n^1	0.125	level	1	$(1/8) \cdot n - 1$	$0.0625/n$	write B[i0]
$n^{0.5}$	0.5	level	$(1/4) \cdot n + (7/4)$	1	$0.5 \cdot n^{-2}$	write B[i0] (i0=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n$	1	$0.5 \cdot n^{-2}$	write B[i0] (i0=0)
n^0	13.4	level	5	6	$3 \cdot n^{-2}$	read B[i1] (i1=0)
n^0	11.3	level	2	8	$4 \cdot n^{-2}$	write B[i0] (i0=0); read B[i1] (i0=1, i1=0) (+1)
n^0	10.4	level	3	6	$3 \cdot n^{-2}$	read A[i0]
n^0	9	level	1	9	$4.5 \cdot n^{-2}$	read B[i0] (i0=0); read B[i0] (i0=1, i1=0) (+2)
n^0	2.45	level	6	1	$0.5 \cdot n^{-2}$	read B[i1] (i0=16, i1=0)
n^0	2.24	level	5	1	$0.5 \cdot n^{-2}$	read B[i1] (i0=9, i1=0)
n^0	2	level	4	1	$0.5 \cdot n^{-2}$	write B[i0] (i0=0); read C[i0, i1] (i0=1, i1=0) (+1)

The triangular matrix returns across passes: levels at distance $(1/16)n^2 + (3/4)n$ — the triangular footprint in lines — with population $\sim n^2/16$, giving the n^3 order (0.0156). trisolve is the cleanest case of a kernel whose only asymptotic locality lives across invocations.

trisolve — single-shot [exact]

Accesses $A(n) = 2 \cdot n^2 + 3 \cdot n$ (exact on $n \equiv 0 \pmod{8}$); DMD order $n^{2.5}$, headroom **+0.5**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0226 \cdot n^{2.5} + 2.88 \cdot n^2 + 1.02 \cdot n^{1.5} + 34.5 \cdot n^1 + 40.7 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.0171	ramp	$9 \rightarrow (1/4) \cdot n + 1$	$(3/64) \cdot n^2 + (-15/8) \cdot n + 18$	0.0234	read B[i1]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.00285	ramp	$9 \rightarrow (1/4) \cdot n + 1$	$(1/128) \cdot n^2 + (-5/16) \cdot n + 3$	0.00391	read B[i1]
$n^{2.5}$	0.00269	ramp	$10 \rightarrow (1/4) \cdot n$	$(1/128) \cdot n^2 + (-7/16) \cdot n + 6$	0.00391	read B[i1]
n^2	2.27	level	3	$(21/16) \cdot n^2 - 21 \cdot n + 84$	0.656	read C[i0, i1]; read B[i1] (+1)
n^2	0.5	level	1	$(1/2) \cdot n^2 + (1/2) \cdot n$	0.25	read B[i0] (i0=0); read B[i0] (i1=0) (+1)
n^2	0.108	level	3	$(1/16) \cdot n^2 + (-5/8) \cdot n + 1$	0.0312	write B[i0]
$n^{1.5}$	0.257	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(3/4) \cdot n - 6$	0.375/n	read A[i0]
$n^{1.5}$	0.254	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(3/4) \cdot n - 12$	0.375/n	read B[i1] (i1=0)
$n^{1.5}$	0.254	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(3/4) \cdot n - 12$	0.375/n	read B[i1]
$n^{1.5}$	0.0428	ramp	$5 \rightarrow (1/4) \cdot n + 1$	$(1/8) \cdot n - 1$	0.0625/n	read A[i0]
$n^{1.5}$	0.0424	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(1/8) \cdot n - 2$	0.0625/n	read B[i1] (i1=0)
$n^{1.5}$	0.0424	ramp	$7 \rightarrow (1/4) \cdot n + 1$	$(1/8) \cdot n - 2$	0.0625/n	read B[i1]
$n^{1.5}$	0.0418	ramp	$6 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	0.0625/n	read B[i1]
$n^{1.5}$	0.0414	ramp	$8 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 3$	0.0625/n	read B[i1]
$n^{1.5}$	0.0414	ramp	$8 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 3$	0.0625/n	read B[i1] (i1=0)
n^1	6.06	level	3	$(7/2) \cdot n - 28$	1.75/n	read C[i0, i1]
n^1	4.55	level	3	$(21/8) \cdot n - 21$	1.31/n	read B[i1]
n^1	4.55	level	3	$(21/8) \cdot n - 21$	1.31/n	write B[i0]
n^1	3.71	level	2	$(21/8) \cdot n$	1.31/n	read C[i0, i1]
n^1	3.71	level	2	$(21/8) \cdot n$	1.31/n	read B[i1]
n^1	2.62	level	1	$(21/8) \cdot n$	1.31/n	write B[i0]
n^1	2.47	level	2	$(7/4) \cdot n$	0.875/n	write B[i0]; read C[i0, i0]
n^1	1.52	level	3	$(7/8) \cdot n - 7$	0.438/n	write B[i0]
n^1	1.52	level	3	$(7/8) \cdot n - 7$	0.438/n	write B[i0]
n^1	1.41	level	2	$n + 1$	0.5/n	read B[i1] (i0=7, i1=0); write B[i0]
n^1	1.06	level	2	$(3/4) \cdot n - 6$	0.375/n	read B[i1]

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.75	level	1	$(3/4) \cdot n$	$0.375/n$	write B[i0]
n^1	0.217	level	3	$(1/8) \cdot n - 1$	$0.0625/n$	write B[i0]
n^1	0.177	level	2	$(1/8) \cdot n - 1$	$0.0625/n$	read B[i1]
n^1	0.125	level	1	$(1/8) \cdot n$	$0.0625/n$	write B[i0]
n^0	13.4	level	5	6	$3 \cdot n^{-2}$	read B[i1] (i1=0)
n^0	10.4	level	3	6	$3 \cdot n^{-2}$	read A[i0]
n^0	8.49	level	2	6	$3 \cdot n^{-2}$	read B[i1] (i1=0)
n^0	2.45	level	6	1	$0.5 \cdot n^{-2}$	read B[i1] (i0=16, i1=0)
n^0	2.24	level	5	1	$0.5 \cdot n^{-2}$	read B[i1] (i0=9, i1=0)
n^0	2	level	4	1	$0.5 \cdot n^{-2}$	read B[i1] (i0=8, i1=0)
n^0	1.73	level	3	1	$0.5 \cdot n^{-2}$	read A[i0] (i0=1)

The solution vector read B[i1] is re-read against a growing prefix: ramp families from ~ 9 lines up to $(1/4)n + 1$ (the vector footprint), $d = 2.5$, headroom $+0.5$. The matrix C is streamed once (triangular row by row) and contributes only line-reuse levels at distance 1-3.

trmm — infinite-repeat [exact]

Accesses $A(n) = 2 \cdot n^3$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0169 \cdot n^4 + 1.05 \cdot n^{3.5} + 1.74 \cdot n^3 + 2.67 \cdot n^{2.5} + 3.68 \cdot n^2 + 0.119 \cdot n^{1.5} + 16.8 \cdot n^1 + 5.54 \cdot n^{0.5} + 2 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0148	ramp	$(3/8) \cdot n + 5$ $\rightarrow$ $(1/8) \cdot n^2 +$ $n - 2$	$(7/128) \cdot n^3$ $+ (-$ $63/64) \cdot n^2$ $+ (13/8) \cdot n +$ 2	0.0273	read A[i3, i2] (i0=0, i3=0); read A[i3, i2] (i0=0)
n^4	0.00207	ramp	$n + 17 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 18$	$(1/128) \cdot n^3$ $+ (-$ $13/64) \cdot n^2$ $+ (11/8) \cdot n -$ 2	0.00391	read A[i3, i2] (i0=0, i3=0); read A[i3, i2] (i0=0)
$n^{3.5}$	0.494	ramp	$5 \rightarrow 2 \cdot n - 1$	$(7/16) \cdot n^3$ $+ (-7/16) \cdot n^2$ $+ (-7/8) \cdot n$	0.219	read B[i3, i1] (i0=0, i3=0); read B[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.487	ramp	$5 \rightarrow 2 \cdot n - 3$	$(7/16) \cdot n^3$ + $(-35/16) \cdot n^2$ + $(21/8) \cdot n$	0.219	read A[i3, i2] (i0=0, i3=0); read A[i3, i2] (i0=0)
$n^{3.5}$	0.0684	ramp	$6 \rightarrow 2 \cdot n$	$(1/16) \cdot n^3$ + $(-11/16) \cdot n^2$ + $(13/8) \cdot n - 1$	0.0312	read B[i3, i1] (i0=0, i3=0); read B[i3, i1] (i0=0)
n^3	0.758	level	3	$(7/16) \cdot n^3$ + $(-7/16) \cdot n^2$	0.219	read A[i3, i2] (i0=0, i1=0, i3=0); write A[i1, i2] (i0=0)
n^3	0.438	level	1	$(7/16) \cdot n^3$ + $(7/16) \cdot n^2$ + $(7/8) \cdot n$	0.219	read A[i1, i2] (i0=0, i3=0); read A[i1, i2] (i0=0) (+1)
n^3	0.125	ramp	$(1/4) \cdot n + 2$ $\rightarrow$ $(1/8) \cdot n^2 +$ $(7/8) \cdot n - 1$	$(7/16) \cdot n^2$ + $(1/8) \cdot n - 3$	0.219/n	read A[i1, i2] (i0=0, i3=0); read A[i3, i2] (i0=0, i3=0) (+1)
n^3	0.125	ramp	$(1/2) \cdot n + 3$ $\rightarrow$ $(1/8) \cdot n^2 +$ $n - 1$	$(7/16) \cdot n^2$ + $(-7/8) \cdot n - 1$	0.219/n	read A[i3, i2] (i0=0, i2=0, i3=0); read A[i3, i2] (i0=0, i2=0)
n^3	0.108	level	3	$(1/16) \cdot n^3$ + $(-1/16) \cdot n^2$ + $(7/8) \cdot n$	0.0312	read A[i3, i2] (i0=0, i3=0); write A[i1, i2] (i0=0, i3=0) (+1)
n^3	0.0625	level	1	$(1/16) \cdot n^3$ + $(-1/16) \cdot n^2$ + $(1/8) \cdot n$	0.0312	read A[i1, i2] (i0=0); write A[i1, i2] (i0=0)
n^3	0.0329	ramp	$(27/8) \cdot n - 5$ $\rightarrow$ $(3/16) \cdot n^2$ + $(-5/8) \cdot n - 7$	$(7/64) \cdot n^2$ + $(-29/8) \cdot n$ + 30	0.0547/n	read A[i3, i2] (i0=0, i1=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^3	0.0271	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	$(1/16) \cdot n^2 + (-3/2) \cdot n + 9$	$0.0312/n$	read B[i3, i1] (i0=0, i2=0, i3=0); read B[i3, i1] (i0=0, i2=0, i3=7) (+1)
n^3	0.0248	ramp	$(1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + n - 2$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	$0.0547/n$	read A[i1, i2] (i0=0, i3=0)
n^3	0.0172	ramp	$(9/8) \cdot n + 10 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 18$	$(1/16) \cdot n^2 + (-7/8) \cdot n + 3$	$0.0312/n$	read A[i3, i2] (i0=0, i2=0)
n^3	0.0172	ramp	$n + 9 \rightarrow (1/8) \cdot n^2 + n - 19$	$(1/16) \cdot n^2 + (-7/8) \cdot n + 3$	$0.0312/n$	read A[i3, i2] (i0=0)
n^3	0.0046	ramp	$(33/8) \cdot n - 4 \rightarrow (3/16) \cdot n^2 + (-5/8) \cdot n - 21$	$(1/64) \cdot n^2 + (-5/8) \cdot n + 6$	$0.00781/n$	read A[i3, i2] (i0=0, i1=0)
n^3	0.00352	ramp	$n + 17 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 18$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	$0.00781/n$	read A[i1, i2] (i0=0, i3=0)
$n^{2.5}$	1.24	level	$2 \cdot n - 1$	$(7/8) \cdot n^2 + (-7/4) \cdot n$	$0.438/n$	read A[i3, i2] (i0=0, i1=0, i3=0); read A[i3, i2] (i0=0, i1=0)
$n^{2.5}$	0.825	ramp	$5 \rightarrow 2 \cdot n - 1$	$(7/8) \cdot n^2 + (-7/4) \cdot n$	$0.438/n$	read A[i3, i2] (i0=0, i1=0); read A[i3, i2] (i0=0)
$n^{2.5}$	0.49	ramp	$6 \rightarrow 2 \cdot n - 2$	$(7/16) \cdot n^2 + (-7/4) \cdot n + 1$	$0.219/n$	read A[i1, i2] (i0=0, i2=0, i3=0); read B[i3, i1] (i0=0, i2=0, i3=0) (+1)
$n^{2.5}$	0.118	ramp	$6 \rightarrow 2 \cdot n$	$(1/8) \cdot n^2 + (-3/8) \cdot n$	$0.0625/n$	read B[i3, i1] (i0=0, i2=0); read B[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	1	level	1	$n^2 - n$	$0.5/n$	read A[i1, i2] (i0=0, i1=0, i2=0); read A[i1, i2] (i0=0)
n^2	0.433	level	$(3/16) \cdot n^2 + (-5/8) \cdot n + 9$	$n - 9$	$0.5 \cdot n^{-2}$	read B[i3, i1] (i0=0, i1=0, i2=0)
n^2	0.375	ramp	$(1/8) \cdot n^2 + n \rightarrow (3/16) \cdot n^2 + (3/8) \cdot n$	$n - 9$	$0.5 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0)
n^2	0.359	ramp	$(3/16) \cdot n^2 + (-11/8) \cdot n + 22 \rightarrow (3/16) \cdot n^2 + (3/8) \cdot n - 8$	$(7/8) \cdot n - 14$	$0.438 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0)
n^2	0.325	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	$(3/4) \cdot n - 7$	$0.375 \cdot n^{-2}$	read B[i3, i1] (i0=0, i2=0)
n^2	0.257	ramp	$(3/16) \cdot n^2 + (-3/8) \cdot n + 12 \rightarrow (3/16) \cdot n^2 + (3/8) \cdot n - 2$	$(5/8) \cdot n - 10$	$0.312 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0)
n^2	0.239	ramp	$(3/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + n - 1$	$n - 2$	$0.5 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0, i3=0)
n^2	0.226	ramp	$(19/8) \cdot n - 4 \rightarrow (3/16) \cdot n^2 + (-29/8) \cdot n + 42$	$(7/8) \cdot n - 15$	$0.438 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^2	0.0541	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read B[i3, i1] (i0=0, i2=0)
n^2	0.0541	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=0, i3=0)
n^2	0.0515	ramp	$(3/16) \cdot n^2 + (1/4) \cdot n + 2 \rightarrow (3/16) \cdot n^2 + (3/8) \cdot n - 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0, i3=0)
n^2	0.0513	ramp	$(3/16) \cdot n^2 + (-1/2) \cdot n + 14 \rightarrow (3/16) \cdot n^2 + (3/8) \cdot n - 7$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0, i3=6)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^2	0.0511	ramp	$(3/16) \cdot n^2 + (-3/2) \cdot n + 23 \rightarrow$ $(3/16) \cdot n^2 + (3/8) \cdot n - 22$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0, i3=14)
n^2	0.0317	ramp	$(25/8) \cdot n - 3 \rightarrow$ $(3/16) \cdot n^2 + (-9/2) \cdot n + 49$	$(1/8) \cdot n - 3$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^2	0.0297	ramp	$(9/8) \cdot n + 15 \rightarrow$ $(1/8) \cdot n^2 + (9/8) \cdot n - 17$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.0297	ramp	$(13/4) \cdot n - 4 \rightarrow$ $(1/8) \cdot n^2 + 2$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0)
n^2	0.0296	ramp	$(25/8) \cdot n - 3 \rightarrow$ $(1/8) \cdot n^2$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i1=0)
n^2	0.0296	ramp	$n + 15 \rightarrow$ $(1/8) \cdot n^2 + n - 17$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0296	ramp	$n + 14 \rightarrow$ $(1/8) \cdot n^2 + n - 18$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0294	ramp	$(9/8) \cdot n + 9 \rightarrow$ $(1/8) \cdot n^2 + (1/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0, i3=0)
$n^{1.5}$	0.067	ramp	$(1/4) \cdot n + 4 \rightarrow$ $(3/8) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
$n^{1.5}$	0.0518	ramp	$(1/8) \cdot n + 3 \rightarrow$ $(1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=0, i1=1); read A[i1, i2] (i0=0)
n^1	2.17	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	5	$2.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read B[i3, i1] (i0=0, i3=0)
n^1	0.866	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	2	$1 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i3=0); read A[i3, i2] (i0=0, i1=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.707	level	$(1/8) \cdot n^2 + n - 1$	2	$1 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (9/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i2=0, i3=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-5/2) \cdot n + 24$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0, i3=14)
n^1	0.433	level	$(3/16) \cdot n^2 + (-19/8) \cdot n + 23$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-9/4) \cdot n + 22$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-17/8) \cdot n + 21$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 - 2 \cdot n + 20$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-15/8) \cdot n + 19$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-7/4) \cdot n + 18$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-13/8) \cdot n + 17$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-1/2) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0, i3=6)
n^1	0.433	level	$(3/16) \cdot n^2 + (-3/8) \cdot n + 6$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-1/4) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-1/8) \cdot n + 4$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + 3$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (1/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^1	0.433	level	$(3/16) \cdot n^2 + (1/4) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0, i3=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-5/8) \cdot n + 9$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0, i3=7)
n^1	0.433	level	$(3/16) \cdot n^2 + (-1/2) \cdot n + 8$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-3/8) \cdot n + 7$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-1/4) \cdot n + 6$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (-1/8) \cdot n + 5$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + 4$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (1/8) \cdot n + 3$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (1/4) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i1=0, i2=0, i3=0)
n^1	0.433	level	$(3/16) \cdot n^2 + (3/8) \cdot n$	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i2=0, i3=0)
n^1	0.25	level	4	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read B[i3, i1] (i0=0, i3=0)
$n^{0.5}$	1.5	level	$(9/4) \cdot n - 3$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
$n^{0.5}$	1.46	level	$(17/8) \cdot n - 2$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i1=0, i2=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + (21/4)$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=0, i3=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=0, i2=0, i3=0); read A[i1, i2] (i0=0, i1=1, i2=0) (+1)
$n^{0.5}$	0.354	level	$(1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i1=1); read A[i1, i2] (i0=0)
n^0	2	level	4	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i2=0, i3=0)

Triangular multiply: read A[i3,i2] ramps from $(3/8)n + 5$ to $(1/8)n^2 + n$ lines (population $n^3/16$, $0.0148 \cdot n^4$). The wide $n^{3.5}$ band ($0.49 + 0.49$) is the B-panel and A-row reuse ramping to $2n$ lines — trmm reaches its matrix boundary with a much larger runner-up coefficient than syrk, which is why its n^4 term anchors earlier.

trmm — single-shot [exact]

Accesses $A(n) = 2 \cdot n^3$ (exact on $n \equiv 0 \pmod{8}$); DMD order n^4 , headroom **+1**; conservation $\Sigma_{\text{mass/warm}} = 1$ at $n=256$, 1 at $n=264$.

DMD spectrum: $0.0169 \cdot n^4 + 1.05 \cdot n^{3.5} + 1.68 \cdot n^3 + 1.31 \cdot n^{2.5} + 1.23 \cdot n^2 + 0.943 \cdot n^{1.5} + 1.77 \cdot n^1 + 2.58 \cdot n^{0.5} + 2 \cdot n^0$

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
n^4	0.0148	ramp	$(3/8) \cdot n + 5 \rightarrow$ $(1/8) \cdot n^2 +$ $n - 2$	$(7/128) \cdot n^3$ + (- $63/64) \cdot n^2$ + $(13/8) \cdot n +$ 2	0.0273	read A[i3, i2] (i0=0, i3=0); read A[i3, i2] (i0=0)
n^4	0.00207	ramp	$n + 17 \rightarrow$ $(1/8) \cdot n^2 +$ $(9/8) \cdot n - 18$	$(1/128) \cdot n^3$ + (- $13/64) \cdot n^2$ + $(11/8) \cdot n -$ 2	0.00391	read A[i3, i2] (i0=0, i3=0); read A[i3, i2] (i0=0)
$n^{3.5}$	0.494	ramp	$5 \rightarrow 2 \cdot n - 1$	$(7/16) \cdot n^3$ + $(-7/16) \cdot n^2$ + $(-7/8) \cdot n$	0.219	read B[i3, i1] (i0=0, i3=0); read B[i3, i1] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{3.5}$	0.491	ramp	$5 \rightarrow 2 \cdot n - 1$	$(7/16) \cdot n^3 + (-21/16) \cdot n^2 + (7/8) \cdot n$	0.219	read A[i3, i2] (i0=0, i3=0); read A[i3, i2] (i0=0)
$n^{3.5}$	0.0688	ramp	$6 \rightarrow 2 \cdot n$	$(1/16) \cdot n^3 + (-9/16) \cdot n^2 + (3/8) \cdot n + 1$	0.0312	read A[i3, i2] (i0=0, i2=0, i3=0); read A[i3, i2] (i0=0, i2=0) (+2)
n^3	0.866	level	3	$(1/2) \cdot n^3 + (-1/2) \cdot n^2 + (7/8) \cdot n$	0.25	read B[i3, i1] (i0=0, i3=0); write A[i1, i2] (i0=0)
n^3	0.5	level	1	$(1/2) \cdot n^3 + (1/2) \cdot n^2$	0.25	read A[i1, i2] (i0=0); write A[i1, i2] (i0=0)
n^3	0.125	ramp	$(1/4) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + (7/8) \cdot n - 1$	$(7/16) \cdot n^2 + (1/8) \cdot n - 3$	0.219/n	read A[i1, i2] (i0=0, i3=0); read A[i3, i2] (i0=0, i3=0) (+1)
n^3	0.125	ramp	$(1/2) \cdot n + 3 \rightarrow (1/8) \cdot n^2 + n - 1$	$(7/16) \cdot n^2 + (-7/8) \cdot n - 1$	0.219/n	read A[i3, i2] (i0=0, i2=0, i3=0); read A[i3, i2] (i0=0, i2=0)
n^3	0.0248	ramp	$(1/4) \cdot n + 4 \rightarrow (1/8) \cdot n^2 + n - 2$	$(7/64) \cdot n^2 + (-15/8) \cdot n + 2$	0.0547/n	read A[i1, i2] (i0=0, i3=0)
n^3	0.0172	ramp	$(9/8) \cdot n + 10 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 18$	$(1/16) \cdot n^2 + (-7/8) \cdot n + 3$	0.0312/n	read A[i3, i2] (i0=0, i2=0)
n^3	0.0172	ramp	$n + 9 \rightarrow (1/8) \cdot n^2 + n - 19$	$(1/16) \cdot n^2 + (-7/8) \cdot n + 3$	0.0312/n	read A[i3, i2] (i0=0)
n^3	0.00352	ramp	$n + 17 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 18$	$(1/64) \cdot n^2 + (-3/8) \cdot n + 2$	0.00781/n	read A[i1, i2] (i0=0, i3=0)
$n^{2.5}$	0.825	ramp	$5 \rightarrow 2 \cdot n - 1$	$(7/8) \cdot n^2 + (-7/4) \cdot n$	0.438/n	read A[i3, i2] (i0=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{2.5}$	0.49	ramp	$6 \rightarrow 2 \cdot n - 2$	$(7/16) \cdot n^2 + (-7/4) \cdot n + 1$	$0.219/n$	read B[i3, i1] (i0=0, i2=0, i3=0); read B[i3, i1] (i0=0, i2=0)
n^2	0.875	level	1	$(7/8) \cdot n^2$	$0.438/n$	read A[i1, i2] (i0=0, i3=0); read A[i1, i2] (i0=0)
n^2	0.239	ramp	$(3/8) \cdot n + 2 \rightarrow (1/8) \cdot n^2 + n - 1$	$n - 2$	$0.5 \cdot n^{-2}$	read A[i1, i2] (i0=0, i2=0, i3=0)
n^2	0.0297	ramp	$(9/8) \cdot n + 15 \rightarrow (1/8) \cdot n^2 + (9/8) \cdot n - 17$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0)
n^2	0.0296	ramp	$n + 15 \rightarrow (1/8) \cdot n^2 + n - 17$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=0, i3=0)
n^2	0.0296	ramp	$n + 14 \rightarrow (1/8) \cdot n^2 + n - 18$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^2	0.0294	ramp	$(9/8) \cdot n + 9 \rightarrow (1/8) \cdot n^2 + (1/8) \cdot n - 7$	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0, i3=0)
$n^{1.5}$	0.824	ramp	$6 \rightarrow 2 \cdot n - 2$	$(7/8) \cdot n - 2$	$0.438 \cdot n^{-2}$	read B[i3, i1] (i0=0, i2=0)
$n^{1.5}$	0.067	ramp	$(1/4) \cdot n + 4 \rightarrow (3/8) \cdot n + 1$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
$n^{1.5}$	0.0518	ramp	$(1/8) \cdot n + 3 \rightarrow (1/4) \cdot n$	$(1/8) \cdot n - 2$	$0.0625 \cdot n^{-2}$	read A[i1, i2] (i0=0)
n^1	1.52	level	3	$(7/8) \cdot n$	$0.438 \cdot n^{-2}$	read A[i3, i2] (i0=0, i3=0)
n^1	0.25	level	4	$(1/8) \cdot n - 1$	$0.0625 \cdot n^{-2}$	read A[i3, i2] (i0=0, i2=0, i3=0); read B[i3, i1] (i0=0, i3=0)
$n^{0.5}$	0.612	level	$(3/8) \cdot n + (21/4)$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=0, i3=0)

order	coeff	kind	distance (lines)	population (accesses)	portion	source access
$n^{0.5}$	0.612	level	$(3/8) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i2=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 2$	1	$0.5 \cdot n^{-3}$	read A[i3, i2] (i0=0, i3=0)
$n^{0.5}$	0.5	level	$(1/4) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0, i2=0)
$n^{0.5}$	0.354	level	$(1/8) \cdot n + 1$	1	$0.5 \cdot n^{-3}$	read A[i1, i2] (i0=0)
n^0	2	level	4	1	$0.5 \cdot n^{-3}$	read B[i3, i1] (i0=0, i2=0, i3=0)

Triangular multiply: read A[i3,i2] ramps from $(3/8)n + 5$ to $(1/8)n^2 + n$ lines (population $n^{3/16}$, $0.0148 \cdot n^4$). The wide $n^{3.5}$ band ($0.49 + 0.49$) is the B-panel and A-row reuse ramping to $2n$ lines — trmm reaches its matrix boundary with a much larger runner-up coefficient than syrk, which is why its n^4 term anchors earlier.